MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project **Mansourah Massarah Gold Project**
Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **CALCULATION**
3380 ACIDULATION
DESIGN OF BUILDING FOUNDATION

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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	'EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities

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1. INTENT

This document is to prepare design of foundation for **Acidulation structure**.The analysis & design is carried out in accordance with project specifications and codes & standards. The design calculation ensures the safety and serviceability of the foundation system.

2.EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

The analysis has been carried out in STAAD PRO. For design of raft force values are directly taken from STAAD & designed in excel sheet. Column is designed in STAAD.

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4. DESIGN REQUIREMENTS

4.1 Design Load

All Design Loads are indicated in Attachment - 1. Also referred Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.2.

5. CONCRETE FOUNDATIONS

5.1 Design

Design of concrete structures and foundations shall be in accordance with SBC-304, ACI 318M.

5.2 Materials

Concrete Strength of 28 days

- Foundation and Structural Members: 35 MPa
- Grade slab and pavement: 30 MPa
- Blinding Concrete below foundation bases: 15 Mpa

Reinforcement Grade

- Deformed Bar : ASTM A 615 / A 615M Grade 60 (fy = 420MPa)
- Welded Wire Fabric: ASTM A 185(Plain) / A 497(Deformed)

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5.3 Minimum Concrete Cover

Concrete Exposure	Member	Minimum Cover(mm)
Exposed to weather or contact with ground	Foundation	50
Exposed to weather or contact with ground	Column	50

5.4 Allowable Soil Bearing Capacity

The net safe bearing capacity of soil for foundation design has been considered 350 Kn/m^2 at 1.25 m below EGL. (Refer Table 5 - BH-9 of Geotechnical Report : MD-216-3000-EG-CV-RPT-1008).

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6. DESIGN CALCULATION FOR FOUNDATION

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DESIGN CALCULATION OF FOOTING F1

Size of Columns at node	Along-X	Along-Z
2	0.40m	0.60m
522	0.40m	0.60m
Size of fdn. raft:	3.25m	4.90m
Thickness of raft:	0.50	m
Original Gr. Level:	0.00	m
Finished Gr. Level:	0.00	m
U/S of B/Pl Above GL:	0.70	m
U/S level of fdn:	-1.25	m
Unit weight of conc:	24	kN/m^3
Unit weight of soil:	20	kN/m^3

Design Input:

Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	50mm

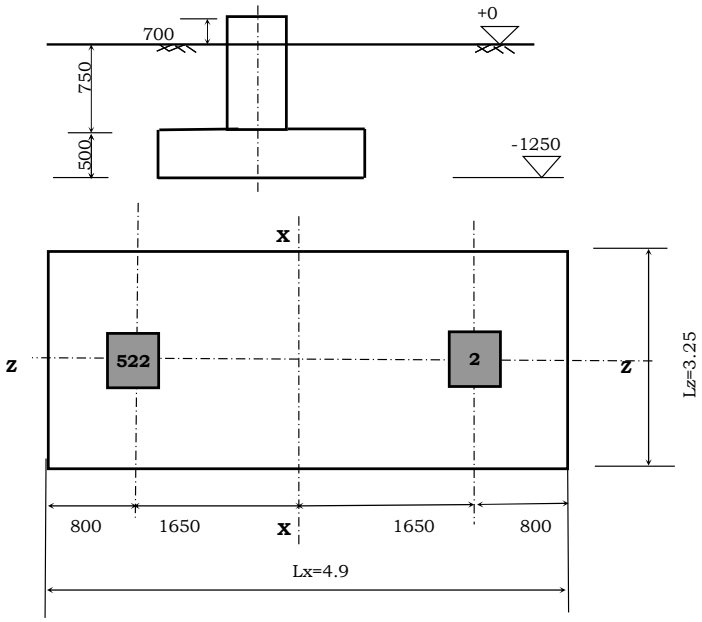
Calculation of Weight of Foundation :

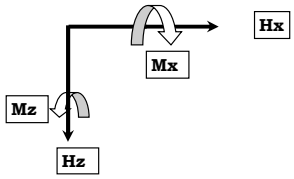
Weight of raft = (3.25*4.9*0.5*24) = 191.10KN

Weight of soil = ((3.25*4.9)-2*(0.4*0.6))*0.75*20 = 331.68KN

Total = 522.78KN

323.72KN





H_x=Force along X-X direction

Hz=Force along Z-Z direction

M_x=Moment about X-X direction

Mz=Moment about Z-Z direction

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Check for overturning and sliding of footing F1

Load at bottom of fooring	Node 2					Node 522				
L/C No.	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
201 1.0DL	163.70	0.00	0.00	-0.01	-0.02	127.16	-0.01	0.01	-1.38	-2.65
202 1.0DL + 1.0LL	336.93	-0.02	0.04	-2.33	-4.55	211.55	-0.02	0.04	-3.25	-6.27
203 1.0DL + 1.0WLX1	159.22	-2.34	4.57	-0.86	-1.69	130.95	-2.37	4.62	-2.14	-4.13
204 1.0DL + 1.0WLX2	168.18	-2.34	4.57	-0.86	-1.69	123.37	-2.37	4.62	-2.14	-4.14
205 1.0DL + 1.0WLZ1	-38.87	0.00	0.01	-51.37	-100.17	329.73	-0.06	0.11	-50.75	-98.84
206 1.0DL + 1.0WLZ2	366.10	0.00	0.01	-51.34	-100.10	-75.24	-0.06	0.11	-50.75	-98.99
207 1.0DL + 0.75WLX1 + 0.75LL	290.26	-1.77	3.45	-2.39	-4.67	193.29	-1.79	3.48	-3.36	-6.47
208 1.0DL + 0.75WLX2 + 0.75LL	296.98	-1.77	3.45	-2.39	-4.67	187.61	-1.79	3.48	-3.36	-6.48
209 1.0DL + 0.75WLZ1 + 0.75LL	141.69	-0.02	0.03	-40.27	-78.53	342.38	-0.05	0.11	-39.81	-77.50
210 1.0DL + 0.75WLZ2 + 0.75LL	445.42	-0.02	0.03	-40.25	-78.48	38.65	-0.05	0.11	-39.81	-77.62
211 0.6DL + 1.0WLX1	93.74	-2.34	4.56	-0.86	-1.68	80.08	-2.36	4.61	-1.59	-3.07
212 0.6DL + 1.0WLX2	102.70	-2.34	4.56	-0.86	-1.68	72.51	-2.36	4.61	-1.59	-3.08
213 0.6DL + 1.0WLZ1	-104.35	0.00	0.00	-51.37	-100.16	278.87	-0.05	0.11	-50.20	-97.78
214 0.6DL + 1.0WLZ2	300.62	0.00	0.00	-51.33	-100.09	-126.10	-0.05	0.11	-50.20	-97.93

Check for overturning and sliding:

	Resultant Load at bottom of footing								
L/C No.	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	Overturning		Sliding	
						F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
201 1.0DL	813.6	-0.01	-60.3	-1.4	-2.7	21.9	745.5	36161.7	233.3
202 1.0DL + 1.0LL	1071.3	-0.04	-206.8	-5.6	-10.8	8.4	242.6	10987.2	76.7
203 1.0DL + 1.0WLX1	812.9	-4.71	-37.5	-3.0	-5.8	35.3	342.3	69.1	108.1
204 1.0DL + 1.0WLX2	814.3	-4.71	-64.8	-3.0	-5.8	20.4	342.7	69.2	108.3
205 1.0DL + 1.0WLZ1	813.6	-0.06	608.3	-102.1	-199.0	2.2	10.0	5424.3	3.2
206 1.0DL + 1.0WLZ2	813.6	-0.06	-728.1	-102.1	-199.1	1.8	10.0	5424.3	3.2
207 1.0DL + 0.75WLX1 + 0.75LL	1006.3	-3.56	-153.1	-5.8	-11.1	10.7	221.3	113.2	70.0
208 1.0DL + 0.75WLX2 + 0.75LL	1007.4	-3.56	-173.5	-5.8	-11.1	9.4	221.5	113.3	70.1
209 1.0DL + 0.75WLZ1 + 0.75LL	1006.8	-0.07	331.3	-80.1	-156.0	4.9	15.8	5753.4	5.0
210 1.0DL + 0.75WLZ2 + 0.75LL	1006.8	-0.07	-671.0	-80.1	-156.1	2.4	15.8	5753.4	5.0
211 0.6DL + 1.0WLX1	696.6	-4.70	-13.4	-2.5	-4.7	84.8	359.5	59.2	113.7
212 0.6DL + 1.0WLX2	698.0	-4.70	-40.7	-2.5	-4.8	27.9	359.9	59.4	114.0
213 0.6DL + 1.0WLZ1	697.3	-0.06	632.4	-101.6	-197.9	1.8	8.6	4980.7	2.7
214 0.6DL + 1.0WLZ2	697.3	-0.06	-704.0	-101.5	-198.0	1.6	8.6	4980.7	2.7

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Base pressure check of footing F1

	Node	L/C	Horizontal Fx kN/m2	Vertical Fy kN/m2	Horizontal Fz kN/m2
Max Px	2	201	0	73.61	0
Min Px	2	201	0	73.61	0
Max Py	335	210	0	155.46	0
Min Py	494	206	0	0.00	0
Max Pz	2	201	0	73.61	0
Min Pz	2	201	0	73.61	0

Gross Maximum Base pressure, p_{max} = 155.5 KN/m^2

Net Allowable Safe Bearing Capacity = 350 KN/m^2
Depth of soil above footing = 0.75 m
Unit weight of soil = 20 KN/m^3

Gross Safe Bearing Capacity = 365 KN/m^2
> 155.5 KN/m^2
Hence OK

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Design of foundation raft F1

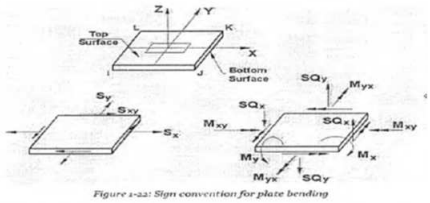
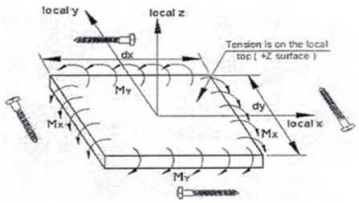


Figure 1-22: Sign convention for plate bending



Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:
PLATE CENTER STRESS : FDN_F1_MOMENT At the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	333	0.575	0.125	0	0	0	-79.04	-37.147	6.404	-85.441	-43.551	106:1.2DL + 1.6WLZ2 + 1.0 LL
Min Qx	327	-0.575	0.125	0	0	0	-79.01	-37.144	0	-79.013	-37.144	106:1.2DL + 1.6WLZ2 + 1.0 LL
Max Qy	416	0.065	0.381	0	0	0	-79.27	-2.453	9.434	-88.707	-11.887	102:1.4DL + 1.7LL
Min Qy	346	0.212	-0.611	0	0	0	-134.2	-58.565	0.871	-135.11	-59.436	106:1.2DL + 1.6WLZ2 + 1.0 LL
Max Sx	668	-0.009	-0.01	0	0	0	-0.461	3.181	1.052	-1.513	4.233	101:1.4DL
Min Sx	668	-0.009	-0.01	0	0	0	-0.461	3.181	1.052	-1.513	4.233	101:1.4DL
Max Sy	668	-0.009	-0.01	0	0	0	-0.461	3.181	1.052	-1.513	4.233	101:1.4DL
Min Sy	668	-0.009	-0.01	0	0	0	-0.461	3.181	1.052	-1.513	4.233	101:1.4DL
Max Sxy	668	-0.009	-0.01	0	0	0	-0.461	3.181	1.052	-1.513	4.233	101:1.4DL
Min Sxy	668	-0.009	-0.01	0	0	0	-0.461	3.181	1.052	-1.513	4.233	101:1.4DL
Max Mx	507	0.134	-0.308	0	0	0	51.952	36.92	-6.226	58.178	43.146	110:0.9DL + 1.6WLZ2
Min Mx	345	-0.212	-0.611	0	0	0	-134.2	-58.563	-0.861	-135.11	-59.424	106:1.2DL + 1.6WLZ2 + 1.0 LL
Max My	416	-0.125	-0.347	0	0	0	30.515	84.601	3.989	34.504	88.59	109:0.9DL + 1.6WLZ1
Min My	468	0.074	-0.467	0	0	0	-67.1	-70.043	-3.855	-70.95	-73.898	109:0.9DL + 1.6WLZ1
Max Mxy	417	0.099	0.215	0	0	0	-49.61	26.511	38.78	-88.389	65.291	106:1.2DL + 1.6WLZ2 + 1.0 LL
Min Mxy	414	-0.099	0.215	0	0	0	-49.6	26.513	-38.773	-88.37	65.286	106:1.2DL + 1.6WLZ2 + 1.0 LL

PLATE CENTER STRESS : FDN_F1_SHEAR At a distance "d" from the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	334	0.338	0.058	0	0	0	-39.35	-22.936	14.878	-54.223	-37.814	106:1.2DL + 1.6WLZ2 + 1.0 LL
Min Qx	326	-0.338	0.058	0	0	0	-39.34	-22.934	0	-39.335	-22.934	106:1.2DL + 1.6WLZ2 + 1.0 LL
Max Qy	406	0.044	0.231	0	0	0	-56.71	20.892	5.665	-62.375	26.557	102:1.4DL + 1.7LL
Min Qy	476	0.07	-0.345	0	0	0	-64.85	-23.443	-5.921	-70.768	-29.364	105:1.2DL + 1.6WLZ1 + 1.0 LL
Max Sx	668	-0.009	-0.01	0	0	0	-0.461	3.181	1.052	-1.513	4.233	101:1.4DL
Min Sx	668	-0.009	-0.01	0	0	0	-0.461	3.181	1.052	-1.513	4.233	101:1.4DL
Max Sy	668	-0.009	-0.01	0	0	0	-0.461	3.181	1.052	-1.513	4.233	101:1.4DL
Min Sy	668	-0.009	-0.01	0	0	0	-0.461	3.181	1.052	-1.513	4.233	101:1.4DL
Max Sxy	668	-0.009	-0.01	0	0	0	-0.461	3.181	1.052	-1.513	4.233	101:1.4DL
Min Sxy	668	-0.009	-0.01	0	0	0	-0.461	3.181	1.052	-1.513	4.233	101:1.4DL
Max Mx	499	0.072	-0.094	0	0	0	50.074	5.787	-0.92	50.994	6.707	110:0.9DL + 1.6WLZ2
Min Mx	356	0.123	-0.186	0	0	0	-119.8	-9.108	-1.003	-120.76	-10.111	106:1.2DL + 1.6WLZ2 + 1.0 LL
Max My	406	-0.029	-0.096	0	0	0	-0.248	61.395	0.214	-0.462	61.609	105:1.2DL + 1.6WLZ1 + 1.0 LL
Min My	476	0.012	-0.012	0	0	0	1.37	-53.946	2.858	4.228	-56.804	110:0.9DL + 1.6WLZ2
Max Mxy	418	0.147	0.164	0	0	0	-31.66	13.75	29.448	-61.105	43.198	106:1.2DL + 1.6WLZ2 + 1.0 LL
Min Mxy	413	-0.146	0.164	0	0	0	-31.65	13.747	-29.437	-61.087	43.184	106:1.2DL + 1.6WLZ2 + 1.0 LL

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	50	mm
	top face	50	mm
Thicknes of slab =		500	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

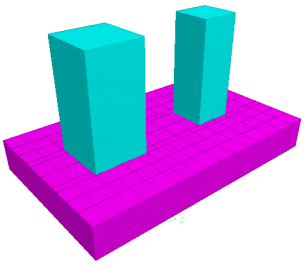
Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 / φ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2)	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
110:0.9D	58.18	top face	50	444	58.178	0.39	0.0009	418.8	532.8	12	212.36	175	646.6	9.12	11.4	0.114	tension control
106:1.2D	-135.11	bottom face	50	442	135.11	0.92	0.0022	985.9		16	204.02	175	1149.4	16.2	20.26	0.062	tension control
106:1.2D	-88.39	bottom face	50	442	88.389	0.60	0.0015	641.4		16	313.59	175	1149.4	16.2	20.26	0.062	tension control
106:1.2D	-88.37	bottom face	50	442	88.37	0.60	0.0015	641.3		16	313.66	175	1149.4	16.2	20.26	0.062	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$	Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN		KN		
Load condition	Max Qx	0.338	1	169	54.223	0.001	0.444	435.49	431.271	761.754	431.27	1624 > 169	ok	323.5	φVc>Vu	not required
	Min Qx	0.338	1	169	39.335			441.247			431.27	1624 > 169	ok	323.5	φVc>Vu	not required

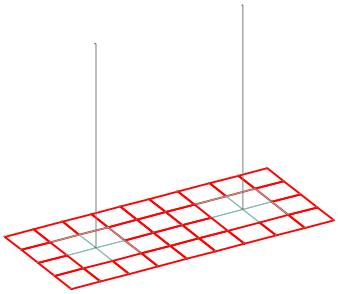
Reinforcement Along Z:

Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, My	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 / φ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2)	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
109:0.9D	88.59	top face	50	444	88.59	0.60	0.0014	639.9	532.8	12	176.81	175	646.6	9.12	11.4	0.114	tension control
109:0.9D	-73.90	bottom face	50	442	73.898	0.50	0.0012	535.3		16	375.73	175	1149.4	16.2	20.26	0.062	tension control
106:1.2D	65.29	top face	50	442	65.291	0.45	0.0011	472.5		16	377.52	175	1149.4	16.2	20.26	0.062	tension control
106:1.2D	65.29	top face	50	442	65.286	0.45	0.0011	472.5		16	377.52	175	1149.4	16.2	20.26	0.062	tension control

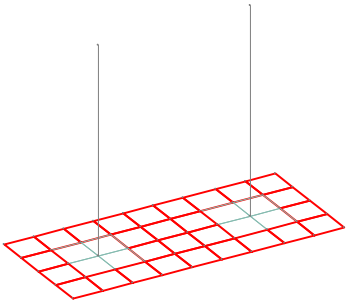
Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$	Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	T-m/m		m^2	KN	KN	KN	KN	KN		KN		
Load condition	102:1.4D	0.231	1	115.5	26.557	0.001	0.444	441.504	431.271	fc'761.8	431.27	453 > 115.5	ok	323.5	φVc>Vu	not required
	105:1.2D	0.345	1	172.5	29.364			448.949			431.27	453 > 172.5	ok	323.5	φVc>Vu	not required



3D rendered view



Highlighted plate for Moment at face of column



Highlighted plate for Shear at 'd' distance from face of column

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :			
At Bottom	**16ϕT**	**175**	c/c
At Top	**12ϕT**	**175**	c/c
Along Z Direction :			
At Bottom	**16ϕT**	**175**	c/c
At Top	**12ϕT**	**175**	c/c

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DESIGN CALCULATION OF FOOTING F2

Size of Columns at node	Along-Z	Along-X
2	0.60m	0.40m

Size of fdn. raft:	2.50m	1.50m
Thickness of raft:	0.50	m
Original Gr. Level:	0.00	m
Finished Gr. Level:	0.00	m
U/S of B/Pl Above GL:	0.70	m
U/S level of fdn:	-1.25	m
Unit weight of conc:	24	kN/m^3
Unit weight of soil:	20	kN/m^3

Calculation of Weight of Foundation :

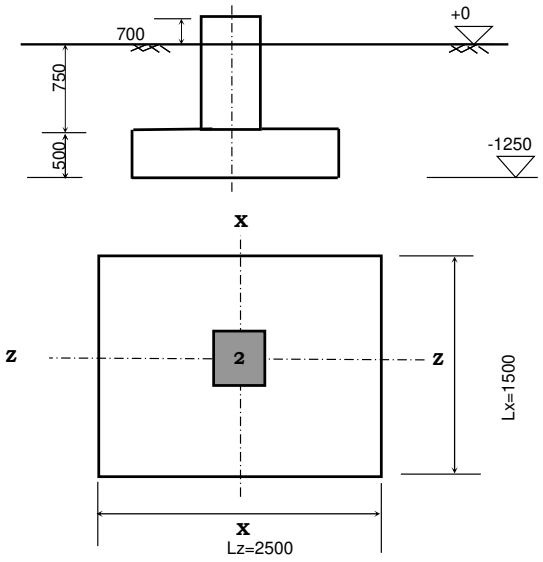
Weight of raft = (2.5*1.5*0.5*24) = 45.00KN

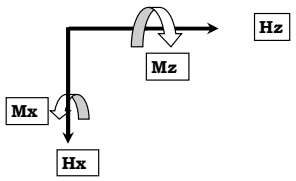
Weight of soil = ((2.5*1.5)-(0.6*0.4))*0.75*20 = 52.65KN

Total = 97.65KN

Design Input:

Characteristic strength of concrete, f'_c=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	50mm





H_x=Force along X-X direction

Hz=Force along Z-Z direction

M_x=Moment about X-X direction

Mz=Moment about Z-Z direction

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Check for overturning and sliding of footing F2

Load at bottom of fooring	Node 2				
L/C No.	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
201 1.0DL	88.10	0.00	0.00	-0.28	-0.55
202 1.0DL + 1.0LL	145.13	0.01	-0.01	0.98	1.90
203 1.0DL + 1.0WLX1	88.97	-1.33	2.60	-0.39	-0.77
204 1.0DL + 1.0WLX2	87.24	-1.33	2.60	-0.39	-0.77
205 1.0DL + 1.0WLZ1	78.36	-0.02	0.05	-3.51	-6.84
206 1.0DL + 1.0WLZ2	97.85	-0.02	0.05	-3.51	-6.84
207 1.0DL + 0.75WLX1 + 0.75LL	131.52	-1.00	1.94	0.58	1.13
208 1.0DL + 0.75WLX2 + 0.75LL	130.22	-1.00	1.94	0.58	1.13
209 1.0DL + 0.75WLZ1 + 0.75LL	123.56	-0.01	0.03	-1.76	-3.43
210 1.0DL + 0.75WLZ2 + 0.75LL	138.18	-0.01	0.03	-1.76	-3.43
211 0.6DL + 1.0WLX1	53.73	-1.33	2.60	-0.28	-0.55
212 0.6DL + 1.0WLX2	52.00	-1.33	2.60	-0.28	-0.55
213 0.6DL + 1.0WLZ1	43.11	-0.02	0.05	-3.39	-6.62
214 0.6DL + 1.0WLZ2	62.61	-0.02	0.05	-3.39	-6.62

Check for overturning and sliding:

L/C No.	Resultant Load at bottom of footing								
	V (kN)	Hz (kN)	Mx (kN-m)	Hx (kN)	Mz (kN-m)	Overturning		Sliding	
						F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
201 1.0DL	185.8	-0.28	-0.6	0.0	0.0	69657.8	422.2	74301.6	263.5
202 1.0DL + 1.0LL	242.8	0.98	1.9	0.0	0.0	15173.6	159.6	16185.1	99.6
203 1.0DL + 1.0WLX1	186.6	-0.39	-0.8	-1.3	2.6	53.9	304.5	56.0	189.9
204 1.0DL + 1.0WLX2	184.9	-0.39	-0.8	-1.3	2.6	53.4	301.7	55.5	188.2
205 1.0DL + 1.0WLZ1	176.0	-3.51	-6.8	0.0	0.0	2808.6	32.2	2933.4	20.1
206 1.0DL + 1.0WLZ2	195.5	-3.51	-6.8	0.0	0.0	3119.6	35.7	3258.3	22.3
207 1.0DL + 0.75WLX1 + 0.75LL	229.2	0.58	1.1	-1.0	1.9	88.5	254.4	92.0	158.6
208 1.0DL + 0.75WLX2 + 0.75LL	227.9	0.58	1.1	-1.0	1.9	88.0	253.0	91.5	157.7
209 1.0DL + 0.75WLZ1 + 0.75LL	221.2	-1.76	-3.4	0.0	0.0	6144.7	80.7	6320.3	50.3
210 1.0DL + 0.75WLZ2 + 0.75LL	235.8	-1.76	-3.4	0.0	0.0	6550.8	86.0	6738.0	53.7
211 0.6DL + 1.0WLX1	151.4	-0.28	-0.5	-1.3	2.6	43.7	346.6	45.4	216.3
212 0.6DL + 1.0WLX2	149.6	-0.28	-0.5	-1.3	2.6	43.2	342.6	44.9	213.8
213 0.6DL + 1.0WLZ1	140.8	-3.39	-6.6	0.0	0.0	2199.4	26.6	2346.1	16.6
214 0.6DL + 1.0WLZ2	160.3	-3.39	-6.6	0.0	0.0	2504.0	30.3	2670.9	18.9

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Base pressure check of footing F2

	Node	L/C	Horizontal Fx kN/m2	Vertical Fy kN/m2	Horizontal Fz kN/m2
Max Px	2	201	0	72.36	0
Min Px	2	201	0	72.36	0
Max Py	2	202	0	88.45	0
Min Py	368	213	0	35.27	0
Max Pz	2	201	0	72.36	0
Min Pz	2	201	0	72.36	0

Gross Maximum Base pressure, p_{max} = 88.4 KN/m^2

Net Allowable Safe Bearing Capacity = 350 KN/m^2
Depth of soil above footing = 0.75 m
Unit weight of soil = 20 KN/m^3

Gross Safe Bearing Capacity = 365 KN/m^2
> 88.4 KN/m^2
Hence OK

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Design of foundation raft F2

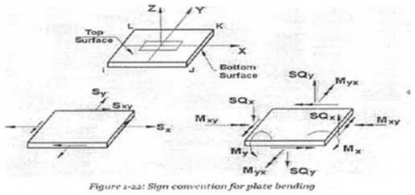
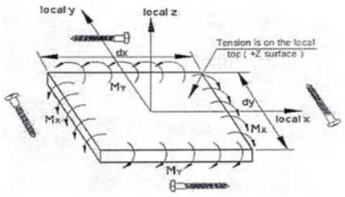


Figure 1-22: Sign convention for plate bending



Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:
PLATE CENTER STRESS : FDN_F2_MOMENT At the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm²)	(N/mm²)	(N/mm²)	(N/mm²)	(N/mm²)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	427	0.121	0.03	0	0	0	-6.218	-35.694	2.814	-9.032	-38.508	102:1.4DL + 1.7LL
Min Qx	424	-0.121	0.03	0	0	0	-6.227	-35.698	-2.812	-9.039	-38.51	102:1.4DL + 1.7LL
Max Qy	415	-0.022	0.163	0	0	0	-16.4	-22.909	-2.808	-19.205	-25.717	102:1.4DL + 1.7LL
Min Qy	346	0.023	-0.154	0	0	0	-14.92	-22.088	-2.278	-17.2	-24.366	106:1.4DL + 1.6WLZ2 + 1.0 LL
Max Sx	428	0.018	0.013	0	0	0	-0.352	-16.919	0.587	-0.939	-17.506	101:1.4DL
Min Sx	428	0.018	0.013	0	0	0	-0.352	-16.919	0.587	-0.939	-17.506	101:1.4DL
Max Sy	428	0.018	0.013	0	0	0	-0.352	-16.919	0.587	-0.939	-17.506	101:1.4DL
Min Sy	428	0.018	0.013	0	0	0	-0.352	-16.919	0.587	-0.939	-17.506	101:1.4DL
Max Sxy	428	0.018	0.013	0	0	0	-0.352	-16.919	0.587	-0.939	-17.506	101:1.4DL
Min Sxy	428	0.018	0.013	0	0	0	-0.352	-16.919	0.587	-0.939	-17.506	101:1.4DL
Max Mx	423	-0.009	0.007	0	0	0	-0.059	-10.251	-0.424	-0.483	-10.675	108:0.9DL + 1.6WLX2
Min Mx	416	0.022	0.163	0	0	0	-16.4	-22.908	2.81	-19.208	-25.718	102:1.4DL + 1.7LL
Max My	386	0	0.003	0	0	0	-0.791	-0.099	0.064	-0.855	-0.163	109:0.9DL + 1.6WLZ1
Min My	327	-0.119	-0.036	0	0	0	-6.035	-35.803	3.03	-9.065	-38.833	102:1.4DL + 1.7LL
Max Mxy	344	-0.061	-0.098	0	0	0	-5.867	-21.864	5.612	-11.479	-27.476	102:1.4DL + 1.7LL
Min Mxy	347	0.061	-0.098	0	0	0	-5.863	-21.861	-5.615	-11.478	-27.476	102:1.4DL + 1.7LL

PLATE CENTER STRESS : FDN_F2_SHEAR At a distance "d" from the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm²)	(N/mm²)	(N/mm²)	(N/mm²)	(N/mm²)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	428	0.032	0.021	0	0	0	-0.633	-30.478	0.956	-1.589	-31.434	102:1.4DL + 1.7LL
Min Qx	423	-0.032	0.021	0	0	0	-0.635	-30.481	-0.955	-1.59	-31.436	102:1.4DL + 1.7LL
Max Qy	405	-0.013	0.095	0	0	0	-8.873	-11.011	-1.419	-10.292	-12.43	102:1.4DL + 1.7LL
Min Qy	356	0.012	-0.09	0	0	0	-8.208	-10.521	-1.264	-9.472	-11.785	106:1.4DL + 1.6WLZ2 + 1.0 LL
Max Sx	428	0.018	0.013	0	0	0	-0.352	-16.919	0.587	-0.939	-17.506	101:1.4DL
Min Sx	428	0.018	0.013	0	0	0	-0.352	-16.919	0.587	-0.939	-17.506	101:1.4DL
Max Sy	428	0.018	0.013	0	0	0	-0.352	-16.919	0.587	-0.939	-17.506	101:1.4DL
Min Sy	428	0.018	0.013	0	0	0	-0.352	-16.919	0.587	-0.939	-17.506	101:1.4DL
Max Sxy	428	0.018	0.013	0	0	0	-0.352	-16.919	0.587	-0.939	-17.506	101:1.4DL
Min Sxy	428	0.018	0.013	0	0	0	-0.352	-16.919	0.587	-0.939	-17.506	101:1.4DL
Max Mx	423	-0.009	0.007	0	0	0	-0.059	-10.251	-0.424	-0.483	-10.675	108:0.9DL + 1.6WLX2
Min Mx	405	-0.013	0.095	0	0	0	-8.873	-11.011	-1.419	-10.292	-12.43	102:1.4DL + 1.7LL
Max My	386	0	0.003	0	0	0	-0.791	-0.099	0.064	-0.855	-0.163	109:0.9DL + 1.6WLZ1
Min My	423	-0.032	0.021	0	0	0	-0.635	-30.481	-0.955	-1.59	-31.436	102:1.4DL + 1.7LL
Max Mxy	354	-0.019	-0.073	0	0	0	-5.111	-11.218	3.164	-8.275	-14.382	102:1.4DL + 1.7LL
Min Mxy	357	0.019	-0.073	0	0	0	-5.11	-11.217	-3.167	-8.277	-14.384	102:1.4DL + 1.7LL

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	50	mm
	top face	50	mm
Thicknes of slab =		500	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

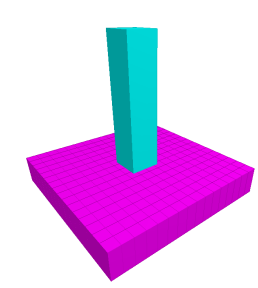
Force Values			Depth		Reinforcement Calculation													
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
108:0.9D	-0.48	bottom face	50	442	0.483	0.00	0.0000	3.5	795.6	16	252.82	200	1005.8	14.18	17.73	0.072	tension control
102:1.4D	-19.21	bottom face	50	442	19.208	0.13	0.0003	138.3		16	252.82	200	1005.8	14.18	17.73	0.072	tension control
102:1.4D	-11.48	bottom face	50	442	11.479	0.08	0.0002	82.6		16	252.82	200	1005.8	14.18	17.73	0.072	tension control
102:1.4D	-11.48	bottom face	50	442	11.478	0.08	0.0002	82.5		16	252.82	200	1005.8	14.18	17.73	0.072	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Rema rks	φV_c	Remar ks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qx	0.032	1	16	1.589	0.002	0.442	494.484	435.484	758.323	435.48	1621	> 16	ok	326.6	φVc>Vu	not required
	Min Qx	0.032	1	16	1.59			494.436			435.48	1621	> 16	ok	326.6	φVc>Vu	not required

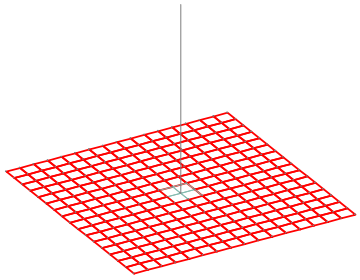
Reinforcement Along Z:

Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, My	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 ϕ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,\ provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
109:0.9D	-0.16	bottom face	50	442	0.163	0.00	0.0000	1.2	795.6	16	252.82	200	1005.8	14.18	17.73	0.072	tension control
102:1.4D	-38.83	bottom face	50	442	38.833	0.27	0.0006	280.2		16	252.82	200	1005.8	14.18	17.73	0.072	tension control
102:1.4D	-27.48	bottom face	50	442	27.476	0.19	0.0004	198.0		16	252.82	200	1005.8	14.18	17.73	0.072	tension control
102:1.4D	-27.48	bottom face	50	442	27.476	0.19	0.0004	198.0		16	252.82	200	1005.8	14.18	17.73	0.072	tension control

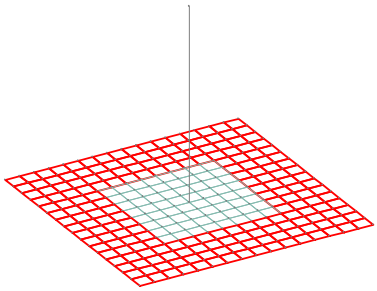
Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Rema rks	φV_c	Remar ks	Remarks :Min shear reinf
		Mpa		KN	T-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qy	0.095	1	47.5	12.43	0.002	0.442	447.266	435.484	fc'758.3	435.48	456	> 47.5	ok	326.6	φVc>Vu	not required
	Min Qy	0.09	1	45	11.785			447.243			435.48	456	> 45	ok	326.6	φVc>Vu	not required



3D rendered view



Highlighted plate for Moment at face of column



Highlighted plate for Shear at 'd' distance from face of column

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :
At Bottom **16ϕT** **200** c/c
At Top **10ϕT** **200** c/c
Along Z Direction :
At Bottom **16ϕT** **200** c/c
At Top **10ϕT** **200** c/c

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DESIGN CALCULATION OF FOOTING F3

Size of Columns at node	Along-X	Along-Z
765	0.40m	0.60m
522	0.40m	0.60m
Size of fdn. raft:	5.00m	2.50m
Thickness of raft:	0.50	m
Original Gr. Level:	0.00	m
Finished Gr. Level:	0.00	m
U/S of B/Pl Above GL:	0.70	m
U/S level of fdn:	-1.25	m
Unit weight of conc:	24	kN/m^3
Unit weight of soil:	20	kN/m^3

Design Input:

Characteristic strength of concrete, f'_c=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	50mm

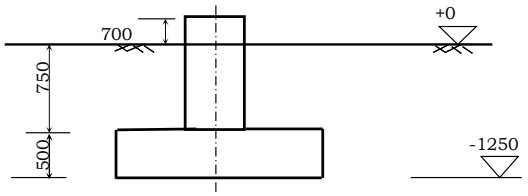
Calculation of Weight of Foundation :

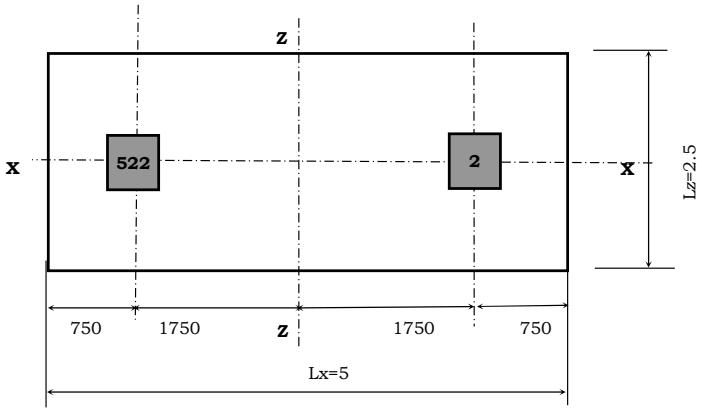
Weight of raft = (5*2.5*0.5*24) = 150.00KN

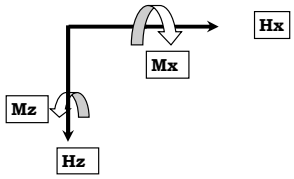
Weight of soil = ((5*2.5)-2*(0.4*0.6))*0.75*20 = 180.30KN

Total = 330.30KN

174.05KN







H_x=Force along X-X direction

Hz=Force along Z-Z direction

M_x=Moment about X-X direction

Mz=Moment about Z-Z direction

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Check for overturning and sliding of footing F3

Load at bottom of fooring	Node 522					Node 765				
L/C No.	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
201 1.0DL	47.59	-3.10	6.05	-0.03	-0.04	38.53	-3.14	6.12	-0.07	-0.12
202 1.0DL + 1.0LL	122.29	-11.80	23.01	-0.08	-0.11	87.36	-11.88	23.17	-0.25	-0.45
203 1.0DL + 1.0WLX1	-34.05	-25.80	50.31	-0.03	-0.07	120.80	-25.87	50.44	-0.07	-0.10
204 1.0DL + 1.0WLX2	129.22	-25.82	50.35	-0.03	-0.01	-43.74	-25.87	50.44	-0.07	-0.16
205 1.0DL + 1.0WLZ1	73.93	-5.53	10.79	-0.50	-0.96	23.79	-5.48	10.68	-0.69	-1.33
206 1.0DL + 1.0WLZ2	21.29	-5.53	10.79	-0.50	-0.98	53.25	-5.47	10.67	-0.69	-1.32
207 1.0DL + 0.75WLX1 + 0.75LL	42.39	-26.65	51.97	-0.07	-0.12	136.86	-26.74	52.14	-0.20	-0.35
208 1.0DL + 0.75WLX2 + 0.75LL	164.84	-26.67	52.00	-0.07	-0.07	13.45	-26.74	52.14	-0.20	-0.40
209 1.0DL + 0.75WLZ1 + 0.75LL	123.37	-11.45	22.33	-0.42	-0.78	64.10	-11.45	22.32	-0.66	-1.27
210 1.0DL + 0.75WLZ2 + 0.75LL	83.89	-11.45	22.33	-0.42	-0.79	86.19	-11.45	22.32	-0.67	-1.27
211 0.6DL + 1.0WLX1	-53.08	-24.56	47.89	-0.02	-0.06	105.39	-24.61	47.99	-0.05	-0.05
212 0.6DL + 1.0WLX2	110.19	-24.58	47.93	-0.02	0.01	-59.15	-24.61	47.99	-0.05	-0.11
213 0.6DL + 1.0WLZ1	54.89	-4.29	8.37	-0.49	-0.94	8.38	-4.22	8.23	-0.66	-1.28
214 0.6DL + 1.0WLZ2	2.25	-4.29	8.37	-0.49	-0.96	37.83	-4.22	8.22	-0.66	-1.27

Check for overturning and sliding:

L/C No.	Resultant Load at bottom of footing					Overturning		Sliding	
	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
201 1.0DL	416.4	-6.24	-3.7	-0.1	-0.2	282.6	3116.9	26.7	1665.7
202 1.0DL + 1.0LL	539.9	-23.68	-14.9	-0.3	-0.6	90.3	1203.1	9.1	662.5
203 1.0DL + 1.0WLX1	417.1	-51.67	371.7	-0.1	-0.2	2.8	3013.4	3.2	1619.6
204 1.0DL + 1.0WLX2	415.8	-51.69	-201.9	-0.1	-0.2	5.1	3004.2	3.2	1614.7
205 1.0DL + 1.0WLZ1	428.0	-11.01	-66.3	-1.2	-2.3	16.1	234.0	15.6	144.0
206 1.0DL + 1.0WLZ2	404.8	-11.01	77.4	-1.2	-2.3	13.1	220.1	14.7	136.0
207 1.0DL + 0.75WLX1 + 0.75LL	509.5	-53.39	269.4	-0.3	-0.5	4.7	1366.8	3.8	749.3
208 1.0DL + 0.75WLX2 + 0.75LL	508.6	-53.41	-160.8	-0.3	-0.5	7.9	1364.2	3.8	747.9
209 1.0DL + 0.75WLZ1 + 0.75LL	517.8	-22.90	-59.1	-1.1	-2.1	21.9	315.4	9.0	190.7
210 1.0DL + 0.75WLZ2 + 0.75LL	500.4	-22.90	48.7	-1.1	-2.1	25.7	303.5	8.7	184.1
211 0.6DL + 1.0WLX1	382.6	-49.17	373.2	-0.1	-0.1	2.6	4511.9	3.1	2429.3
212 0.6DL + 1.0WLX2	381.3	-49.19	-200.4	-0.1	-0.1	4.8	4454.8	3.1	2421.2
213 0.6DL + 1.0WLZ1	393.6	-8.51	-64.8	-1.1	-2.2	15.2	221.7	18.5	137.0
214 0.6DL + 1.0WLZ2	370.4	-8.51	78.9	-1.2	-2.2	11.7	207.4	17.4	128.7

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Base pressure check of footing F3

	Node	L/C	Horizontal Fx kN/m2	Vertical Fy kN/m2	Horizontal Fz kN/m2
Max Px	422	201	0	55.76	0
Min Px	422	201	0	55.76	0
Max Py	796	207	0	94.63	0
Min Py	423	211	0	0.00	0
Max Pz	422	201	0	55.76	0
Min Pz	422	201	0	55.76	0

Gross Maximum Base pressure, p_{max} = 94.6 KN/m^2

Net Allowable Safe Bearing Capacity = 350 KN/m^2
Depth of soil above footing = 0.75 m
Unit weight of soil = 20 KN/m^3

Gross Safe Bearing Capacity = 365 KN/m^2
> 94.6 KN/m^2
Hence OK

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Design of foundation raft F3

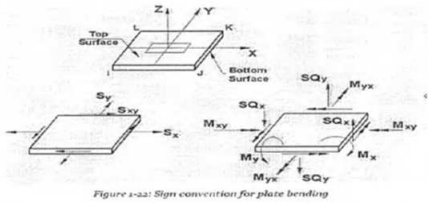
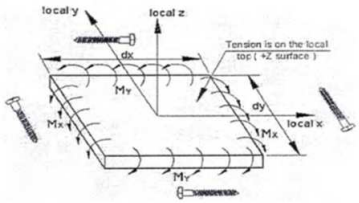


Figure 1-22: Sign convention for plate bending



Sign convention :
Local Z axis of plate is perpendicular to surface & positive towards upward direction.

Moment :
Mx : BM along X direction, +ve means tension on +Z surface of plate
My : BM along Y direction, +ve means tension on +Z surface of plate
For -ve Mx & My it is vice-virsa

Shear force :
SQx : Shear at face perpendicular to X axis
SQy : Shear at face perpendicular to Y axis

Axial force :
+ve Sx means tension along local X axis of plate & -ve Sx means compression along local X axis. for Sy this is similar.

Reinforcement in Foundation:
PLATE CENTER STRESS : FDN_F3_MOMENT At the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	517	0.335	0.1	0	0	0	-49.07	-37.721	-0.075	-49.141	-37.796	104:1.2DL + 1.6WLX2 + 1.0 LL
Min Qx	741	-0.139	0.108	0	0	0	-1.524	-14.161	0	-1.524	-14.161	103:1.2DL + 1.6WLX1 + 1.0 LL
Max Qy	508	-0.047	0.224	0	0	0	-32.1	-21.207	0.077	-32.172	-21.284	104:1.2DL + 1.6WLX2 + 1.0 LL
Min Qy	469	0.129	-0.203	0	0	0	-32.81	-20.463	-3.204	-36.014	-23.667	104:1.2DL + 1.6WLX2 + 1.0 LL
Max Sx	817	-0.001	0.011	0	0	0	-0.101	-1.001	-0.03	-0.131	-1.031	101:1.4DL
Min Sx	817	-0.001	0.011	0	0	0	-0.101	-1.001	-0.03	-0.131	-1.031	101:1.4DL
Max Sy	817	-0.001	0.011	0	0	0	-0.101	-1.001	-0.03	-0.131	-1.031	101:1.4DL
Min Sy	817	-0.001	0.011	0	0	0	-0.101	-1.001	-0.03	-0.131	-1.031	101:1.4DL
Max Sxy	817	-0.001	0.011	0	0	0	-0.101	-1.001	-0.03	-0.131	-1.031	101:1.4DL
Min Sxy	817	-0.001	0.011	0	0	0	-0.101	-1.001	-0.03	-0.131	-1.031	101:1.4DL
Max Mx	744	0.154	-0.07	0	0	0	53.547	9.675	-4.014	57.561	13.689	104:1.2DL + 1.6WLX2 + 1.0 LL
Min Mx	517	0.335	0.1	0	0	0	-49.07	-37.721	-0.075	-49.141	-37.796	104:1.2DL + 1.6WLX2 + 1.0 LL
Max My	514	0.15	-0.066	0	0	0	16.453	20.143	-3.308	19.761	23.451	107:0.9DL + 1.6WLX1
Min My	767	0.245	0.101	0	0	0	-23.46	-42.563	-2.597	-26.059	-45.16	103:1.2DL + 1.6WLX1 + 1.0 LL
Max Mxy	720	-0.059	0.013	0	0	0	26.898	-7.175	14.478	41.376	-21.653	103:1.2DL + 1.6WLX1 + 1.0 LL
Min Mxy	740	-0.058	-0.019	0	0	0	28.931	-8.439	-14.689	43.62	-23.128	103:1.2DL + 1.6WLX1 + 1.0 LL

PLATE CENTER STRESS : FDN_F3_SHEAR At a distance "d" from the face of the column

		Shear		Membrane			Bending Moment			M + I M_{xy} I		
	Plate	Qx	Qy	Sx	Sy	Sxy	Mx	My	Mxy	Mx	My	L/C
		(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(N/mm^2)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	(kN-m/m)	
Max Qx	518	0.193	0.04	0	0	0	-21.06	-27.49	2.552	-23.607	-30.042	104:1.2DL + 1.6WLX2 + 1.0 LL
Min Qx	737	-0.077	0.064	0	0	0	11.804	-6.688	0	11.804	-6.688	103:1.2DL + 1.6WLX1 + 1.0 LL
Max Qy	775	0.037	0.123	0	0	0	-2.231	-27.335	-0.592	-2.823	-27.927	103:1.2DL + 1.6WLX1 + 1.0 LL
Min Qy	770	0.03	-0.127	0	0	0	-1.996	-24.579	0.757	-2.753	-25.336	103:1.2DL + 1.6WLX1 + 1.0 LL
Max Sx	817	-0.001	0.011	0	0	0	-0.101	-1.001	-0.03	-0.131	-1.031	101:1.4DL
Min Sx	817	-0.001	0.011	0	0	0	-0.101	-1.001	-0.03	-0.131	-1.031	101:1.4DL
Max Sy	817	-0.001	0.011	0	0	0	-0.101	-1.001	-0.03	-0.131	-1.031	101:1.4DL
Min Sy	817	-0.001	0.011	0	0	0	-0.101	-1.001	-0.03	-0.131	-1.031	101:1.4DL
Max Sxy	817	-0.001	0.011	0	0	0	-0.101	-1.001	-0.03	-0.131	-1.031	101:1.4DL
Min Sxy	817	-0.001	0.011	0	0	0	-0.101	-1.001	-0.03	-0.131	-1.031	101:1.4DL
Max Mx	743	0.079	-0.022	0	0	0	38.019	8.719	0.297	38.316	9.016	104:1.2DL + 1.6WLX2 + 1.0 LL
Min Mx	500	-0.015	0.119	0	0	0	-27.87	-7.57	-0.985	-28.853	-8.555	104:1.2DL + 1.6WLX2 + 1.0 LL
Max My	513	0.044	-0.035	0	0	0	2.351	19.453	-0.525	2.876	19.978	107:0.9DL + 1.6WLX1
Min My	776	0.071	0.055	0	0	0	-3.258	-39.063	-0.156	-3.414	-39.219	103:1.2DL + 1.6WLX1 + 1.0 LL
Max Mxy	725	-0.069	-0.058	0	0	0	12.749	-5.71	9.466	22.215	-15.176	103:1.2DL + 1.6WLX1 + 1.0 LL
Min Mxy	737	-0.077	0.064	0	0	0	11.804	-6.688	-9.755	21.559	-16.443	103:1.2DL + 1.6WLX1 + 1.0 LL

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Material Property :

Grade of concrete =		fc'35	
Grade of reinf. =		fy420	
Clear cover =	bottom face	50	mm
	top face	50	mm
Thicknes of slab =		500	mm

Design Constants :

φ for Moment =	0.9
φ for shear =	0.75
β_1	0.8
ε_t limit (for tension controlled section) =	0.005
Concrete compressive strain, ε_{cu} =	0.003

Reinforcement Along X:

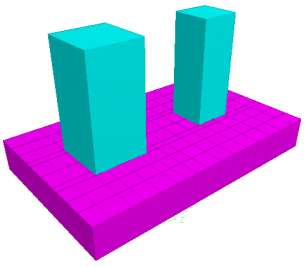
Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, Mx	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 / φ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2)	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
104:1.2D	57.56	top face	50	444	57.561	0.39	0.0009	414.3	532.8	12	212.36	200	565.8	7.98	9.972	0.131	tension control
104:1.2D	-49.14	bottom face	50	442	49.141	0.34	0.0008	355.0		16	377.52	200	1005.8	14.2	17.73	0.072	tension control
103:1.2D	41.38	top face	50	444	41.376	0.28	0.0007	297.2		12	212.36	200	565.8	7.98	9.972	0.131	tension control
103:1.2D	43.62	top face	50	444	43.62	0.30	0.0007	313.4		12	212.36	200	565.8	7.98	9.972	0.131	tension control

Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	KN-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	Max Qx	0.193	1	96.5	23.607	0.001	0.444	437.736	429.897	761.754	429.9	1623	> 96.5	ok	322.4	φVc>Vu	not required
	Min Qx	0.077	1	38.5	11.804			434.208			429.9	1623	> 38.5	ok	322.4	φVc>Vu	not required

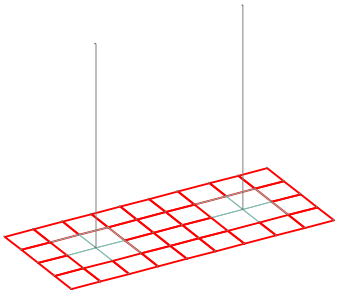
Reinforcement Along Z:

Force Values			Depth		Reinforcement Calculation												
Load condition	Factored Mt, My	Tension face due to bending	clear cover	d	M_u	Kn = Mu1/bd2 / φ	ρ	$A_{s\ required}$	$A_{s,min}$	Dia of bar	$S_{required}$	$S_{provided}$	$A_{s,provided}$	a	c = a/β_1	ε_t = ε_{cu}/c *(d-c)	Remarks
	KN-m/m		mm	mm	KN-m/m	(N/mm2)	mm	mm^2	mm^2	mm	mm c/c	mm c/c	mm^2	mm	mm		
107:0.9D	23.45	top face	50	444	23.451	0.16	0.0004	168.1	532.8	12	212.36	200	565.8	7.98	9.972	0.131	tension control
103:1.2D	-45.16	bottom face	50	442	45.16	0.31	0.0007	326.1		16	377.52	200	1005.8	14.2	17.73	0.072	tension control
103:1.2D	-21.65	bottom face	50	442	21.653	0.15	0.0004	155.9		16	377.52	200	1005.8	14.2	17.73	0.072	tension control
103:1.2D	-23.13	bottom face	50	442	23.128	0.16	0.0004	166.6		16	377.52	200	1005.8	14.2	17.73	0.072	tension control

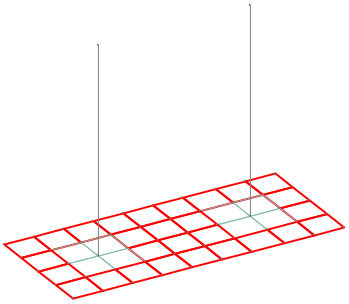
Shear Check		Qx	FOS	Vu	Mu	$\rho_{w,pro}$	bd	V_{ca}	V_{cb}	V_{cc}	$V_{c,design}$	$\varphi(V_c+0.66\sqrt{f_c'}bd)\geq V_u$		Remarks	φV_c	Remarks	Remarks :Min shear reinf
		Mpa		KN	T-m/m		m^2	KN	KN	KN	KN	KN			KN		
Load condition	103:1.2D	0.123	1	61.5	27.927	0.001	0.444	429.683	429.897	fc'761.8	429.68	452	> 61.5	ok	322.3	φVc>Vu	not required
	103:1.2D	0.127	1	63.5	25.336			430.982			429.9	452	> 63.5	ok	322.4	φVc>Vu	not required



3D rendered view



Highlighted plate for Moment at face of column



Highlighted plate for Shear at 'd' distance from face of column

SUMMARY OF REINFORCEMENT IN RAFT:

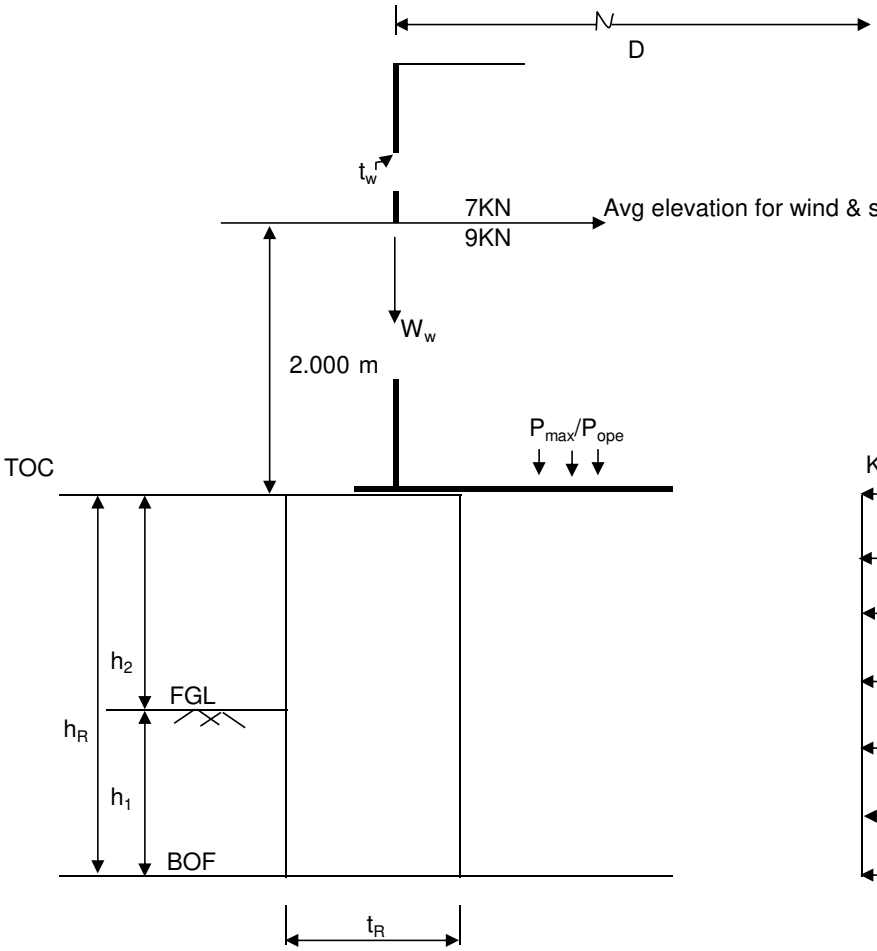
Along X Direction :			
At Bottom	**16ϕT**	**200**	c/c
At Top	**12ϕT**	**200**	c/c
Along Z Direction :			
At Bottom	**16ϕT**	**200**	c/c
At Top	**12ϕT**	**200**	c/c

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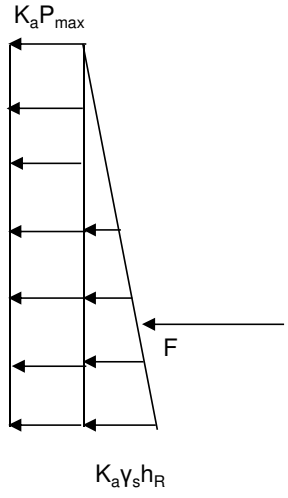
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Design Calculation of Foundation for 3380-TNK-002



Typical Section Through Tank



Pressure on ring wall

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Tank Data :

Diameter of tank ,D =	1.71	m
Self weight of tank =	57.0	KN
Self weight of tank per m of circumference, W_w =	10.6	KN/m
Design height of liquid, h_L =	4.92	m
Maximum weight of Fluid=	130	KN
Pressure at base of tank , P_{max}	56.5	KN/m^2
Operating weight of Fluid=	110	KN
Pressure at base of tank , P_{ope}	47.8	KN/m^2

Foundation Material :

Concrete, f_c' =	35	Mpa
Reinforcement, f_y =	420	Mpa
Unit weight of concrete, γ_c =	24	Kn/m^3

Soil Data :

Unit weight of soil, γ_s =	20	Kn/m^3
Allowable bearing pressure, q_{all}=	350	Kn/m^2
Coefficient of active earth pressure, K_a =	0.33	

Design Data of Ringwall :

Ringwall depth, $h_R = h_1 + h_2$ =	1.95	m
FGL to Bottom of foundation , h_1 =	1.25	m
FGL to Top of concrete , h_2 =	0.7	m
Ringwall thickness, t_R =	450	mm
Inner dia of ringwall, D_i =	1.412	m
Outer dia of ringwall, D_o =	2.312	m
$S_{xx} = \pi(D_o^4-D_i^4)/32D_o$ =	1.04	m^3

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Check of soil bearing pressure (Max Fluid Load): For wind Load

Wind force = 9 kn acting at a height of 3.95 m from BOF

M_{OT} = Overturning moment = 35.6 KN-m

M_{OT} / S_{xx} = 34 KN/m^2

Load Combination : Dead + Fluid

$q_{RF} = W_w/t_R + P_{max}/2 + h_R\gamma_c$ = 98.6 KN/m^2

Load Combination : Dead + Fluid +/- Wind

$q_{max} = q_{RF} + M_{OT}/S_{xx}$ = 133 KN/m^2 < 350.0 Kn/m^2 OK

$q_{min} = q_{RF} - M_{OT}/S_{xx}$ = 64.6 KN/m^2

Check of soil bearing pressure (Empty Tank): For wind Load

Soil bearing pressure under ringwall :

Load Combination : Dead +/- Wind

$q_{RE} = W_w/t_R + h_R\gamma_c$ = 70.4 KN/m^2

$q_{max} = q_{RE} + M_{OT}/S_{xx}$ = 104.4 KN/m^2

$q_{min} = q_{RE} - M_{OT}/S_{xx}$ = 36.32 KN/m^2 no -ve pressure

Check for overturning (Empty Tank): For wind Load

M_{OT} = Overturning moment = 35.6 KN-m

M_{OR} = Restoring moment = (57+PI()*1.712*0.45*1.95*25)*1.712/2 = 149.8 KN-m

FOS = M_{OR} / M_{OT} 4.21 > 1.5 ok

Check for soil bearing pressure (Operating Fluid Load): For Seismic Load

Seismic force at center of mass of equipment = 7 KN

acting at a height of 3.95 m from BOF

M_{OT} = Overturning moment = 27.7 KN-m

M_{OT} / S_{xx} = 26.5 KN/m^2

Load Combination : Dead + Fluid

$q_{RF} = W_w /t_R + P_{ope}/2 + h_R\gamma_c$ = 94.2 KN/m^2

Load Combination : Dead + Fluid +/- 0.7Earthquake load

$q_{max} = q_{RF} + 0.7M_{OT}/S_{xx}$ = 113 KN/m^2 < 350.0 Kn/m^2 OK

$q_{min} = q_{RF} - 0.7M_{OT}/S_{xx}$ = 75.7 KN/m^2

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Check for differential settlement (Max Slurry Load): For Static condition

Soil bearing pressure under ringwall :

$q_{RF} = W_w/t_R + 0.35P_{max} + h_R\gamma_c$ = 90.1 Kn/m^2 < 350.0 KN/m^2 Ok

Soil bearing pressure under tank :

$q_{TF} = \gamma_p h_L + h_R\gamma_s$ = 95.5 Kn/m^2 < 350.0 KN/m^2 Ok

As the tank encroaches the ring wall by 150mm out of total 450mm , the pressure under the tank is reduced to 150/450 = 0.35 times.

Soil bearing pressure under ring wall and tank bottom is almost equal, therefore, no settlement problem will occur.

Ringwall Reinforcement :

Radial outward force on ringwall, $F = K_a h_R(P_{max} + \gamma_s h_R/2)$ = 48.9 Kn/m

For Hydrotest :

Radial outward force on ringwall, $F = K_a h_R(\gamma_w h_L + \gamma_s h_R)$ = 44.2 Kn/m

Hoop tension in ringwall, T = 1/2FD = 41.8 Kn

Factored design tension, T_u= 58.6 Kn FOS = 1.4

$A_{s,reqd} = T_u/\phi f_y$ = 155 mm^2 ϕ = 0.9

$A_{s,min} = 0.0025 h_R t_R$ = 1755 mm^2

$A_{s,reqd}$ (per face of ring wall)= 878 mm^2

Required : 12 dia bar @ 252 c/c (per face)

Provide : 12 dia bar @ 150 c/c (per face)

$A_{s,\ provided}$(per face) = 1471 mm^2 ok

Vertical Reinforcement :

$A_{s,min} = 0.0015 t_R$ = 540 mm^2/m

$A_{s,reqd}$ (per face of ring wall)= 270 mm^2

Required : 10 dia bar @ 291 c/c (per face)

Provide : 10 dia bar @ 200 c/c (per face)

$A_{s,\ provided}$(per face) = 393 mm^2 ok

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REFERENCES

Reference is made to codes, standards and specifications as indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.1.

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ATTACHMENT 1 INPUT DRAWINGS

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Project: Mansourah Massarah Gold Project

Customer: MA'ADEN GOLD AND BASE METALS (MA'ADEN)

Title: CALCULATION

3380: ACIDULATION BUILDING

DESIGN CALCULATION

AS BUILT

MD-216-3380-EG-ST-CAL-1001

Rev	Date	Revision Text	Prepared By	Reviewed By	Approved By	Employer
D01	18-11-2023	AS BUILT	SGI	SYC	PRI	
C01	21-07-2020	ISSUED FOR CONSTRUCTION	SGI	SYC	PRI	
A02	22-06-2020	Revised as per Employer Comments dated 11-06-2020	SGI	SYC	PRI	
A01	08-05-2020	FIRST ISSUE	SGI	SYC	PRI	

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Ma'aden Document Number:

MD-216-3380-EG-ST-CAL-1001 D01

Outotec Document Number:

900707 GAU01 HL15 CED01 33802 OU100674671 3

L&T Document Number:

O19045 450 S K 26201 001 3

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MA'ADEN معادن شركة التعدين العربية السعودية Saudi Arabian Mining Company	Outotec	LARSEN & TOUBRO	**DESIGN DOCUMENT**						PAGE
			Ma'aden Document Number:	**MD-216-3380-EG-ST-CAL-1001**					2
			Outotec Document Number:	**900707_GAU01_HL15_CED01_33802_OU100674671**					of
			L&T Document Number:	**O19045_450_S_K_26201_001**					126

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	A02	C01	D01	Outotec / L&T Revision No.:	0	1	2	3
Date	08-05-2020	22-06-2020	21-07-2020	18-11-2023	Date	08-05-2020	22-06-2020	21-07-2020	18-11-2023
Revision Text	FIRST ISSUE	Revised as per Employer Comments dated 11-06-2020	ISSUED FOR CONSTRUCTION	AS BUILT	Revision Text	FIRST ISSUE	Revised as per Employer Comments dated 11-06-2020	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	SGI	SGI	SGI	SGI	Prepared By	SGI	SGI	SGI	SGI
Checked	SYC	SYC	SYC	SYC	Checked	SYC	SYC	SYC	SYC
Approved	PRI	PRI	PRI	PRI	Approved	PRI	PRI	PRI	PRI

REFERENCE DOCUMENTS

DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria
MD-216-3000-EG-CV-RPT-1008	Geotechnical Report
SBC 301 :2007	Saudi Structural - Loading & Forces Code
SBC 304 :2018	Saudi Concrete Structures Code
ACI 318 :2014	Building Code Requirement of Structural Concrete

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Checked	SYC	SYC	SYC	SYC	Checked	SYC	SYC	SYC	SYC
Approved	PRI	PRI	PRI	PRI	Approved	PRI	PRI	PRI	PRI

DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
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Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	MansourahMassarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities

Global Project ID	**900707**
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Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

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Checked	SYC	SYC	SYC	SYC
Approved	PRI	PRI	PRI	PRI

1. INTENT

The document is prepared for the design calculation of Equipment Supporting Structure of Acidulation area of Plant.
The design calculation ensures the safety and serviceability of structural members.

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah&Massarah (MMGP) Gold Project.
MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.
The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

The design document is prepared for Equipment Supporting Structure of Acidulation area of Plant. Structures are modelled in Staad-Pro and analysis & design has been done for Supporting Structure supporting structure in Staad

a) MD-216-1000-EG-ST-CRT-1001 _Civil, Structural & Architectural Design Criteria

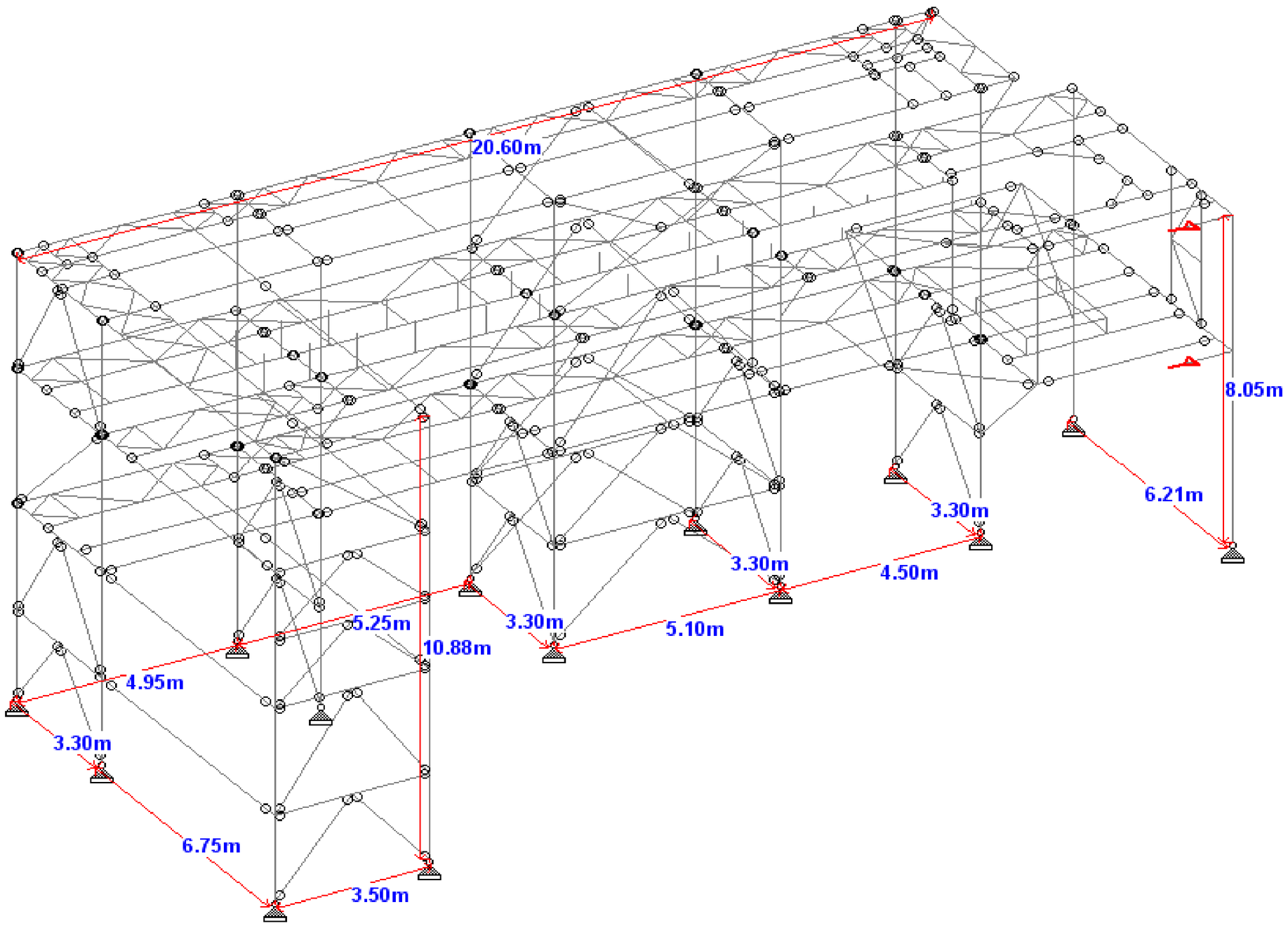
Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

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Prepared By	SGI	SGI	SGI	SGI	Prepared By	SGI	SGI	SGI	SGI
Checked	SYC	SYC	SYC	SYC	Checked	SYC	SYC	SYC	SYC
Approved	PRI	PRI	PRI	PRI	Approved	PRI	PRI	PRI	PRI

4. DESIGN OF EQUIPMENT SUPPORTING STRUCTURE OF ACIDULATION AREA

4.1 Structural Geometry of Equipment Supporting Structure of Acidulation area

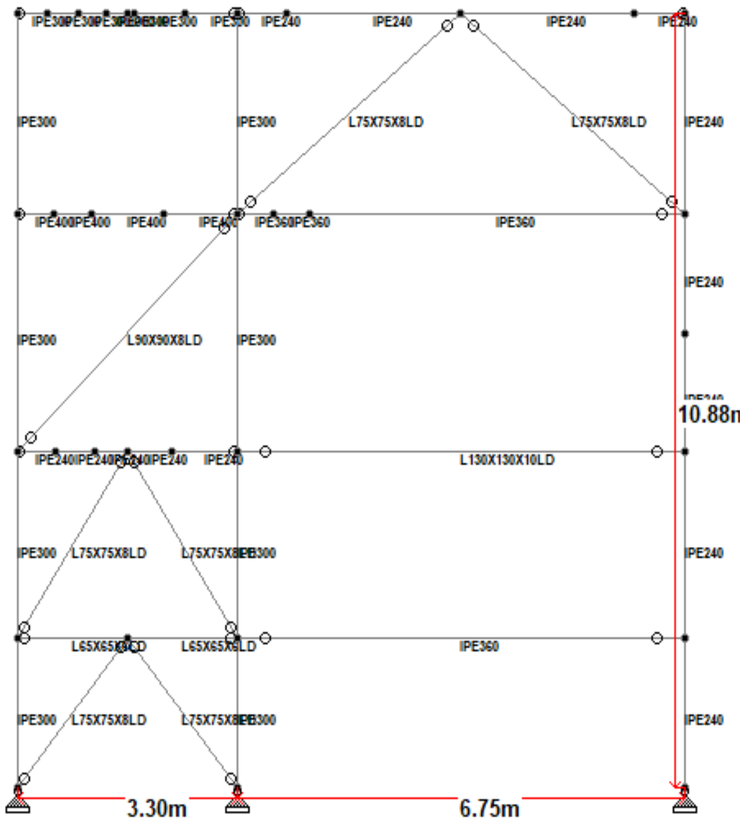
Three Dimensional view of equipment supporting structure from Staad Model is shown below:

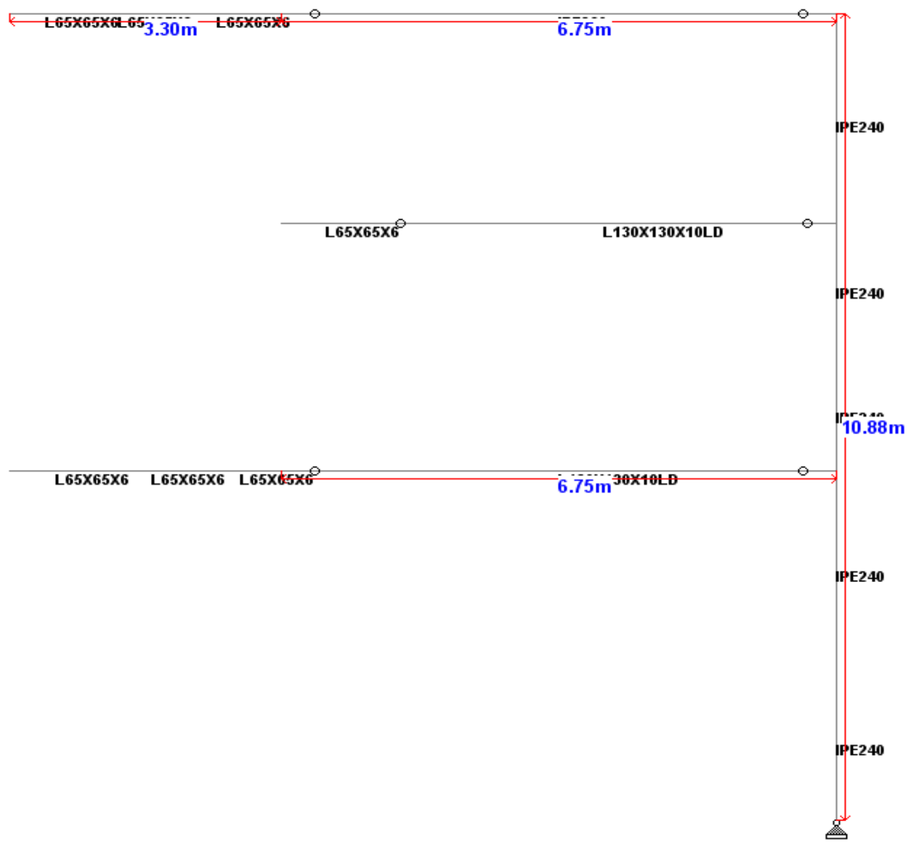


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Project Country	**KINGDOM OF SAUDI ARABIA**

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Approved	PRI	PRI	PRI	PRI	Approved	PRI	PRI	PRI	PRI

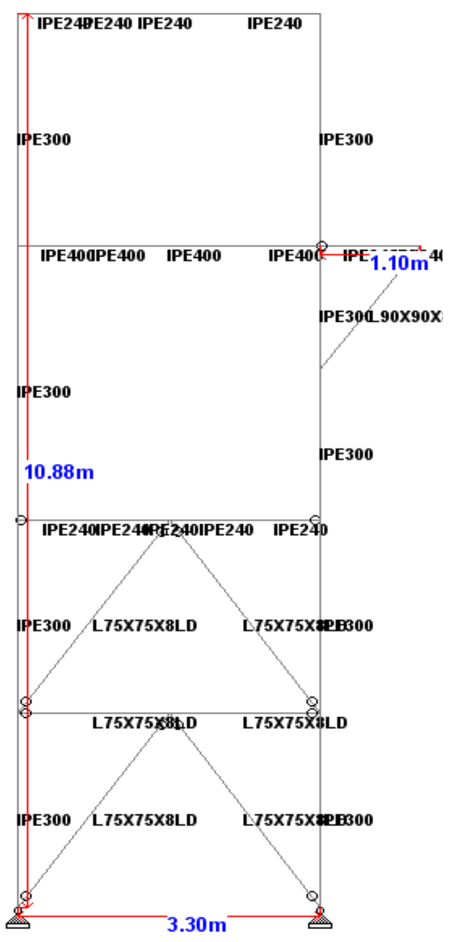
4.2 Section Profile of Equipment Supporting Structure of Acidulation area

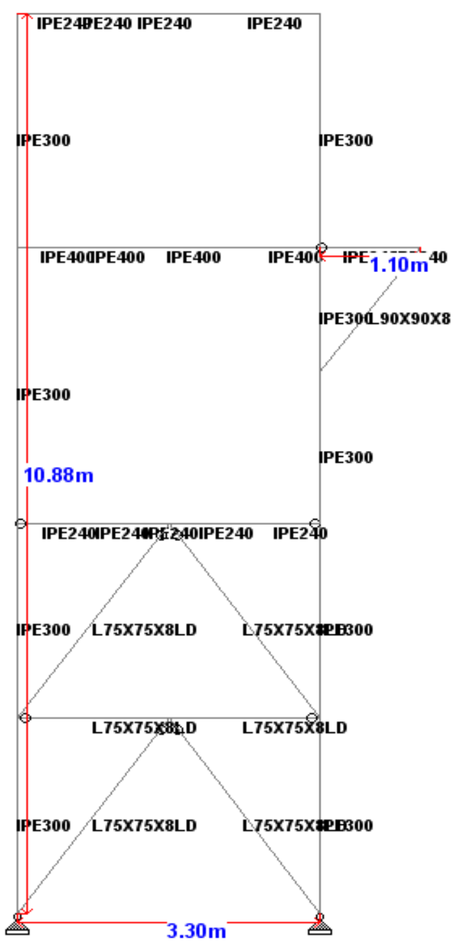


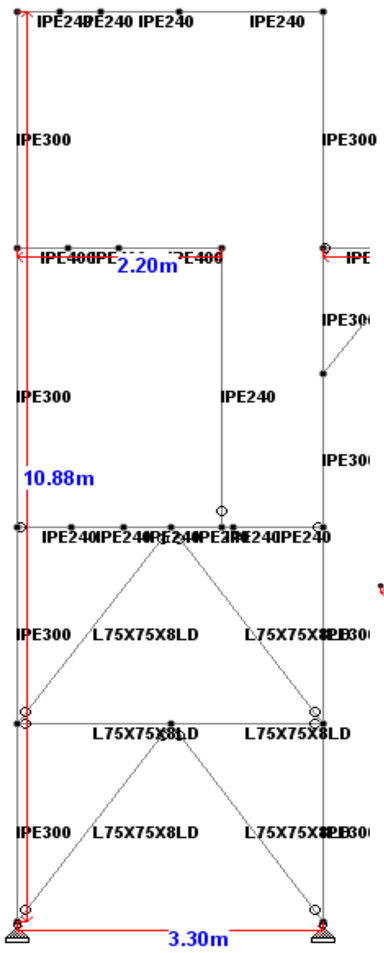


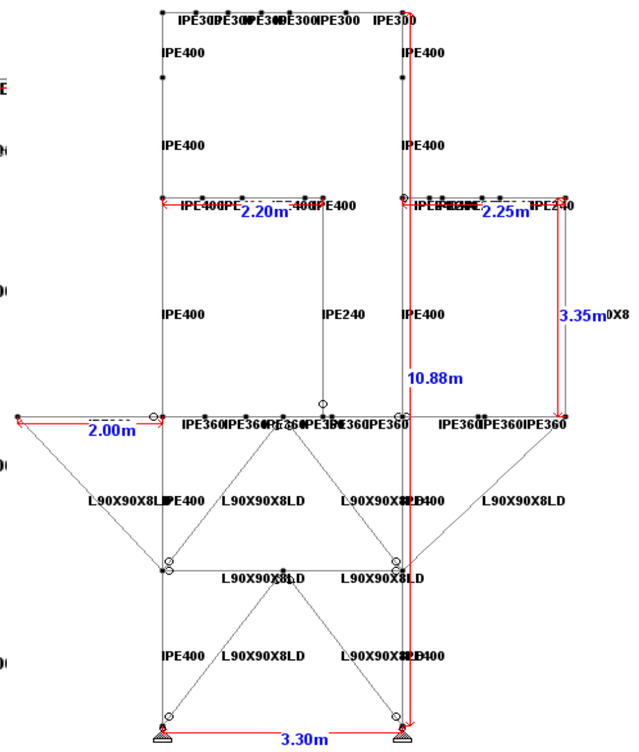
CROSS SECTION ON LINE-3410/1a

CROSS SECTION ON LINE-3410/1b









CROSS SECTION ON LINE-3410/1c

CROSS SECTION ON LINE-3410/2a

CROSS SECTION ON LINE-3410/2b

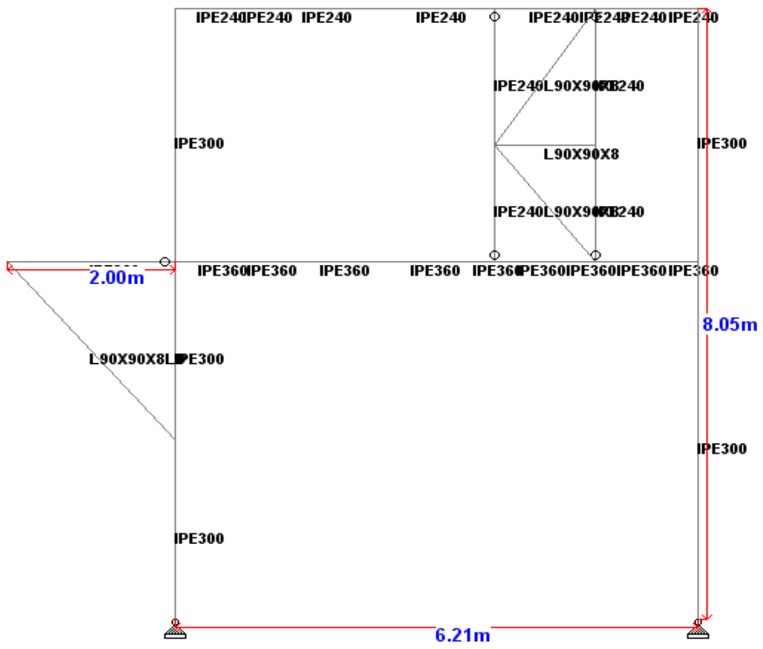
CROSS SECTION ON LINE-3410/3a

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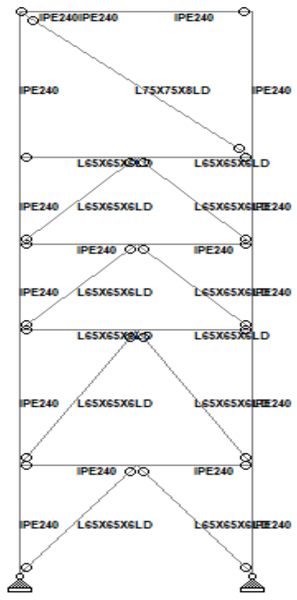
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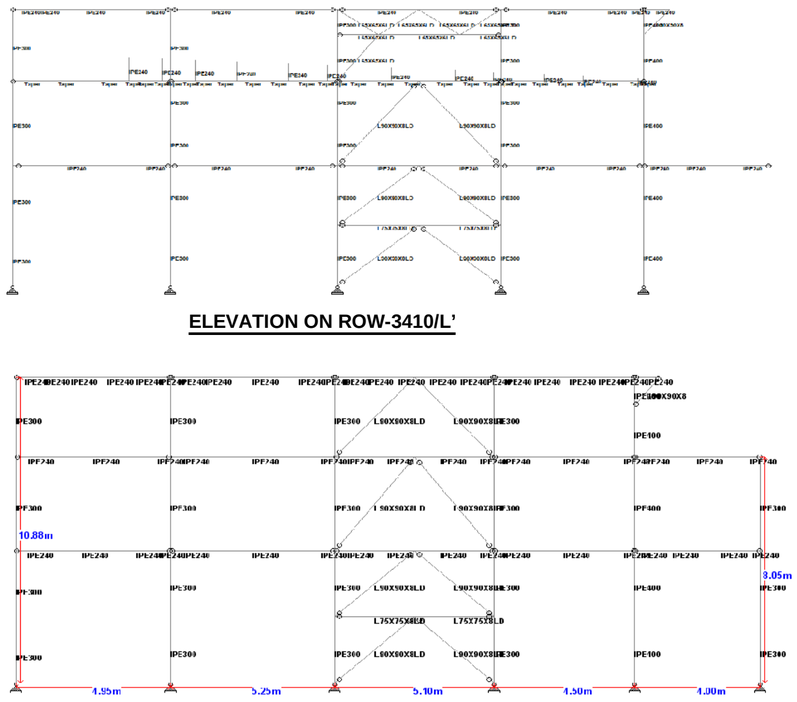
4.2 Section Profiles of Equipment Supporting Structure of Acidulation area



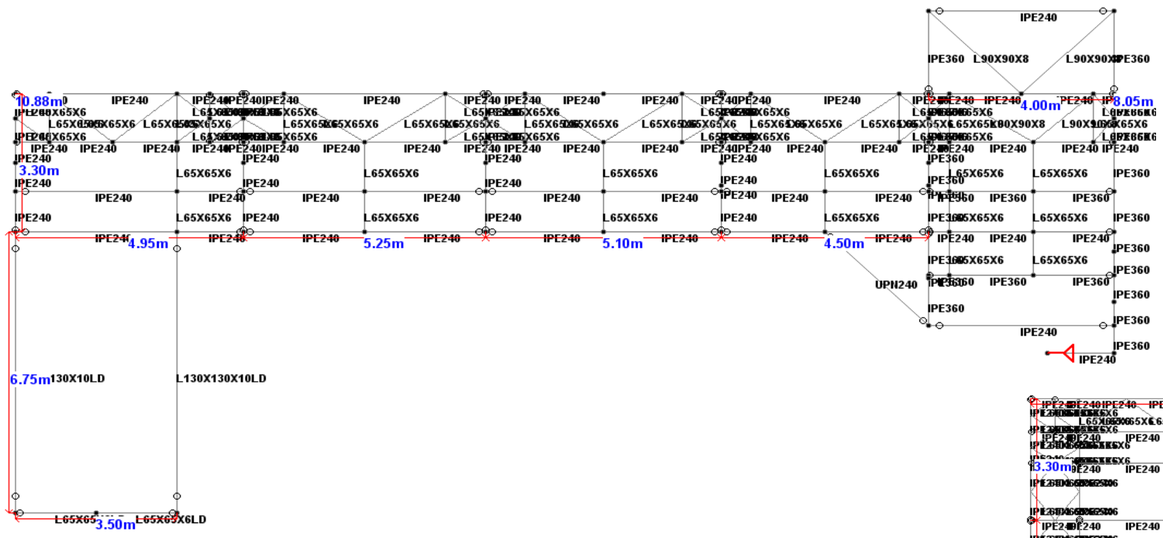
CROSS SECTION ON LINE-3410/3b



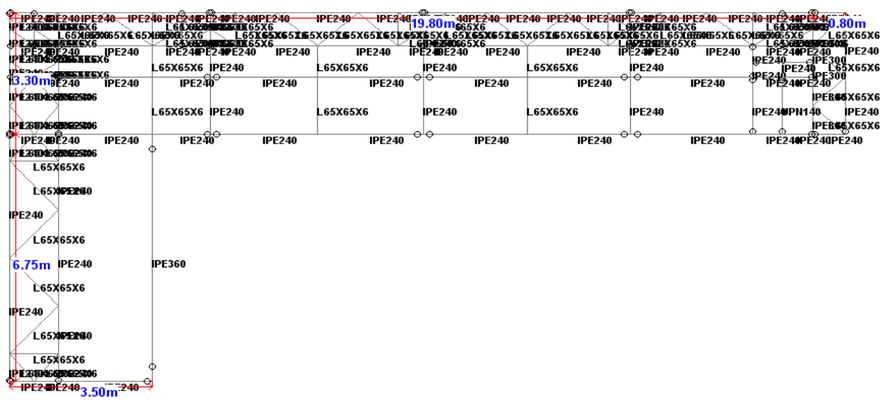
ELEVATION ON ROW-3410/J'



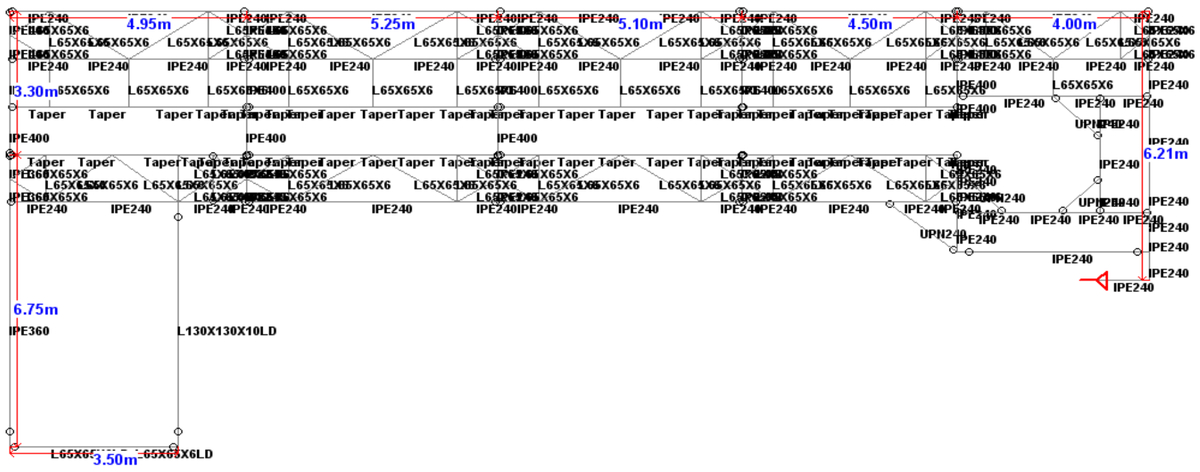
ELEVATION ON ROW-3410/L



PLATFORM AT EL. 918.300M LEVEL



PLATFORM AT EL. 924.480M LEVEL



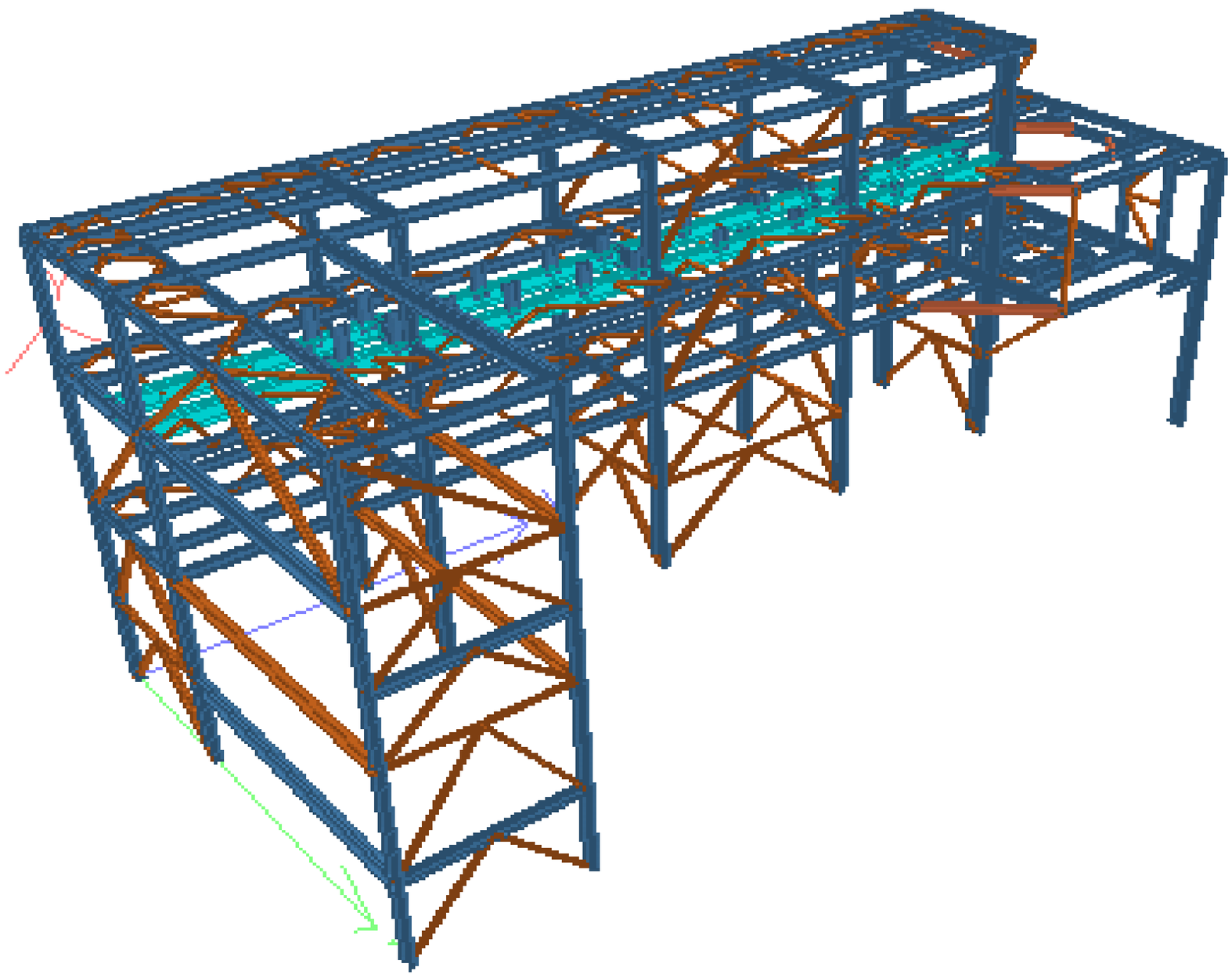
PLATFORM AT EL. 921.650M LEVEL

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4.2a 3D MODEL Equipment Supporting Structure of Acidulation area



3D MODEL

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Approved	PRI	PRI	PRI	PRI	Approved	PRI	PRI	PRI	PRI

4.3 Load Calculation forEquipment Supporting Structure of Acidulation area

All the structural members are designed in Staad using AISC LRFD code.
Max. Elevation considered to be 15m above Base Plate Level.
All the structural members are designed in Staad using AISC LRFD code.

L/C 1. Dead Load :

a) Load due to self wt of stringer channel & grating :
Weight of Grating @ 35kg/m2 = 0.35 kN/m2
Self weight of stringer = 0.33 kN/m (UPN 240)
Stair width = 1.2 m
Stringer Length = 6.75 m
Conc. Load on beam supporting stringer = {(1.2/2 *0.35+0.33+0.15)*6.75/2} kN = 2.33 kN = 0.25 T
Self weight of handrail = 0.15 kN/m
weight of monorail beam (IPE300) @ 42.2kg/m = 0422 kN/m
Maximum span of monorail = 5.55m
Load on each supporting beam =5.55*0.422 = 2.5 kN

Dead load of equipment:

P1 = 5.1 kN, P2 = 4.6 kN, P3 = 8.8 kN, P4 = 7.4 kN & P5 = 2.1 kN
P32 & P33 = 30 kN & P45 = 20kN

Pipe & Cable Load:

Considered pipe & cable load 50 kN on each column load = 50 kN

L/C 2. Live Load-1 :

Live Load on platform @ 500 kg/m2 = 5 kN/m2
Live Load on stair @ 400 kg/m2 = 4 kN/m2
Conc. Load on beam supporting stringer = {(1.2/2 *4)*6.75/2} kN = 8.1 kN = 0.85 T

Monorail Load:

Monorail capacity = 10 kN
Considered weight og pulley bloock = 10 kN
Impact factor = 1.25
Load on each supporting beam = (10+10) * 1.25 = 25 kN

L/C 3. Live Load-2 : Equipment Dynamic Load

P32 & P33 = 40 kN

L/C 4. Live Load-3 : Equipment Test Load

P45 = 80 kN,

L/C 5. Live Load-4 : Equipment Live Load

P45 = 125 kN,

L/C 6. Wind load : (As per SBC 301 : 2007)

Wind load is calculated using the following parameters:
Basic Wind Speed = 155 Km/hr (as per Civil, Structural Design Criteria)
Exposure Category = C (as per Civil, Structural Design Criteria)
Topographic factor, Kzt = 1 (Section 6.4.3, SBC 301)
Importance Factor, I = 1 (Section 6.5, SBC 301)
Velocity Pressure Co-efficient, Kz = 1.09 (Table 7.2-2 , SBC 301)
Wind Directionality Factor, Kd = 0.85 (Section 6.4.1, SBC 301)
Velocity pressure, qz = 0.0473 x 10-3 x Kz x Kzt x Kd x V2 x I = 1.06 kN/m2

Wind Load in +X Dir.

Sections of the members exposed to Wind :
For Column Section IPE400, IPE300 & IPE240
IPE400 Column Depth = 400 mm, IPE400 Flange width = 180 mm, IPE300 Column Depth = 300 mm, IPE300 Flange width = 150 mm, IPE240 Column Depth = 240 mm, IPE240 Flange width = 120 mm
Beam Depth =360 mm, Beam Depth = 300 mm, Beam Depth = 240 mm, Stair Stringer Depth = 240 mm
Bracing Depth = 65 mm, 75 mm, 90 mm, 130 mm
Force Coeff. , Cf for Flat Section = 2 (Fig. 7.2-17 , SBC 301)
Wind load on Col IPE400 = 1.06*400/1000*2 = 0.085 T/m, IPE300 = 1.06*300/1000*2 = 0.065 T/m, IPE240 = 1.06*240/1000*2 = 0.052 T/m
Wind load on 360 mm Beam = 1.06*360/1000*2 = 0.077 T/m, Wind load on 300 mm Beam = 1.06*300/1000*2 = 0.065 T/m, Wind load on 240 mm Beam = 1.06*240/1000*2 = 0.052 T/m
Conc. Load on beam supporting stringer for Stair = 0.052 * 7.5 / 2 = 0.2 T
Wind load on 65 mm Bracing = 1.06*65/1000*2 = 0.014 T/m, Wind load on 75 mm Bracing = 1.06*75/1000*2 = 0.016 T/m
Wind load on 90 mm Bracing = 1.06*90/1000*2 = 0.020 T/m, Wind load on 130 mm Bracing = 1.06*130/1000*2 = 0.028 T/m
Wind load on 300 mm Monorail = 1.06*300/1000*2*5.55 = 0.36 T

Wind load on equipment:

P32 & P33 = 0.2 kN, P45 = 5/4 kN = 1.25kN

Wind load on Launder:

P1 & P5 = 0.063kN

Wind Load in +Z Dir.

Sections of the members exposed to Wind :
For Column Section IPE400, IPE300 & IPE240
IPE400 Column Depth = 400 mm, IPE400 Flange width = 180 mm, IPE300 Column Depth = 300 mm, IPE300 Flange width = 150 mm, IPE240 Column Depth = 240 mm, IPE240 Flange width = 120 mm
Beam Depth =360 mm, Beam Depth = 300 mm, Beam Depth = 240 mm, Stair Stringer Depth = 240 mm
Bracing Depth = 65 mm, 75 mm, 90 mm, 130 mm
Force Coeff. , Cf for Flat Section = 2 (Fig. 7.2-17 , SBC 301)
Wind load on Col IPE400 = 1.06*400/1000*2 = 0.085 T/m, IPE300 = 1.06*300/1000*2 = 0.065 T/m, IPE240 = 1.06*240/1000*2 = 0.052 T/m
Wind load on 360 mm Beam = 1.06*360/1000*2 = 0.077 T/m, Wind load on 300 mm Beam = 1.06*300/1000*2 = 0.065 T/m, Wind load on 240 mm Beam = 1.06*240/1000*2 = 0.052 T/m
Conc. Load on beam supporting stringer for Stair = 0.052 * 7.5 / 2 = 0.2 T
Wind load on 65 mm Bracing = 1.06*65/1000*2 = 0.014 T/m, Wind load on 75 mm Bracing = 1.06*75/1000*2 = 0.016 T/m
Wind load on 90 mm Bracing = 1.06*90/1000*2 = 0.020 T/m, Wind load on 130 mm Bracing = 1.06*130/1000*2 = 0.028 T/m
Wind load on 300 mm Monorail = 1.06*300/1000*2*5.55 = 0.36 T

Wind load on equipment:

P32 & P33 = 0.2 kN, P45 = 5/4 kN = 1.25kN

Wind load on Launder:

P1 = 0.6kN, P2 = 0.6kN, P3 = 1.1kN, P4 = 0.9kN & P5 = 0.2kN

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Seismic Load (SL) for platform supporting Structure:

Seismic Parameters :

Seismic Design Category =	B		
Site Class =	B		
Seismic Coeffecient, Ss =	0.1	g	(From Civil-Structural Design Criteria)
S_1 =	0.02	g	
Occupancy importance factor, I =	1		(From Table 9.5, SBC 301:2007)
Response modification factor, R =	4		(From Table 10.2, SBC 301:2007)
Height of the building, h =	20.57	m	
For Moment resisting steel frames,			
Time Period, T =	0.76	sec	(eqn. 10.9.3.2-1, SBC 301:2007)
Factors, Fa =	1		(From Table 9.4.3a, SBC 301:2007)
Fv =	1		(From Table 9.4.3b, SBC 301:2007)
Maximum spectral acceleration, S_{MS} = Fa * Ss =	0.1	g	(eqn. 9.4.3-1, SBC 301:2007)
Maximum spectral acceleration, S_{M1} = Fa * S_1 =	0.02	g	(eqn. 9.4.3-2, SBC 301:2007)
Design spectral acceleration, S_{DS} = 2/3 * S_{MS} =	0.067	g	(eqn. 9.4.4-1, SBC 301:2007)
Design spectral acceleration, S_{D1} = 2/3 * S_{M1} =	0.013	g	(eqn. 9.4.4-2, SBC 301:2007)
Seismic base Shear, V =	Cs * W		(eqn. 10.9.2-1, SBC 301:2007)
Seismic Response co-eff, Cs = S_{DS} * I /R =	0.017	g	(eqn. 10.9.2.1-1, SBC 301:2007)
Max. value of C_S = S_{D1}*I /(T * R) =	0.004	g	(eqn. 10.9.2.1-2, SBC 301:2007)
Min. value of C_S = 0.044 * S_{DS} * I =	0.003	g	(eqn. 10.9.2.1-3, SBC 301:2007)
Total Dead + Live Load , W =	304	T	(From Staad Output File)
Seismic base Shear, V =	13.01	T	
Total base - shear due to wind =	25.00	T	(From Staad Output File)

> Seismic Base Shear

Hence, wind analysis governs the design.

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4.4a LOAD COMBINATIONS (Refer SBC 301 : 2007 for Strength Design)

100 1.4*DL
101 1.2DL+1.6*LL
102 1.2DL+0.8WLX
103 1.2DL+0.5LL+1.6WLX
104 0.9DL+1.6WLX
105 1.2DL+0.8WLZ
106 1.2DL+0.5LL+1.6WLZ
107 0.9DL+1.6WLZ

4.4b LOAD COMBINATIONS (Refer SBC 301 : 2007 for Servicibility)

201 1.0*DL + 1.0*LL
202 1.0*DL + 1.0*LL + 1.0*WLX
203 1.0*DL + 1.0*WLX
204 1.0*DL + 1.0*LL + 1.0*WLZ
205 1.0*DL + 1.0*WLZ

MA'ADEN معادن شركة التعدين العربية السعودية Saudi Arabian Mining Company	Outotec	LARSEN & TOUBRO

DESIGN DOCUMENT

Ma'aden Document Number:	**MD-216-3380-EG-ST-CAL-1001**
Outotec Document Number:	**900707_GAU01_HL15_CED01_33802_OU100674671**
L&T Document Number:	**O19045_450_S_K_26201_001**

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4.5 Foundation Load Data

(+ve Fy means Vertical Downward Load)

Col. Mkd.	Load Comb.	HORZ. Fx (Mton)	Vertical Fy(Mton)	HORZ. Fz (Mton)
3410/L - 3410/1a	DL	0.0	7.7	0.0
	LL-1	0.0	6.7	-0.1
	LL-2	0.0	0.1	0.0
	LL-3	0.0	0.0	0.0
	LL-4	0.0	0.0	0.0
	WL(+)X	-0.2	-0.6	-0.1
	WL(-)X	0.2	0.6	0.1
	WL(+)Z	0.0	-13.4	-3.3
	WL(-)Z	0.0	13.4	3.3
3410/L - 3410/1c	DL	0.0	9.8	0.1
	LL-1	0.0	12.9	0.1
	LL-2	0.0	0.0	0.0
	LL-3	0.0	0.0	0.0
	LL-4	0.0	0.0	0.0
	WL(+)X	-0.2	-0.1	-0.1
	WL(-)X	0.2	0.1	0.1
	WL(+)Z	0.0	-9.5	-2.3
	WL(-)Z	0.0	9.5	2.3
3410/L - 3410/2a	DL	0.1	9.6	0.1
	LL-1	0.0	12.2	0.0
	LL-2	0.0	-0.1	0.0
	LL-3	0.0	-0.1	0.0
	LL-4	0.0	-0.1	0.0
	WL(+)X	-4.9	-12.9	0.0
	WL(-)X	4.9	12.9	0.0
	WL(+)Z	-0.5	-12.8	-2.4
	WL(-)Z	0.5	12.8	2.4
3410/L - 3410/2b	DL	-0.1	10.4	0.1
	LL-1	0.0	12.4	0.1
	LL-2	0.0	0.4	0.0
	LL-3	0.0	-0.1	0.0
	LL-4	0.0	-0.2	0.0
	WL(+)X	-4.9	13.3	0.2
	WL(-)X	4.9	-13.3	-0.2
	WL(+)Z	-0.5	-8.4	-2.4
	WL(-)Z	0.5	8.4	2.4
3410/L - 3410/3a	DL	0.0	12.4	0.1
	LL-1	0.0	9.0	0.2
	LL-2	0.0	3.9	0.0
	LL-3	0.0	0.4	0.1
	LL-4	0.0	0.6	0.1
	WL(+)X	-0.3	0.4	0.1
	WL(-)X	0.3	-0.4	-0.1
	WL(+)Z	0.0	-21.7	-5.3
	WL(-)Z	0.0	21.7	5.3
3410/L - 3410/3b	DL	0.0	8.2	0.0
	LL-1	0.0	6.1	-0.1
	LL-2	0.0	0.1	0.0
	LL-3	0.0	1.2	0.0
	LL-4	0.0	1.9	0.1
	WL(+)X	-0.1	0.1	0.0
	WL(-)X	0.1	-0.1	0.0
	WL(+)Z	0.0	-1.0	-0.3
	WL(-)Z	0.0	1.0	0.3
3410/L' - 3410/1a	DL	0.0	11.2	-0.1
	LL-1	0.0	15.2	-0.1
	LL-2	0.0	-0.1	0.0
	LL-3	0.0	0.0	0.0
	LL-4	0.0	0.0	0.0
	WL(+)X	-0.3	0.6	-0.1
	WL(-)X	0.3	-0.6	0.1
	WL(+)Z	0.0	12.2	-3.3
	WL(-)Z	0.0	-12.2	3.3
3410/L' - 3410/1c	DL	0.0	15.0	0.0
	LL-1	0.0	16.4	0.1
	LL-2	0.0	0.0	0.0
	LL-3	0.0	0.0	0.0
	LL-4	0.0	0.0	0.0
	WL(+)X	-0.2	0.1	-0.1
	WL(-)X	0.2	-0.1	0.1
	WL(+)Z	0.0	9.5	-2.3
	WL(-)Z	0.0	-9.5	2.3
3410/L' - 3410/2a	DL	0.1	13.4	0.0
	LL-1	-0.1	12.9	0.0
	LL-2	0.0	0.0	0.0
	LL-3	0.0	0.0	0.0
	LL-4	0.0	0.0	0.0
	WL(+)X	-4.9	-12.3	0.0
	WL(-)X	4.9	12.3	0.0
	WL(+)Z	0.2	11.7	-2.4
	WL(-)Z	-0.2	-11.7	2.4

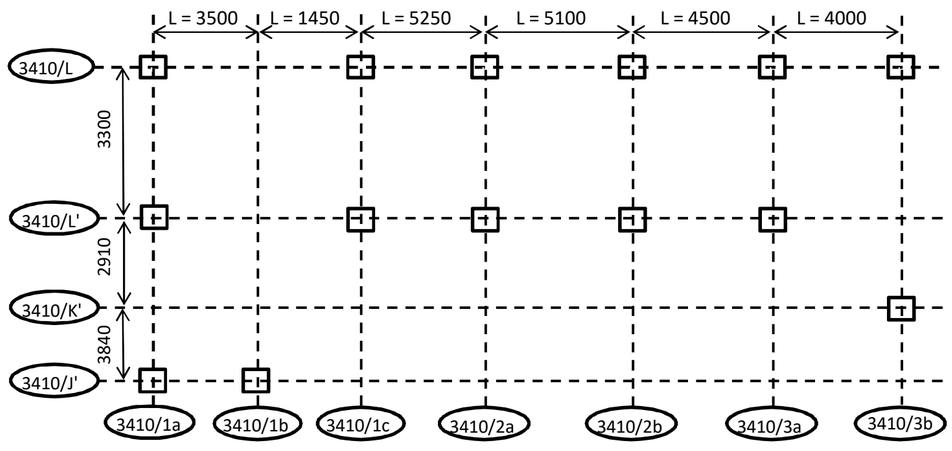
4.5 Foundation Load Data

(+ve Fy means Vertical Downward Load)

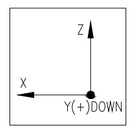
Col. Mkd.	Load Comb.	HORZ. Fx (Mton)	Vertical Fy(Mton)	HORZ. Fz (Mton)
3410/L' - 3410/2b	DL	-0.1	12.9	0.0
	LL-1	-0.1	11.9	0.0
	LL-2	0.0	0.6	0.0
	LL-3	0.0	0.2	0.0
	LL-4	0.0	0.2	0.0
	WL(+)X	-4.9	11.9	0.2
	WL(-)X	4.9	-11.9	-0.2
	WL(+)Z	0.2	9.6	-2.3
	WL(-)Z	-0.2	-9.5	2.3
3410/L' - 3410/3a	DL	0.0	16.3	0.0
	LL-1	0.0	18.5	0.2
	LL-2	0.0	3.5	0.0
	LL-3	0.0	5.2	0.1
	LL-4	0.0	8.3	0.1
	WL(+)X	-0.3	-0.5	0.1
	WL(-)X	0.3	0.5	-0.1
	WL(+)Z	0.0	21.7	-5.5
	WL(-)Z	0.0	-21.7	5.5
3410/J' - 3410/1a	DL	0.3	3.8	0.0
	LL-1	0.9	8.0	0.0
	LL-2	0.0	0.0	0.0
	LL-3	0.0	0.0	0.0
	LL-4	0.0	0.0	0.0
	WL(+)X	-2.4	-8.7	0.0
	WL(-)X	2.4	8.7	0.0
	WL(+)Z	0.3	2.8	-0.1
	WL(-)Z	-0.3	-2.8	0.1
3410/K' - 3410/3b	DL	0.0	8.2	-0.1
	LL-1	0.0	6.9	-0.3
	LL-2	0.0	-0.1	0.0
	LL-3	0.0	1.7	-0.1
	LL-4	0.0	2.7	-0.2
	WL(+)X	-0.1	0.0	0.0
	WL(-)X	0.1	0.0	0.0
	WL(+)Z	0.0	1.0	-0.3
	WL(-)Z	0.0	-1.0	0.3
3410/J' - 3410/1b	DL	-0.3	2.9	0.0
	LL-1	-0.9	5.2	0.0
	LL-2	0.0	0.0	0.0
	LL-3	0.0	0.0	0.0
	LL-4	0.0	0.0	0.0
	WL(+)X	-2.4	8.8	0.0
	WL(-)X	2.4	-8.8	0.0
	WL(+)Z	0.2	-1.5	-0.1
	WL(-)Z	-0.2	1.5	0.1

NOTE:

1. WL(+)X :- WIND LOAD FROM GRID-3410/1a TO GRID-3410/2a
2. WL(+)Z :- WIND LOAD FROM LINE-3410/L TO LINE-3410/L'
3. WIND LOAD IN X AND Z DIRECTION WILL NOT ACT SIMULTANEOUSLY.
4. LL-1: SUPER IMPOSED LOAD
5. LL-2: EQUIPMENT DYNAMIC LOAD
6. LL-3: EQUIPMENT TEST LOAD
7. LL-1: EQUIPMENT LIVE LOAD



Equipment Supporting Structure of Acidulation area



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Project Country	**KINGDOM OF SAUDI ARABIA**

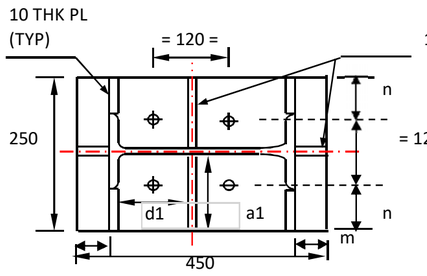
Ma'aden Revision No:	A01	A02	C01	D01	Outotec / L&T Revision No.:	0	1	2	3
Date	08-05-2020	22-06-2020	21-07-2020	18-11-2023	Date	08-05-2020	22-06-2020	21-07-2020	18-11-2023
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Approved	PRI	PRI	PRI	PRI	Approved	PRI	PRI	PRI	PRI

4.6a Design of Base Plates for Columns

DESIGN OF BASE PLATE FOR COLUMN AT GRID 3410/L_3410/3a & 3410/L'_3410/3a

(AS PER SBC - 306 & AISC : 2018 LRFD Specification)

Here, Column Section =	IPE 400			
Grade of Conc. =	35	N/mm2		
Grade of Steel, Fy =	275	N/mm2		
Maximum axial load, Pu =	66.09	Ton	L/C	103
& Minimum axial Load =	-22.45	Ton (Ten.)	(Refer Staad File for Load Combs)	
Length of base plate (N)=	450	mm		
Width of base plate (B) =	250	mm		
Area of base plate, A1 =	0.1125	m2		
Length of concrete pedestal =	600	mm		
Width of concrete pedestal =	600	mm		
Area of concrete pedestal, A2 =	0.36	m2		
Design Bearing strength = 0.85*fc' =	29.75	N/mm2	(As per Section 10.9 of SBC 306)	
Pp = 0.85*fc' *A1*SQRT(A2/A1) =	5987.1	KN	(Eqn. 10.9-2 of SBC 306)	
ϕc * Pp = 3592.2 KN	> 660.88KN - OK		(ϕ_c = 0.6)	



m =	(N - 0.95d)/ 2 =	35	mm
n =	(B - 0.85bf)/2 =	48.5	mm
l =	2 (X) / (1+ (1-X)) =	0.41	
l n'=		27.8	mm
l =	max(m, n, ln')	48.5	mm
tmin, l*sqrt((2*Pu)/(0.9*Fy*B*N))=		10.6 mm	(As per AISC-2018 : LRFD)

Actual area of base plate =	1125	cm2		
Uniform base pressure below base plate,fpu =	5.87	N/mm2		
Moment capacity of Base Plate, M_{pu} = $f_{pu} * l^2/2$ =	6904	N-mm/mm width		(As per AISC-2018 : LRFD)
Required thk. of Base Plate, t_{reqd} = $\sqrt{\frac{4 * Mpu}{\phi * F_y}}$	10.6	mm	< tmin	ϕ = 0.9

Now, a1 = 12.07 cm and d1 = 18.15 cm
Therefore, a1/d1 = 12.07/18.15 = 0.665
Hence, α_3 = 0.083 as per Mukhanov's chart
Now BM per cm = α3 X q X $d1^2$
= 0.083 X 5.87 X 18.15^2 = 1608.32 kg-cm/cm

Considering 4 nos bolt, tension per bolt = -22.452 / 4 = - 5.62 Ton

Considering 50% of tension will be carried to the web, BM at the face of the web (thickness = 8.6 mm),
=0.5 X 5.62 X 1000 X 5.57 = 15651.7 kg-cm

Moment per unit width =	15651.7 / 11.14 = 1405 kg-cm/cm			i.e. < 1608.32 kg-cm/cm
Hence, thickness of baseplate required, treqd = $\sqrt{4*M/(\phi * fy)}$ =		1.612	cm	f = 0.9

Hence, provide 450mm X 250 mm X 20 mm thk. baseplate

Dia of Bolt =	24	mm		
Ultimate Tensile strength of Bolts, Fu =	410	N/mm2		
Nominal Tensile Capcity of bolt, Rn =	13.91	Ton	= 0.75*f_u * A_b	[AISC Spec]
Design Tensile Capcity of bolt, ϕ * Rn =	10.43	Ton	*> 5.62 Ton, OK*	ϕ = 0.75

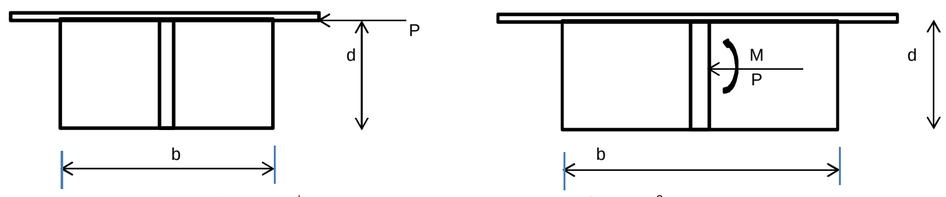
Hence, 4 nos 24mm dia. Bolts are required
However provide, 4 nos 25mm dia. Bolts

Global Project ID	**900707**
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Project Country	**KINGDOM OF SAUDI ARABIA**

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Design of Shear Key (As per AISC 2018 Specification - LSD)



Concrete Compressive Strength , f_c' = 35 N/mm^2
Concrete Pedestal Length = 600 mm
Concrete Pedestal Width = 600 mm
Max. Factored shear, P_u = 90 KN (From STAAD Output - L/C 109)
Equating the load to the bearing capacity of concrete, Area reqd. for shear key,
$A_{treqd.}$ = $P_u/(0.8 * fc')$ = 32.14 cm^2

Shear Key Width , b = 150 mm d_{reqd} = 21.43 mm
d_{provd} for shear Key = 150 mm (excluding Grout thk. = 25mm)
Thickness of shear Key, t = 36 mm
Projected Depth of Shear Key at Pier edge = 282 mm
Projected area excl. the area of Shear Key, A_v = 146700 mm^2
Shear cap. of conc. In front of shear key = $4*f*sqrt(fc')*A_v$ = 2083 KN > 90
Using Cantilever Model for Mid Plate,
Moment @ shear key base, M = V (G + d/2) = 11.25 KN-m
Z, $bt^2/4$ = M/ (f * Fy)
t_{reqd} = 35 mm < t provided - OK

For Wing Plate, Cantilever Moment = 1.6875 KN-m
t_{reqd} = 14 mm < t provided - OK

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Approved	PRI	PRI	PRI	PRI	Approved	PRI	PRI	PRI	PRI

4.6b Design of Base Plates for Columns

DESIGN OF BASE PLATE FOR COLUMN AT GRID 3410/D_3410/1a, 3410/D_3410/1d, 3410/F_3410/1a & 3410/F_3410/1d

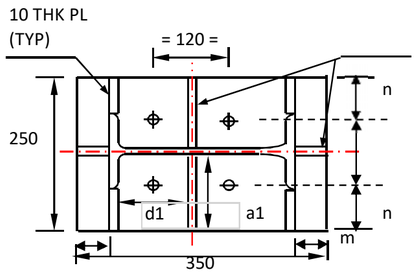
(AS PER SBC - 306 & AISC : 2018 LRFD Specification)

Here, Column Section =	IPE 300			
Grade of Conc. =	35	N/mm2		
Grade of Steel, Fy =	275	N/mm2		
Maximum axial load, Pu =	42.28	Ton	L/C	101
& Minimum axial Load =	-13.90	Ton (Ten.)	(Refer Staad File for Load Combs)	
Length of base plate (N)=	350	mm		
Width of base plate (B) =	250	mm		
Area of base plate, A1 =	0.0875	m2		
Length of concrete pedestal =	500	mm		
Width of concrete pedestal =	500	mm		
Area of concrete pedestal, A2 =	0.25	m2		

Design Bearing strength = 0.85*fc' = 29.75 N/mm2 (As per Section 10.9 of SBC 306)

Pp = 0.85*fc' *A1*SQRT(A2/A1) = 4400.1 KN (Eqn. 10.9-2 of SBC 306)

ϕc * Pp = 2640.1 KN > 422.84KN - OK (ϕ_c = 0.6)



m =	(N - 0.95d)/ 2 =	32.5	mm
n =	(B - 0.85bf)/2 =	61.25	mm
l =	2 (X) / (1+ (1-X)) =	0.39	
l n'=		20.8	mm
l =	max(m, n, ln')	61.3	mm

tmin, l*sqrt((2*Pu)/(0.9*Fy*B*N))= 12.1 mm (As per AISC-2018 : LRFD)

Actual area of base plate = 875 cm2

Uniform base pressure below base plate,fpu = 4.83 N/mm2

Moment capacity of Base Plate, M_{pu} = $f_{pu} * l^2/2$ = 9060 N-mm/mm width (As per AISC-2018 : LRFD)

Required thk. of Base Plate, t_{reqd} = $\sqrt{\frac{4 * Mpu}{\phi * F_y}}$ = 12.1 mm < tmin ϕ = 0.9

Now, a1 = 12.145 cm and d1 = 13.43 cm

Therefore, a1/d1 = 12.145/13.43 = 0.904

Hence, α_3 = 0.107 as per Mukhanov's chart

Now BM per cm = α3 X q X $d1^2$

= 0.107 X 4.83 X 13.43^2 = 934.61 kg-cm/cm

Considering 4 nos bolt, tension per bolt = -13.9 / 4 = - 3.48 Ton

Considering 50% of tension will be carried to the web, BM at the face of the web (thickness = 7.1 mm),

=0.5 X 3.48 X 1000 X 5.645 = 9822.3 kg-cm

Moment per unit width = 9822.3 / 11.29 = 870 kg-cm/cm i.e. < 934.61 kg-cm/cm

Hence, thickness of baseplate required, treqd = $\sqrt{4*M/(\phi * fy)}$ = 1.229 cm f = 0.9

Hence, provide 350mm X 250 mm X 20 mm thk. baseplate

Dia of Bolt =	24	mm		
Ultimate Tensile strength of Bolts, Fu =	410	N/mm2		
Nominal Tensile Capcity of bolt, Rn =	13.91	Ton	= 0.75*f_u * A_b	[AISC Spec]
Design Tensile Capcity of bolt, ϕ * Rn =	10.43	Ton	*> 3.48 Ton, OK*	ϕ = 0.75

Hence, 4 nos 24mm dia. Bolts are required

However provide, 4 nos 25mm dia. Bolts

MA'ADEN معادن شركة التعدين العربية السعودية Saudi Arabian Mining Company	Outotec	LARSEN & TOUBRO

DESIGN DOCUMENT

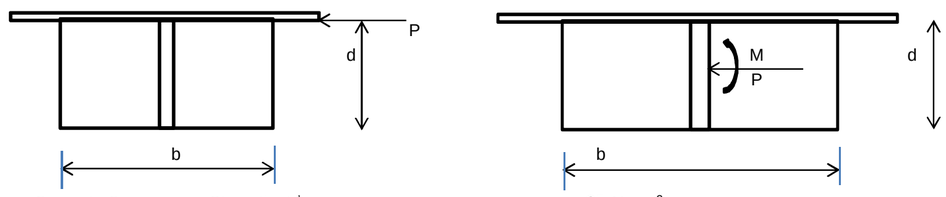
Ma'aden Document Number:	**MD-216-3380-EG-ST-CAL-1001**
Outotec Document Number:	**900707_GAU01_HL15_CED01_33802_OU100674671**
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Design of Shear Key (As per AISC 2018 Specification - LSD)



Concrete Compressive Strength , f_c' =		35	N/mm^2	
Concrete Pedestal Length =		500	mm	
Concrete Pedestal Width =		500	mm	
Max. Factored shear, P_u =		80	KN	(From STAAD Output - L/C 109)
Equating the load to the bearing capacity of concrete, Area reqd. for shear key,				
$A_{treqd.}$ =	$P_u/(0.8 * fc')$ =	28.57	cm^2	
Shear Key Width , b =		150	mm	d_{reqd} = 19.05 mm
d_{provd} for shear Key =		150	mm	(excluding Grout thk. = 25mm)
Thickness of shear Key, t =		36	mm	
Projected Depth of Shear Key at Pier edge =		232	mm	
Projected area excl. the area of Shear Key, A_v =		93500	mm^2	
Shear cap. of conc. In front of shear key =	$4*f*sqrt(fc')*A_v$ =	1328	KN	> 80
Using Cantilever Model for Mid Plate,				
Moment @ shear key base, M = V (G + d/2) =		10	KN-m	
	Z, $bt^2/4$ =	M/ (f * Fy)		
	t_{reqd} =	33	mm	< t provided - OK
For Wing Plate, Cantilever Moment =		1.5	KN-m	
	t_{reqd} =	13	mm	< t provided - OK

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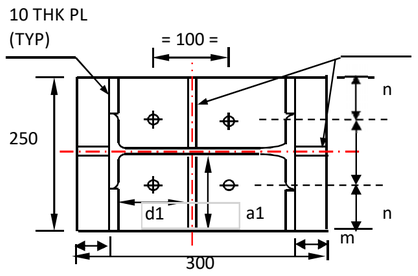
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Approved	PRI	PRI	PRI	PRI	Approved	PRI	PRI	PRI	PRI

4.6c Design of Base Plates for Columns

DESIGN OF BASE PLATE FOR COLUMN AT GRID 3410/D_3410/3a, 3410/F_3410/3a & 3410/D'_3410/2a

(AS PER SBC - 306 & AISC : 2018 LRFD Specification)

Here, Column Section =	IPE 240			
Grade of Conc. =	35	N/mm2		
Grade of Steel, Fy =	275	N/mm2		
Maximum axial load, Pu =	21.38	Ton	L/C	101
& Minimum axial Load =	-10.90	Ton (Ten.)	(Refer Staad File for Load Combs)	
Length of base plate (N)=	300	mm		
Width of base plate (B) =	250	mm		
Area of base plate, A1 =	0.075	m2		
Length of concrete pedestal =	500	mm		
Width of concrete pedestal =	500	mm		
Area of concrete pedestal, A2 =	0.25	m2		
Design Bearing strength =	0.85*fc' =	29.75 N/mm2	(As per Section 10.9 of SBC 306)	
Pp = 0.85*fc' *A1*SQRT(A2/A1) =		4073.7 KN	(Eqn. 10.9-2 of SBC 306)	
ϕc * Pp =	2444.2 KN	> 213.84KN - OK	(ϕ_c = 0.6)	



m =	(N - 0.95d)/ 2 =	36	mm
n =	(B - 0.85bf)/2 =	74	mm
l =	2 (X) / (1+ (1-X)) =	0.28	
l n'=		12.1	mm
l =	max(m, n, ln')	74.0	mm
tmin, l*sqrt((2*Pu)/(0.9*Fy*B*N))=		11.2	mm (As per AISC-2018 : LRFD)

Actual area of base plate =		750 cm2		
Uniform base pressure below base plate,fpu =		2.85 N/mm2		
Moment capacity of Base Plate, M_{pu} =	$f_{pu} * l^2/2$ =	7803 N-mm/mm width		(As per AISC-2018 : LRFD)
Required thk. of Base Plate, t_{reqd} =	$\sqrt{\frac{4 * Mpu}{\phi * F_y}}$	11.2 mm	< tmin	ϕ = 0.9

Now, a1 = 12.19 cm and d1 = 10.52 cm
Therefore, a1/d1 = 12.19/10.52 = 1.158
Hence, α_3 = 0.118 as per Mukhanov's chart
Now BM per cm = α3 X q X $d1^2$
= 0.118 X 2.85 X 10.52^2 = 373.55 kg-cm/cm

Considering 4 nos bolt, tension per bolt = -10.897 / 4 = - 2.73 Ton

Considering 50% of tension will be carried to the web, BM at the face of the web (thickness = 6.2 mm),
=0.5 X 2.73 X 1000 X 4.69 = 6401.85 kg-cm

Moment per unit width =	6401.85 / 9.38 = 682.5 kg-cm/cm			i.e. > 373.55 kg-cm/cm
Hence, thickness of baseplate required, treqd =	$\sqrt{4*M/(\phi * fy)}$ =	1.123 cm		f = 0.9

Hence, provide 300mm X 250 mm X 16 mm thk. baseplate

Dia of Bolt =	20 mm		
Ultimate Tensile strength of Bolts, Fu =	410 N/mm2		
Nominal Tensile Capcity of bolt, Rn =	9.66 Ton	= 0.75*f_u * A_b	[AISC Spec]
Design Tensile Capcity of bolt, ϕ * Rn =	7.25 Ton	*> 2.73 Ton, OK*	ϕ = 0.75

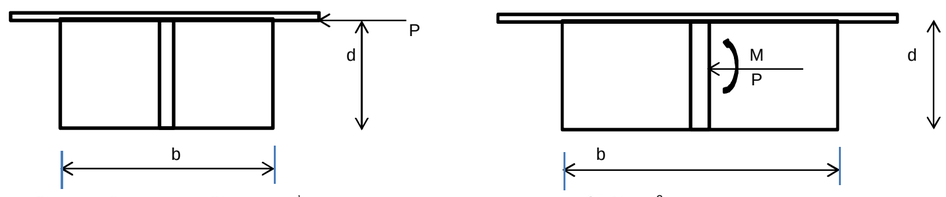
Hence, 4 nos 20mm dia. Bolts are required

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Design of Shear Key (As per AISC 2018 Specification - LSD)



Concrete Compressive Strength , f_c' =		35	N/mm^2	
Concrete Pedestal Length =		500	mm	
Concrete Pedestal Width =		500	mm	
Max. Factored shear, P_u =		50	KN	(From STAAD Output - L/C 109)
Equating the load to the bearing capacity of concrete, Area reqd. for shear key,				
$A_{treqd.}$ =	$P_u/(0.8 * fc')$ =	17.86	cm^2	
Shear Key Width , b =		125	mm	d_{reqd} = 14.29 mm
d_{provd} for shear Key =		150	mm	(excluding Grout thk. = 25mm)
Thickness of shear Key, t =		32	mm	
Projected Depth of Shear Key at Pier edge =		234	mm	
Projected area excl. the area of Shear Key, A_v =		98250	mm^2	
Shear cap. of conc. In front of shear key =	$4*f*sqrt(fc')*A_v$ =	1395	KN	> 50
Using Cantilever Model for Mid Plate,				
Moment @ shear key base, M = V (G + d/2) =		6.25	KN-m	
	Z, $bt^2/4$ =	M/ (f * Fy)		
	t_{reqd} =	29	mm	< t provided - OK
For Wing Plate, Cantilever Moment =		0.78125	KN-m	
	t_{reqd} =	11	mm	< t provided - OK

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4.7 Design of Staircase Stringer for Stair [As per SBC-Limit State Method]: (SBC 306 : 2007)

Design parameters:

Axial compression, P_u =	0.0	kN	
Axial tension, T =	0.0	kN	
Moment about z-axis, M_{uz} =	25.0	kN-m	
Moment about y-axis, M_{uy} =	5.0	kN-m	(assumed nominal moment)
Shear along y-axis, V_y =	15.0	kN	
Shear along z-axis, V_z =	0.0	kN	
Effective length about z-axis, k_zL_z =	6.75	m	
Effective length about y-axis, k_yL_y =	2	m	
Eff. lat. torsional buck. length, L_{LT} =	2	m	
Allowable slenderness ratio =	200		

Grade of steel =	S275 (Ult 450)		E =	200000	N/mm^2
Lateral support condition =	Unsupported	("1" for simply supported/"2" for cantilever)	G =	77200	N/mm^2
Loading applied at =	Shear centre		F_u =	450	N/mm^2
Loading direction w.r.t. shear centre =	Towards		f_{yf} =	275	N/mm^2
Beam/column type =	Simply suppo	("1" for simply supported/"2" for cantilever)	F_{yw} =	275	N/mm^2
Provision of tranv. stiff. =	At supports		Fr=	69	N/mm^2
Stiffener spacing, c =	2000 mm		$F_L(F_{yf}-F_r)$ =	206	N/mm^2

Sectional properties:

Section: UPN 240

D =	240	mm	A =	42.3	cm^2
b_{top} =	85	mm	I_y =	248	cm^4
t_{top} =	13	mm	r_z =	9.22	cm
b_{bot} =	85	mm	r_y =	2.42	cm
t_{bot} =	13	mm	Z_{ezT} =	300	cm^3
t_w =	9.5	mm	Z_{ey} =	39.6	cm^3
d =	184	mm	Z_{pz} =	358	cm^3
I_z =	3600	cm^4	Z_{py} =	75.7	cm^3
			J =	19.7	cm^4
			Cw =	2.21E+10	cm^6

Unstiffened Element	Flanges	Compact		*(Table 2.5-1)*		
		λ=	b/t =	6.5		
Compact Limit	λp=	$0.38(E/F_y)^{1/2}$	=		10.2	
Non Compact Limit	λr=	$0.83(E/F_L)^{1/2}$	=		25.9	
Unstiffened Element	Web	Compact				
		λ=	d/t_w =	19.4		
Compact Limit	λp=	$3.76(E/F_y)^{1/2}$	=		101.4	
Non Compact Limit	λr=	$5.70(E/F_y)^{1/2}$	=		153.7	

Hence the section is compact

Design Compressive Strength for Flexural Buckling *(Section 5.2)*

Design Compressive Strength	=	$f_c * P_n$				f_c =	0.85
	λc =	$(Kl/(rp))*(F_y/E)^{1/2}$ =				1.2	< 1.5
	F_{cr} =	$(0.658^{lc2})*Fy$		if		λc	< 1.5
	=	$(0.877/lc^2)*F$		if		λc	> 1.5
	F_{cr} =	147.59	N/mm^2				
	P_n =	624.33	KN				
	$\phi_c * P_n$ =	530.68	KN				

Check Bending about the Z-Z Axis *(Section 6.1)*

Actual Unbraced Length	L_b =	2500	mm			
Limit. lat. unbraced length,	$L_{p,}$ 1.76 *ry sqrt(E/Fy* ϕ) =	1149	mm		<	L_b
Limiting Buckling moment =	$M_{r=}$ $F_L * Z_{EZ}$ =	61.8	KN-m			
	X_1 =	26547.3				
	X_2 =	13.870				
Lr = (r_y* X_1/F_L)*SQRT(1+SQRT(1+ X_2 F_L^2)) =		86437.89	mm		>	L_b
Plastic moment of section =	M_p=	98.45	KN-m		<	1.5*M_y = 123.75
	C_b =	1.00				ϕ_b = 0.9
	M_n =	97.87	KN-m		<	M_p
Flexural Design Strength about the Z-Z Axis = $\varphi_b * M_n$ =				88.1	KN-m	

Check Bending about the Y-Y Axis

	M_p=	20.82	KN-m	>	1.5*M_y = 16.34
	M_n=	16.34	KN-m		
Flexural Design Strength about the Y-Y Axis = $\varphi_b * M_n$ =				14.7015	KN-m

Check For Biaxial Bending with Compression *(Section 8.1)*

$P_u/(\phi*P_n)$ =	0.00	< 0.2
(Pu/(2*Pn) + Muz/φb * Mnz+ Muy/φb * Mny) =	0.62	< 1

Check Shear *(Section 6.2)*

Clear distance between stiffeners	a =		0.001	mm	φv =	0.9
	h =		184	mm		
	a/h =		5.43478E-06			
	kv =	5				
	h/t_w =		19.37	< 1.1*(kvE/F_{yw})^0.5 =		66.332496
	V_n =		376.2	KN		
Design shear strength = $\varphi_v * V_n$ =	338.58	KN		>	15.0	

Check for Deflection

UDL on beam, w =	4.5	Kn/m		
Actual Deflection,	d_{act} = 5/384 * 4.5 * 6.75^4 / (200000 * 3600) =		16.89	mm
Allowable Deflection,	d_{allw} = L/ 360		= 18.75	mm
			> Actual deflection- Ok	

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4.8 Design of Monorail Beam Capacity 1T

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Design parameters:

Capacity =	1.0	T	
Self -wt =	0.5	T	
Wheel load per wheel set =	1.5	T	
Inpact load =	25 % of Wheel load for Electric Hoist		(Cl. 10.3.2 of SBC 301 : 2007)
Hence, Design wheel load =	1.9	T	
=	3.0	T	(approx)
Effective length , k_zL_z =	5.55	m	
Factored Maximum Bending moment, M_u =	6.3	T-m	= 1.5 x 3 x 5.55/ 4
Allowable slenderness ratio =	200		
Distance between wheels =	200	mm	

Material Properties :

Grade of steel =	S275 (Ult 450)	
E =	200000	N/mm^2
G =	77200	N/mm^2
F_u =	450	N/mm^2
f_{yf} =	275	N/mm^2
F_{yw} =	275	N/mm^2
Fr=	69	N/mm^2
$F_L(F_{yf}-F_r)$ =	206	N/mm^2

Sectional properties:

Section: IPE 300

D =	300	mm	A =	53.81	cm^2
b_{top} =	150	mm	I_y =	604	cm^4
t_{top} =	10.7	mm	r_z =	12.46	cm
b_{bot} =	150	mm	r_y =	3.35	cm
t_{bot} =	10.7	mm	Z_{ezT} =	557.1	cm^3
t_w =	7.1	mm	Z_{ey} =	80.5	cm^3
d =	248.6	mm	Z_{pz} =	628.4	cm^3
I_z =	8356	cm^4	Z_{py} =	125.2	cm^3
			J =	20.1	cm^4
			Cw =	1.26E+11	cm^6

Unstiffened Element	Flanges	Compact		*(Table 2.5-1)*	
		λ=	b/t = 7.0		
Compact Limit	λp=	$0.38(E/F_y)^{1/2}$	=	10.2	
Non Compact Limit	λr=	$0.83(E/F_L)^{1/2}$	=	25.9	
Unstiffened Element	Web	Compact			
		λ=	d/t_w = 35.0		
Compact Limit	λp=	$3.76(E/F_y)^{1/2}$	=	101.4	
Non Compact Limit	λr=	$5.70(E/F_y)^{1/2}$	=	153.7	

Hence the section is compact

Check Bending about the Z-Z Axis *(Section 6.1)*

Actual Unbraced Length	L_b =	6500	mm				
Limit. lat. unbraced length,	L_p, 1.76 *ry sqrt(E/Fyϕ) =	1590	mm	<		L_b	
Limiting Buckling moment =	$M_r = F_L * Z_{EZ}$ =	114.8	KN-m				
	X_1 =	16286.8	N/mm2				
	X_2 =	107.555643	mm4/N				
Lr = (r_y* X_1/F_L)*SQRT(1+SQRT(1+ X_2 F_L^2)) =		122449.16	mm	>		L_b	
Plastic moment of section =	M_p=	172.81	KN-m	<		1.5*M_y =	229.80
	C_b =	1.00				ϕ_b =	0.9
	M_n =	170.45	KN-m	<		M_p	
Flexural Design Strength about the Z-Z Axis = $\varphi_b * M_n$ =		153.4	KN-m		#VALUE!	M_u	
Design Flexural Compressive stress = $\varphi_b * M_n/Z_{pz}$ =		**244.2**	**N/mm2**				
Actual Flexural Compressive stress,	**SGI*10^4/=**	**100.3**	**N/mm2**		**< 244.2**	**N/mm2**	**OK**

Check for flange thickness :-

load per wheel =	3 /4 =	0.75	T	
Position of wheel from cl of web. =		50.00	mm	
Moment at the flange =	0.75x(50-/2)/1000x1.5 =	0.06	T-m	
eff. width of the fl.pl.(considering 30 deg dispersion) =		160.91	mm	
thickness of bott. Flg near cl .of web =		10.70	mm	
Calculated tensile stress at bottom flange =	*4/ (160.91*10.7^2) =			
		130.30	**N/mm2**	
Combined stress =	**100.3 + 130.3 =**	**230.60**	**N/mm2**	**< 244.2** **N/mm2** **OK**

Check for Deflection

Conc. load on beam, P =		50.00	kN	
Actual Deflection,	d_{act} = 1/48 * 50 * 5.55^3 / (200000 * 8356) =			10.66 mm
Allowable Deflection,	**d_{allw} = L/ 500 =**			**11.10 mm**
				> Actual deflection- Ok

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REFERENCES

[1] Structural Loading and Forces ; SBC Code 301: 2007
[2] Saudi Concrete Structures Code ; SBC 304 : 2018
[3] Civil,Structural and Architectural Design Criteria ; MD-216-1000-EG-ST-CRT-1001
[4] ACI 318 : 2014 Building Code Requirement of Structural Concrete
[5] Geotechnical Report No. MD-216-3000-EG-CV-RPT-1008

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ATTACHMENT 1 STAAD INPUT FOR EQUIPMENT SURROTING STRUCTURE OF ACIDULATION AREA

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 08-Apr-20
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER MTON
JOINT COORDINATES
1 0 0 0; 2 4.95 0 0; 3 10.2 0 0; 4 15.3 0 0; 5 19.8 0 0; 6 23.8 0 0; 7 0 0 3.3;
8 4.95 0 3.3; 9 10.2 0 3.3; 10 15.3 0 3.3; 11 19.8 0 3.3; 12 0 0 10.05;
13 23.8 0 6.21; 14 0 10.88 0; 15 4.95 10.88 0; 16 10.2 10.88 0;
17 15.3 10.88 0; 18 19.8 10.88 0; 19 0 10.88 3.3; 20 4.95 10.88 3.3;
21 10.2 10.88 3.3; 22 15.3 10.88 3.3; 23 19.8 10.88 3.3; 24 0 10.88 10.05;
25 3.5 10.88 3.3; 26 3.5 10.88 10.05; 27 0 10.88 1.75; 28 4.95 10.88 1.75;
29 10.2 10.88 1.75; 30 15.3 10.88 1.75; 31 19.8 10.88 1.75; 32 18.3 10.88 1.75;
33 19.05 10.88 1.75; 34 20.6 10.88 0; 35 20.6 10.88 3.3; 36 18.3 10.88 1.35;
37 19.8 10.88 1.35; 38 18.3 10.88 3.3; 39 19.05 10.88 3.3; 40 1.2 10.88 3.3;
41 1.2 10.88 10.05; 42 1.2 10.88 1.75; 43 0 8.05 0; 44 4.95 8.05 0;
45 10.2 8.05 0; 46 15.3 8.05 0; 47 19.8 8.05 0; 48 0 8.05 3.3;
49 4.95 8.05 3.3; 50 10.2 8.05 3.3; 51 15.3 8.05 3.3; 52 0 8.05 2.2;
53 0 8.05 10.05; 54 4.95 8.05 2.2; 55 10.2 8.05 2.2; 56 15.3 8.05 2.2;
57 19.8 8.05 2.2; 58 3.5 8.05 10.05; 59 19.8 8.05 3.3; 60 23.8 8.05 0;
61 23.8 8.05 5.55; 62 0 8.05 4.4; 63 4.95 8.05 4.4; 64 10.2 8.05 4.4;
65 15.3 8.05 4.4; 66 3.5 8.05 4.4; 67 19.8 8.05 1.95; 68 23.8 8.05 1.95;
69 21.8 8.05 1.95; 70 22.78 8.05 1.95; 71 23.8 8.05 4.65; 72 22.78 8.05 4.65;
73 18.3 8.05 4.4; 74 19.8 8.05 4.65; 75 19.8 8.05 3.67; 76 20.78 8.05 4.65;
77 21.965 8.05 4.65; 78 22.78 8.05 3.835; 79 22.78 8.05 2.895; 80 0 4.7 0;
81 4.95 4.7 0; 82 10.2 4.7 0; 83 15.3 4.7 0; 84 19.8 4.7 0; 85 0 4.7 3.3;
86 4.95 4.7 3.3; 87 10.2 4.7 3.3; 88 15.3 4.7 3.3; 89 0 4.7 1.15;
90 4.95 4.7 1.15; 91 10.2 4.7 1.15; 92 15.3 4.7 1.15; 93 0 4.7 10.05;
94 3.5 4.7 10.05; 95 19.8 4.7 3.3; 96 23.8 4.7 0; 97 23.8 4.7 5.55;
98 3.5 4.7 3.3; 99 19.8 4.7 1.15; 100 19.8 4.7 5.55; 101 23.8 4.7 3.3;
102 19.8 4.7 -2; 103 23.8 4.7 -2; 104 19.8 4.7 2.2; 105 19.8 4.7 4.425;
106 19.8 4.7 2.325; 107 22.06 4.7 2.325; 108 19.8 4.7 4.335;
109 22.06 4.7 4.335; 110 20.245 4.7 2.325; 111 20.245 4.7 4.335;
112 22.06 5.1 2.325; 113 22.06 5.1 4.335; 114 20.245 5.1 2.325;
115 20.245 5.1 4.335; 116 3.5 0 10.05; 117 0 6.375 10.05; 118 3.5 6.375 10.05;
119 0 2.085 10.05; 120 3.5 2.085 10.05; 121 0 2.085 3.3; 122 23.8 8.05 4.988;
123 23.8 4.7 4.988; 124 23.8 4.7 3.788; 125 23.8 8.05 3.788;
126 23.8 6.25 4.988; 127 23.8 6.25 3.788; 128 23.8 2.35 0; 129 0 10.88 6.675;
130 3.655 8.05 3.3; 131 3.655 8.05 2.2; 132 3.655 9 2.2; 133 0 10.88 0.9;
134 4.95 10.88 0.9; 135 10.2 10.88 0.9; 136 15.3 10.88 0.9; 137 19.8 10.88 0.9;
138 18.3 10.88 0.9; 139 20.6 10.88 1.75; 140 19.8 10.88 2.525;
141 19.05 10.88 0; 142 19.05 10.88 0.9; 143 18.3 10.88 0; 144 17.175 10.88 0;
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163 164 166 TO 169 173 TO 182 203 481 TO 495 497 498 500 501 503 TO 505 516 -
518 519 521 522 524 525 527 528 530 TO 533 546 547 549 550 552 553 560 561 -
563 565 568 572 573 579 582 584 586 588 590 592 594 597 604 645 648 651 654 -
710 793
_FLOOR-2_1 57 59 61 63 65 74 77 TO 80 84 86 87 89 TO 95 107 TO 124 204 371 -
372 TO 383 385 386 388 389 416 417 419 TO 421 427 430 431 433 434 436 441 -
444 445 447 448 450 453 458 459 461 462 464 469 472 475 TO 477 480 658 661 -
664 668 670 672 674 709 722 TO 733 762
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693 699 722 TO 724 746 TO 762
_FLOOR-3 14 TO 19 23 TO 27 29 31 33 35 37 TO 51 54 55 217 224 TO 236 239 242 -
243 245 TO 249 251 252 262 263 265 266 268 TO 271 280 284 285 287 288 290 -
295 297 299 302 306 308 310 312 TO 318 320 321 333 TO 339 348 349 351 353 -
354 TO 355 362 607 708
JOINT
END GROUP DEFINITION
DEFINE MATERIAL START
ISOTROPIC STEEL
E 2.09042e+007
POISSON 0.3
DENSITY 7.83341
ALPHA 1.2e-005
DAMP 0.03
TYPE STEEL
STRENGTH FY 25819.2 FU 41584 RY 1.5 RT 1.2
END DEFINE MATERIAL
MEMBER PROPERTY EUROPEAN
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98 TO 100 102 TO 106 108 TO 118 120 121 123 126 128 130 132 134 137 139 141 -
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461 462 464 469 473 TO 478 480 481 486 TO 495 497 498 500 501 503 TO 505 -
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708 TO 710 746 749 751 752 754 756 757 759 762 793 TABLE ST IPE240
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208 TO 210 369 370 720 790 TO 792 TABLE ST L90X90X8
183 TO 190 TABLE ST IPE240
202 205 TO 207 TABLE ST IPE240
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340 TO 347 350 352 356 TO 361 363 TO 366 384 387 390 TO 395 398 401 403 406 -
407 TO 412 414 415 418 422 TO 426 428 429 432 435 437 TO 440 442 443 446 449 -
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5 11 64 88 133 160 213 367 368 628 TABLE ST IPE400
3 4 9 10 60 62 69 71 129 131 138 140 623 625 TO 627 788 789 796 -
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629 633 637 638 646 647 TABLE LD L75X75X8 SP 0.01
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632 636 643 644 655 656 TABLE LD L90X90X8 SP 0.01
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700 TO 703 705 706 712 TO 719 TABLE LD L90X90X8 SP 0.01
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67 70 72 73 84 86 87 89 91 TO 95 219 TO 222 396 399 402 413 657 658 661 664 -
668 670 672 674 680 682 685 687 689 691 693 699 722 TO 733 747 748 750 753 -
755 758 760 761 TAPERED 0.4 0.01 0.4 0.25 0.025 0.25 0.025
223 659 660 662 663 665 TO 667 669 671 673 675 TO 679 681 683 684 686 688 -
690 692 694 TABLE ST IPE240
CONSTANTS
BETA 90 MEMB 1 TO 13 56 58 60 62 64 66 68 69 71 75 88 125 127 129 131 133 -
135 136 138 140 149 160 162 165 191 TO 195 197 199 200 211 213 215 367 368 -
598 599 622 TO 628 787 TO 789 796 797
MATERIAL STEEL ALL
SUPPORTS
1 TO 13 116 PINNED
245 299 FIXED BUT FY FZ MX MY MZ
MEMBER RELEASE
14 TO 24 29 30 33 35 37 39 40 43 44 46 50 51 54 55 57 61 63 65 67 70 -
72 TO 74 76 81 84 86 87 89 91 93 95 97 TO 101 103 TO 106 108 110 115 116 -
119 122 124 126 128 130 132 134 137 139 141 TO 147 150 153 155 157 TO 159 -
161 168 TO 172 178 180 196 198 201 202 205 216 218 225 227 229 231 235 244 -
307 319 320 370 372 374 376 378 380 401 475 479 481 490 TO 493 499 594 600 -
602 603 605 606 608 609 614 TO 621 629 TO 632 637 TO 644 646 647 649 650 -
652 653 655 656 696 698 700 TO 703 705 TO 707 712 TO 719 START MY MZ
28 30 42 45 46 49 TO 51 53 82 96 101 114 119 121 122 124 144 159 166 168 172 -
177 201 206 207 215 216 218 236 242 243 253 265 266 280 281 287 288 302 304 -
306 308 309 322 335 337 339 348 349 370 385 386 400 406 417 430 431 444 445 -
458 459 475 479 484 TO 489 497 498 506 516 518 519 527 528 546 547 560 561 -
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573 582 584 586 588 590 592 594 598 601 TO 603 605 606 608 TO 621 -
633 TO 638 641 TO 644 646 647 652 653 655 TO 657 674 682 687 693 700 TO 706 -
711 TO 719 746 752 757 799 END MY MZ
MEMBER TRUSS
212 214 237 238 240 241 246 250 254 TO 261 264 267 272 TO 279 282 283 286 -
289 291 TO 294 296 298 300 301 303 305 311 316 323 TO 332 340 TO 347 350 -
352 356 TO 361 363 TO 366 369 384 387 390 TO 395 398 403 407 TO 412 414 415 -
418 422 TO 426 428 429 432 435 437 TO 440 442 443 446 449 451 452 -
454 TO 457 460 463 465 TO 468 470 TO 472 496 502 507 TO 515 517 520 523 526 -
529 534 TO 545 548 551 554 TO 559 562 564 566 567 569 TO 571 574 TO 578 580 -
581 583 585 587 589 591 593 695 697 734 TO 745 763 TO 786 790 TO 792 794 -
795 800 TO 802
LOAD 1 LOADTYPE Dead  TITLE DEAD LOAD
SELFWEIGHT Y -1.1
*Grating Weight
FLOOR LOAD
_FLOOR-1 FLOAD -0.035 GY
_FLOOR-2_1 FLOAD -0.035 GY
_FLOOR-2_2 FLOAD -0.035 GY
_FLOOR-3 FLOAD -0.035 GY
*Hand rail weight
MEMBER LOAD
14 TO 18 23 24 29 31 35 37 39 40 43 44 48 TO 50 54 55 57 59 61 63 65 74 76 -
82 90 97 TO 100 102 111 TO 113 126 128 130 132 134 137 139 141 TO 143 151 -
163 164 166 168 174 176 203 204 217 224 236 242 245 247 249 262 265 268 270 -
280 284 287 295 299 302 306 310 313 315 317 321 333 335 337 338 348 349 351 -
353 TO 355 362 371 383 385 388 397 400 404 416 420 430 433 444 447 458 461 -
464 475 479 480 482 483 486 494 497 500 504 518 521 524 527 546 549 560 563 -
568 584 588 594 604 607 708 TO 710 746 749 752 754 759 793 UNI GY -0.015
*Stair Load
MEMBER LOAD
201 CON GY -0.25 6.3
201 CON GY -0.25 5.1
97 137 196 198 CON GY -0.25 0.35
137 196 198 CON GY -0.25 1.55
612 613 CON GY -0.25 0.2
612 613 CON GY -0.25 1.4
137 CON GY -0.25 1.95
137 CON GY -0.25 3.15
397 CON GY -0.25 0.8
397 CON GY -0.25 1.2
397 CON GY -0.25 2.4
53 CON GY -0.25 0.75
53 CON GY -0.25 1.95
204 CON GY -0.25 0
113 CON GY -0.25 0.338
*Monorail Load
31 32 34 36 38 CON GY -0.25 0.25
168 172 CON GY -0.25 1.27
563 CON GY -0.25 0.825
JOINT LOAD
*Pipe & Cable Load
14 TO 23 60 244 FY -5
MEMBER LOAD
*Equipment P32 & P33
46 CON GY -3 1
42 CON GY -3 0.25
*Equipment P45
JOINT LOAD
112 TO 115 FY -0.5
*Launder Load
343 344 346 348 349 355 357 361 365 367 FY -0.51
132 342 345 347 350 353 359 363 FY -0.46
340 351 FY -0.88
341 352 FY -0.74
339 369 FY -0.21
LOAD 2 LOADTYPE Live  TITLE LIVE-1 LOAD
FLOOR LOAD
_FLOOR-1 FLOAD -0.5 GY
_FLOOR-2_1 FLOAD -0.5 GY
_FLOOR-2_2 FLOAD -0.5 GY
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_FLOOR-3 FLOAD -0.5 GY
*Stair Load
MEMBER LOAD
201 CON GY -0.85 6.3
201 CON GY -0.85 5.1
97 137 196 198 CON GY -0.85 0.35
137 196 198 CON GY -0.85 1.55
612 613 CON GY -0.85 0.2
612 613 CON GY -0.85 1.4
137 CON GY -0.85 1.95
137 CON GY -0.85 3.15
397 CON GY -0.85 0.8
397 CON GY -0.85 1.2
397 CON GY -0.85 2.4
53 CON GY -0.85 0.75
53 CON GY -0.85 1.95
204 CON GY -0.85 0
113 CON GY -0.85 0.338
*Monorail Load
31 32 34 36 38 CON GY -2.5 0.25
168 172 CON GY -2.5 1.27
563 CON GY 2.5 0.825
LOAD 3 LOADTYPE None  TITLE LIVE-2 LOAD
MEMBER LOAD
*Equipment P32 & P33
46 CON GY -4 1
42 CON GY -4 0.25
LOAD 4 LOADTYPE None  TITLE LIVE-3 LOAD
*Equipment P45
JOINT LOAD
112 TO 115 FY -2
LOAD 5 LOADTYPE None  TITLE LIVE-4 LOAD
*Equipment P45
JOINT LOAD
112 TO 115 FY -3.2
LOAD 6 LOADTYPE Wind  TITLE WIND LOAD (+X DIR)
*Stair Load
MEMBER LOAD
97 137 196 198 CON GX 0.2 0.35
137 196 198 CON GX 0.2 1.55
612 613 CON GX 0.2 0.2
612 613 CON GX 0.2 1.4
137 CON GX 0.2 1.95
137 CON GX 0.2 3.15
397 CON GX 0.2 0.8
397 CON GX 0.2 1.2
397 CON GX 0.2 2.4
53 CON GX 0.2 0.75
53 CON GX 0.2 1.95
*Launder Load
JOINT LOAD
343 344 346 348 349 355 357 361 365 367 FX 0.007
132 342 345 347 350 353 359 363 FX 0.007
340 351 FX 0.007
341 352 FX 0.007
339 369 FX 0.007
MEMBER LOAD
*Equipment P32 & P33
46 CON GX 0.02 1
42 CON GX 0.02 0.25
*Equipment P45
JOINT LOAD
112 TO 115 FX 0.125
MEMBER LOAD
*Monorail Load
168 172 CON GX 0.36 1.27
563 CON GX 0.36 0.825
MEMBER LOAD
5 11 64 88 133 160 213 367 368 628 UNI GX 0.085
30 76 96 148 161 163 164 167 170 171 173 TO 177 179 201 203 404 482 483 553 -
568 595 654 UNI GX 0.077
```

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1 TO 4 6 TO 10 13 23 25 TO 27 31 32 34 36 38 47 56 58 60 62 66 68 69 71 74 -
77 TO 80 82 83 85 107 125 127 129 131 135 136 138 140 162 165 200 211 224 -
226 228 230 232 233 239 251 252 271 290 321 335 338 371 373 375 377 379 388 -
389 427 436 453 599 607 622 TO 627 787 TO 789 796 797 UNI GX 0.065
12 24 43 45 50 54 55 75 103 TO 106 111 TO 113 115 117 119 121 TO 124 143 145 -
146 TO 147 149 151 152 154 156 187 190 TO 195 197 199 204 215 217 235 236 -
337 348 349 351 353 TO 355 383 405 464 473 476 477 479 480 486 TO 489 504 -
505 530 531 594 597 598 604 645 648 651 751 756 762 UNI GX 0.052
51 183 TO 186 UNI GX 0.03
101 144 159 UNI GX 0.028
202 205 TO 207 223 659 660 662 663 665 TO 667 669 671 673 675 TO 679 681 683 -
684 686 688 690 692 694 UNI GX 0.026
208 TO 210 212 214 218 632 636 643 644 655 656 720 721 790 TO 792 UNI GX 0.02
602 603 605 606 608 609 629 TO 631 633 TO 635 637 TO 642 646 647 649 650 652 -
653 UNI GX 0.016
600 601 UNI GX 0.014
LOAD 7 LOADTYPE Wind  TITLE WIND LOAD (-X DIR)
*Stair Load
MEMBER LOAD
97 137 196 198 CON GX -0.2 0.35
137 196 198 CON GX -0.2 1.55
612 613 CON GX -0.2 0.2
612 613 CON GX -0.2 1.4
137 CON GX -0.2 1.95
137 CON GX -0.2 3.15
397 CON GX -0.2 0.8
397 CON GX -0.2 1.2
397 CON GX -0.2 2.4
53 CON GX -0.2 0.75
53 CON GX -0.2 1.95
*Launder Load
JOINT LOAD
343 344 346 348 349 355 357 361 365 367 FX -0.007
132 342 345 347 350 353 359 363 FX -0.007
340 351 FX -0.007
341 352 FX -0.007
339 369 FX -0.007
MEMBER LOAD
*Equipment P32 & P33
46 CON GX -0.02 1
42 CON GX -0.02 0.25
*Equipment P45
JOINT LOAD
112 TO 115 FX -0.125
MEMBER LOAD
*Monorail Load
168 172 CON GX -0.36 1.27
563 CON GX -0.36 0.825
MEMBER LOAD
5 11 64 88 133 160 213 367 368 628 UNI GX -0.085
30 76 96 148 161 163 164 167 170 171 173 TO 177 179 201 203 404 482 483 553 -
568 595 654 UNI GX -0.077
1 TO 4 6 TO 10 13 23 25 TO 27 31 32 34 36 38 47 56 58 60 62 66 68 69 71 74 -
77 TO 80 82 83 85 107 125 127 129 131 135 136 138 140 162 165 200 211 224 -
226 228 230 232 233 239 251 252 271 290 321 335 338 371 373 375 377 379 388 -
389 427 436 453 599 607 622 TO 627 787 TO 789 796 797 UNI GX -0.065
12 24 43 45 50 54 55 75 103 TO 106 111 TO 113 115 117 119 121 TO 124 143 145 -
146 TO 147 149 151 152 154 156 187 190 TO 195 197 199 204 215 217 235 236 -
337 348 349 351 353 TO 355 383 405 464 473 476 477 479 480 486 TO 489 504 -
505 530 531 594 597 598 604 645 648 651 751 756 762 UNI GX -0.052
51 183 TO 186 UNI GX -0.03
101 144 159 UNI GX -0.028
202 205 TO 207 223 659 660 662 663 665 TO 667 669 671 673 675 TO 679 681 683 -
684 686 688 690 692 694 UNI GX -0.026
208 TO 210 212 214 218 632 636 643 644 655 656 720 721 790 TO 792 UNI GX -0.02
602 603 605 606 608 609 629 TO 631 633 TO 635 637 TO 642 646 647 649 650 652 -
653 UNI GX -0.016
600 601 UNI GX -0.014
LOAD 8 LOADTYPE Wind  TITLE WIND LOAD(+Z DIR)
*Stair Load
MEMBER LOAD
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201 CON GZ 0.2 6.3
201 CON GZ 0.2 5.1
204 CON GZ 0.2 0
113 CON GZ 0.2 0.338
JOINT LOAD
*Launder Load
343 344 346 348 349 355 357 361 365 367 FZ 0.06
132 342 345 347 350 353 359 363 FZ 0.06
340 351 FZ 0.11
341 352 FZ 0.09
339 369 FZ 0.02
MEMBER LOAD
*Equipment P32 & P33
46 CON GZ 0.02 1
42 CON GZ 0.02 0.25
*Equipment P45
JOINT LOAD
112 TO 115 FZ 0.125
MEMBER LOAD
*Monorail Load
31 32 34 36 38 CON GZ 0.36 0.25
MEMBER LOAD
178 180 TO 182 484 485 UNI GZ 0.077
67 70 72 73 84 86 87 89 91 TO 95 219 TO 222 396 399 402 413 657 658 661 664 -
668 670 672 674 680 682 685 687 689 691 693 699 722 TO 733 747 748 750 753 -
755 758 760 761 UNI GZ 0.065
14 TO 22 28 29 33 35 37 39 TO 42 44 46 48 49 52 53 57 59 61 63 65 90 -
97 TO 100 102 108 TO 110 114 116 118 TO 120 122 124 126 128 130 132 134 137 -
139 141 142 153 155 157 158 166 168 169 172 173 179 183 TO 186 188 189 196 -
198 202 205 TO 207 223 225 227 229 231 234 242 243 245 247 TO 249 262 263 -
265 266 268 TO 270 280 281 284 285 287 288 295 297 299 302 304 306 308 310 -
312 TO 315 317 318 333 334 336 362 372 374 376 378 380 TO 382 385 386 397 -
400 416 417 419 TO 421 430 431 433 434 441 444 445 447 448 450 458 459 461 -
462 469 474 475 478 479 481 490 TO 495 497 498 500 501 503 516 518 519 521 -
522 524 525 527 528 532 533 546 547 549 550 552 560 561 563 565 572 573 579 -
582 584 586 588 590 592 594 596 597 607 612 613 659 660 662 663 665 TO 667 -
669 671 673 675 TO 679 681 683 684 686 688 690 692 694 708 TO 710 746 749 -
752 754 757 759 793 UNI GZ 0.052
5 11 64 88 133 160 213 367 368 628 UNI GZ 0.039
1 TO 4 6 TO 10 13 56 58 60 62 66 68 69 71 125 127 129 131 135 136 138 140 -
162 165 200 211 599 622 TO 627 787 TO 789 796 797 UNI GZ 0.032
12 75 149 191 TO 195 197 199 215 598 UNI GZ 0.026
369 370 700 TO 703 705 706 712 TO 719 UNI GZ 0.02
216 698 704 707 711 UNI GZ 0.016
81 150 610 611 614 TO 621 695 TO 697 798 TO 802 UNI GZ 0.014
LOAD 9 LOADTYPE Wind  TITLE WIND LOAD(-Z DIR)
*Stair Load
MEMBER LOAD
201 CON GZ -0.2 6.3
201 CON GZ -0.2 5.1
204 CON GZ -0.2 0
113 CON GZ -0.2 0.338
JOINT LOAD
*Launder Load
343 344 346 348 349 355 357 361 365 367 FZ -0.06
132 342 345 347 350 353 359 363 FZ -0.06
340 351 FZ -0.11
341 352 FZ -0.09
339 369 FZ -0.02
MEMBER LOAD
*Equipment P32 & P33
46 CON GZ -0.02 1
42 CON GZ -0.02 0.25
*Equipment P45
JOINT LOAD
112 TO 115 FZ -0.125
MEMBER LOAD
*Monorail Load
31 32 34 36 38 CON GZ -0.36 0.25
MEMBER LOAD
178 180 TO 182 484 485 UNI GZ -0.077
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67 70 72 73 84 86 87 89 91 TO 95 219 TO 222 396 399 402 413 657 658 661 664 -
668 670 672 674 680 682 685 687 689 691 693 699 722 TO 733 747 748 750 753 -
755 758 760 761 UNI GZ -0.065
14 TO 22 28 29 33 35 37 39 TO 42 44 46 48 49 52 53 57 59 61 63 65 90 -
97 TO 100 102 108 TO 110 114 116 118 TO 120 122 124 126 128 130 132 134 137 -
139 141 142 153 155 157 158 166 168 169 172 173 179 183 TO 186 188 189 196 -
198 202 205 TO 207 223 225 227 229 231 234 242 243 245 247 TO 249 262 263 -
265 266 268 TO 270 280 281 284 285 287 288 295 297 299 302 304 306 308 310 -
312 TO 315 317 318 333 334 336 362 372 374 376 378 380 TO 382 385 386 397 -
400 416 417 419 TO 421 430 431 433 434 441 444 445 447 448 450 458 459 461 -
462 469 474 475 478 479 481 490 TO 495 497 498 500 501 503 516 518 519 521 -
522 524 525 527 528 532 533 546 547 549 550 552 560 561 563 565 572 573 579 -
582 584 586 588 590 592 594 596 597 607 612 613 659 660 662 663 665 TO 667 -
669 671 673 675 TO 679 681 683 684 686 688 690 692 694 708 TO 710 746 749 -
752 754 757 759 793 UNI GZ -0.052
5 11 64 88 133 160 213 367 368 628 UNI GZ -0.039
1 TO 4 6 TO 10 13 56 58 60 62 66 68 69 71 125 127 129 131 135 136 138 140 -
162 165 200 211 599 622 TO 627 787 TO 789 796 797 UNI GZ -0.032
12 75 149 191 TO 195 197 199 215 598 UNI GZ -0.026
369 370 700 TO 703 705 706 712 TO 719 UNI GZ -0.02
216 698 704 707 711 UNI GZ -0.016
81 150 610 611 614 TO 621 695 TO 697 798 TO 801 UNI GZ -0.014
**********LOAD COMBS FOR DESIGN*******************************************
LOAD COMB 101 1.4*DL
1 1.4
LOAD COMB 102 1.2DL+1.6*LL-3
1 1.2 4 1.6
LOAD COMB 103 1.2DL+1.6*LL-1+1.6*LL-2+1.6*LL-4
1 1.2 2 1.6 3 1.6 5 1.6
LOAD COMB 104 1.2DL+0.8WL+X
1 1.2 6 0.8
LOAD COMB 105 1.2DL+0.8WL-X
1 1.2 7 0.8
LOAD COMB 106 1.2DL+0.5LL-1+0.5LL-2+0.5LL-4+1.6WL+X
1 1.2 2 0.5 3 0.5 5 0.5 6 1.6
LOAD COMB 107 1.2DL+0.5LL-1+0.5LL-2+0.5LL-4+1.6WL-X
1 1.2 2 0.5 3 0.5 5 0.5 7 1.6
LOAD COMB 108 0.9DL+1.6WL+X
1 0.9 6 1.6
LOAD COMB 109 0.9DL+1.6WL-X
1 0.9 7 1.6
LOAD COMB 110 1.2DL+0.8WL+Z
1 1.2 8 0.8
LOAD COMB 111 1.2DL+0.8WL-Z
1 1.2 9 0.8
LOAD COMB 112 1.2DL+0.5LL-1+0.5LL-2+0.5LL-4+1.6WL+Z
1 1.2 2 0.5 3 0.5 5 0.5 8 1.6
LOAD COMB 113 1.2DL+0.5LL-1+0.5LL-2+0.5LL-4+1.6WL-Z
1 1.2 2 0.5 3 0.5 5 0.5 9 1.6
LOAD COMB 114 0.9DL+1.6WL+Z
1 0.9 8 1.6
LOAD COMB 115 0.9DL+1.6WL-Z
1 0.9 9 1.6
**************************************************
*********LOAD COMBS FOR DEFLECTION*****************************************
LOAD COMB 201 DL+LL-3
1 1.0 4 1.0
LOAD COMB 202 DL+LL-1+LL-2+LL-4
1 1.0 2 1.0 3 1.0 5 1.0
LOAD COMB 203 DL+LL-1+LL-2+LL-4+WL+X
1 1.0 2 1.0 3 1.0 5 1.0 6 1.0
LOAD COMB 204 DL+LL-1+LL-2+LL-4+WL-X
1 1.0 2 1.0 3 1.0 5 1.0 7 1.0
LOAD COMB 205 DL+WL+X
1 1.0 6 1.0
LOAD COMB 206 DL+WL-X
1 1.0 7 1.0
LOAD COMB 207 DL+LL-1+LL-2+LL-4+WL+Z
1 1.0 2 1.0 3 1.0 5 1.0 8 1.0
LOAD COMB 208 DL+LL-1+LL-2+LL-4+WL-Z
1 1.0 2 1.0 3 1.0 5 1.0 9 1.0
```

```
LOAD COMB 209 DL+WL+Z
1 1.0 8 1.0
LOAD COMB 210 DL+WL-Z
1 1.0 9 1.0
**************************************************
PERFORM ANALYSIS PRINT STATICS CHECK
LOAD LIST 1 TO 7 101 TO 115
PRINT SUPPORT REACTION LIST 1 TO 369 371 TO 388 406 TO 408
LOAD LIST 1 TO 7 201 TO 210
PRINT JOINT DISPLACEMENTS LIST 14 TO 24 26 60 244
LOAD LIST 101 TO 115
PARAMETER 1
CODE LRFD
FYLD 27500 ALL
FU 41000 ALL
MAIN 200 MEMB 208 TO 210 212 214 216 218 237 238 240 241 244 246 250 -
253 TO 261 264 267 272 TO 279 282 283 286 289 291 TO 294 296 298 300 301 -
303 305 307 309 311 316 319 320 322 TO 332 339 TO 347 350 352 356 TO 361 -
363 TO 366 369 370 384 387 390 TO 395 398 401 403 406 TO 412 414 415 418 -
422 TO 426 428 429 432 435 437 TO 440 442 443 446 449 451 452 454 TO 457 -
460 463 465 TO 468 470 TO 472 496 499 502 506 TO 515 517 520 523 526 529 -
534 TO 545 548 551 554 TO 559 562 564 566 567 569 TO 571 574 TO 578 580 581 -
583 585 587 589 591 593 600 TO 603 605 606 608 609 614 TO 621 629 TO 644 -
646 647 649 650 652 653 655 656 695 TO 698 700 TO 707 711 TO 721 734 TO 745 -
763 TO 786 790 TO 792 798 TO 802
LZ 3.35 MEMB 149 193 TO 195 202 205 TO 207
LY 4.5 MEMB 1 2 5 TO 8 11 200 211 213 599 622 624 628
LZ 4.7 MEMB 6 191 199 211
LY 2.83 MEMB 64 88 367 368
LZ 2.83 MEMB 64 88 367 368
LZ 4.95 MEMB 14 19 28 33 40 52 57 67 84 90 91 97 102 126 137 153 166 219 225 -
306 308 310 312 TO 315 317 318 333 334 336 372 381 382 385 386 396 397 399 -
400 402 413 490 494 495 497 498 500 501 503 516 657 658 722 TO 724
LY 1.65 MEMB 67 84 91 219 396 399 402 413 657 658 722 TO 724
LZ 5.25 MEMB 15 20 35 59 70 86 92 93 128 139 155 220 221 227 284 285 287 288 -
295 297 299 302 304 374 416 417 419 TO 421 491 518 519 521 522 532 582 584 -
661 664 680 682 725 TO 727 747 748 750
LZ 5.1 MEMB 21 37 87 94 95 157 229 263 266 269 280 281 376 431 434 441 492 -
525 528 533 586 597 668 728 TO 730
LY 2 MEMB 70 86 92 93 220 221 661 664 680 682 725 TO 727 747 748 750
LY 2 MEMB 72 87 94 95 222 668 685 687 699 728 TO 730 753 755
LY 1.2 MEMB 151 152 154 156 164 167 176 203 483 604 645 648 651 654
LZ 2.6 MEMB 16 61 72 79 80 106 107 115 117 121 123 130 141 222 262 265 268 -
270 377 379 430 433 436 453 476 477 524 527 588 685 687 699 708 TO 710 753 -
755 762
LZ 4.5 MEMB 17 22 39 41 42 48 49 63 73 89 100 132 142 158 231 234 242 243 -
245 247 TO 249 378 444 445 447 448 450 478 493 546 547 549 550 552 590 592 -
670 672 674 689 691 693 731 TO 733 757 TO 761
LY 2.35 MEMB 73 89 100 478 670 672 674 689 691 693 731 TO 733 757 TO 761
LZ 4 MEMB 65 108 TO 110 114 116 118 120 134 169 173 178 180 TO 182 380 458 -
459 461 462 469 481 484 485 560 561 563 565 572 573 579 793
LZ 1.65 MEMB 143 145 TO 148 151 152 154 156 167 173 177 486 TO 489 504 505 -
530 531 553 597 604 645 648 651 654
LZ 6.25 MEMB 111 TO 113 163 164 174 176 203 204 383 464 473 480 482 483 568 595
LZ 3.3 MEMB 23 25 TO 27 31 32 34 36 38 43 47 74 76 TO 78 82 83 85 103 104 -
224 226 228 230 232 233 236 239 251 252 271 290 321 335 338 371 373 375 388 -
389 404 405 427 607 751
LY 2.7 MEMB 112 204
LZ 6.75 MEMB 24 54 217 348 349 351 353 TO 355
LZ 6.18 MEMB 71 88 140 160 368 789 797
LY 5.1 MEMB 696 798 799
KZ 2 MEMB 223 659 660 662 663 665 TO 667 669 671 673 675 TO 679 681 683 684 -
686 688 690 692 694
KY 2 MEMB 81 150 196 198 223 268 433 524 600 601 610 TO 613 629 TO 636 659 -
660 662 663 665 TO 667 669 671 673 675 TO 679 681 683 684 686 688 690 692 -
694 698 704 707 TO 711
KY 2.52 MEMB 244 253 307 309 319 322 401 406 499 506
TRACK 0 ALL
CHECK CODE ALL
STEEL TAKE OFF ALL
FINISH
```

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	A02	C01	D01
Date	08-05-2020	22-06-2020	21-07-2020	18-11-2023
Revision Text	FIRST ISSUE	Revised as per Employer Comments dated 11-06-2020	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	SGI	SGI	SGI	SGI
Checked	SYC	SYC	SYC	SYC
Approved	PRI	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2	3
Date	08-05-2020	22-06-2020	21-07-2020	18-11-2023
Revision Text	FIRST ISSUE	Revised as per Employer Comments dated 11-06-2020	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	SGI	SGI	SGI	SGI
Checked	SYC	SYC	SYC	SYC
Approved	PRI	PRI	PRI	PRI

ATTACHMENT 2 STAAD OUTPUT FOR EQUIPMENT SURROTING STRUCTURE OF ACIDULATION AREA

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**************************************************
*                                                *
*           STAAD.Pro V8i SELECTseries4          *
*           Version  20.07.09.31                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    JUN 16, 2020                *
*           Time=    15:39:28                    *
*                                                *
*      USER ID: Microsoft                        *
**************************************************
```

```
     1. STAAD SPACE
INPUT FILE: MD-216-3380-EG-ST-CAL-1001-RA02.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 08-APR-20
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. UNIT METER MTON
     7. JOINT COORDINATES
     8. 1 0 0 0; 2 4.95 0 0; 3 10.2 0 0; 4 15.3 0 0; 5 19.8 0 0; 6 23.8 0 0; 7 0 0 3.3
     9. 8 4.95 0 3.3; 9 10.2 0 3.3; 10 15.3 0 3.3; 11 19.8 0 3.3; 12 0 0 10.05
    10. 13 23.8 0 6.21; 14 0 10.88 0; 15 4.95 10.88 0; 16 10.2 10.88 0
    11. 17 15.3 10.88 0; 18 19.8 10.88 0; 19 0 10.88 3.3; 20 4.95 10.88 3.3
    12. 21 10.2 10.88 3.3; 22 15.3 10.88 3.3; 23 19.8 10.88 3.3; 24 0 10.88 10.05
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117. MEMBER INCIDENCES
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242. 765 392 391; 766 391 389; 767 392 217; 768 217 393; 769 389 394; 770 394 390
243. 771 395 396; 772 397 398; 773 397 394; 774 394 398; 775 395 399; 776 399 396
244. 777 397 371; 778 371 395; 779 400 401; 780 402 403; 781 402 399; 782 399 403
245. 783 400 405; 784 405 401; 785 402 404; 786 404 400; 787 406 49; 788 407 50
246. 789 408 51; 790 64 407; 791 63 406; 792 65 408; 793 409 284; 794 102 409
247. 795 103 409; 796 410 21; 797 411 22; 798 412 413; 799 413 410; 800 413 159
248. 801 159 412; 802 413 50
249. START GROUP DEFINITION
250. FLOOR
251. _FLOOR-1 126 128 130 132 134 137 139 141 TO 143 145 TO 148 151 TO 158 161 -
252. 163 164 166 TO 169 173 TO 182 203 481 TO 495 497 498 500 501 503 TO 505 516 -
253. 518 519 521 522 524 525 527 528 530 TO 533 546 547 549 550 552 553 560 561 -
254. 563 565 568 572 573 579 582 584 586 588 590 592 594 597 604 645 648 651 654 -
255. 710 793
256. _FLOOR-2_1 57 59 61 63 65 74 77 TO 80 84 86 87 89 TO 95 107 TO 124 204 371 -
257. 372 TO 383 385 386 388 389 416 417 419 TO 421 427 430 431 433 434 436 441 -
258. 444 445 447 448 450 453 458 459 461 462 464 469 472 475 TO 477 480 658 661 -
259. 664 668 670 672 674 709 722 TO 733 762
260. _FLOOR-2_2 67 70 72 73 76 82 TO 84 91 97 TO 100 102 TO 106 117 219 TO 222 -
261. 396 397 399 400 402 404 405 413 476 TO 479 657 658 680 682 685 687 689 691 -
262. 693 699 722 TO 724 746 TO 762
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263. _FLOOR-3 14 TO 19 23 TO 27 29 31 33 35 37 TO 51 54 55 217 224 TO 236 239 242 -
264. 243 245 TO 249 251 252 262 263 265 266 268 TO 271 280 284 285 287 288 290 -
265. 295 297 299 302 306 308 310 312 TO 318 320 321 333 TO 339 348 349 351 353 -
266. 354 TO 355 362 607 708
267. JOINT
268. END GROUP DEFINITION
269. DEFINE MATERIAL START
270. ISOTROPIC STEEL
271. E 2.09042E+007
272. POISSON 0.3
273. DENSITY 7.83341
274. ALPHA 1.2E-005
275. DAMP 0.03
276. TYPE STEEL
277. STRENGTH FY 25819.2 FU 41584 RY 1.5 RT 1.2
278. END DEFINE MATERIAL
279. MEMBER PROPERTY EUROPEAN
280. 12 75 149 191 TO 195 197 199 215 598 TABLE ST IPE240
281. 14 TO 22 24 28 29 33 35 37 39 TO 46 48 TO 50 52 TO 55 57 59 61 63 65 90 97 -
282. 98 TO 100 102 TO 106 108 TO 118 120 121 123 126 128 130 132 134 137 139 141 -
283. 142 TO 143 145 TO 147 151 TO 158 166 168 169 172 196 198 204 217 225 227 229 -
284. 231 234 TO 236 242 243 245 247 TO 249 262 263 265 266 268 TO 270 280 281 -
285. 284 285 287 288 295 297 299 302 304 306 308 310 312 TO 315 317 318 333 334 -
286. 336 337 348 349 351 353 TO 355 362 372 374 376 378 380 TO 383 385 386 397 -
287. 400 405 416 417 419 TO 421 430 431 433 434 441 444 445 447 448 450 458 459 -
288. 461 462 464 469 473 TO 478 480 481 486 TO 495 497 498 500 501 503 TO 505 -
289. 516 518 519 521 522 524 525 527 528 530 TO 533 546 547 549 550 552 560 561 -
290. 563 565 572 573 579 582 584 586 588 590 592 596 597 604 612 613 645 648 651 -
291. 708 TO 710 746 749 751 752 754 756 757 759 762 793 TABLE ST IPE240
292. 51 TABLE ST UPN140
293. 208 TO 210 369 370 720 790 TO 792 TABLE ST L90X90X8
294. 183 TO 190 TABLE ST IPE240
295. 202 205 TO 207 TABLE ST IPE240
296. 237 238 240 241 244 246 250 253 TO 261 264 267 272 TO 279 282 283 286 289 -
297. 291 TO 294 296 298 300 301 303 305 307 309 311 316 319 320 322 TO 332 339 -
298. 340 TO 347 350 352 356 TO 361 363 TO 366 384 387 390 TO 395 398 401 403 406 -
299. 407 TO 412 414 415 418 422 TO 426 428 429 432 435 437 TO 440 442 443 446 449 -
300. 451 452 454 TO 457 460 463 465 TO 468 470 TO 472 496 499 502 506 TO 515 517 -
301. 520 523 526 529 534 TO 545 548 551 554 TO 559 562 564 566 567 569 TO 571 -
302. 576 TO 578 580 581 583 585 587 589 591 593 734 TO 745 763 TO 785 -
303. 786 TABLE ST L65X65X6
304. 119 122 124 479 594 TABLE ST UPN240
305. 74 77 TO 80 82 83 85 107 371 373 375 377 379 388 389 427 436 -
306. 453 TABLE ST IPE400
307. 1 2 6 TO 8 13 56 58 66 68 125 127 135 136 162 165 200 211 599 622 624 -
308. 787 TABLE ST IPE300
309. 5 11 64 88 133 160 213 367 368 628 TABLE ST IPE400
310. 3 4 9 10 60 62 69 71 129 131 138 140 623 625 TO 627 788 789 796 -
311. 797 TABLE ST IPE300
312. 600 601 TABLE LD L65X65X6 SP 0.01
313. 629 633 637 638 646 647 TABLE LD L75X75X8 SP 0.01
314. 630 634 639 640 649 650 TABLE LD L75X75X8 SP 0.01
315. 631 635 641 642 652 653 TABLE LD L75X75X8 SP 0.01
316. 632 636 643 644 655 656 TABLE LD L90X90X8 SP 0.01
317. 81 150 610 611 614 TO 621 695 TO 697 798 TO 802 TABLE LD L65X65X6 SP 0.01
318. 698 704 707 711 TABLE LD L75X75X8 SP 0.01
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319. 700 TO 703 705 706 712 TO 719 TABLE LD L90X90X8 SP 0.01
320. 30 76 96 201 404 TABLE ST IPE360
321. 216 602 603 605 606 608 609 TABLE LD L75X75X8 SP 0.01
322. 218 TABLE LD L90X90X8 SP 0.01
323. 101 144 159 TABLE LD L130X130X10 SP 0.01
324. 212 214 721 TABLE LD L90X90X8 SP 0.01
325. 574 575 794 795 TABLE ST L90X90X8
326. 148 161 163 164 167 170 171 173 TO 182 203 482 TO 485 553 568 595 -
327. 654 TABLE ST IPE360
328. 23 25 TO 27 31 32 34 36 38 47 224 226 228 230 232 233 239 251 252 271 290 -
329. 321 335 338 607 TABLE ST IPE300
330. MEMBER PROPERTY EUROPEAN
331. 67 70 72 73 84 86 87 89 91 TO 95 219 TO 222 396 399 402 413 657 658 661 664 -
332. 668 670 672 674 680 682 685 687 689 691 693 699 722 TO 733 747 748 750 753 -
333. 755 758 760 761 TAPERED 0.4 0.01 0.4 0.25 0.025 0.25 0.025
334. 223 659 660 662 663 665 TO 667 669 671 673 675 TO 679 681 683 684 686 688 -
335. 690 692 694 TABLE ST IPE240
336. CONSTANTS
337. BETA 90 MEMB 1 TO 13 56 58 60 62 64 66 68 69 71 75 88 125 127 129 131 133 -
338. 135 136 138 140 149 160 162 165 191 TO 195 197 199 200 211 213 215 367 368 -
339. 598 599 622 TO 628 787 TO 789 796 797
340. MATERIAL STEEL ALL
341. SUPPORTS
342. 1 TO 13 116 PINNED
343. 245 299 FIXED BUT FY FZ MX MY MZ
344. MEMBER RELEASE
345. 14 TO 24 29 30 33 35 37 39 40 43 44 46 50 51 54 55 57 61 63 65 67 70 -
346. 72 TO 74 76 81 84 86 87 89 91 93 95 97 TO 101 103 TO 106 108 110 115 116 -
347. 119 122 124 126 128 130 132 134 137 139 141 TO 147 150 153 155 157 TO 159 -
348. 161 168 TO 172 178 180 196 198 201 202 205 216 218 225 227 229 231 235 244 -
349. 307 319 320 370 372 374 376 378 380 401 475 479 481 490 TO 493 499 594 600 -
350. 602 603 605 606 608 609 614 TO 621 629 TO 632 637 TO 644 646 647 649 650 -
351. 652 653 655 656 696 698 700 TO 703 705 TO 707 712 TO 719 START MY MZ
352. 28 30 42 45 46 49 TO 51 53 82 96 101 114 119 121 122 124 144 159 166 168 172 -
353. 177 201 206 207 215 216 218 236 242 243 253 265 266 280 281 287 288 302 304 -
354. 306 308 309 322 335 337 339 348 349 370 385 386 400 406 417 430 431 444 445 -
355. 458 459 475 479 484 TO 489 497 498 506 516 518 519 527 528 546 547 560 561 -
356. 573 582 584 586 588 590 592 594 598 601 TO 603 605 606 608 TO 621 -
357. 633 TO 638 641 TO 644 646 647 652 653 655 TO 657 674 682 687 693 700 TO 706 -
358. 711 TO 719 746 752 757 799 END MY MZ
359. MEMBER TRUSS
360. 212 214 237 238 240 241 246 250 254 TO 261 264 267 272 TO 279 282 283 286 -
361. 289 291 TO 294 296 298 300 301 303 305 311 316 323 TO 332 340 TO 347 350 -
362. 352 356 TO 361 363 TO 366 369 384 387 390 TO 395 398 403 407 TO 412 414 415 -
363. 418 422 TO 426 428 429 432 435 437 TO 440 442 443 446 449 451 452 -
364. 454 TO 457 460 463 465 TO 468 470 TO 472 496 502 507 TO 515 517 520 523 526 -
365. 529 534 TO 545 548 551 554 TO 559 562 564 566 567 569 TO 571 574 TO 578 580 -
366. 581 583 585 587 589 591 593 695 697 734 TO 745 763 TO 786 790 TO 792 794 -
367. 795 800 TO 802
368. LOAD 1 LOADTYPE DEAD  TITLE DEAD LOAD
369. SELFWEIGHT Y -1.1
370. *GRATING WEIGHT
371. FLOOR LOAD
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  372. _FLOOR-1 FLOAD -0.035 GY
**NOTE** about Floor/OneWay Loads/Weights.
  Please note that depending on the shape of the floor you may
  have to break up the FLOOR/ONEWAY LOAD into multiple commands.
  For details please refer to Technical Reference Manual
  Section 5.32.4 Note 6.

  373. _FLOOR-2_1 FLOAD -0.035 GY
  374. _FLOOR-2_2 FLOAD -0.035 GY
  375. _FLOOR-3 FLOAD -0.035 GY
  376. *HAND RAIL WEIGHT
  377. MEMBER LOAD
  378. 14 TO 18 23 24 29 31 35 37 39 40 43 44 48 TO 50 54 55 57 59 61 63 65 74 76 -
  379. 82 90 97 TO 100 102 111 TO 113 126 128 130 132 134 137 139 141 TO 143 151 -
  380. 163 164 166 168 174 176 203 204 217 224 236 242 245 247 249 262 265 268 270 -
  381. 280 284 287 295 299 302 306 310 313 315 317 321 333 335 337 338 348 349 351 -
  382. 353 TO 355 362 371 383 385 388 397 400 404 416 420 430 433 444 447 458 461 -
  383. 464 475 479 480 482 483 486 494 497 500 504 518 521 524 527 546 549 560 563 -
  384. 568 584 588 594 604 607 708 TO 710 746 749 752 754 759 793 UNI GY -0.015
  385. *STAIR LOAD
  386. MEMBER LOAD
  387. 201 CON GY -0.25 6.3
  388. 201 CON GY -0.25 5.1
  389. 97 137 196 198 CON GY -0.25 0.35
  390. 137 196 198 CON GY -0.25 1.55
  391. 612 613 CON GY -0.25 0.2
  392. 612 613 CON GY -0.25 1.4
  393. 137 CON GY -0.25 1.95
  394. 137 CON GY -0.25 3.15
  395. 397 CON GY -0.25 0.8
  396. 397 CON GY -0.25 1.2
  397. 397 CON GY -0.25 2.4
  398. 53 CON GY -0.25 0.75
  399. 53 CON GY -0.25 1.95
  400. 204 CON GY -0.25 0
  401. 113 CON GY -0.25 0.338
  402. *MONORAIL LOAD
  403. 31 32 34 36 38 CON GY -0.25 0.25
  404. 168 172 CON GY -0.25 1.27
  405. 563 CON GY -0.25 0.825
  406. JOINT LOAD
  407. *PIPE _CABLE LOAD
  408. 14 TO 23 60 244 FY -5
  409. MEMBER LOAD
  410. *EQUIPMENT P32 _P33
  411. 46 CON GY -3 1
  412. 42 CON GY -3 0.25
  413. *EQUIPMENT P45
  414. JOINT LOAD
  415. 112 TO 115 FY -0.5
  416. *LAUNDER LOAD
  417. 343 344 346 348 349 355 357 361 365 367 FY -0.51
  418. 132 342 345 347 350 353 359 363 FY -0.46
  419. 340 351 FY -0.88
  420. 341 352 FY -0.74
  421. 339 369 FY -0.21
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422. LOAD 2 LOADTYPE LIVE  TITLE LIVE-1 LOAD
423. FLOOR LOAD
424. _FLOOR-1 FLOAD -0.5 GY
425. _FLOOR-2_1 FLOAD -0.5 GY
426. _FLOOR-2_2 FLOAD -0.5 GY
427. _FLOOR-3 FLOAD -0.5 GY
428. *STAIR LOAD
429. MEMBER LOAD
430. 201 CON GY -0.85 6.3
431. 201 CON GY -0.85 5.1
432. 97 137 196 198 CON GY -0.85 0.35
433. 137 196 198 CON GY -0.85 1.55
434. 612 613 CON GY -0.85 0.2
435. 612 613 CON GY -0.85 1.4
436. 137 CON GY -0.85 1.95
437. 137 CON GY -0.85 3.15
438. 397 CON GY -0.85 0.8
439. 397 CON GY -0.85 1.2
440. 397 CON GY -0.85 2.4
441. 53 CON GY -0.85 0.75
442. 53 CON GY -0.85 1.95
443. 204 CON GY -0.85 0
444. 113 CON GY -0.85 0.338
445. *MONORAIL LOAD
446. 31 32 34 36 38 CON GY -2.5 0.25
447. 168 172 CON GY -2.5 1.27
448. 563 CON GY 2.5 0.825
449. LOAD 3 LOADTYPE NONE  TITLE LIVE-2 LOAD
450. MEMBER LOAD
451. *EQUIPMENT P32 _P33
452. 46 CON GY -4 1
453. 42 CON GY -4 0.25
454. LOAD 4 LOADTYPE NONE  TITLE LIVE-3 LOAD
455. *EQUIPMENT P45
456. JOINT LOAD
457. 112 TO 115 FY -2
458. LOAD 5 LOADTYPE NONE  TITLE LIVE-4 LOAD
459. *EQUIPMENT P45
460. JOINT LOAD
461. 112 TO 115 FY -3.2
462. LOAD 6 LOADTYPE WIND  TITLE WIND LOAD (+X DIR)
463. *STAIR LOAD
464. MEMBER LOAD
465. 97 137 196 198 CON GX 0.2 0.35
466. 137 196 198 CON GX 0.2 1.55
467. 612 613 CON GX 0.2 0.2
468. 612 613 CON GX 0.2 1.4
469. 137 CON GX 0.2 1.95
470. 137 CON GX 0.2 3.15
471. 397 CON GX 0.2 0.8
472. 397 CON GX 0.2 1.2
473. 397 CON GX 0.2 2.4
474. 53 CON GX 0.2 0.75
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475. 53 CON GX 0.2 1.95
476. *LAUNDER LOAD
477. JOINT LOAD
478. 343 344 346 348 349 355 357 361 365 367 FX 0.007
479. 132 342 345 347 350 353 359 363 FX 0.007
480. 340 351 FX 0.007
481. 341 352 FX 0.007
482. 339 369 FX 0.007
483. MEMBER LOAD
484. *EQUIPMENT P32 _P33
485. 46 CON GX 0.02 1
486. 42 CON GX 0.02 0.25
487. *EQUIPMENT P45
488. JOINT LOAD
489. 112 TO 115 FX 0.125
490. MEMBER LOAD
491. *MONORAIL LOAD
492. 168 172 CON GX 0.36 1.27
493. 563 CON GX 0.36 0.825
494. MEMBER LOAD
495. 5 11 64 88 133 160 213 367 368 628 UNI GX 0.085
496. 30 76 96 148 161 163 164 167 170 171 173 TO 177 179 201 203 404 482 483 553 -
497. 568 595 654 UNI GX 0.077
498. 1 TO 4 6 TO 10 13 23 25 TO 27 31 32 34 36 38 47 56 58 60 62 66 68 69 71 74 -
499. 77 TO 80 82 83 85 107 125 127 129 131 135 136 138 140 162 165 200 211 224 -
500. 226 228 230 232 233 239 251 252 271 290 321 335 338 371 373 375 377 379 388 -
501. 389 427 436 453 599 607 622 TO 627 787 TO 789 796 797 UNI GX 0.065
502. 12 24 43 45 50 54 55 75 103 TO 106 111 TO 113 115 117 119 121 TO 124 143 145 -
503. 146 TO 147 149 151 152 154 156 187 190 TO 195 197 199 204 215 217 235 236 -
504. 337 348 349 351 353 TO 355 383 405 464 473 476 477 479 480 486 TO 489 504 -
505. 505 530 531 594 597 598 604 645 648 651 751 756 762 UNI GX 0.052
506. 51 183 TO 186 UNI GX 0.03
507. 101 144 159 UNI GX 0.028
508. 202 205 TO 207 223 659 660 662 663 665 TO 667 669 671 673 675 TO 679 681 683 -
509. 684 686 688 690 692 694 UNI GX 0.026
510. 208 TO 210 212 214 218 632 636 643 644 655 656 720 721 790 TO 792 UNI GX 0.02
511. 602 603 605 606 608 609 629 TO 631 633 TO 635 637 TO 642 646 647 649 650 652 -
512. 653 UNI GX 0.016
513. 600 601 UNI GX 0.014
514. LOAD 7 LOADTYPE WIND  TITLE WIND LOAD (-X DIR)
515. *STAIR LOAD
516. MEMBER LOAD
517. 97 137 196 198 CON GX -0.2 0.35
518. 137 196 198 CON GX -0.2 1.55
519. 612 613 CON GX -0.2 0.2
520. 612 613 CON GX -0.2 1.4
521. 137 CON GX -0.2 1.95
522. 137 CON GX -0.2 3.15
523. 397 CON GX -0.2 0.8
524. 397 CON GX -0.2 1.2
525. 397 CON GX -0.2 2.4
526. 53 CON GX -0.2 0.75
527. 53 CON GX -0.2 1.95
528. *LAUNDER LOAD
529. JOINT LOAD
530. 343 344 346 348 349 355 357 361 365 367 FX -0.007
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531. 132 342 345 347 350 353 359 363 FX -0.007
532. 340 351 FX -0.007
533. 341 352 FX -0.007
534. 339 369 FX -0.007
535. MEMBER LOAD
536. *EQUIPMENT P32 _P33
537. 46 CON GX -0.02 1
538. 42 CON GX -0.02 0.25
539. *EQUIPMENT P45
540. JOINT LOAD
541. 112 TO 115 FX -0.125
542. MEMBER LOAD
543. *MONORAIL LOAD
544. 168 172 CON GX -0.36 1.27
545. 563 CON GX -0.36 0.825
546. MEMBER LOAD
547. 5 11 64 88 133 160 213 367 368 628 UNI GX -0.085
548. 30 76 96 148 161 163 164 167 170 171 173 TO 177 179 201 203 404 482 483 553 -
549. 568 595 654 UNI GX -0.077
550. 1 TO 4 6 TO 10 13 23 25 TO 27 31 32 34 36 38 47 56 58 60 62 66 68 69 71 74 -
551. 77 TO 80 82 83 85 107 125 127 129 131 135 136 138 140 162 165 200 211 224 -
552. 226 228 230 232 233 239 251 252 271 290 321 335 338 371 373 375 377 379 388 -
553. 389 427 436 453 599 607 622 TO 627 787 TO 789 796 797 UNI GX -0.065
554. 12 24 43 45 50 54 55 75 103 TO 106 111 TO 113 115 117 119 121 TO 124 143 145 -
555. 146 TO 147 149 151 152 154 156 187 190 TO 195 197 199 204 215 217 235 236 -
556. 337 348 349 351 353 TO 355 383 405 464 473 476 477 479 480 486 TO 489 504 -
557. 505 530 531 594 597 598 604 645 648 651 751 756 762 UNI GX -0.052
558. 51 183 TO 186 UNI GX -0.03
559. 101 144 159 UNI GX -0.028
560. 202 205 TO 207 223 659 660 662 663 665 TO 667 669 671 673 675 TO 679 681 683 -
561. 684 686 688 690 692 694 UNI GX -0.026
562. 208 TO 210 212 214 218 632 636 643 644 655 656 720 721 790 TO 792 UNI GX -0.02
563. 602 603 605 606 608 609 629 TO 631 633 TO 635 637 TO 642 646 647 649 650 652 -
564. 653 UNI GX -0.016
565. 600 601 UNI GX -0.014
566. LOAD 8 LOADTYPE WIND  TITLE WIND LOAD(+Z DIR)
567. *STAIR LOAD
568. MEMBER LOAD
569. 201 CON GZ 0.2 6.3
570. 201 CON GZ 0.2 5.1
571. 204 CON GZ 0.2 0
572. 113 CON GZ 0.2 0.338
573. JOINT LOAD
574. *LAUNDER LOAD
575. 343 344 346 348 349 355 357 361 365 367 FZ 0.06
576. 132 342 345 347 350 353 359 363 FZ 0.06
577. 340 351 FZ 0.11
578. 341 352 FZ 0.09
579. 339 369 FZ 0.02
580. MEMBER LOAD
581. *EQUIPMENT P32 _P33
582. 46 CON GZ 0.02 1
583. 42 CON GZ 0.02 0.25
584. *EQUIPMENT P45
585. JOINT LOAD
586. 112 TO 115 FZ 0.125
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587. MEMBER LOAD
588. *MONORAIL LOAD
589. 31 32 34 36 38 CON GZ 0.36 0.25
590. MEMBER LOAD
591. 178 180 TO 182 484 485 UNI GZ 0.077
592. 67 70 72 73 84 86 87 89 91 TO 95 219 TO 222 396 399 402 413 657 658 661 664 -
593. 668 670 672 674 680 682 685 687 689 691 693 699 722 TO 733 747 748 750 753 -
594. 755 758 760 761 UNI GZ 0.065
595. 14 TO 22 28 29 33 35 37 39 TO 42 44 46 48 49 52 53 57 59 61 63 65 90 -
596. 97 TO 100 102 108 TO 110 114 116 118 TO 120 122 124 126 128 130 132 134 137 -
597. 139 141 142 153 155 157 158 166 168 169 172 173 179 183 TO 186 188 189 196 -
598. 198 202 205 TO 207 223 225 227 229 231 234 242 243 245 247 TO 249 262 263 -
599. 265 266 268 TO 270 280 281 284 285 287 288 295 297 299 302 304 306 308 310 -
600. 312 TO 315 317 318 333 334 336 362 372 374 376 378 380 TO 382 385 386 397 -
601. 400 416 417 419 TO 421 430 431 433 434 441 444 445 447 448 450 458 459 461 -
602. 462 469 474 475 478 479 481 490 TO 495 497 498 500 501 503 516 518 519 521 -
603. 522 524 525 527 528 532 533 546 547 549 550 552 560 561 563 565 572 573 579 -
604. 582 584 586 588 590 592 594 596 597 607 612 613 659 660 662 663 665 TO 667 -
605. 669 671 673 675 TO 679 681 683 684 686 688 690 692 694 708 TO 710 746 749 -
606. 752 754 757 759 793 UNI GZ 0.052
607. 5 11 64 88 133 160 213 367 368 628 UNI GZ 0.039
608. 1 TO 4 6 TO 10 13 56 58 60 62 66 68 69 71 125 127 129 131 135 136 138 140 -
609. 162 165 200 211 599 622 TO 627 787 TO 789 796 797 UNI GZ 0.032
610. 12 75 149 191 TO 195 197 199 215 598 UNI GZ 0.026
611. 369 370 700 TO 703 705 706 712 TO 719 UNI GZ 0.02
612. 216 698 704 707 711 UNI GZ 0.016
613. 81 150 610 611 614 TO 621 695 TO 697 798 TO 802 UNI GZ 0.014
614. LOAD 9 LOADTYPE WIND  TITLE WIND LOAD(-Z DIR)
615. *STAIR LOAD
616. MEMBER LOAD
617. 201 CON GZ -0.2 6.3
618. 201 CON GZ -0.2 5.1
619. 204 CON GZ -0.2 0
620. 113 CON GZ -0.2 0.338
621. JOINT LOAD
622. *LAUNDER LOAD
623. 343 344 346 348 349 355 357 361 365 367 FZ -0.06
624. 132 342 345 347 350 353 359 363 FZ -0.06
625. 340 351 FZ -0.11
626. 341 352 FZ -0.09
627. 339 369 FZ -0.02
628. MEMBER LOAD
629. *EQUIPMENT P32 _P33
630. 46 CON GZ -0.02 1
631. 42 CON GZ -0.02 0.25
632. *EQUIPMENT P45
633. JOINT LOAD
634. 112 TO 115 FZ -0.125
635. MEMBER LOAD
636. *MONORAIL LOAD
637. 31 32 34 36 38 CON GZ -0.36 0.25
638. MEMBER LOAD
639. 178 180 TO 182 484 485 UNI GZ -0.077
640. 67 70 72 73 84 86 87 89 91 TO 95 219 TO 222 396 399 402 413 657 658 661 664 -
641. 668 670 672 674 680 682 685 687 689 691 693 699 722 TO 733 747 748 750 753 -
642. 755 758 760 761 UNI GZ -0.065
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643. 14 TO 22 28 29 33 35 37 39 TO 42 44 46 48 49 52 53 57 59 61 63 65 90 -
644. 97 TO 100 102 108 TO 110 114 116 118 TO 120 122 124 126 128 130 132 134 137 -
645. 139 141 142 153 155 157 158 166 168 169 172 173 179 183 TO 186 188 189 196 -
646. 198 202 205 TO 207 223 225 227 229 231 234 242 243 245 247 TO 249 262 263 -
647. 265 266 268 TO 270 280 281 284 285 287 288 295 297 299 302 304 306 308 310 -
648. 312 TO 315 317 318 333 334 336 362 372 374 376 378 380 TO 382 385 386 397 -
649. 400 416 417 419 TO 421 430 431 433 434 441 444 445 447 448 450 458 459 461 -
650. 462 469 474 475 478 479 481 490 TO 495 497 498 500 501 503 516 518 519 521 -
651. 522 524 525 527 528 532 533 546 547 549 550 552 560 561 563 565 572 573 579 -
652. 582 584 586 588 590 592 594 596 597 607 612 613 659 660 662 663 665 TO 667 -
653. 669 671 673 675 TO 679 681 683 684 686 688 690 692 694 708 TO 710 746 749 -
654. 752 754 757 759 793 UNI GZ -0.052
655. 5 11 64 88 133 160 213 367 368 628 UNI GZ -0.039
656. 1 TO 4 6 TO 10 13 56 58 60 62 66 68 69 71 125 127 129 131 135 136 138 140 -
657. 162 165 200 211 599 622 TO 627 787 TO 789 796 797 UNI GZ -0.032
658. 12 75 149 191 TO 195 197 199 215 598 UNI GZ -0.026
659. 369 370 700 TO 703 705 706 712 TO 719 UNI GZ -0.02
660. 216 698 704 707 711 UNI GZ -0.016
661. 81 150 610 611 614 TO 621 695 TO 697 798 TO 801 UNI GZ -0.014
662. **********LOAD COMBS FOR DESIGN*******************************************
663. LOAD COMB 101 1.4*DL
664. 1 1.4
665. LOAD COMB 102 1.2DL+1.6*LL-3
666. 1 1.2 4 1.6
667. LOAD COMB 103 1.2DL+1.6*LL-1+1.6*LL-2+1.6*LL-4
668. 1 1.2 2 1.6 3 1.6 5 1.6
669. LOAD COMB 104 1.2DL+0.8WL+X
670. 1 1.2 6 0.8
671. LOAD COMB 105 1.2DL+0.8WL-X
672. 1 1.2 7 0.8
673. LOAD COMB 106 1.2DL+0.5LL-1+0.5LL-2+0.5LL-4+1.6WL+X
674. 1 1.2 2 0.5 3 0.5 5 0.5 6 1.6
675. LOAD COMB 107 1.2DL+0.5LL-1+0.5LL-2+0.5LL-4+1.6WL-X
676. 1 1.2 2 0.5 3 0.5 5 0.5 7 1.6
677. LOAD COMB 108 0.9DL+1.6WL+X
678. 1 0.9 6 1.6
679. LOAD COMB 109 0.9DL+1.6WL-X
680. 1 0.9 7 1.6
681. LOAD COMB 110 1.2DL+0.8WL+Z
682. 1 1.2 8 0.8
683. LOAD COMB 111 1.2DL+0.8WL-Z
684. 1 1.2 9 0.8
685. LOAD COMB 112 1.2DL+0.5LL-1+0.5LL-2+0.5LL-4+1.6WL+Z
686. 1 1.2 2 0.5 3 0.5 5 0.5 8 1.6
687. LOAD COMB 113 1.2DL+0.5LL-1+0.5LL-2+0.5LL-4+1.6WL-Z
688. 1 1.2 2 0.5 3 0.5 5 0.5 9 1.6
689. LOAD COMB 114 0.9DL+1.6WL+Z
690. 1 0.9 8 1.6
691. LOAD COMB 115 0.9DL+1.6WL-Z
692. 1 0.9 9 1.6
693. **************************************************
694. *********LOAD COMBS FOR DEFLECTION*****************************************
695. LOAD COMB 201 DL+LL-3
696. 1 1.0 4 1.0
697. LOAD COMB 202 DL+LL-1+LL-2+LL-4
698. 1 1.0 2 1.0 3 1.0 5 1.0
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699. LOAD COMB 203 DL+LL-1+LL-2+LL-4+WL+X
700. 1 1.0 2 1.0 3 1.0 5 1.0 6 1.0
701. LOAD COMB 204 DL+LL-1+LL-2+LL-4+WL-X
702. 1 1.0 2 1.0 3 1.0 5 1.0 7 1.0
703. LOAD COMB 205 DL+WL+X
704. 1 1.0 6 1.0
705. LOAD COMB 206 DL+WL-X
706. 1 1.0 7 1.0
707. LOAD COMB 207 DL+LL-1+LL-2+LL-4+WL+Z
708. 1 1.0 2 1.0 3 1.0 5 1.0 8 1.0
709. LOAD COMB 208 DL+LL-1+LL-2+LL-4+WL-Z
710. 1 1.0 2 1.0 3 1.0 5 1.0 9 1.0
711. LOAD COMB 209 DL+WL+Z
712. 1 1.0 8 1.0
713. LOAD COMB 210 DL+WL-Z
714. 1 1.0 9 1.0
715. ************************************************
716. PERFORM ANALYSIS PRINT STATICS CHECK
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**412**	**NUMBER OF MEMBERS**	**802**
NUMBER OF PLATES	**0**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**16**

SOLVER USED IS THE IN-CORE ADVANCED SOLVER

TOTAL PRIMARY LOAD CASES = 9, TOTAL DEGREES OF FREEDOM = 2428

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 1
LOADTYPE DEAD TITLE DEAD LOAD

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.118576135E+02
Y = 0.875673829E+01
Z = 0.243790600E+01

***TOTAL APPLIED LOAD (MTON METE) SUMMARY (LOADING 1)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -134.98
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 329.08 MY= 0.00 MZ= -1600.59

***TOTAL REACTION LOAD(MTON METE) SUMMARY (LOADING 1)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 134.98
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -329.08 MY= 0.00 MZ= 1600.59

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 1)

	MAXIMUMS	AT NODE
X =	-3.38046E-02	78
Y =	-5.66249E-01	32
Z =	8.74325E-02	73
RX=	3.35068E-03	36
RY=	5.45378E-04	77
RZ=	2.39545E-03	33

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 2
LOADTYPE LIVE TITLE LIVE-1 LOAD

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.106648992E+02
Y = 0.739438203E+01
Z = 0.273981702E+01

***TOTAL APPLIED LOAD (MTON METE) SUMMARY (LOADING 2)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -147.06
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 402.92 MY= 0.00 MZ= -1568.40

***TOTAL REACTION LOAD(MTON METE) SUMMARY (LOADING 2)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 147.06
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -402.92 MY= 0.00 MZ= 1568.40

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 2)

	MAXIMUMS	AT NODE
X =	-4.06764E-02	78
Y =	-1.11603E+00	193
Z =	1.93734E-01	411
RX=	4.63034E-03	218
RY=	7.84016E-04	77
RZ=	5.81508E-03	214

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 3
LOADTYPE NONE TITLE LIVE-2 LOAD

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.193000002E+02
Y = 0.108800001E+02
Z = 0.154999994E+01

***TOTAL APPLIED LOAD (MTON METE) SUMMARY (LOADING 3)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -8.00
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 12.40 MY= 0.00 MZ= -154.40

***TOTAL REACTION LOAD(MTON METE) SUMMARY (LOADING 3)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 8.00
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -12.40 MY= 0.00 MZ= 154.40

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 3)

	MAXIMUMS	AT NODE
X =	-2.89124E-02	78
Y =	-5.78093E-01	32
Z =	7.43244E-02	76
RX=	4.23375E-03	36
RY=	4.57858E-04	77
RZ=	2.60855E-03	33

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 4
LOADTYPE NONE TITLE LIVE-3 LOAD

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.211525001E+02
Y = 0.509999988E+01
Z = 0.332999999E+01

***TOTAL APPLIED LOAD (MTON METE) SUMMARY (LOADING 4)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -8.00
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 26.64 MY= 0.00 MZ= -169.22

***TOTAL REACTION LOAD(MTON METE) SUMMARY (LOADING 4)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 8.00
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -26.64 MY= 0.00 MZ= 169.22

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 4)

	MAXIMUMS	AT NODE
X =	1.83940E-02	114
Y =	-1.57251E-01	113
Z =	4.63541E-02	299
RX=	-4.55200E-04	61
RY=	1.21544E-04	72
RZ=	-7.28312E-04	110

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 5
LOADTYPE NONE TITLE LIVE-4 LOAD

CENTER OF FORCE BASED ON Y FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.211525001E+02
Y = 0.509999988E+01
Z = 0.332999999E+01

***TOTAL APPLIED LOAD (MTON METE) SUMMARY (LOADING 5)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = -12.80
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 42.62 MY= 0.00 MZ= -270.75

***TOTAL REACTION LOAD(MTON METE) SUMMARY (LOADING 5)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 12.80
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -42.62 MY= 0.00 MZ= 270.75

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 5)

	MAXIMUMS	AT NODE
X =	2.94303E-02	114
Y =	-2.51601E-01	113
Z =	7.41666E-02	299
RX=	-7.28320E-04	61
RY=	1.94470E-04	72
RZ=	-1.16530E-03	110

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 6
LOADTYPE WIND TITLE WIND LOAD (+X DIR)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.103945645E+02
Y = 0.606091562E+01
Z = 0.360590221E+01

***TOTAL APPLIED LOAD (MTON METE) SUMMARY (LOADING 6)
SUMMATION FORCE-X = 26.48
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 0.00 MY= 95.50 MZ= -160.52

***TOTAL REACTION LOAD(MTON METE) SUMMARY (LOADING 6)
SUMMATION FORCE-X = -26.48
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 0.00 MY= -95.50 MZ= 160.52

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 6)

	MAXIMUMS	AT NODE
X =	6.74564E-01	302
Y =	-6.51121E-02	299
Z =	-4.22622E-01	245
RX=	3.97510E-04	73
RY=	-3.05248E-03	244
RZ=	-4.46506E-03	7

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 7
LOADTYPE WIND TITLE WIND LOAD (-X DIR)

CENTER OF FORCE BASED ON X FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.103945645E+02
Y = 0.606091562E+01
Z = 0.360590221E+01

***TOTAL APPLIED LOAD (MTON METE) SUMMARY (LOADING 7)
SUMMATION FORCE-X = -26.48
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 0.00 MY= -95.50 MZ= 160.52

***TOTAL REACTION LOAD(MTON METE) SUMMARY (LOADING 7)
SUMMATION FORCE-X = 26.48
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = 0.00

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 0.00 MY= 95.50 MZ= -160.52

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 7)

	MAXIMUMS	AT NODE
X =	-6.74564E-01	302
Y =	6.51121E-02	299
Z =	4.22622E-01	245
RX=	-3.97510E-04	73
RY=	3.05248E-03	244
RZ=	4.46506E-03	7

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 8
LOADTYPE WIND TITLE WIND LOAD(+Z DIR)

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.117824952E+02
Y = 0.719201254E+01
Z = 0.263724839E+01

***TOTAL APPLIED LOAD (MTON METE) SUMMARY (LOADING 8)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = 30.72

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 220.97 MY= -362.01 MZ= 0.00

***TOTAL REACTION LOAD(MTON METE) SUMMARY (LOADING 8)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = -30.72

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -220.97 MY= 362.01 MZ= 0.00

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 8)

	MAXIMUMS	AT NODE
X =	2.85708E-01	102
Y =	-2.82974E-01	65
Z =	1.68685E+00	412
RX=	8.42375E-03	353
RY=	-3.78743E-03	413
RZ=	4.62696E-04	402

STATIC LOAD/REACTION/EQUILIBRIUM SUMMARY FOR CASE NO. 9
LOADTYPE WIND TITLE WIND LOAD(-Z DIR)

CENTER OF FORCE BASED ON Z FORCES ONLY (METE).
(FORCES IN NON-GLOBAL DIRECTIONS WILL INVALIDATE RESULTS)

X = 0.117834566E+02
Y = 0.719020880E+01
Z = 0.263657414E+01

***TOTAL APPLIED LOAD (MTON METE) SUMMARY (LOADING 9)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = -30.69

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= -220.69 MY= 361.67 MZ= 0.00

***TOTAL REACTION LOAD(MTON METE) SUMMARY (LOADING 9)
SUMMATION FORCE-X = 0.00
SUMMATION FORCE-Y = 0.00
SUMMATION FORCE-Z = 30.69

SUMMATION OF MOMENTS AROUND THE ORIGIN-
MX= 220.69 MY= -361.67 MZ= 0.00

MAXIMUM DISPLACEMENTS (CM /RADIANS) (LOADING 9)

	MAXIMUMS	AT NODE
X =	-2.85646E-01	102
Y =	2.82406E-01	65
Z =	-1.62051E+00	412
RX=	-8.42232E-03	353
RY=	3.35873E-03	413
RZ=	-4.61585E-04	402

********** END OF DATA FROM INTERNAL STORAGE **********

717. LOAD LIST 1 TO 7 101 TO 115
718. PRINT SUPPORT REACTION LIST 1 TO 369 371 TO 388 406 TO 408

SUPPORT REACTIONS -UNIT MTON METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
1	1	0.00	7.29	-0.02	0.00	0.00	0.00
	2	0.00	6.38	-0.10	0.00	0.00	0.00
	3	0.00	0.06	0.01	0.00	0.00	0.00
	4	0.00	0.00	0.00	0.00	0.00	0.00
	5	0.00	0.00	0.00	0.00	0.00	0.00
	6	-0.18	-0.60	-0.12	0.00	0.00	0.00
	7	0.18	0.60	0.12	0.00	0.00	0.00
	101	0.00	10.20	-0.03	0.00	0.00	0.00
	102	0.00	8.75	-0.02	0.00	0.00	0.00
	103	0.00	19.06	-0.17	0.00	0.00	0.00
	104	-0.14	8.27	-0.12	0.00	0.00	0.00
	105	0.14	9.23	0.08	0.00	0.00	0.00
	106	-0.28	11.01	-0.26	0.00	0.00	0.00
	107	0.28	12.93	0.13	0.00	0.00	0.00
	108	-0.28	5.60	-0.21	0.00	0.00	0.00
	109	0.28	7.52	0.18	0.00	0.00	0.00
	110	0.00	-1.50	-2.52	0.00	0.00	0.00
	111	0.00	18.98	2.47	0.00	0.00	0.00
	112	0.00	-8.52	-5.06	0.00	0.00	0.00
	113	0.00	32.44	4.92	0.00	0.00	0.00
	114	0.00	-13.93	-5.01	0.00	0.00	0.00
	115	0.00	27.03	4.97	0.00	0.00	0.00
2	1	0.00	9.37	0.08	0.00	0.00	0.00
	2	-0.01	12.26	0.09	0.00	0.00	0.00
	3	0.00	-0.01	0.00	0.00	0.00	0.00
	4	0.00	-0.02	0.00	0.00	0.00	0.00
	5	0.00	-0.03	0.00	0.00	0.00	0.00
	6	-0.17	-0.14	-0.05	0.00	0.00	0.00
	7	0.17	0.14	0.05	0.00	0.00	0.00
	101	0.00	13.11	0.12	0.00	0.00	0.00
	102	0.00	11.21	0.09	0.00	0.00	0.00
	103	-0.01	30.79	0.24	0.00	0.00	0.00
	104	-0.14	11.13	0.06	0.00	0.00	0.00
	105	0.14	11.35	0.14	0.00	0.00	0.00
	106	-0.28	17.12	0.06	0.00	0.00	0.00
	107	0.27	17.57	0.22	0.00	0.00	0.00
	108	-0.28	8.21	-0.01	0.00	0.00	0.00
	109	0.28	8.65	0.16	0.00	0.00	0.00
	110	0.00	4.03	-1.68	0.00	0.00	0.00
	111	0.00	18.44	1.88	0.00	0.00	0.00
	112	0.00	2.92	-3.42	0.00	0.00	0.00
	113	-0.01	31.76	3.70	0.00	0.00	0.00
	114	0.00	-6.00	-3.49	0.00	0.00	0.00
	115	0.00	22.84	3.64	0.00	0.00	0.00
3	1	0.06	9.13	0.05	0.00	0.00	0.00
	2	0.01	11.65	0.03	0.00	0.00	0.00

SUPPORT REACTIONS -UNIT MTON METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
	3	0.01	-0.07	-0.01	0.00	0.00	0.00
	4	0.00	-0.05	0.00	0.00	0.00	0.00
	5	-0.01	-0.08	-0.01	0.00	0.00	0.00
	6	-4.71	-12.25	-0.04	0.00	0.00	0.00
	7	4.71	12.25	0.04	0.00	0.00	0.00
	101	0.08	12.78	0.07	0.00	0.00	0.00
	102	0.06	10.87	0.06	0.00	0.00	0.00
	103	0.08	29.36	0.08	0.00	0.00	0.00
	104	-3.70	1.15	0.03	0.00	0.00	0.00
	105	3.84	20.75	0.09	0.00	0.00	0.00
	106	-7.47	-2.90	0.00	0.00	0.00	0.00
	107	7.62	36.30	0.13	0.00	0.00	0.00
	108	-7.49	-11.39	-0.02	0.00	0.00	0.00
	109	7.60	27.82	0.11	0.00	0.00	0.00
	110	-0.34	1.21	-1.74	0.00	0.00	0.00
	111	0.49	20.67	1.86	0.00	0.00	0.00
	112	-0.76	-2.77	-3.54	0.00	0.00	0.00
	113	0.90	36.14	3.67	0.00	0.00	0.00
	114	-0.78	-11.26	-3.56	0.00	0.00	0.00
	115	0.88	27.65	3.65	0.00	0.00	0.00
4	1	-0.10	9.89	0.07	0.00	0.00	0.00
	2	0.01	11.80	0.08	0.00	0.00	0.00
	3	0.01	0.39	-0.01	0.00	0.00	0.00
	4	0.00	-0.10	-0.03	0.00	0.00	0.00
	5	-0.01	-0.17	-0.05	0.00	0.00	0.00
	6	-4.71	12.64	0.15	0.00	0.00	0.00
	7	4.71	-12.64	-0.15	0.00	0.00	0.00
	101	-0.14	13.85	0.10	0.00	0.00	0.00
	102	-0.13	11.70	0.04	0.00	0.00	0.00
	103	-0.11	31.10	0.11	0.00	0.00	0.00
	104	-3.89	21.98	0.20	0.00	0.00	0.00
	105	3.65	1.76	-0.03	0.00	0.00	0.00
	106	-7.66	38.10	0.33	0.00	0.00	0.00
	107	7.42	-2.34	-0.14	0.00	0.00	0.00
	108	-7.63	29.12	0.30	0.00	0.00	0.00
	109	7.45	-11.32	-0.17	0.00	0.00	0.00
	110	-0.53	5.45	-1.72	0.00	0.00	0.00
	111	0.29	18.27	1.88	0.00	0.00	0.00
	112	-0.94	5.05	-3.51	0.00	0.00	0.00
	113	0.71	30.69	3.69	0.00	0.00	0.00
	114	-0.91	-3.93	-3.54	0.00	0.00	0.00
	115	0.73	21.71	3.66	0.00	0.00	0.00
5	1	0.00	11.77	0.14	0.00	0.00	0.00
	2	0.00	8.61	0.19	0.00	0.00	0.00
	3	0.00	3.76	0.05	0.00	0.00	0.00
	4	0.00	0.38	0.06	0.00	0.00	0.00

SUPPORT REACTIONS -UNIT MTON METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
	5	0.00	0.61	0.10	0.00	0.00	0.00
	6	-0.24	0.39	0.08	0.00	0.00	0.00
	7	0.24	-0.39	-0.08	0.00	0.00	0.00
	101	0.00	16.48	0.20	0.00	0.00	0.00
	102	0.00	14.74	0.27	0.00	0.00	0.00
	103	0.00	34.88	0.71	0.00	0.00	0.00
	104	-0.19	14.44	0.23	0.00	0.00	0.00
	105	0.19	13.82	0.11	0.00	0.00	0.00
	106	-0.38	21.23	0.46	0.00	0.00	0.00
	107	0.39	20.00	0.21	0.00	0.00	0.00
	108	-0.38	11.21	0.25	0.00	0.00	0.00
	109	0.39	9.98	0.00	0.00	0.00	0.00
	110	0.00	-2.40	-3.86	0.00	0.00	0.00
	111	0.00	30.64	4.19	0.00	0.00	0.00
	112	-0.01	-12.44	-7.72	0.00	0.00	0.00
	113	0.01	53.64	8.39	0.00	0.00	0.00
	114	-0.01	-22.45	-7.93	0.00	0.00	0.00
	115	0.01	43.62	8.18	0.00	0.00	0.00
6	1	0.00	7.79	-0.03	0.00	0.00	0.00
	2	0.00	5.81	-0.13	0.00	0.00	0.00
	3	0.00	0.06	0.01	0.00	0.00	0.00
	4	0.00	1.12	0.04	0.00	0.00	0.00
	5	0.00	1.79	0.06	0.00	0.00	0.00
	6	-0.14	0.09	-0.01	0.00	0.00	0.00
	7	0.14	-0.09	0.01	0.00	0.00	0.00
	101	0.00	10.91	-0.04	0.00	0.00	0.00
	102	0.00	11.14	0.03	0.00	0.00	0.00
	103	0.00	21.61	-0.13	0.00	0.00	0.00
	104	-0.11	9.42	-0.04	0.00	0.00	0.00
	105	0.11	9.28	-0.03	0.00	0.00	0.00
	106	-0.22	13.32	-0.08	0.00	0.00	0.00
	107	0.22	13.04	-0.05	0.00	0.00	0.00
	108	-0.22	7.15	-0.04	0.00	0.00	0.00
	109	0.22	6.87	-0.01	0.00	0.00	0.00
	110	0.00	8.55	-0.30	0.00	0.00	0.00
	111	0.00	10.14	0.23	0.00	0.00	0.00
	112	0.00	11.59	-0.59	0.00	0.00	0.00
	113	0.00	14.77	0.46	0.00	0.00	0.00
	114	0.00	5.42	-0.55	0.00	0.00	0.00
	115	0.00	8.60	0.50	0.00	0.00	0.00
7	1	0.00	10.68	-0.09	0.00	0.00	0.00
	2	0.00	14.43	-0.14	0.00	0.00	0.00
	3	0.00	-0.06	0.01	0.00	0.00	0.00
	4	0.00	0.00	0.00	0.00	0.00	0.00
	5	0.00	0.00	0.00	0.00	0.00	0.00
	6	-0.29	0.52	-0.12	0.00	0.00	0.00

SUPPORT REACTIONS -UNIT MTON METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
	7	0.29	-0.52	0.12	0.00	0.00	0.00
	101	0.00	14.96	-0.13	0.00	0.00	0.00
	102	0.00	12.82	-0.11	0.00	0.00	0.00
	103	0.00	35.81	-0.32	0.00	0.00	0.00
	104	-0.24	13.24	-0.21	0.00	0.00	0.00
	105	0.24	12.40	-0.01	0.00	0.00	0.00
	106	-0.47	20.84	-0.37	0.00	0.00	0.00
	107	0.47	19.17	0.02	0.00	0.00	0.00
	108	-0.47	10.45	-0.28	0.00	0.00	0.00
	109	0.47	8.78	0.11	0.00	0.00	0.00
	110	0.00	22.12	-2.63	0.00	0.00	0.00
	111	0.00	3.53	2.41	0.00	0.00	0.00
	112	0.00	38.60	-5.22	0.00	0.00	0.00
	113	0.00	1.43	4.87	0.00	0.00	0.00
	114	0.00	28.21	-5.13	0.00	0.00	0.00
	115	0.00	-8.96	4.96	0.00	0.00	0.00
8	1	0.00	14.30	0.00	0.00	0.00	0.00
	2	0.00	15.66	0.06	0.00	0.00	0.00
	3	0.00	0.01	0.00	0.00	0.00	0.00
	4	0.00	0.02	0.00	0.00	0.00	0.00
	5	0.00	0.03	0.00	0.00	0.00	0.00
	6	-0.18	0.11	-0.05	0.00	0.00	0.00
	7	0.18	-0.11	0.05	0.00	0.00	0.00
	101	0.00	20.02	-0.01	0.00	0.00	0.00
	102	0.00	17.19	-0.01	0.00	0.00	0.00
	103	0.00	42.28	0.08	0.00	0.00	0.00
	104	-0.14	17.25	-0.05	0.00	0.00	0.00
	105	0.14	17.08	0.04	0.00	0.00	0.00
	106	-0.28	25.18	-0.06	0.00	0.00	0.00
	107	0.28	24.84	0.10	0.00	0.00	0.00
	108	-0.28	13.04	-0.08	0.00	0.00	0.00
	109	0.28	12.70	0.08	0.00	0.00	0.00
	110	0.00	24.37	-1.79	0.00	0.00	0.00
	111	0.00	9.97	1.78	0.00	0.00	0.00
	112	0.00	39.43	-3.55	0.00	0.00	0.00
	113	0.00	10.62	3.59	0.00	0.00	0.00
	114	0.00	27.29	-3.57	0.00	0.00	0.00
	115	0.00	-1.52	3.56	0.00	0.00	0.00
9	1	0.08	12.74	-0.04	0.00	0.00	0.00
	2	-0.06	12.26	-0.02	0.00	0.00	0.00
	3	-0.02	0.04	-0.01	0.00	0.00	0.00
	4	-0.02	0.01	0.00	0.00	0.00	0.00
	5	-0.03	0.01	-0.01	0.00	0.00	0.00
	6	-4.69	-11.66	-0.03	0.00	0.00	0.00
	7	4.69	11.66	0.03	0.00	0.00	0.00
	101	0.12	17.83	-0.06	0.00	0.00	0.00

SUPPORT REACTIONS -UNIT MTON METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
	102	0.07	15.30	-0.06	0.00	0.00	0.00
	103	-0.07	34.97	-0.11	0.00	0.00	0.00
	104	-3.65	5.96	-0.08	0.00	0.00	0.00
	105	3.86	24.61	-0.03	0.00	0.00	0.00
	106	-7.46	2.78	-0.13	0.00	0.00	0.00
	107	7.56	40.09	-0.02	0.00	0.00	0.00
	108	-7.43	-7.19	-0.10	0.00	0.00	0.00
	109	7.59	30.12	0.02	0.00	0.00	0.00
	110	0.27	24.17	-1.85	0.00	0.00	0.00
	111	-0.07	6.42	1.75	0.00	0.00	0.00
	112	0.38	39.21	-3.68	0.00	0.00	0.00
	113	-0.29	3.70	3.53	0.00	0.00	0.00
	114	0.41	29.23	-3.64	0.00	0.00	0.00
	115	-0.26	-6.27	3.56	0.00	0.00	0.00
10	1	-0.08	12.32	-0.05	0.00	0.00	0.00
	2	-0.07	11.38	-0.01	0.00	0.00	0.00
	3	-0.02	0.53	-0.03	0.00	0.00	0.00
	4	-0.02	0.15	-0.03	0.00	0.00	0.00
	5	-0.03	0.24	-0.05	0.00	0.00	0.00
	6	-4.69	11.32	0.14	0.00	0.00	0.00
	7	4.69	-11.32	-0.14	0.00	0.00	0.00
	101	-0.11	17.24	-0.07	0.00	0.00	0.00
	102	-0.12	15.01	-0.10	0.00	0.00	0.00
	103	-0.27	34.21	-0.19	0.00	0.00	0.00
	104	-3.85	23.83	0.06	0.00	0.00	0.00
	105	3.66	5.72	-0.17	0.00	0.00	0.00
	106	-7.66	38.96	0.13	0.00	0.00	0.00
	107	7.36	2.74	-0.33	0.00	0.00	0.00
	108	-7.58	29.19	0.19	0.00	0.00	0.00
	109	7.44	-7.02	-0.27	0.00	0.00	0.00
	110	0.07	22.05	-1.84	0.00	0.00	0.00
	111	-0.26	7.51	1.73	0.00	0.00	0.00
	112	0.19	35.40	-3.67	0.00	0.00	0.00
	113	-0.48	6.32	3.47	0.00	0.00	0.00
	114	0.26	25.63	-3.62	0.00	0.00	0.00
	115	-0.40	-3.44	3.53	0.00	0.00	0.00
11	1	0.00	15.50	0.00	0.00	0.00	0.00
	2	0.00	17.66	0.24	0.00	0.00	0.00
	3	0.00	3.35	-0.01	0.00	0.00	0.00
	4	0.00	4.92	0.07	0.00	0.00	0.00
	5	0.00	7.86	0.11	0.00	0.00	0.00
	6	-0.24	-0.46	0.09	0.00	0.00	0.00
	7	0.24	0.46	-0.09	0.00	0.00	0.00
	101	0.00	21.70	0.00	0.00	0.00	0.00
	102	0.00	26.46	0.11	0.00	0.00	0.00
	103	0.00	64.81	0.53	0.00	0.00	0.00

SUPPORT REACTIONS -UNIT MTON METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
	104	-0.19	18.23	0.07	0.00	0.00	0.00
	105	0.19	18.97	-0.07	0.00	0.00	0.00
	106	-0.38	32.31	0.31	0.00	0.00	0.00
	107	0.38	33.77	0.03	0.00	0.00	0.00
	108	-0.38	13.22	0.14	0.00	0.00	0.00
	109	0.38	14.68	-0.14	0.00	0.00	0.00
	110	0.00	35.13	-4.19	0.00	0.00	0.00
	111	0.00	2.09	4.19	0.00	0.00	0.00
	112	0.00	66.09	-8.21	0.00	0.00	0.00
	113	0.00	0.02	8.54	0.00	0.00	0.00
	114	0.00	47.00	-8.38	0.00	0.00	0.00
	115	0.00	-19.07	8.37	0.00	0.00	0.00
12	1	0.31	3.66	0.00	0.00	0.00	0.00
	2	0.89	7.61	0.00	0.00	0.00	0.00
	3	0.00	0.00	0.00	0.00	0.00	0.00
	4	0.00	0.01	0.00	0.00	0.00	0.00
	5	0.00	0.01	0.00	0.00	0.00	0.00
	6	-2.32	-8.32	0.00	0.00	0.00	0.00
	7	2.32	8.32	0.00	0.00	0.00	0.00
	101	0.44	5.12	0.00	0.00	0.00	0.00
	102	0.37	4.40	0.00	0.00	0.00	0.00
	103	1.81	16.59	-0.01	0.00	0.00	0.00
	104	-1.48	-2.27	0.00	0.00	0.00	0.00
	105	2.23	11.05	0.00	0.00	0.00	0.00
	106	-2.89	-5.12	-0.01	0.00	0.00	0.00
	107	4.53	21.52	-0.01	0.00	0.00	0.00
	108	-3.43	-10.03	0.00	0.00	0.00	0.00
	109	3.99	16.61	0.00	0.00	0.00	0.00
	110	0.57	6.54	-0.04	0.00	0.00	0.00
	111	0.18	2.25	0.04	0.00	0.00	0.00
	112	1.22	12.50	-0.08	0.00	0.00	0.00
	113	0.42	3.91	0.07	0.00	0.00	0.00
	114	0.68	7.59	-0.08	0.00	0.00	0.00
	115	-0.12	-1.00	0.08	0.00	0.00	0.00
13	1	0.00	7.81	-0.12	0.00	0.00	0.00
	2	0.00	6.57	-0.29	0.00	0.00	0.00
	3	0.00	-0.06	0.00	0.00	0.00	0.00
	4	0.00	1.58	-0.10	0.00	0.00	0.00
	5	0.00	2.53	-0.16	0.00	0.00	0.00
	6	-0.12	-0.01	-0.02	0.00	0.00	0.00
	7	0.12	0.01	0.02	0.00	0.00	0.00
	101	0.00	10.94	-0.16	0.00	0.00	0.00
	102	0.00	11.91	-0.30	0.00	0.00	0.00
	103	0.00	23.85	-0.86	0.00	0.00	0.00
	104	-0.10	9.37	-0.16	0.00	0.00	0.00
	105	0.10	9.39	-0.12	0.00	0.00	0.00

SUPPORT REACTIONS -UNIT MTON METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
	106	-0.20	13.88	-0.40	0.00	0.00	0.00
	107	0.19	13.92	-0.33	0.00	0.00	0.00
	108	-0.20	7.01	-0.14	0.00	0.00	0.00
	109	0.19	7.06	-0.07	0.00	0.00	0.00
	110	0.00	10.17	-0.37	0.00	0.00	0.00
	111	0.00	8.58	0.09	0.00	0.00	0.00
	112	0.00	15.49	-0.82	0.00	0.00	0.00
	113	0.00	12.31	0.09	0.00	0.00	0.00
	114	0.00	8.62	-0.56	0.00	0.00	0.00
	115	0.00	5.44	0.35	0.00	0.00	0.00
116	1	-0.32	2.74	0.01	0.00	0.00	0.00
	2	-0.89	4.98	0.02	0.00	0.00	0.00
	3	0.00	0.00	0.00	0.00	0.00	0.00
	4	0.00	0.00	0.00	0.00	0.00	0.00
	5	0.00	-0.01	0.00	0.00	0.00	0.00
	6	-2.32	8.39	0.00	0.00	0.00	0.00
	7	2.32	-8.39	0.00	0.00	0.00	0.00
	101	-0.45	3.83	0.01	0.00	0.00	0.00
	102	-0.38	3.28	0.01	0.00	0.00	0.00
	103	-1.81	11.24	0.04	0.00	0.00	0.00
	104	-2.24	9.99	0.01	0.00	0.00	0.00
	105	1.47	-3.43	0.01	0.00	0.00	0.00
	106	-4.54	19.19	0.02	0.00	0.00	0.00
	107	2.88	-7.65	0.02	0.00	0.00	0.00
	108	-4.00	15.89	0.01	0.00	0.00	0.00
	109	3.42	-10.96	0.01	0.00	0.00	0.00
	110	-0.19	2.08	-0.04	0.00	0.00	0.00
	111	-0.57	4.48	0.06	0.00	0.00	0.00
	112	-0.45	3.36	-0.08	0.00	0.00	0.00
	113	-1.21	8.17	0.12	0.00	0.00	0.00
	114	0.09	0.06	-0.09	0.00	0.00	0.00
	115	-0.67	4.86	0.11	0.00	0.00	0.00
245	1	0.03	0.00	0.00	0.00	0.00	0.00
	2	-0.02	0.00	0.00	0.00	0.00	0.00
	3	0.02	0.00	0.00	0.00	0.00	0.00
	4	0.00	0.00	0.00	0.00	0.00	0.00
	5	0.00	0.00	0.00	0.00	0.00	0.00
	6	-0.42	0.00	0.00	0.00	0.00	0.00
	7	0.42	0.00	0.00	0.00	0.00	0.00
	101	0.04	0.00	0.00	0.00	0.00	0.00
	102	0.04	0.00	0.00	0.00	0.00	0.00
	103	0.04	0.00	0.00	0.00	0.00	0.00
	104	-0.30	0.00	0.00	0.00	0.00	0.00
	105	0.38	0.00	0.00	0.00	0.00	0.00
	106	-0.64	0.00	0.00	0.00	0.00	0.00
	107	0.72	0.00	0.00	0.00	0.00	0.00

SUPPORT REACTIONS -UNIT MTON METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
	108	-0.65	0.00	0.00	0.00	0.00	0.00
	109	0.71	0.00	0.00	0.00	0.00	0.00
	110	-0.05	0.00	0.00	0.00	0.00	0.00
	111	0.12	0.00	0.00	0.00	0.00	0.00
	112	-0.13	0.00	0.00	0.00	0.00	0.00
	113	0.20	0.00	0.00	0.00	0.00	0.00
	114	-0.14	0.00	0.00	0.00	0.00	0.00
	115	0.19	0.00	0.00	0.00	0.00	0.00
299	1	0.02	0.00	0.00	0.00	0.00	0.00
	2	0.13	0.00	0.00	0.00	0.00	0.00
	3	0.00	0.00	0.00	0.00	0.00	0.00
	4	0.04	0.00	0.00	0.00	0.00	0.00
	5	0.07	0.00	0.00	0.00	0.00	0.00
	6	-1.05	0.00	0.00	0.00	0.00	0.00
	7	1.05	0.00	0.00	0.00	0.00	0.00
	101	0.02	0.00	0.00	0.00	0.00	0.00
	102	0.09	0.00	0.00	0.00	0.00	0.00
	103	0.33	0.00	0.00	0.00	0.00	0.00
	104	-0.82	0.00	0.00	0.00	0.00	0.00
	105	0.86	0.00	0.00	0.00	0.00	0.00
	106	-1.56	0.00	0.00	0.00	0.00	0.00
	107	1.80	0.00	0.00	0.00	0.00	0.00
	108	-1.67	0.00	0.00	0.00	0.00	0.00
	109	1.70	0.00	0.00	0.00	0.00	0.00
	110	0.21	0.00	0.00	0.00	0.00	0.00
	111	-0.17	0.00	0.00	0.00	0.00	0.00
	112	0.49	0.00	0.00	0.00	0.00	0.00
	113	-0.26	0.00	0.00	0.00	0.00	0.00
	114	0.39	0.00	0.00	0.00	0.00	0.00
	115	-0.36	0.00	0.00	0.00	0.00	0.00

********** END OF LATEST ANALYSIS RESULT **********

719. LOAD LIST 1 TO 7 201 TO 210
720. PRINT JOINT DISPLACEMENTS LIST 14 TO 24 26 60 244

JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN
14	1	0.0072	-0.0633	0.0301	-0.0001	0.0000	0.0000
	2	0.0046	-0.0466	0.0675	-0.0001	0.0000	0.0000
	3	0.0033	-0.0001	-0.0016	0.0000	0.0000	0.0000
	4	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	5	-0.0001	0.0000	0.0000	0.0000	0.0000	0.0000
	6	0.2874	0.0014	0.0067	0.0000	0.0007	0.0002
	7	-0.2874	-0.0014	-0.0067	0.0000	-0.0007	-0.0002
	201	0.0072	-0.0633	0.0301	-0.0001	0.0000	0.0000
	202	0.0150	-0.1100	0.0961	-0.0002	0.0000	0.0000
	203	0.3024	-0.1086	0.1028	-0.0002	0.0007	0.0002
	204	-0.2724	-0.1114	0.0894	-0.0001	-0.0007	-0.0002
	205	0.2947	-0.0619	0.0368	-0.0001	0.0007	0.0002
	206	-0.2802	-0.0647	0.0234	-0.0001	-0.0007	-0.0002
	207	0.0846	-0.0833	0.4054	-0.0002	-0.0001	0.0000
	208	-0.0545	-0.1367	-0.2129	-0.0002	0.0001	0.0000
	209	0.0768	-0.0366	0.3394	-0.0001	-0.0001	0.0000
	210	-0.0622	-0.0899	-0.2789	-0.0001	0.0001	0.0000
15	1	0.0068	-0.0761	0.0577	0.0002	0.0000	0.0001
	2	0.0039	-0.0802	0.1158	0.0007	-0.0001	0.0005
	3	0.0034	0.0000	0.0010	0.0000	0.0000	0.0000
	4	0.0001	0.0001	0.0026	0.0000	0.0000	0.0000
	5	0.0001	0.0001	0.0042	0.0000	0.0000	0.0000
	6	0.2869	0.0001	-0.0025	0.0000	0.0000	0.0000
	7	-0.2869	-0.0001	0.0025	0.0000	0.0000	0.0000
	201	0.0068	-0.0761	0.0603	0.0002	0.0000	0.0001
	202	0.0142	-0.1562	0.1786	0.0009	-0.0001	0.0007
	203	0.3011	-0.1561	0.1761	0.0009	-0.0001	0.0007
	204	-0.2727	-0.1563	0.1812	0.0009	-0.0001	0.0006
	205	0.2937	-0.0760	0.0551	0.0001	0.0000	0.0002
	206	-0.2801	-0.0763	0.0602	0.0002	-0.0001	0.0001
	207	0.0855	-0.1333	0.9138	0.0012	-0.0009	0.0006
	208	-0.0569	-0.1790	-0.5551	0.0006	0.0006	0.0007
	209	0.0781	-0.0533	0.7928	0.0005	-0.0008	0.0000
	210	-0.0643	-0.0990	-0.6761	-0.0001	0.0007	0.0002
16	1	0.0047	-0.0727	0.0664	0.0002	0.0000	-0.0002
	2	0.0008	-0.0737	0.1370	0.0007	0.0000	-0.0005
	3	0.0037	0.0003	0.0062	0.0000	0.0000	0.0000
	4	0.0001	0.0002	0.0068	0.0000	0.0000	0.0000
	5	0.0001	0.0003	0.0109	0.0000	0.0000	0.0000
	6	0.2829	0.0275	-0.0035	0.0000	0.0000	0.0000
	7	-0.2829	-0.0275	0.0035	0.0000	0.0000	0.0000
	201	0.0048	-0.0724	0.0732	0.0002	0.0000	-0.0002
	202	0.0093	-0.1457	0.2205	0.0009	-0.0001	-0.0007
	203	0.2923	-0.1182	0.2170	0.0009	0.0000	-0.0007
	204	-0.2736	-0.1732	0.2240	0.0009	-0.0001	-0.0007
	205	0.2877	-0.0451	0.0629	0.0002	0.0000	-0.0002

JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN
	206	-0.2782	-0.1002	0.0699	0.0002	0.0000	-0.0001
	207	0.0634	-0.1076	1.3002	0.0015	-0.0005	-0.0008
	208	-0.0446	-0.1838	-0.8566	0.0003	0.0004	-0.0006
	209	0.0588	-0.0345	1.1461	0.0008	-0.0005	-0.0002
	210	-0.0493	-0.1107	-1.0107	-0.0004	0.0005	-0.0001
17	1	0.0034	-0.0734	0.0641	0.0004	0.0000	0.0000
	2	-0.0021	-0.0707	0.1419	0.0009	0.0000	0.0000
	3	0.0041	-0.0044	0.0158	0.0002	0.0000	0.0000
	4	0.0000	0.0002	0.0110	0.0000	0.0000	0.0000
	5	0.0000	0.0004	0.0176	0.0000	0.0000	0.0000
	6	0.2818	-0.0282	-0.0226	0.0000	0.0000	0.0001
	7	-0.2818	0.0282	0.0226	0.0000	0.0000	-0.0001
	201	0.0034	-0.0732	0.0751	0.0004	0.0000	0.0000
	202	0.0054	-0.1481	0.2395	0.0015	0.0000	0.0000
	203	0.2872	-0.1763	0.2169	0.0015	0.0001	0.0001
	204	-0.2764	-0.1199	0.2621	0.0015	0.0000	-0.0001
	205	0.2852	-0.1016	0.0415	0.0004	0.0001	0.0001
	206	-0.2784	-0.0452	0.0867	0.0004	0.0000	-0.0001
	207	0.0423	-0.1199	1.4639	0.0023	-0.0001	-0.0001
	208	-0.0315	-0.1763	-0.9826	0.0007	0.0002	0.0000
	209	0.0403	-0.0452	1.2885	0.0012	-0.0001	-0.0001
	210	-0.0335	-0.1016	-1.1580	-0.0004	0.0002	0.0000
18	1	0.0032	-0.0610	0.0496	0.0007	0.0000	-0.0002
	2	-0.0028	-0.0354	0.1045	0.0007	0.0001	-0.0004
	3	0.0041	-0.0232	0.0285	0.0007	0.0000	0.0000
	4	-0.0001	-0.0010	0.0135	0.0000	0.0000	0.0000
	5	-0.0001	-0.0016	0.0216	0.0001	0.0000	0.0000
	6	0.2830	-0.0009	-0.0308	-0.0001	0.0000	0.0001
	7	-0.2830	0.0009	0.0308	0.0001	0.0000	-0.0001
	201	0.0031	-0.0620	0.0630	0.0007	0.0000	-0.0002
	202	0.0044	-0.1212	0.2041	0.0021	0.0001	-0.0006
	203	0.2874	-0.1221	0.1733	0.0021	0.0001	-0.0005
	204	-0.2787	-0.1203	0.2349	0.0022	0.0000	-0.0007
	205	0.2862	-0.0619	0.0188	0.0006	0.0001	-0.0001
	206	-0.2798	-0.0601	0.0804	0.0008	0.0000	-0.0003
	207	0.0364	-0.0836	1.4154	0.0033	0.0000	-0.0007
	208	-0.0276	-0.1587	-1.0056	0.0010	0.0001	-0.0006
	209	0.0352	-0.0234	1.2608	0.0018	0.0000	-0.0002
	210	-0.0288	-0.0985	-1.1602	-0.0004	0.0001	-0.0001
19	1	0.0051	-0.0819	0.0299	-0.0001	0.0000	0.0000
	2	-0.0051	-0.0935	0.0673	-0.0001	0.0000	0.0000
	3	0.0020	0.0002	-0.0015	0.0000	0.0000	0.0000
	4	-0.0004	0.0000	0.0001	0.0000	0.0000	0.0000
	5	-0.0006	0.0000	0.0001	0.0000	0.0000	0.0000
	6	0.3202	-0.0015	0.0060	0.0000	0.0001	0.0001
	7	-0.3202	0.0015	-0.0060	0.0000	-0.0001	-0.0001

JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN
	201	0.0048	-0.0819	0.0300	-0.0001	0.0000	0.0000
	202	0.0015	-0.1752	0.0958	-0.0002	0.0000	0.0000
	203	0.3217	-0.1767	0.1018	-0.0002	0.0000	0.0001
	204	-0.3187	-0.1737	0.0898	-0.0001	-0.0001	-0.0002
	205	0.3253	-0.0835	0.0359	-0.0001	0.0000	0.0001
	206	-0.3150	-0.0804	0.0240	-0.0001	-0.0001	-0.0002
	207	-0.0333	-0.2066	0.4078	-0.0002	-0.0002	0.0001
	208	0.0362	-0.1438	-0.2159	-0.0002	0.0001	-0.0001
	209	-0.0296	-0.1134	0.3419	-0.0001	-0.0001	0.0001
	210	0.0399	-0.0505	-0.2818	-0.0001	0.0001	-0.0002
20	1	0.0060	-0.1098	0.0570	0.0000	0.0000	0.0000
	2	-0.0041	-0.0960	0.1136	-0.0006	0.0000	0.0000
	3	0.0020	0.0000	0.0010	0.0000	0.0000	0.0000
	4	-0.0003	-0.0001	0.0026	0.0000	0.0000	0.0000
	5	-0.0005	-0.0001	0.0042	0.0000	0.0000	0.0000
	6	0.3197	0.0001	-0.0025	0.0000	0.0000	0.0000
	7	-0.3197	-0.0001	0.0025	0.0000	0.0000	0.0000
	201	0.0057	-0.1098	0.0597	0.0000	0.0000	0.0000
	202	0.0034	-0.2059	0.1758	-0.0006	0.0000	0.0000
	203	0.3231	-0.2058	0.1732	-0.0006	0.0000	0.0000
	204	-0.3163	-0.2060	0.1783	-0.0006	0.0000	0.0000
	205	0.3257	-0.1097	0.0545	0.0000	0.0000	-0.0001
	206	-0.3137	-0.1098	0.0596	0.0000	0.0000	0.0000
	207	-0.0300	-0.2287	0.9114	-0.0003	-0.0002	0.0001
	208	0.0367	-0.1831	-0.5585	-0.0009	0.0002	-0.0001
	209	-0.0274	-0.1325	0.7927	0.0003	-0.0002	0.0001
	210	0.0394	-0.0870	-0.6772	-0.0003	0.0002	-0.0001
21	1	0.0071	-0.0945	0.0656	0.0000	0.0000	0.0000
	2	-0.0031	-0.0726	0.1345	-0.0006	0.0000	0.0000
	3	0.0020	-0.0001	0.0061	0.0000	0.0000	0.0000
	4	-0.0002	-0.0001	0.0068	0.0000	0.0000	0.0000
	5	-0.0004	-0.0002	0.0109	0.0000	0.0000	0.0000
	6	0.3172	0.0224	-0.0035	0.0000	0.0000	-0.0001
	7	-0.3172	-0.0224	0.0035	0.0000	0.0000	0.0001
	201	0.0068	-0.0946	0.0724	0.0000	0.0000	0.0000
	202	0.0056	-0.1675	0.2172	-0.0006	0.0000	0.0001
	203	0.3228	-0.1450	0.2137	-0.0006	0.0000	-0.0001
	204	-0.3116	-0.1899	0.2207	-0.0006	0.0000	0.0002
	205	0.3243	-0.0721	0.0621	0.0000	0.0000	-0.0001
	206	-0.3101	-0.1169	0.0692	0.0000	0.0000	0.0002
	207	-0.0262	-0.2017	1.2966	0.0000	-0.0001	0.0001
	208	0.0375	-0.1333	-0.8595	-0.0011	0.0002	0.0000
	209	-0.0248	-0.1287	1.1450	0.0006	-0.0001	0.0001
	210	0.0389	-0.0603	-1.0111	-0.0006	0.0002	0.0000
22	1	0.0076	-0.0855	0.0640	-0.0003	0.0000	0.0000
	2	-0.0019	-0.0598	0.1411	-0.0009	0.0000	0.0000

JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN
	3	0.0021	-0.0042	0.0158	-0.0002	0.0000	0.0000
	4	-0.0002	-0.0002	0.0110	0.0000	0.0000	0.0000
	5	-0.0002	-0.0003	0.0176	0.0000	0.0000	0.0000
	6	0.3122	-0.0250	-0.0226	0.0000	0.0000	0.0000
	7	-0.3122	0.0250	0.0226	0.0000	0.0000	0.0000
	201	0.0074	-0.0857	0.0750	-0.0003	0.0000	0.0000
	202	0.0075	-0.1498	0.2385	-0.0013	0.0000	0.0000
	203	0.3198	-0.1749	0.2159	-0.0013	0.0000	0.0001
	204	-0.3047	-0.1248	0.2611	-0.0013	0.0001	0.0000
	205	0.3198	-0.1106	0.0414	-0.0003	0.0000	0.0001
	206	-0.3046	-0.0605	0.0866	-0.0003	0.0001	0.0000
	207	-0.0225	-0.1694	1.4643	-0.0009	0.0000	0.0001
	208	0.0376	-0.1303	-0.9850	-0.0017	0.0001	0.0000
	209	-0.0225	-0.1051	1.2897	0.0002	-0.0001	0.0000
	210	0.0376	-0.0660	-1.1595	-0.0007	0.0001	0.0000
23	1	0.0078	-0.0680	0.0483	-0.0006	0.0000	-0.0002
	2	-0.0011	-0.0506	0.1035	-0.0005	0.0000	-0.0004
	3	0.0020	-0.0205	0.0270	-0.0006	0.0000	0.0000
	4	-0.0002	-0.0096	0.0135	0.0000	0.0000	0.0000
	5	-0.0003	-0.0154	0.0216	0.0000	0.0000	0.0000
	6	0.3135	0.0008	-0.0308	0.0000	0.0001	-0.0001
	7	-0.3135	-0.0008	0.0308	0.0000	-0.0001	0.0001
	201	0.0076	-0.0776	0.0618	-0.0005	0.0000	-0.0002
	202	0.0084	-0.1544	0.2004	-0.0017	0.0000	-0.0006
	203	0.3219	-0.1536	0.1696	-0.0017	0.0001	-0.0007
	204	-0.3051	-0.1552	0.2312	-0.0016	-0.0001	-0.0005
	205	0.3213	-0.0672	0.0174	-0.0006	0.0001	-0.0003
	206	-0.3057	-0.0688	0.0791	-0.0005	-0.0001	-0.0001
	207	-0.0211	-0.1798	1.4121	-0.0009	-0.0003	-0.0006
	208	0.0379	-0.1290	-1.0098	-0.0025	0.0002	-0.0006
	209	-0.0217	-0.0934	1.2599	0.0003	-0.0002	-0.0002
	210	0.0373	-0.0426	-1.1619	-0.0014	0.0003	-0.0003
24	1	-0.0101	-0.0237	0.0288	-0.0001	0.0000	0.0000
	2	-0.0331	-0.0491	0.0661	-0.0001	-0.0001	0.0001
	3	0.0000	0.0000	-0.0015	0.0000	0.0000	0.0000
	4	-0.0002	0.0000	0.0001	0.0000	0.0000	0.0000
	5	-0.0003	-0.0001	0.0001	0.0000	0.0000	0.0000
	6	0.3035	0.0340	-0.0005	-0.0001	-0.0002	0.0002
	7	-0.3035	-0.0340	0.0005	0.0001	0.0002	-0.0002
	201	-0.0104	-0.0238	0.0289	-0.0001	0.0000	0.0000
	202	-0.0437	-0.0729	0.0935	-0.0002	-0.0001	0.0001
	203	0.2598	-0.0388	0.0930	-0.0003	-0.0003	0.0003
	204	-0.3471	-0.1069	0.0939	-0.0002	0.0001	-0.0002
	205	0.2933	0.0103	0.0283	-0.0002	-0.0002	0.0003
	206	-0.3136	-0.0577	0.0293	0.0000	0.0001	-0.0002
	207	-0.1263	-0.0937	0.4055	-0.0003	-0.0004	0.0002

JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN
	208	0.0388	-0.0521	-0.2182	-0.0002	0.0002	0.0000
	209	-0.0927	-0.0445	0.3408	-0.0001	-0.0003	0.0001
	210	0.0723	-0.0029	-0.2829	-0.0001	0.0003	-0.0001
26	1	-0.0101	-0.0197	0.0487	0.0000	0.0001	0.0000
	2	-0.0331	-0.0339	0.1008	0.0000	0.0001	0.0001
	3	0.0000	0.0000	0.0001	0.0000	0.0000	0.0000
	4	-0.0002	0.0000	0.0016	0.0000	0.0000	0.0000
	5	-0.0004	0.0000	0.0026	0.0000	0.0000	0.0000
	6	0.3065	-0.0311	-0.0026	0.0000	0.0000	0.0003
	7	-0.3065	0.0311	0.0026	0.0000	0.0000	-0.0003
	201	-0.0104	-0.0197	0.0504	0.0000	0.0001	0.0000
	202	-0.0437	-0.0536	0.1522	0.0000	0.0001	0.0001
	203	0.2628	-0.0846	0.1496	0.0000	0.0001	0.0004
	204	-0.3501	-0.0225	0.1548	0.0000	0.0001	-0.0002
	205	0.2963	-0.0508	0.0462	0.0000	0.0000	0.0003
	206	-0.3166	0.0113	0.0513	0.0000	0.0001	-0.0002
	207	-0.1266	-0.0457	0.7566	0.0001	0.0002	0.0002
	208	0.0392	-0.0615	-0.4513	-0.0001	0.0000	0.0000
	209	-0.0931	-0.0118	0.6532	0.0001	0.0002	0.0001
	210	0.0727	-0.0276	-0.5547	-0.0001	0.0000	-0.0001
60	1	-0.0001	-0.0486	-0.0652	-0.0001	0.0001	0.0000
	2	-0.0101	-0.0297	-0.1182	0.0001	0.0002	0.0001
	3	0.0036	-0.0003	-0.0250	0.0000	0.0000	0.0000
	4	-0.0035	-0.0053	-0.0277	-0.0002	0.0001	0.0000
	5	-0.0056	-0.0085	-0.0444	-0.0003	0.0001	0.0000
	6	0.2405	-0.0005	0.0162	0.0000	0.0000	-0.0001
	7	-0.2405	0.0005	-0.0162	0.0000	0.0000	0.0001
	201	-0.0035	-0.0539	-0.0930	-0.0002	0.0002	0.0000
	202	-0.0122	-0.0870	-0.2529	-0.0003	0.0004	0.0001
	203	0.2283	-0.0875	-0.2367	-0.0004	0.0004	0.0000
	204	-0.2526	-0.0866	-0.2691	-0.0003	0.0005	0.0001
	205	0.2404	-0.0491	-0.0490	-0.0001	0.0000	-0.0001
	206	-0.2406	-0.0482	-0.0814	0.0000	0.0001	0.0001
	207	0.0124	-0.0822	0.5536	-0.0001	0.0004	0.0002
	208	-0.0367	-0.0918	-1.0590	-0.0006	0.0004	-0.0001
	209	0.0245	-0.0438	0.7412	0.0002	0.0001	0.0002
	210	-0.0246	-0.0534	-0.8714	-0.0003	0.0001	-0.0002
244	1	-0.0001	-0.0491	-0.0691	-0.0003	0.0001	0.0002
	2	0.0000	-0.0328	-0.1298	-0.0009	-0.0001	0.0000
	3	0.0000	0.0003	-0.0247	0.0000	0.0001	0.0000
	4	0.0000	-0.0079	-0.0311	-0.0003	0.0000	0.0000
	5	0.0000	-0.0127	-0.0498	-0.0005	0.0000	0.0000
	6	0.0007	0.0002	0.0166	0.0000	-0.0031	0.0002
	7	-0.0007	-0.0002	-0.0166	0.0000	0.0031	-0.0002
	201	-0.0001	-0.0570	-0.1002	-0.0006	0.0001	0.0002
	202	-0.0001	-0.0943	-0.2734	-0.0018	0.0001	0.0002

JOINT DISPLACEMENT (CM RADIANS) STRUCTURE TYPE = SPACE

JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN
203		0.0007	-0.0941	-0.2567	-0.0018	-0.0030	0.0004
204		-0.0008	-0.0944	-0.2900	-0.0017	0.0031	0.0000
205		0.0007	-0.0489	-0.0524	-0.0004	-0.0030	0.0004
206		-0.0008	-0.0493	-0.0857	-0.0003	0.0031	0.0000
207		0.0001	-0.0989	0.5329	-0.0015	0.0004	0.0002
208		-0.0002	-0.0897	-1.0793	-0.0021	-0.0003	0.0002
209		0.0001	-0.0537	0.7372	0.0000	0.0004	0.0002
210		-0.0002	-0.0445	-0.8750	-0.0006	-0.0002	0.0002

********** END OF LATEST ANALYSIS RESULT **********

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721. LOAD LIST 101 TO 115
722. PARAMETER 1
723. CODE LRFD
724. FYLD 27500 ALL
725. FU 41000 ALL
726. MAIN 200 MEMB 208 TO 210 212 214 216 218 237 238 240 241 244 246 250 -
727. 253 TO 261 264 267 272 TO 279 282 283 286 289 291 TO 294 296 298 300 301 -
728. 303 305 307 309 311 316 319 320 322 TO 332 339 TO 347 350 352 356 TO 361 -
729. 363 TO 366 369 370 384 387 390 TO 395 398 401 403 406 TO 412 414 415 418 -
730. 422 TO 426 428 429 432 435 437 TO 440 442 443 446 449 451 452 454 TO 457 -
731. 460 463 465 TO 468 470 TO 472 496 499 502 506 TO 515 517 520 523 526 529 -
732. 534 TO 545 548 551 554 TO 559 562 564 566 567 569 TO 571 574 TO 578 580 581 -
733. 583 585 587 589 591 593 600 TO 603 605 606 608 609 614 TO 621 629 TO 644 -
734. 646 647 649 650 652 653 655 656 695 TO 698 700 TO 707 711 TO 721 734 TO 745 -
735. 763 TO 786 790 TO 792 798 TO 802
736. LZ 3.35 MEMB 149 193 TO 195 202 205 TO 207
737. LY 4.5 MEMB 1 2 5 TO 8 11 200 211 213 599 622 624 628
738. LZ 4.7 MEMB 6 191 199 211
739. LY 2.83 MEMB 64 88 367 368
740. LZ 2.83 MEMB 64 88 367 368
741. LZ 4.95 MEMB 14 19 28 33 40 52 57 67 84 90 91 97 102 126 137 153 166 219 225 -
742. 306 308 310 312 TO 315 317 318 333 334 336 372 381 382 385 386 396 397 399 -
743. 400 402 413 490 494 495 497 498 500 501 503 516 657 658 722 TO 724
744. LY 1.65 MEMB 67 84 91 219 396 399 402 413 657 658 722 TO 724
745. LZ 5.25 MEMB 15 20 35 59 70 86 92 93 128 139 155 220 221 227 284 285 287 288 -
746. 295 297 299 302 304 374 416 417 419 TO 421 491 518 519 521 522 532 582 584 -
747. 661 664 680 682 725 TO 727 747 748 750
748. LZ 5.1 MEMB 21 37 87 94 95 157 229 263 266 269 280 281 376 431 434 441 492 -
749. 525 528 533 586 597 668 728 TO 730
750. LY 2 MEMB 70 86 92 93 220 221 661 664 680 682 725 TO 727 747 748 750
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751. LY 2 MEMB 72 87 94 95 222 668 685 687 699 728 TO 730 753 755
752. LY 1.2 MEMB 151 152 154 156 164 167 176 203 483 604 645 648 651 654
753. LZ 2.6 MEMB 16 61 72 79 80 106 107 115 117 121 123 130 141 222 262 265 268 -
754. 270 377 379 430 433 436 453 476 477 524 527 588 685 687 699 708 TO 710 753 -
755. 755 762
756. LZ 4.5 MEMB 17 22 39 41 42 48 49 63 73 89 100 132 142 158 231 234 242 243 -
757. 245 247 TO 249 378 444 445 447 448 450 478 493 546 547 549 550 552 590 592 -
758. 670 672 674 689 691 693 731 TO 733 757 TO 761
759. LY 2.35 MEMB 73 89 100 478 670 672 674 689 691 693 731 TO 733 757 TO 761
760. LZ 4 MEMB 65 108 TO 110 114 116 118 120 134 169 173 178 180 TO 182 380 458 -
761. 459 461 462 469 481 484 485 560 561 563 565 572 573 579 793
762. LZ 1.65 MEMB 143 145 TO 148 151 152 154 156 167 173 177 486 TO 489 504 505 -
763. 530 531 553 597 604 645 648 651 654
764. LZ 6.25 MEMB 111 TO 113 163 164 174 176 203 204 383 464 473 480 482 483 568 595
765. LZ 3.3 MEMB 23 25 TO 27 31 32 34 36 38 43 47 74 76 TO 78 82 83 85 103 104 -
766. 224 226 228 230 232 233 236 239 251 252 271 290 321 335 338 371 373 375 388 -
767. 389 404 405 427 607 751
768. LY 2.7 MEMB 112 204
769. LZ 6.75 MEMB 24 54 217 348 349 351 353 TO 355
770. LZ 6.18 MEMB 71 88 140 160 368 789 797
771. LY 5.1 MEMB 696 798 799
772. KZ 2 MEMB 223 659 660 662 663 665 TO 667 669 671 673 675 TO 679 681 683 684 -
773. 686 688 690 692 694
774. KY 2 MEMB 81 150 196 198 223 268 433 524 600 601 610 TO 613 629 TO 636 659 -
775. 660 662 663 665 TO 667 669 671 673 675 TO 679 681 683 684 686 688 690 692 -
776. 694 698 704 707 TO 711
777. KY 2.52 MEMB 244 253 307 309 319 322 401 406 499 506
778. TRACK 0 ALL
779. CHECK CODE ALL
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STAAD.Pro CODE CHECKING - (LRFD 3RD EDITION) v1.0
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ALL UNITS ARE - MTON METE (UNLESS OTHERWISE Noted)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
1 ST	IPE300		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.576	113
		26.07 C	0.00	0.11	2.09
2 ST	IPE300		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.675	103
		30.35 C	-0.02	-0.11	2.35
3 ST	IPE300		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.348	113
		30.33 C	0.00	-0.48	2.35
4 ST	IPE300		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.343	106
		30.55 C	0.06	-0.11	2.35
5 ST	IPE400		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.446	113
		41.69 C	0.02	-0.24	2.35
6 ST	IPE300		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.488	103
		21.48 C	0.00	0.30	2.35
7 ST	IPE300		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.772	103
		35.30 C	0.00	0.05	2.09
8 ST	IPE300		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.924	103
		42.38 C	0.00	0.00	0.00
9 ST	IPE300		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.385	112
		33.87 C	0.00	0.50	2.35
10 ST	IPE300		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.369	103
		33.64 C	0.00	0.28	2.35
11 ST	IPE400		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.706	103
		64.70 C	0.01	-1.03	2.35
12 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.275	107
		16.09 C	-0.06	0.01	2.09
13 ST	IPE300		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.895	103
		23.59 C	-0.01	4.02	4.70
14 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.063	103
		0.00 C	0.00	-0.57	0.60
15 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.077	103
		1.20 C	0.00	-0.63	0.62

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MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
16	ST IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.035	112
		4.84 C	0.00	-0.05	0.50
17	ST IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.075	103
		0.98 C	0.00	-0.61	0.75
18	ST IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.012	112
		1.26 T	-0.01	-0.01	0.40
19	ST IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.145	103
		0.00 T	0.00	-1.30	0.60
20	ST IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.054	112
		0.67 T	0.05	-0.15	2.63
21	ST IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.040	112
		0.98 T	0.05	0.06	2.55
22	ST IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.090	103
		0.23 T	0.00	-0.65	3.00
23	ST IPE300		(EUROPEAN SECTIONS)		
		PASS	SHEAR-Y	0.131	103
		0.18 C	0.00	0.00	0.00
24	ST IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.050	106
		1.83 C	-0.03	-0.16	0.75
25	ST IPE300		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.203	103
		1.88 C	0.00	3.03	0.00
26	ST IPE300		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.186	113
		1.68 C	0.00	2.77	0.00
27	ST IPE300		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.227	113
		1.85 C	0.00	3.41	0.00
28	ST IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.143	103
		0.50 T	-0.01	-1.20	0.00
29	ST IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.156	103
		0.00 T	0.00	-1.41	0.60
30	ST IPE360		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.187	106
		0.04 T	0.70	-0.43	3.37

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MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
31 ST	IPE300		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.350	103
		0.09 C	0.00	-5.43	0.19
32 ST	IPE300		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.252	103
		1.42 C	0.01	-3.79	0.00
33 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.089	103
		0.04 T	0.00	-0.80	0.60
34 ST	IPE300		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.249	103
		1.71 C	0.00	-3.74	0.00
35 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.187	103
		0.19 T	0.00	-1.44	2.63
36 ST	IPE300		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.364	103
		0.44 C	-0.01	-5.55	0.00
37 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.175	103
		0.22 T	0.00	-1.35	2.55
38 ST	IPE300		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.732	103
		1.93 C	-0.04	-11.06	0.00
39 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.895	103
		0.17 T	0.00	-6.59	3.00
40 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.141	103
		0.12 T	0.01	-1.21	1.45
41 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.727	103
		0.17 T	0.00	-6.59	0.00
42 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.667	103
		0.17 T	0.00	-6.04	0.00
43 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.064	103
		0.13 T	0.00	-0.54	1.60
44 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.007	112
		0.35 T	-0.01	-0.01	0.40
45 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	SHEAR-Y	0.092	103
		0.00 C	0.00	0.00	0.40

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MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
46	ST IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.386	103
		0.00 C	0.00	-3.43	1.00
47	ST IPE300		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.732	103
		1.93 C	-0.04	-11.06	0.40
48	ST IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.073	112
		0.28 T	0.06	-0.33	0.00
49	ST IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.061	103
		0.23 T	0.00	-0.53	0.00
50	ST IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.020	106
		0.00 T	0.02	-0.05	0.78
51	ST UPN140		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.077	103
		0.01 C	0.00	-0.18	0.78
52	ST IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.303	103
		0.50 T	0.02	-2.34	0.00
53	ST IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.355	103
		0.00 T	0.01	-2.82	0.77
54	ST IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.139	103
		0.04 T	0.00	-1.25	0.75
55	ST IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.043	112
		5.57 C	0.01	-0.06	0.77
56	ST IPE300		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.106	103
		11.34 C	0.00	0.51	0.00
57	ST IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.097	103
		0.01 T	0.00	-0.88	0.83
58	ST IPE300		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.343	103
		14.33 C	-0.27	2.28	0.00
59	ST IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.080	112
		2.63 C	0.07	-0.23	0.87
60	ST IPE300		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.298	103
		13.37 C	0.28	1.71	0.00

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MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
61 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.099	112
		4.78 C	0.11	-0.09	0.85
62 ST	IPE300		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.313	113
		10.75 C	0.00	-3.41	2.83
63 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.081	103
		1.63 C	0.00	-0.64	0.65
64 ST	IPE400		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.307	103
		21.54 C	0.32	-5.64	1.83
65 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.078	103
		2.63 C	-0.01	-0.52	0.60
66 ST	IPE300		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.119	103
		13.48 C	0.00	0.53	0.00
67	TAP ERED		(EUROPEAN SECTIONS)		
		PASS	SHEAR-Y	0.045	103
		0.07 C	0.00	0.00	0.00
68 ST	IPE300		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.259	103
		11.82 C	0.00	2.58	2.83
69 ST	IPE300		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.201	112
		9.10 C	0.00	-2.39	0.00
70	TAP ERED		(EUROPEAN SECTIONS)		
		PASS	SHEAR-Y	0.049	103
		0.09 T	0.00	0.00	0.00
71 ST	IPE300		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.231	112
		10.04 C	-0.01	2.76	1.83
72	TAP ERED		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.014	113
		5.47 T	0.13	-0.04	0.85
73	TAP ERED		(EUROPEAN SECTIONS)		
		PASS	SHEAR-Y	0.034	103
		1.18 C	0.00	0.00	0.00
74 ST	IPE400		(EUROPEAN SECTIONS)		
		PASS	SHEAR-Y	0.057	103
		0.43 T	0.00	0.00	0.00
75 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.082	107
		3.16 C	-0.06	0.10	0.00

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MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
76 ST	IPE360		(EUROPEAN SECTIONS)		
		PASS	SHEAR-Y	0.120	103
		1.57 T	0.00	0.00	0.00
77 ST	IPE400		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.168	113
		0.48 C	-0.01	5.33	0.00
78 ST	IPE400		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.223	113
		0.57 C	0.01	7.12	0.00
79 ST	IPE400		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.223	113
		0.64 C	0.00	7.16	0.00
80 ST	IPE400		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.321	113
		2.31 C	0.00	10.17	0.00
81 LD	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.185	112
		0.02 T	-0.08	0.01	1.75
82 ST	IPE400		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.113	103
		0.29 T	0.08	-3.15	0.00
83 ST	IPE400		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.202	112
		0.30 T	0.00	6.50	1.10
84	TAP ERED		(EUROPEAN SECTIONS)		
		PASS	SHEAR-Y	0.045	103
		0.19 T	0.00	0.00	0.00
85 ST	IPE400		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.266	112
		0.80 T	0.00	8.55	1.10
86	TAP ERED		(EUROPEAN SECTIONS)		
		PASS	SHEAR-Y	0.049	103
		0.35 T	0.00	0.00	0.00
87	TAP ERED		(EUROPEAN SECTIONS)		
		PASS	SHEAR-Y	0.035	103
		0.29 T	0.00	0.00	0.00
88 ST	IPE400		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.319	103
		21.07 C	0.32	6.09	1.83
89	TAP ERED		(EUROPEAN SECTIONS)		
		PASS	SHEAR-Y	0.034	103
		0.10 T	0.00	0.00	0.00
90 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.228	103
		0.27 C	0.00	-1.59	1.65

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MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
91	TAP ERED		(EUROPEAN SECTIONS)		
		PASS	SHEAR-Y	0.066	103
		0.19 T	0.00	0.00	0.00
92	TAP ERED		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.058	103
		0.35 T	0.02	-3.73	0.00
93	TAP ERED		(EUROPEAN SECTIONS)		
		PASS	SHEAR-Y	0.050	103
		0.35 T	0.00	0.00	0.00
94	TAP ERED		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.048	103
		0.29 T	-0.04	-3.01	0.80
95	TAP ERED		(EUROPEAN SECTIONS)		
		PASS	SHEAR-Y	0.045	103
		0.29 T	0.00	0.00	0.00
96 ST	IPE360		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.334	103
		1.70 T	0.00	-5.44	0.00
97 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.374	103
		0.01 C	0.00	-3.40	0.75
98 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.110	103
		0.22 C	0.00	-0.99	0.87
99 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.114	103
		1.60 C	0.00	-0.93	0.85
100 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.097	103
		1.30 T	0.02	-0.70	0.65
101 LD	L130X130X10		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.137	106
		0.10 C	0.18	-0.21	2.82
102 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.540	103
		0.30 C	0.00	-4.88	0.00
103 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.029	103
		3.32 T	0.00	-0.10	0.55
104 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.028	103
		3.10 T	0.00	-0.10	0.55
105 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.018	103
		1.13 T	0.00	-0.10	0.55

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MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
106	ST IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.108	103
		0.43 C	-0.07	-0.60	0.37
107	ST IPE400		(EUROPEAN SECTIONS)		
		PASS	SHEAR-Y	0.118	112
		0.48 C	0.00	1.61	0.25
108	ST IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.259	103
		0.26 C	0.15	-1.42	2.00
109	ST IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.347	103
		0.25 T	-0.34	-1.37	0.98
110	ST IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.347	103
		0.17 T	0.34	-1.37	1.02
111	ST IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.580	103
		3.65 C	0.00	5.04	0.00
112	ST IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.250	103
		4.82 C	-0.01	-1.70	1.23
113	ST IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.282	103
		4.91 C	0.01	-2.21	0.00
114	ST IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.209	103
		0.04 C	-0.10	-1.39	0.00
115	ST IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.103	103
		0.84 T	0.08	-0.50	0.95
116	ST IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.293	103
		1.16 T	-0.31	-0.99	0.98
117	ST IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.117	103
		0.34 T	0.04	-0.86	0.18
118	ST IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.289	103
		0.39 T	-0.31	-0.99	0.00
119	ST UPN240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.018	113
		0.88 C	0.01	-0.02	0.69
120	ST IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.289	103
		0.44 C	0.22	-1.48	0.00

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MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
121 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.239	103
		0.48 C	-0.33	-0.45	0.00
122 ST	UPN240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.011	106
		0.65 T	-0.01	-0.01	0.58
123 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.238	103
		0.35 T	-0.33	-0.45	0.94
124 ST	UPN240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.015	107
		0.53 C	-0.01	-0.02	0.68
125 ST	IPE300		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.251	103
		15.41 C	0.00	0.51	3.35
126 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.093	103
		0.00 C	0.00	-0.84	0.75
127 ST	IPE300		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.523	113
		16.49 C	0.14	-3.68	3.35
128 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.116	103
		0.43 T	0.00	-1.03	0.87
129 ST	IPE300		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.604	113
		17.81 C	-0.10	-4.75	3.35
130 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.075	112
		1.17 T	0.10	-0.09	0.85
131 ST	IPE300		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.579	113
		16.92 C	0.00	-4.97	3.35
132 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.093	113
		5.14 C	-0.05	-0.28	0.65
133 ST	IPE400		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.456	113
		23.82 C	0.04	10.15	0.00
134 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.106	115
		12.04 C	-0.04	-0.13	0.44
135 ST	IPE300		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.386	103
		25.13 C	0.00	0.53	3.35

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MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
136 ST	IPE300		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.380	103
		31.87 C	0.00	-1.36	1.85
137 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.943	103
		0.01 T	0.00	-6.47	2.04
138 ST	IPE300		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.361	112
		21.05 C	0.00	-3.98	0.00
139 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.207	103
		0.01 T	0.00	-1.60	2.63
140 ST	IPE300		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.297	112
		13.55 C	-0.01	-3.51	0.00
141 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.089	106
		6.08 C	0.00	0.26	2.55
142 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.251	103
		0.31 T	0.00	-2.02	2.25
143 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.053	112
		7.33 C	-0.01	-0.05	0.57
144 LD	L130X130X10		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.194	106
		0.15 T	0.26	-0.30	3.37
145 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.050	103
		0.22 C	0.00	-0.44	0.57
146 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.056	103
		0.48 C	0.00	-0.48	0.57
147 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.053	114
		4.39 C	0.01	0.23	0.57
148 ST	IPE360		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.262	113
		8.78 T	0.00	5.97	0.00
149 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.085	112
		6.67 C	0.00	-0.33	0.00
150 LD	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.235	112
		1.35 T	-0.10	0.01	1.75

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MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
151	ST IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.110	103
		0.16 C	0.00	0.98	0.50
152	ST IPE240		(EUROPEAN SECTIONS)		
		PASS	SHEAR-Y	0.211	103
		0.47 C	0.00	1.61	0.50
153	ST IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.173	103
		0.03 T	0.00	-1.57	0.75
154	ST IPE240		(EUROPEAN SECTIONS)		
		PASS	SHEAR-Y	0.211	103
		0.22 C	0.00	1.54	0.50
155	ST IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.213	103
		0.30 C	0.00	-1.91	0.87
156	ST IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.245	103
		0.13 C	-0.01	2.18	0.50
157	ST IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.200	103
		0.06 C	0.00	-1.80	0.85
158	ST IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.146	103
		0.97 C	0.01	-1.23	0.65
159	LD L130X130X10		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.193	106
		0.08 T	-0.26	-0.30	3.37
160	ST IPE400		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.311	112
		18.21 C	0.00	-6.83	0.00
161	ST IPE360		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.270	103
		11.18 T	-0.06	-5.70	1.03
162	ST IPE300		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.641	103
		11.31 C	-0.01	7.21	0.00
163	ST IPE360		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.398	103
		3.76 T	0.00	9.78	0.00
164	ST IPE360		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.276	103
		1.28 T	0.12	-6.19	0.56
165	ST IPE300		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.387	103
		10.86 C	0.03	-3.87	0.00

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MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
166 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.534	103
		0.01 T	0.00	-4.76	0.00
167 ST	IPE360		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.198	113
		16.58 T	0.36	1.83	0.00
168 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.730	103
		0.56 T	0.00	-4.62	1.33
169 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.086	103
		0.17 C	0.02	-0.67	0.44
170 ST	IPE360		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.019	106
		1.55 T	-0.06	-0.04	1.00
171 ST	IPE360		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.018	107
		1.06 T	0.06	-0.04	1.00
172 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.584	103
		0.01 C	0.00	-3.71	1.33
173 ST	IPE360		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.224	103
		12.95 T	-0.01	-4.68	0.12
174 ST	IPE360		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.317	103
		4.06 T	-0.10	-7.17	0.00
175 ST	IPE360		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.253	103
		11.41 T	-0.05	-5.30	0.00
176 ST	IPE360		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.332	103
		4.07 T	0.01	-8.03	0.55
177 ST	IPE360		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.225	103
		13.01 T	-0.01	-4.68	0.00
178 ST	IPE360		(EUROPEAN SECTIONS)		
		PASS	SHEAR-Y	0.220	103
		0.02 C	0.00	0.00	0.00
179 ST	IPE360		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.270	103
		11.41 T	-0.06	-5.69	0.00
180 ST	IPE360		(EUROPEAN SECTIONS)		
		PASS	SHEAR-Y	0.222	103
		0.10 C	0.00	0.00	0.00

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MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
181 ST	IPE360		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.251	103
		5.36 T	0.00	-5.93	1.82
182 ST	IPE360		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.253	103
		5.33 T	0.00	-5.98	1.82
183 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.340	103
		5.50 C	0.00	2.80	0.00
184 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.344	103
		5.50 C	0.00	2.83	0.00
185 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.282	103
		6.13 C	0.00	-2.23	0.00
186 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.283	103
		6.13 C	0.00	-2.25	0.00
187 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.085	113
		0.06 C	0.13	0.08	0.00
188 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.121	113
		2.16 C	0.13	-0.27	1.82
189 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.117	112
		2.20 C	-0.12	-0.27	1.82
190 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.119	113
		0.07 C	0.17	0.20	0.00
191 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.242	106
		13.75 C	0.06	-0.04	2.09
192 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.100	112
		1.78 C	0.00	0.62	0.00
193 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.102	112
		1.81 C	0.00	-0.77	0.00
194 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.079	107
		4.55 C	-0.06	0.10	1.68
195 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.081	112
		1.38 C	0.00	0.62	1.68

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MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
196 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.119	112
		0.43 T	-0.17	0.15	1.75
197 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.127	107
		10.16 C	-0.06	0.01	0.00
198 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.135	112
		0.42 T	-0.18	0.26	1.75
199 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.128	113
		5.30 C	0.00	0.60	2.61
200 ST	IPE300		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.710	103
		31.38 C	0.00	-0.40	2.61
201 ST	IPE360		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.251	106
		0.06 C	0.68	-1.30	3.94
202 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.032	112
		2.66 C	0.03	0.00	1.55
203 ST	IPE360		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.317	103
		4.06 T	-0.10	-7.17	0.49
204 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.294	103
		4.82 C	0.01	-2.21	0.86
205 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.046	112
		4.92 T	0.03	0.01	1.55
206 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.042	112
		2.63 C	0.04	0.00	0.00
207 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.036	112
		2.15 C	0.03	-0.01	0.00
208 ST	L90X90X8		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.033	112
		0.18 T	0.00	-0.01	0.00
209 ST	L90X90X8		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.129	103
		4.98 T	0.00	-0.03	0.00
210 ST	L90X90X8		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.309	103
		4.55 C	0.00	-0.02	1.96

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MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
211	ST IPE300		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.590	103
		19.72 C	-0.01	-2.57	2.35
212	LD L90X90X8		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.065	103
		2.12 C	0.00	0.00	3.09
213	ST IPE400		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.418	113
		27.92 C	-0.04	4.17	0.00
214	LD L90X90X8		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.128	103
		4.18 C	0.00	0.00	3.09
215	ST IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.324	114
		6.31 C	0.00	1.70	0.00
216	LD L75X75X8		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.344	106
		2.66 C	0.00	-0.05	2.25
217	ST IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.099	103
		0.08 T	0.00	0.76	0.00
218	LD L90X90X8		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.724	113
		10.08 C	0.00	-0.06	2.35
219	TAP ERED		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.052	103
		0.23 T	0.01	-3.32	0.00
220	TAP ERED		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.059	113
		4.79 T	-0.22	-2.76	0.00
221	TAP ERED		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.060	103
		1.21 T	-0.03	-3.76	1.08
222	TAP ERED		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.040	112
		3.85 C	0.17	1.68	1.13
223	ST IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.078	112
		0.59 C	0.13	0.00	0.00
224	ST IPE300		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.296	103
		0.33 T	-0.01	-4.52	0.43
225	ST IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.114	103
		0.13 C	0.00	-1.01	0.60

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MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
226 ST	IPE300		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.252	103
		1.41 C	0.01	-3.79	0.85
227 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.125	103
		0.41 T	0.00	-1.11	0.62
228 ST	IPE300		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.248	103
		1.71 C	0.00	-3.74	0.85
229 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.107	103
		0.15 T	0.00	-0.96	0.55
230 ST	IPE300		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.363	103
		0.44 C	-0.01	-5.55	0.85
231 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.170	103
		1.11 T	0.00	-1.48	0.75
232 ST	IPE300		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.495	113
		3.05 C	0.00	7.49	0.00
233 ST	IPE300		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.632	103
		1.93 C	0.00	-9.69	0.45
234 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.373	103
		0.10 C	0.00	-3.38	0.00
235 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.088	103
		0.00 C	0.00	-0.80	0.45
236 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.062	103
		0.03 C	0.00	-0.53	0.00
237 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.133	112
		1.49 C	0.00	0.00	0.00
238 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.059	112
		0.68 C	0.00	0.00	0.00
239 ST	IPE300		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.521	103
		1.88 C	0.00	7.97	0.78
240 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.046	115
		0.56 C	0.00	0.00	0.00

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MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
241 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.010	108
		0.12 C	0.00	0.00	0.00
242 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.070	103
		0.34 T	0.00	-0.62	0.00
243 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.206	103
		0.37 C	0.00	-1.85	0.00
244 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.034	112
		0.67 T	0.00	0.00	0.45
245 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.113	103
		0.16 T	0.00	-1.00	0.00
246 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.044	115
		0.62 C	0.00	0.00	0.00
247 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.123	103
		0.08 C	0.00	-1.12	0.38
248 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.418	103
		0.49 T	0.00	-3.38	0.00
249 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.125	103
		0.77 C	0.00	-1.09	1.12
250 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.015	112
		0.21 C	0.00	0.00	0.00
251 ST	IPE300		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.243	103
		0.44 C	0.02	-3.64	0.45
252 ST	IPE300		(EUROPEAN SECTIONS)		
		PASS	SHEAR-Y	0.428	103
		1.88 C	0.00	1.09	0.00
253 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.043	112
		0.89 C	0.00	0.00	0.00
254 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	TENSION	0.016	112
		0.29 T	0.00	0.00	0.00
255 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.038	112
		0.54 C	0.00	0.00	0.00

ALL UNITS ARE - MTON METE (UNLESS OTHERWISE Noted)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
256 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.085	115
		0.77 C	0.00	0.00	0.00
257 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.135	112
		1.21 C	0.00	0.00	0.00
258 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.106	112
		1.51 C	0.00	0.00	0.00
259 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.095	115
		1.35 C	0.00	0.00	0.00
260 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.099	112
		1.42 C	0.00	0.00	0.00
261 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.115	115
		1.64 C	0.00	0.00	0.00
262 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.051	114
		4.98 C	0.03	0.02	1.00
263 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.304	103
		0.41 T	0.00	-2.53	2.00
264 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.020	115
		0.28 C	0.00	0.00	0.00
265 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.027	112
		3.46 C	0.00	-0.05	0.09
266 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.112	103
		1.02 T	0.00	-0.96	0.00
267 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.025	113
		0.35 C	0.00	0.00	0.00
268 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.114	112
		6.27 C	-0.07	0.25	1.00
269 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.306	103
		0.75 T	0.00	-2.53	0.00
270 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.042	114
		3.58 C	0.03	0.02	0.00

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MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
271 ST	IPE300		(EUROPEAN SECTIONS)		
		PASS	SHEAR-Y	0.178	103
		1.70 C	0.00	0.01	0.00
272 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.034	112
		0.53 C	0.00	0.00	0.00
273 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.038	115
		0.59 C	0.00	0.00	0.00
274 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.086	112
		0.85 C	0.00	0.00	0.00
275 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.042	115
		0.41 C	0.00	0.00	0.00
276 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.071	113
		0.70 C	0.00	0.00	0.00
277 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.094	114
		0.92 C	0.00	0.00	0.00
278 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.034	114
		0.54 C	0.00	0.00	0.00
279 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.050	113
		0.78 C	0.00	0.00	0.00
280 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.175	103
		0.22 T	0.00	-1.35	0.00
281 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.037	112
		0.36 T	0.05	0.06	0.00
282 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.037	113
		0.54 C	0.00	0.00	0.00
283 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.036	113
		0.29 C	0.00	0.00	0.00
284 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.151	103
		1.17 C	0.00	-1.30	1.00
285 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.322	103
		0.34 T	0.00	-2.68	2.00

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MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
286 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.023	113
		0.32 C	0.00	0.00	0.00
287 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.076	103
		0.95 C	0.00	-0.63	0.00
288 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.124	103
		0.22 T	0.00	-1.11	0.00
289 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.014	115
		0.19 C	0.00	0.00	0.00
290 ST	IPE300		(EUROPEAN SECTIONS)		
		PASS	SHEAR-Y	0.185	103
		1.40 C	0.00	0.33	0.00
291 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.043	114
		0.65 C	0.00	0.00	0.00
292 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.069	113
		1.04 C	0.00	0.00	0.00
293 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.031	112
		0.46 C	0.00	0.00	0.00
294 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.017	106
		0.26 C	0.00	0.00	0.00
295 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.189	103
		1.08 C	0.00	-1.52	1.00
296 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.114	114
		1.12 C	0.00	0.00	0.00
297 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.321	103
		0.25 T	0.00	-2.68	0.00
298 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.098	113
		0.97 C	0.00	0.00	0.00
299 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.150	103
		1.00 C	0.00	-1.30	0.00
300 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.020	108
		0.20 C	0.00	0.00	0.00

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MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
301 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.032	112
		0.32 C	0.00	0.00	0.00
302 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.187	103
		0.19 T	0.00	-1.44	0.00
303 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.036	113
		0.52 C	0.00	0.00	0.00
304 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.054	112
		0.67 T	0.05	-0.15	0.00
305 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.033	113
		0.26 C	0.00	0.00	0.00
306 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.082	103
		1.09 C	0.00	-0.67	0.00
307 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.043	112
		0.74 C	0.00	0.00	0.45
308 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.135	103
		0.59 T	0.00	-1.19	0.00
309 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.047	112
		0.98 T	0.00	0.00	0.00
310 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.132	103
		0.81 C	-0.01	-1.12	0.00
311 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.061	112
		0.86 C	0.00	0.00	0.00
312 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.222	103
		0.19 T	-0.01	-1.98	0.00
313 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.153	103
		0.45 C	0.00	-1.35	0.00
314 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.303	103
		0.74 C	0.00	-2.38	1.15
315 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.153	103
		0.68 T	0.00	-1.35	1.15

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MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
316 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.135	112
		1.89 C	0.00	0.00	0.00
317 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.120	103
		0.38 T	0.01	-1.01	0.60
318 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.210	103
		1.00 C	0.01	-1.78	0.60
319 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.084	112
		1.85 C	-0.01	0.00	0.45
320 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.518	112
		6.23 C	0.01	-0.02	0.43
321 ST	IPE300		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.247	103
		0.33 T	-0.04	-3.63	0.45
322 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.065	112
		1.38 T	-0.01	0.00	0.00
323 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.190	115
		2.90 C	0.00	0.00	0.00
324 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.180	112
		2.76 C	0.00	0.00	0.00
325 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.156	115
		2.39 C	0.00	0.00	0.00
326 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.177	112
		2.71 C	0.00	0.00	0.00
327 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.424	112
		3.74 C	0.00	0.00	0.00
328 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.345	115
		3.04 C	0.00	0.00	0.00
329 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.089	115
		1.28 C	0.00	0.00	0.00
330 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.114	112
		1.65 C	0.00	0.00	0.00

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MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
331 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	TENSION	0.064	112
		1.20 T	0.00	0.00	0.00
332 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.123	112
		1.77 C	0.00	0.00	0.00
333 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.196	103
		0.12 T	-0.01	-1.55	1.34
334 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.271	103
		0.50 T	0.02	-2.34	0.60
335 ST	IPE300		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.219	103
		0.69 C	0.00	-3.36	0.00
336 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.159	103
		0.13 T	0.02	-1.34	0.60
337 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.030	112
		3.35 C	0.01	-0.06	0.00
338 ST	IPE300		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.332	103
		0.09 C	0.00	-5.15	0.33
339 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.444	112
		5.10 C	0.01	-0.02	0.00
340 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.066	112
		1.02 C	0.00	0.00	0.00
341 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	TENSION	0.057	112
		1.06 T	0.00	0.00	0.00
342 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	TENSION	0.051	112
		0.95 T	0.00	0.00	0.00
343 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.065	112
		1.01 C	0.00	0.00	0.00
344 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.077	112
		1.02 C	0.00	0.00	0.00
345 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	TENSION	0.050	112
		0.93 T	0.00	0.00	0.00

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MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
346 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.077	115
		1.02 C	0.00	0.00	0.00
347 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.107	112
		1.42 C	0.00	0.00	0.00
348 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.042	103
		0.10 T	0.00	-0.37	0.00
349 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.138	103
		0.03 T	0.00	-1.25	0.00
350 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.041	115
		0.46 C	0.00	0.00	0.00
351 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.106	103
		0.88 C	0.00	0.76	2.63
352 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.028	114
		0.31 C	0.00	0.00	0.00
353 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.302	103
		0.03 T	0.00	-2.73	1.31
354 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.415	103
		0.03 T	0.00	-3.21	1.31
355 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.301	103
		0.02 T	0.00	-2.73	0.00
356 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.052	112
		0.69 C	0.00	0.00	0.00
357 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.042	113
		0.57 C	0.00	0.00	0.00
358 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.181	113
		1.11 C	0.00	0.00	0.00
359 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.182	114
		1.12 C	0.00	0.00	0.00
360 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.186	113
		1.14 C	0.00	0.00	0.00

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MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
361 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.178	114
		1.09 C	0.00	0.00	0.00
362 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.301	103
		0.00 T	0.01	-2.67	0.60
363 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.049	106
		0.66 C	0.00	0.00	0.00
364 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.058	112
		0.78 C	0.00	0.00	0.00
365 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.029	115
		0.42 C	0.00	0.00	0.00
366 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.042	115
		0.34 C	0.00	0.00	0.00
367 ST	IPE400		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.304	103
		20.84 C	0.32	-5.64	0.00
368 ST	IPE400		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.316	103
		20.37 C	0.32	6.09	0.00
369 ST	L90X90X8		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.035	103
		0.90 C	0.00	0.00	1.28
370 ST	L90X90X8		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.021	103
		0.89 C	0.00	0.00	0.64
371 ST	IPE400		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.113	103
		0.27 T	0.08	-3.15	1.10
372 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.180	103
		0.18 C	0.00	-1.62	0.83
373 ST	IPE400		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.136	113
		0.52 T	0.04	-4.12	1.10
374 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.204	103
		0.37 T	0.00	-1.82	0.87
375 ST	IPE400		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.148	113
		1.04 T	0.10	-4.10	1.10

ALL UNITS ARE - MTON METE (UNLESS OTHERWISE Noted)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
376 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.205	103
		2.47 T	0.00	-1.72	0.85
377 ST	IPE400		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.151	112
		0.10 C	0.29	-3.16	0.00
378 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.136	103
		1.12 T	0.00	-1.18	0.65
379 ST	IPE400		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.211	112
		0.46 C	0.38	-4.54	0.00
380 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.115	103
		2.69 T	-0.01	-0.85	0.60
381 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.338	103
		0.03 C	0.00	-2.95	0.00
382 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.340	103
		0.37 T	0.00	-2.95	1.65
383 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.186	103
		4.48 C	-0.02	-1.29	0.85
384 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.040	112
		0.48 C	0.00	0.00	0.00
385 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.101	103
		0.57 C	0.00	-0.88	0.00
386 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.181	103
		0.51 T	0.00	-1.62	0.00
387 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.022	115
		0.27 C	0.00	0.00	0.00
388 ST	IPE400		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.108	112
		1.04 C	-0.35	-1.31	0.55
389 ST	IPE400		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.135	112
		0.89 T	-0.26	-2.73	0.55
390 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.078	112
		1.03 C	0.00	0.00	0.00

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MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
391 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.119	115
		1.58 C	0.00	0.00	0.00
392 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	TENSION	0.009	113
		0.17 T	0.00	0.00	0.00
393 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.018	112
		0.24 C	0.00	0.00	0.00
394 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.271	115
		1.34 C	0.00	0.00	0.00
395 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.181	112
		0.89 C	0.00	0.00	0.00
396	TAP ERED		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.057	103
		0.02 T	0.01	-3.70	1.38
397 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.866	103
		0.20 C	0.00	-6.58	1.15
398 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.025	113
		0.30 C	0.00	0.00	0.00
399	TAP ERED		(EUROPEAN SECTIONS)		
		PASS	SHEAR-Y	0.046	103
		0.16 T	-0.01	-0.96	0.48
400 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.287	103
		0.22 C	0.00	-2.57	0.00
401 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.033	112
		0.47 C	0.00	0.00	0.55
402	TAP ERED		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.055	103
		0.23 T	0.02	-3.48	0.00
403 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.038	112
		0.46 C	0.00	0.00	0.00
404 ST	IPE360		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.222	103
		1.70 T	0.00	-5.44	0.55
405 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.031	103
		3.80 T	0.00	-0.10	0.00

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MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
406 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.025	106
		0.19 T	0.00	0.00	0.00
407 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.076	114
		1.05 C	0.00	0.00	0.00
408 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.103	113
		1.42 C	0.00	0.00	0.00
409 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.040	115
		0.56 C	0.00	0.00	0.00
410 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.039	112
		0.55 C	0.00	0.00	0.00
411 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.043	115
		0.61 C	0.00	0.00	0.00
412 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.025	112
		0.35 C	0.00	0.00	0.00
413	TAP ERED		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.060	103
		0.24 T	0.02	-3.86	0.57
414 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.219	114
		1.37 C	0.00	0.00	0.00
415 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.215	113
		1.35 C	0.00	0.00	0.00
416 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.089	103
		2.55 C	-0.02	0.56	0.88
417 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.215	103
		2.23 T	0.00	-1.82	0.00
418 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.011	109
		0.13 C	0.00	0.00	0.00
419 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.396	103
		1.92 T	0.00	-3.32	0.00
420 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.203	103
		1.49 C	0.00	-1.24	1.75

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MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
421 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.389	103
		0.74 T	0.00	-3.32	1.75
422 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.016	112
		0.20 C	0.00	0.00	0.00
423 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.133	115
		0.60 C	0.00	0.00	0.00
424 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.165	103
		0.74 C	0.00	0.00	0.00
425 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.089	112
		1.14 C	0.00	0.00	0.00
426 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.088	115
		1.12 C	0.00	0.00	0.00
427 ST	IPE400		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.144	112
		1.36 T	0.00	-4.55	0.00
428 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	TENSION	0.047	113
		0.88 T	0.00	0.00	0.00
429 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.055	113
		0.70 C	0.00	0.00	0.00
430 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.085	112
		1.73 C	0.11	-0.09	0.00
431 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.193	103
		0.52 T	0.00	-1.72	0.00
432 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.028	112
		0.34 C	0.00	0.00	0.00
433 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.156	112
		4.64 C	-0.12	0.21	1.70
434 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.372	103
		2.09 T	0.00	-3.13	1.70
435 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	TENSION	0.008	103
		0.15 T	0.00	0.00	0.00

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MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
436 ST	IPE400		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.151	112
		0.06 C	0.29	-3.16	0.55
437 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	TENSION	0.037	113
		0.70 T	0.00	0.00	0.00
438 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.049	113
		0.64 C	0.00	0.00	0.00
439 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.087	112
		1.13 C	0.00	0.00	0.00
440 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.085	115
		1.10 C	0.00	0.00	0.00
441 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.366	103
		0.95 T	0.00	-3.13	0.00
442 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.151	103
		0.71 C	0.00	0.00	0.00
443 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.140	115
		0.66 C	0.00	0.00	0.00
444 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.090	103
		2.88 C	0.00	-0.64	0.00
445 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.140	103
		1.72 T	0.00	-1.18	0.00
446 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.031	112
		0.37 C	0.00	0.00	0.00
447 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.209	103
		2.16 C	0.00	-1.31	1.60
448 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.284	103
		1.12 T	0.00	-2.43	1.60
449 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.015	115
		0.19 C	0.00	0.00	0.00
450 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.287	103
		1.73 T	0.00	-2.43	0.00

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MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
451 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.024	103
		0.35 C	0.00	0.00	0.00
452 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.027	114
		0.39 C	0.00	0.00	0.00
453 ST	IPE400		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.211	112
		0.46 C	0.38	-4.54	0.55
454 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.089	114
		1.28 C	0.00	0.00	0.00
455 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.089	113
		1.28 C	0.00	0.00	0.00
456 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.194	114
		1.00 C	0.00	0.00	0.00
457 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.314	113
		1.62 C	0.00	0.00	0.00
458 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.061	103
		0.01 C	0.01	-0.52	0.00
459 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.101	103
		0.36 C	0.01	-0.85	0.00
460 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.041	103
		0.50 C	0.00	0.00	0.00
461 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.150	103
		1.40 C	0.00	-1.03	1.40
462 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.204	103
		2.16 T	0.00	-1.72	1.40
463 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	TENSION	0.024	103
		0.45 T	0.00	0.00	0.00
464 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.340	103
		4.50 C	-0.01	2.77	0.00
465 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.033	103
		0.48 C	0.00	0.00	0.00

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MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
466 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	TENSION	0.042	103
		0.78 T	0.00	0.00	0.00
467 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.059	114
		0.87 C	0.00	0.00	0.00
468 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.048	103
		0.72 C	0.00	0.00	0.00
469 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.194	103
		0.21 T	0.00	-1.72	0.00
470 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.195	113
		1.20 C	0.00	0.00	0.00
471 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	TENSION	0.070	103
		1.31 T	0.00	0.00	0.00
472 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.028	115
		0.40 C	0.00	0.00	0.00
473 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.375	103
		2.13 C	0.00	3.27	0.66
474 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.055	113
		0.20 C	-0.09	0.04	1.44
475 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.157	112
		0.18 C	-0.17	-0.38	2.00
476 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.228	103
		0.18 C	0.11	-1.49	0.00
477 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.256	103
		0.14 T	-0.15	-1.55	0.00
478 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.187	103
		1.05 T	-0.04	-1.43	0.00
479 ST	UPN240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.029	113
		0.44 T	0.03	-0.05	0.95
480 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.229	107
		0.93 C	-0.37	0.11	0.56

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MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
481	ST IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.125	112
		11.43 C	0.04	-0.29	0.44
482	ST IPE360		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.293	103
		4.06 T	-0.02	-7.02	0.00
483	ST IPE360		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.332	103
		4.07 T	0.01	-8.04	0.00
484	ST IPE360		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.347	103
		0.01 C	0.00	-8.73	0.00
485	ST IPE360		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.350	103
		0.12 C	0.00	-8.81	0.00
486	ST IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.052	112
		1.47 T	-0.06	-0.10	0.00
487	ST IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.109	103
		0.60 C	-0.01	-0.92	0.00
488	ST IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.114	103
		0.37 T	0.00	-0.99	0.00
489	ST IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.321	112
		3.37 T	-0.04	-2.53	0.00
490	ST IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.416	103
		0.02 C	0.00	-2.85	2.33
491	ST IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.417	103
		0.04 C	0.00	-3.22	2.63
492	ST IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.390	103
		0.05 C	0.00	-3.04	2.55
493	ST IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.292	103
		0.06 C	0.00	-2.36	2.25
494	ST IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.217	103
		0.05 T	0.00	-1.65	1.83
495	ST IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.337	103
		0.00 C	0.00	-3.04	1.38

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MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
496 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.007	114
		0.08 C	0.00	0.00	0.00
497 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.093	103
		0.33 T	0.00	-0.82	0.00
498 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.171	103
		0.33 C	0.00	-1.52	0.00
499 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.051	113
		0.86 C	0.00	0.00	0.57
500 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.153	103
		0.21 T	0.00	-1.37	0.00
501 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.284	103
		0.20 C	0.00	-2.56	0.00
502 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.056	115
		0.65 C	0.00	0.00	0.00
503 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.344	103
		0.09 C	0.00	-3.10	0.34
504 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.086	112
		7.75 C	0.07	-0.03	0.57
505 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.093	112
		5.99 C	0.07	-0.17	0.57
506 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.052	112
		0.89 C	0.00	0.00	0.00
507 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.033	114
		0.45 C	0.00	0.00	0.00
508 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.019	113
		0.26 C	0.00	0.00	0.00
509 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.036	113
		0.22 C	0.00	0.00	0.00
510 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.012	109
		0.07 C	0.00	0.00	0.00

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MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
511 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.107	112
		1.47 C	0.00	0.00	0.00
512 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.101	115
		1.40 C	0.00	0.00	0.00
513 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.113	112
		1.56 C	0.00	0.00	0.00
514 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.112	115
		1.54 C	0.00	0.00	0.00
515 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.131	115
		1.51 C	0.00	0.00	0.00
516 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.265	103
		0.02 C	0.00	-2.36	0.00
517 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.092	115
		1.22 C	0.00	0.00	0.00
518 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.114	103
		0.13 T	0.00	-1.03	0.00
519 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.211	103
		0.07 C	0.00	-1.91	0.00
520 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.018	114
		0.21 C	0.00	0.00	0.00
521 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.272	103
		0.29 T	0.00	-1.86	1.75
522 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.406	103
		0.26 C	0.00	-3.49	1.75
523 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.021	112
		0.25 C	0.00	0.00	0.00
524 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.106	106
		5.73 C	0.00	0.25	1.70
525 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.382	103
		0.19 C	0.00	-3.29	1.70

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MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
526 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.010	115
		0.11 C	0.00	0.00	0.00
527 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.113	112
		6.83 T	0.12	-0.09	0.00
528 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.206	103
		0.96 C	0.00	-1.80	0.00
529 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.065	112
		0.76 C	0.00	0.00	0.00
530 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.082	103
		0.21 C	0.00	-0.72	0.57
531 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.087	114
		5.61 C	0.00	0.48	0.57
532 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.405	103
		0.09 C	0.00	-3.49	0.00
533 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.386	103
		0.81 C	0.00	-3.29	0.00
534 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.066	112
		0.83 C	0.00	0.00	0.00
535 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.052	115
		0.66 C	0.00	0.00	0.00
536 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.165	115
		0.72 C	0.00	0.00	0.00
537 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.123	113
		0.54 C	0.00	0.00	0.00
538 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.045	114
		0.57 C	0.00	0.00	0.00
539 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.050	113
		0.63 C	0.00	0.00	0.00
540 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.038	108
		0.48 C	0.00	0.00	0.00

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MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
541 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.013	107
		0.17 C	0.00	0.00	0.00
542 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.196	112
		0.90 C	0.00	0.00	0.00
543 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.457	115
		2.09 C	0.00	0.00	0.00
544 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.124	112
		1.59 C	0.00	0.00	0.00
545 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.119	115
		1.53 C	0.00	0.00	0.00
546 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.152	113
		11.74 C	-0.08	-0.28	0.00
547 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.145	103
		0.66 C	0.01	-1.23	0.00
548 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.059	112
		0.69 C	0.00	0.00	0.00
549 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.193	103
		0.52 T	0.00	-1.36	1.60
550 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.301	103
		0.78 C	-0.01	-2.55	1.60
551 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.036	113
		0.42 C	0.00	0.00	0.00
552 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.300	103
		0.58 C	-0.01	-2.55	0.00
553 ST	IPE360		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.201	113
		16.69 T	-0.01	3.85	0.00
554 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.040	113
		0.57 C	0.00	0.00	0.00
555 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	TENSION	0.027	113
		0.50 T	0.00	0.00	0.00

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MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
556 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.290	114
		1.45 C	0.00	0.00	0.00
557 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.718	115
		3.59 C	0.00	0.00	0.00
558 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.184	112
		2.63 C	0.00	0.00	0.00
559 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.091	115
		1.30 C	0.00	0.00	0.00
560 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.022	115
		0.20 T	-0.02	-0.09	0.00
561 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.086	103
		0.24 T	0.00	-0.76	0.00
562 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.145	115
		1.69 C	0.00	0.00	0.00
563 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.247	103
		0.34 T	0.00	2.16	0.78
564 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.087	114
		1.02 C	0.00	0.00	0.00
565 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.240	103
		0.56 C	0.00	-2.01	1.51
566 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.248	112
		3.84 C	0.00	0.00	0.00
567 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.215	115
		3.32 C	0.00	0.00	0.00
568 ST	IPE360		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.210	103
		4.06 T	-0.01	4.92	0.00
569 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.078	115
		1.21 C	0.00	0.00	0.00
570 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.144	112
		2.23 C	0.00	0.00	0.00

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571	ST L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.117	115
		1.35 C	0.00	0.00	0.00
572	ST IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.219	103
		0.06 T	0.00	-1.98	0.00
573	ST IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.200	103
		0.17 C	0.00	-1.72	0.00
574	ST L90X90X8		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.480	115
		7.20 C	0.00	0.00	0.00
575	ST L90X90X8		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.214	112
		4.29 C	0.00	0.00	0.00
576	ST L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.082	113
		0.95 C	0.00	0.00	0.00
577	ST L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.039	113
		0.51 C	0.00	0.00	0.00
578	ST L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.027	113
		0.35 C	0.00	0.00	0.00
579	ST IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.207	103
		0.17 C	0.00	-1.75	1.51
580	ST L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.045	115
		0.60 C	0.00	0.00	0.00
581	ST L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.038	115
		0.48 C	0.00	0.00	0.00
582	ST IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.417	103
		0.04 C	0.00	-3.22	0.00
583	ST L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.047	115
		0.54 C	0.00	0.00	0.00
584	ST IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.207	103
		0.01 T	0.00	-1.60	0.00
585	ST L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.020	113
		0.27 C	0.00	0.00	0.00

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MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
586 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.390	103
		0.05 C	0.00	-3.04	0.00
587 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.055	115
		0.63 C	0.00	0.00	0.00
588 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.084	107
		5.63 C	0.00	0.26	0.00
589 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.028	115
		0.37 C	0.00	0.00	0.00
590 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.292	103
		0.06 C	0.00	-2.36	0.00
591 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.091	115
		1.04 C	0.00	0.00	0.00
592 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.251	103
		0.20 C	0.00	-2.02	0.00
593 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.061	115
		0.81 C	0.00	0.00	0.00
594 ST	UPN240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.086	113
		0.41 C	0.07	-0.21	1.59
595 ST	IPE360		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.316	103
		1.28 T	0.00	7.89	0.66
596 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.057	112
		0.49 C	0.09	0.04	1.44
597 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.332	112
		3.54 T	-0.04	-2.65	0.00
598 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.475	112
		8.35 C	0.00	2.04	0.00
599 ST	IPE300		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.407	113
		18.32 C	0.00	0.11	0.00
600 LD	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.272	114
		4.55 C	0.00	0.01	1.65

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601 LD	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.307	115
		5.14 C	0.00	0.01	0.00
602 LD	L75X75X8		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.328	115
		8.03 C	0.00	-0.01	1.33
603 LD	L75X75X8		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.344	112
		8.33 C	0.00	-0.01	1.33
604 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.112	103
		0.68 T	0.00	0.98	0.00
605 LD	L75X75X8		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.488	113
		9.12 C	0.00	-0.02	1.55
606 LD	L75X75X8		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.529	112
		9.92 C	0.00	-0.02	1.55
607 ST	IPE300		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.345	103
		0.09 C	0.00	-5.34	0.10
608 LD	L75X75X8		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.412	113
		3.46 C	0.00	-0.04	2.20
609 LD	L75X75X8		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.484	112
		4.14 C	0.00	-0.04	2.20
610 LD	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.208	112
		0.75 C	-0.08	0.01	0.00
611 LD	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.233	113
		1.34 T	0.10	0.01	0.00
612 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.122	112
		0.34 C	-0.17	0.15	0.00
613 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.138	112
		0.36 C	-0.18	0.26	0.00
614 LD	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.086	107
		2.13 C	0.00	-0.01	1.21
615 LD	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.086	106
		2.14 C	0.00	-0.01	1.21

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616	LD L65X65X6		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.300	107
		4.41 C	0.00	-0.01	1.21
617	LD L65X65X6		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.301	106
		4.42 C	0.00	-0.01	1.21
618	LD L65X65X6		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.521	107
		4.66 C	0.00	-0.01	1.57
619	LD L65X65X6		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.522	106
		4.68 C	0.00	-0.01	1.57
620	LD L65X65X6		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.576	107
		6.95 C	0.00	-0.01	1.36
621	LD L65X65X6		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.577	106
		6.96 C	0.00	-0.01	1.36
622	ST IPE300		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.599	103
		25.72 C	-0.05	-0.38	2.35
623	ST IPE300		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.435	113
		20.26 C	0.02	3.50	2.35
624	ST IPE300		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.832	103
		37.81 C	0.00	-0.12	2.35
625	ST IPE300		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.496	112
		23.98 C	0.00	-3.98	2.35
626	ST IPE300		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.376	113
		17.80 C	0.01	4.04	2.35
627	ST IPE300		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.341	112
		18.19 C	-0.01	-3.51	2.35
628	ST IPE400		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.473	103
		39.39 C	0.01	-2.07	2.35
629	LD L75X75X8		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.160	107
		1.37 T	-0.10	0.02	1.65
630	LD L75X75X8		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.170	107
		1.27 T	-0.10	0.03	1.65

ALL UNITS ARE - MTON METE (UNLESS OTHERWISE Noted)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
631 LD	L75X75X8		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.181	106
		2.46 T	0.10	0.03	1.65
632 LD	L90X90X8		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.148	107
		1.14 C	-0.12	0.03	1.65
633 LD	L75X75X8		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.159	106
		1.23 T	0.10	0.02	0.00
634 LD	L75X75X8		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.169	107
		1.14 T	-0.10	0.03	0.00
635 LD	L75X75X8		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.177	106
		2.00 T	0.10	0.03	0.00
636 LD	L90X90X8		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.280	113
		10.67 C	0.00	0.03	0.00
637 LD	L75X75X8		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.288	113
		6.08 C	0.00	-0.01	1.44
638 LD	L75X75X8		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.278	112
		5.84 C	0.00	-0.01	1.44
639 LD	L75X75X8		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.297	113
		5.84 C	0.00	0.02	2.87
640 LD	L75X75X8		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.297	112
		5.85 C	0.00	0.02	2.87
641 LD	L75X75X8		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.274	113
		5.77 C	0.00	-0.01	1.44
642 LD	L75X75X8		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.275	112
		5.78 C	0.00	-0.01	1.44
643 LD	L90X90X8		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.414	113
		14.29 C	0.00	-0.02	1.44
644 LD	L90X90X8		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.408	112
		14.09 C	0.00	-0.02	1.44
645 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.182	103
		0.60 C	0.00	1.61	0.00

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MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
646 LD	L75X75X8		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.476	113
		10.26 C	0.00	-0.01	1.44
647 LD	L75X75X8		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.468	112
		10.09 C	0.00	-0.01	1.44
648 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	SHEAR-Y	0.181	103
		0.37 T	0.00	1.54	0.00
649 LD	L75X75X8		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.486	113
		10.58 C	0.00	0.01	2.87
650 LD	L75X75X8		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.504	112
		10.95 C	0.00	-0.01	1.44
651 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	SHEAR-Y	0.404	103
		0.27 T	-0.01	2.18	0.00
652 LD	L75X75X8		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.488	113
		10.55 C	0.00	-0.01	1.44
653 LD	L75X75X8		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.675	112
		14.71 C	0.00	-0.01	1.44
654 ST	IPE360		(EUROPEAN SECTIONS)		
		PASS	SHEAR-Y	0.269	103
		12.84 T	-0.04	2.37	0.00
655 LD	L90X90X8		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.431	113
		14.90 C	0.00	-0.02	1.44
656 LD	L90X90X8		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.811	112
		28.46 C	0.00	-0.02	1.44
657	TAP ERED		(EUROPEAN SECTIONS)		
		PASS	SHEAR-Y	0.066	103
		0.16 T	0.00	0.00	0.25
658	TAP ERED		(EUROPEAN SECTIONS)		
		PASS	SHEAR-Y	0.046	103
		0.19 T	0.00	-0.96	0.00
659 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.116	112
		1.09 C	0.19	0.00	0.00
660 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.092	112
		0.92 C	0.15	0.00	0.00

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MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
661	TAP ERED		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.060	103
		0.35 T	-0.04	-3.74	0.51
662 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.060	112
		0.58 C	0.10	0.00	0.00
663 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.053	112
		0.64 C	0.09	0.00	0.00
664	TAP ERED		(EUROPEAN SECTIONS)		
		PASS	SHEAR-Y	0.037	103
		0.35 T	0.02	-0.97	0.00
665 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.047	112
		0.64 C	0.08	0.00	0.00
666 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.038	112
		0.57 C	0.06	0.00	0.00
667 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.030	112
		0.63 C	0.05	0.00	0.00
668	TAP ERED		(EUROPEAN SECTIONS)		
		PASS	SHEAR-Y	0.034	103
		0.29 T	-0.01	-0.57	0.00
669 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.025	112
		0.57 C	0.04	0.00	0.00
670	TAP ERED		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.044	103
		0.10 T	-0.07	-2.62	0.87
671 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.020	112
		0.62 C	0.03	0.00	0.00
672	TAP ERED		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.043	103
		0.10 T	-0.06	-2.61	0.00
673 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.016	112
		0.62 C	0.02	0.00	0.00
674	TAP ERED		(EUROPEAN SECTIONS)		
		PASS	SHEAR-Y	0.035	103
		0.10 T	0.00	0.00	0.20
675 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.004	112
		0.26 C	0.00	0.00	0.00

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MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
676 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.078	112
		0.59 C	0.13	0.00	0.00
677 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.116	112
		1.09 C	0.19	0.00	0.00
678 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.092	112
		0.92 C	0.15	0.00	0.00
679 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.060	112
		0.58 C	0.10	0.00	0.00
680	TAP ERED		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.055	113
		5.70 T	0.22	-2.37	0.00
681 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.053	112
		0.64 C	0.09	0.00	0.00
682	TAP ERED		(EUROPEAN SECTIONS)		
		PASS	SHEAR-Y	0.050	103
		1.50 T	0.00	0.00	0.33
683 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.047	112
		0.64 C	0.08	0.00	0.00
684 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.038	112
		0.57 C	0.06	0.00	0.00
685	TAP ERED		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.016	112
		3.85 C	-0.14	0.23	0.00
686 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.030	112
		0.63 C	0.05	0.00	0.00
687	TAP ERED		(EUROPEAN SECTIONS)		
		PASS	SHEAR-Y	0.013	113
		2.46 T	0.00	0.00	0.21
688 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.025	112
		0.57 C	0.04	0.00	0.00
689	TAP ERED		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.045	103
		1.13 C	-0.05	-2.64	0.87
690 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.020	112
		0.62 C	0.03	0.00	0.00

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MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
691	TAP ERED		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.043	103
		1.07 C	-0.04	-2.63	0.00
692 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.016	112
		0.62 C	0.02	0.00	0.00
693	TAP ERED		(EUROPEAN SECTIONS)		
		PASS	SHEAR-Y	0.035	103
		0.45 C	0.00	0.00	0.20
694 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.004	112
		0.26 C	0.00	0.00	0.00
695 LD	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.047	107
		1.13 C	0.00	0.00	1.62
696 LD	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.239	112
		0.02 C	0.11	0.01	1.27
697 LD	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.080	107
		1.96 C	0.00	0.00	1.62
698 LD	L75X75X8		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.470	108
		7.20 C	0.00	0.02	2.55
699	TAP ERED		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.040	112
		3.85 C	0.17	1.68	0.00
700 LD	L90X90X8		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.447	107
		7.71 C	0.00	-0.04	2.11
701 LD	L90X90X8		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.447	106
		7.71 C	0.00	-0.04	2.11
702 LD	L90X90X8		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.396	107
		10.05 C	0.00	-0.03	1.73
703 LD	L90X90X8		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.401	106
		10.18 C	0.00	-0.03	1.73
704 LD	L75X75X8		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.471	109
		7.21 C	0.00	0.02	0.00
705 LD	L90X90X8		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.398	107
		10.09 C	0.00	-0.03	1.73

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MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
706	LD L90X90X8		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.403	106
		10.23 C	0.00	-0.03	1.73
707	LD L75X75X8		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.472	108
		7.24 C	0.00	0.02	2.55
708	ST IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.103	112
		4.87 C	-0.07	0.25	0.00
709	ST IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.142	112
		3.57 C	-0.12	0.21	0.00
710	ST IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.214	109
		7.66 C	0.00	0.10	0.00
711	LD L75X75X8		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.470	109
		7.22 C	0.00	0.02	0.00
712	LD L90X90X8		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.101	113
		0.97 C	-0.06	-0.03	1.90
713	LD L90X90X8		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.114	112
		1.58 C	-0.06	-0.03	1.90
714	LD L90X90X8		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.371	107
		6.30 C	0.00	-0.04	2.11
715	LD L90X90X8		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.370	106
		6.27 C	0.00	-0.04	2.11
716	LD L90X90X8		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.401	107
		10.17 C	0.00	-0.03	1.73
717	LD L90X90X8		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.404	106
		10.25 C	0.00	-0.03	1.73
718	LD L90X90X8		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.401	107
		10.17 C	0.00	-0.03	1.73
719	LD L90X90X8		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.403	106
		10.23 C	0.00	-0.03	1.73
720	ST L90X90X8		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.578	103
		2.95 C	0.08	0.00	3.35

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MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
721 LD	L90X90X8		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.665	103
		17.24 C	0.00	0.08	3.25
722	TAP ERED		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.059	103
		0.19 T	-0.01	-3.82	1.65
723	TAP ERED		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.051	103
		0.19 T	0.01	-3.31	0.47
724	TAP ERED		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.059	103
		0.19 T	-0.01	-3.84	0.20
725	TAP ERED		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.052	103
		0.35 T	0.06	-3.21	0.67
726	TAP ERED		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.060	103
		0.35 T	-0.04	-3.74	0.00
727	TAP ERED		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.038	103
		0.35 T	-0.05	-2.31	0.00
728	TAP ERED		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.041	103
		0.29 T	-0.01	-2.63	0.57
729	TAP ERED		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.045	103
		0.29 T	-0.05	-2.76	0.82
730	TAP ERED		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.048	103
		0.29 T	-0.04	-3.02	0.00
731	TAP ERED		(EUROPEAN SECTIONS)		
		PASS	SHEAR-Y	0.030	103
		0.10 T	-0.01	-0.40	0.45
732	TAP ERED		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.037	112
		0.32 T	-0.22	-1.67	0.73
733	TAP ERED		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.044	103
		0.10 T	-0.07	-2.63	0.03
734 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.023	115
		0.28 C	0.00	0.00	0.00
735 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.014	113
		0.17 C	0.00	0.00	0.00

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MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
736 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.007	113
		0.08 C	0.00	0.00	0.00
737 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.028	115
		0.34 C	0.00	0.00	0.00
738 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.023	113
		0.28 C	0.00	0.00	0.00
739 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.020	113
		0.24 C	0.00	0.00	0.00
740 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.011	113
		0.13 C	0.00	0.00	0.00
741 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.017	113
		0.21 C	0.00	0.00	0.00
742 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.021	115
		0.26 C	0.00	0.00	0.00
743 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.008	114
		0.09 C	0.00	0.00	0.00
744 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.020	113
		0.24 C	0.00	0.00	0.00
745 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.035	113
		0.43 C	0.00	0.00	0.00
746 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.119	103
		1.60 C	0.00	-0.99	0.00
747	TAP ERED		(EUROPEAN SECTIONS)		
		PASS	SHEAR-Y	0.037	103
		1.50 T	0.01	-0.98	0.54
748	TAP ERED		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.059	113
		4.79 T	-0.22	-2.76	0.51
749 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.274	103
		0.90 C	0.00	-1.79	1.75
750	TAP ERED		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.039	103
		0.09 T	0.07	-2.32	0.00

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MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
751 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.022	103
		2.03 T	0.00	-0.10	0.00
752 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.111	103
		1.30 T	0.00	-0.93	0.00
753	TAP ERED		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.011	113
		2.46 T	-0.10	-0.19	0.00
754 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.244	103
		0.14 C	0.00	-1.69	1.70
755	TAP ERED		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.024	112
		3.85 C	-0.21	0.53	0.82
756 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.022	103
		1.92 T	0.00	-0.10	0.00
757 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.148	107
		0.45 C	0.19	-0.34	0.00
758	TAP ERED		(EUROPEAN SECTIONS)		
		PASS	SHEAR-Y	0.030	103
		0.45 C	0.01	-0.41	0.45
759 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.212	103
		1.22 T	-0.03	-1.51	1.96
760	TAP ERED		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.040	112
		2.93 C	-0.20	-1.68	0.73
761	TAP ERED		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.045	103
		1.07 C	-0.05	-2.64	0.00
762 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.255	103
		0.07 T	-0.15	-1.55	0.55
763 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	TENSION	0.011	113
		0.21 T	0.00	0.00	0.00
764 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.011	107
		0.14 C	0.00	0.00	0.00
765 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.099	114
		0.45 C	0.00	0.00	0.00

ALL UNITS ARE - MTON METE (UNLESS OTHERWISE Noted)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
766 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.121	103
		0.55 C	0.00	0.00	0.00
767 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.095	113
		1.22 C	0.00	0.00	0.00
768 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.057	114
		0.74 C	0.00	0.00	0.00
769 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.046	115
		0.59 C	0.00	0.00	0.00
770 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.032	112
		0.41 C	0.00	0.00	0.00
771 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.016	103
		0.20 C	0.00	0.00	0.00
772 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	TENSION	0.016	113
		0.30 T	0.00	0.00	0.00
773 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	TENSION	0.035	103
		0.65 T	0.00	0.00	0.00
774 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.054	103
		0.71 C	0.00	0.00	0.00
775 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.117	113
		1.52 C	0.00	0.00	0.00
776 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.073	114
		0.95 C	0.00	0.00	0.00
777 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.229	103
		1.08 C	0.00	0.00	0.00
778 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.177	114
		0.83 C	0.00	0.00	0.00
779 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	TENSION	0.029	107
		0.54 T	0.00	0.00	0.00
780 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.015	112
		0.19 C	0.00	0.00	0.00

ALL UNITS ARE - MTON METE (UNLESS OTHERWISE Noted)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
781 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	TENSION	0.015	113
		0.28 T	0.00	0.00	0.00
782 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.032	115
		0.46 C	0.00	0.00	0.00
783 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.096	113
		1.39 C	0.00	0.00	0.00
784 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.086	112
		1.24 C	0.00	0.00	0.00
785 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.192	115
		0.99 C	0.00	0.00	0.00
786 ST	L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.252	114
		1.30 C	0.00	0.00	0.00
787 ST	IPE300		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1A-C	0.429	103
		26.58 C	0.00	3.34	1.50
788 ST	IPE300		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.468	112
		19.64 C	0.00	5.90	1.50
789 ST	IPE300		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.190	112
		12.19 C	0.01	2.04	1.50
790 ST	L90X90X8		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.187	103
		3.45 C	0.00	0.00	1.86
791 ST	L90X90X8		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.350	103
		6.45 C	0.00	0.00	1.86
792 ST	L90X90X8		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.180	103
		3.32 C	0.00	0.00	1.86
793 ST	IPE240		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.146	103
		0.34 T	0.01	1.24	0.00
794 ST	L90X90X8		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.075	108
		0.65 C	0.00	0.00	0.00
795 ST	L90X90X8		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.074	109
		0.64 C	0.00	0.00	0.00

ALL UNITS ARE - MTON METE (UNLESS OTHERWISE Noted)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
796	ST IPE300		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.237	103
		11.08 C	0.00	2.97	1.00
797	ST IPE300		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.233	112
		9.98 C	0.00	2.92	1.00
798	LD L65X65X6		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-C	0.302	112
		0.33 C	0.14	-0.01	1.49
799	LD L65X65X6		(EUROPEAN SECTIONS)		
		PASS	LRFD-H1-1B-T	0.274	112
		0.02 T	0.12	0.01	0.00
800	LD L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.068	107
		1.65 C	0.00	0.00	0.00
801	LD L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.048	108
		1.18 C	0.00	0.00	1.62
802	LD L65X65X6		(EUROPEAN SECTIONS)		
		PASS	COMPRESSION	0.161	107
		2.82 C	0.00	0.00	2.23

********** END OF TABULATED RESULT OF DESIGN **********

780. STEEL TAKE OFF ALL

STEEL TAKE-OFF

	PROFILE		**LENGTH(METE)**	**WEIGHT(MTON)**
ST	IPE300		119.64	5.042
ST	IPE400		36.06	2.387
ST	IPE240		380.39	11.651
ST	IPE360		44.01	2.506
ST	UPN140		1.55	0.025
Tapered	MembNo:	67	39.60	4.963
LD	L65X65X6		45.54	0.537
LD	L130X130X10		19.15	0.756
ST	UPN240		8.97	0.297
ST	L90X90X8		26.35	0.287
LD	L90X90X8		81.11	1.766
LD	L75X75X8		79.37	1.430
ST	L65X65X6		246.65	1.455
			TOTAL =	33.103

********** END OF DATA FROM INTERNAL STORAGE **********

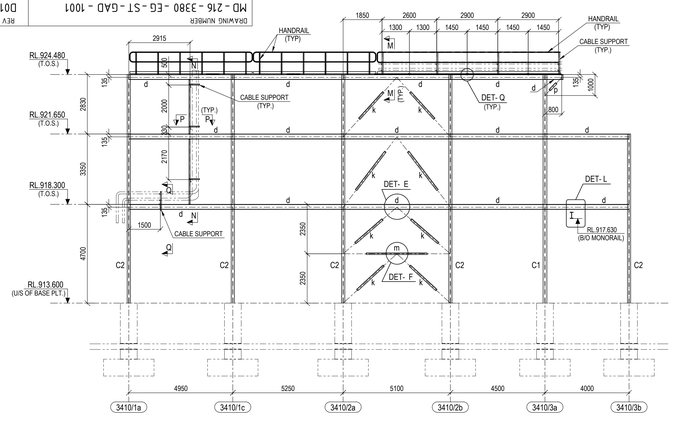
781. FINISH

********** END OF THE STAAD.Pro RUN **********

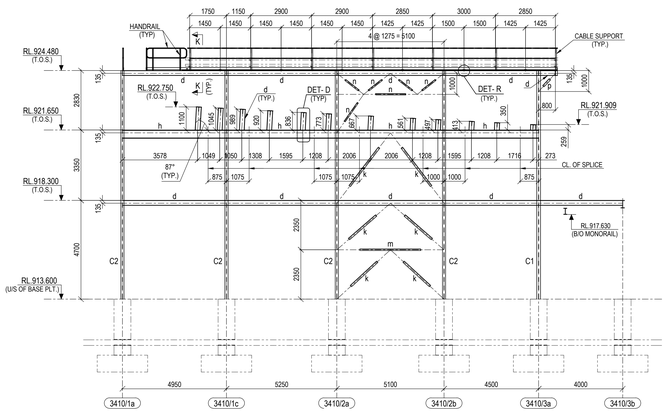
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*************************************************************
*        For questions on STAAD.Pro, please contact         *
*        Bentley Systems or Partner offices                 *
*                                                           *
*            Telephone             Web / Email              *
*  USA       +1  (714) 974-2500                             *
*  UK        +44 (0) 808 101 9246                           *
*  SINGAPORE +65 6225-6158                                  *
*  FRANCE    +33 (0) 1 55238400                             *
*  GERMANY   +49 0931 40468                                 *
*  INDIA     +91 (033) 4006-2021                            *
*  JAPAN     +81 (03)5952-6500    http://www.ctc-g.co.jp    *
*  CHINA     +86 21 6288 4040                               *
*  THAILAND  +66 (0)2645-1018/19 partha.p@reisoftwareth.com*
*                                                           *
*  Worldwide  http://selectservices.bentley.com/en-US/      *
*                                                           *
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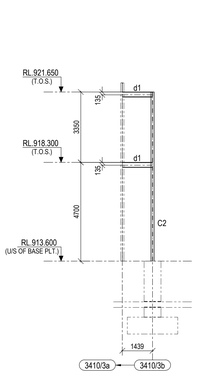
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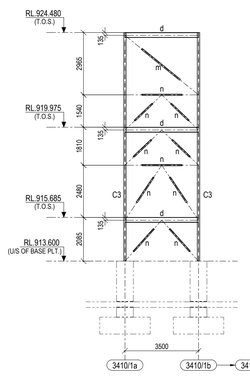
ELEVATION ON GRID- 3410/L
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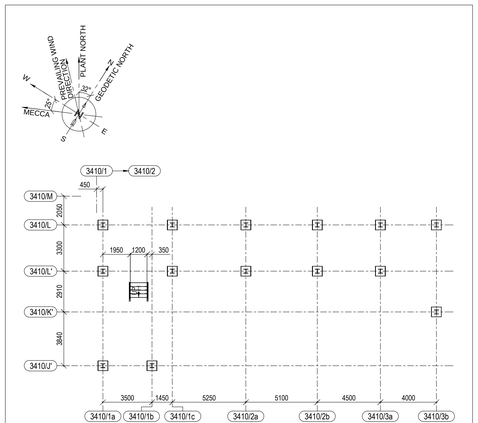
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(SCALE- 1:100)



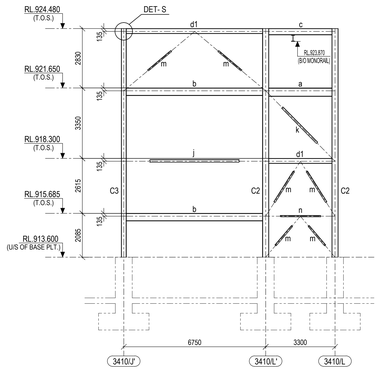
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(SCALE- 1:100)



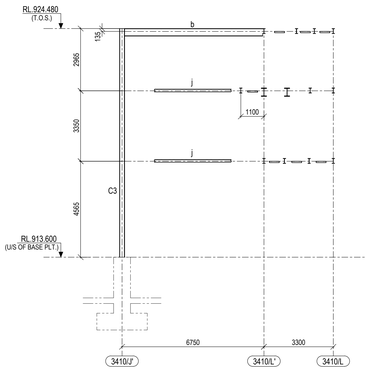
ELEVATION ON GRID- 3410/J'
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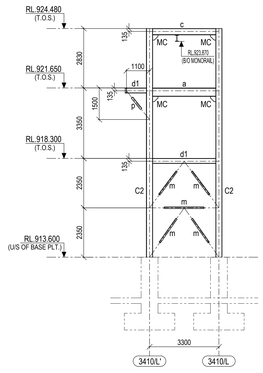
KEY PLAN
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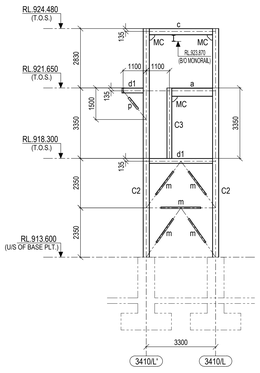
CROSS SECTION AT LINE- 3410/1a
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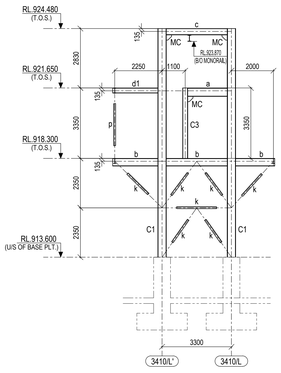
CROSS SECTION AT LINE- 3410/1b
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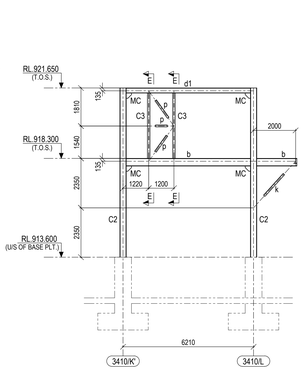
CROSS SECTION AT LINE- 3410/1c & 3410/2a
(SCALE- 1:100)



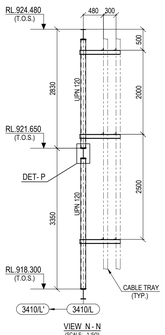
CROSS SECTION AT LINE- 3410/2b
(SCALE- 1:100)



CROSS SECTION AT LINE- 3410/3a
(SCALE- 1:100)



CROSS SECTION AT LINE- 3410/3b
(SCALE- 1:100)

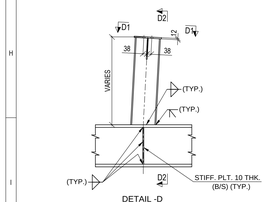


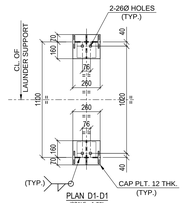
VIEW N - N
(SCALE: 1:50)

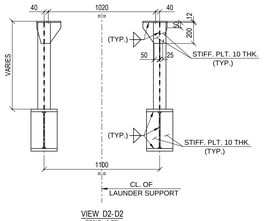
SCHEDULE OF MEMBERS

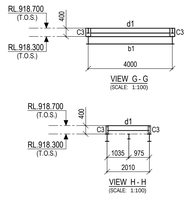
MEMBER MKD.	MEMBER	SECTON	FACTORED AXIAL FORCE(T)	FACTORED SHEAR FORCE(T)	REMARKS
C1	IPE 400	I	-	-	MATERIAL GRADE:S275 JR
C2	IPE 300	I	-	-	
C3	IPE 240	I	-	-	
P1	IPE 240	I	-	-	
a	IPE400	I	5.0	8.0	
b	IPE360	I	17.0	13.0	
b1	IPE360	I	14.0	10.0	
c	IPE300	I	4.0	14.0	
d	IPE240	I	13.0	6.0	
d1	IPE240	I	8.0	9.0	
d2	IPE240	I	12.0	12.0	
e	UPN240	[	-	-	
e1	UPN240	[	-	-	
f	UPN200	[	-	-	
h	1/WEB PLT. 350x10 THK. 2/FLG. PLT. 250x25 THK.		7.0	4.0	
j	L - 130X130X10	⌜⌝	-	-	
k	L - 90X90X8	⌜⌝	29.0	-	
m	L - 75X75X8	⌜⌝	15.0	-	
n	L - 65X65X6	⌜⌝	7.0	-	
p	L - 90X90X8	⌜	7.0	-	
p1	L - 90X90X8	⌜	8.0	-	
q	L - 65X65X6	⌜	5.0	-	
r	L - 50X50X6	⌜	-	-	
s	L - 75X75X8	⌜	-	-	

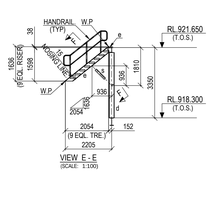
ALL GUSSET PLATE ARE 10mm THK. UNLESS OTHERWISE NOTED.
FOR DEVELOPING CONNECTIONS MEMBER END FORCES MENTIONED IN SCHEDULE OF MEMBERS SHALL BE FOLLOWED.
MINIMUM 6mm FILLET WELD OF 150 mm LENGTH SHALL BE PROVIDED FOR WELDED CONNECTION.
FOR BOLTED CONNECTION, MINIMUM 2 Nos. M20 BOLTS SHALL BE PROVIDED.

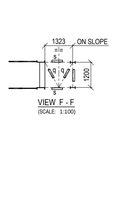


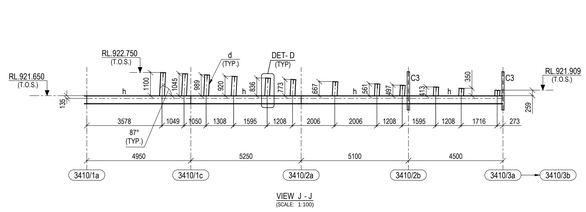


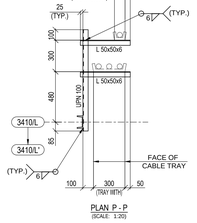


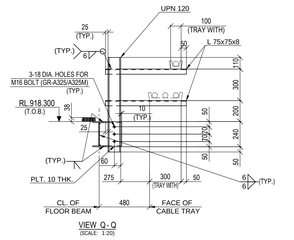


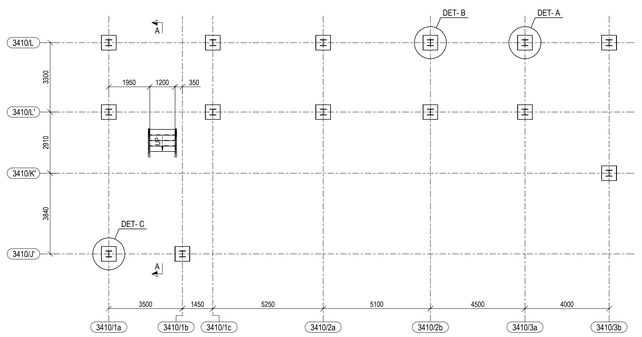




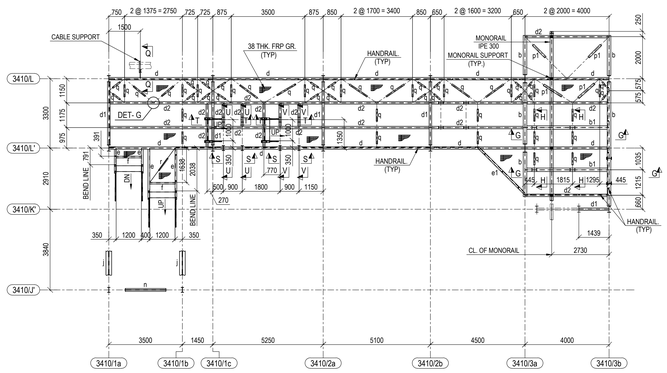




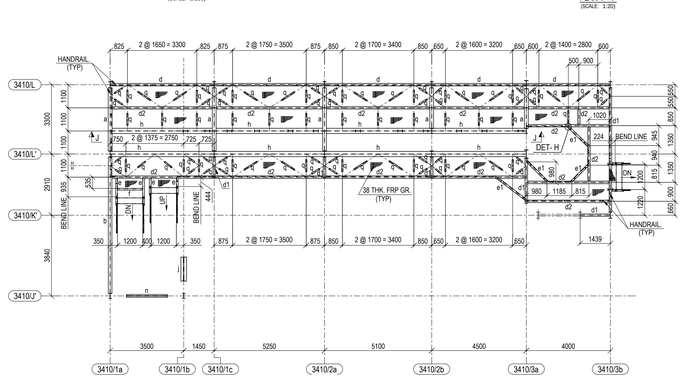




PLAN AT RL 913.600
(SCALE: 1:100)



PLAN AT RL 918.300
(ALL PLAN BRACING ARE BELOW 135 MM. T.O.S)
(SCALE: 1:100)



PLAN AT RL 921.650
(ALL PLAN BRACING ARE BELOW 135 MM. T.O.S)
(SCALE: 1:100)

NOTES:

1. ALL DIMENSIONS & CO-ORDINATES ARE IN MILLIMETER AND LEVELS ARE IN METER UNO.
2. ANCHOR BOLT MATERIAL SHALL BE AS PER ASTM F 1554 Gr.36.
3. HANDRAIL PIPE / POST MATERIAL SHALL BE AS PER ASTM A53 Gr.B
4. SIZE OF WELD SHALL BE DECIDED BASED ON LESSER THICKNESS OF THE JOINED MEMBER (UNLESS NOTED OTHERWISE).
5. ALL WELDS SHALL BE OF SHOP QUALITY & TO BE TESTED AS PER QUALITY ASSURANCE PLAN.
6. ALL PERMANENT STRUCTURAL BOLTS CONFORM TO ASTM A 325M.
7. PAINTING ON STRUCTURES WILL BE AS PER DESIGN BASIS REPORT (DBR).

LEGENDS:-

TOS - TOP OF STEEL
U/S - UNDER SIDE
RL - REDUCED LEVEL
T.O.F.F.L - TOP OF FINISHED FLOOR LEVEL
F/S - FAR SIDE
N/S - NEAR SIDE
DN. - DOWN
W.P. WORKING POINT
GR. - GRATING
MC - MOMENT CONNECTION
TYP - TYPICAL
PLT. - PLATE
THK. - THICKNESS
B/O - BOTTOM OF

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		D01	18/11/2023	AS BUILT	MK	SYC	SGI	PRI	
		C02	29/08/2020	ISSUED FOR CONSTRUCTION & REVISED AS MKD.	MK	SYC	SGI	PRI	
		C01	21/07/2020	ISSUED FOR CONSTRUCTION	MK	SYC	SGI	PRI	
		A02	20/06/2020	REVISED AS PER EMPLOYER'S COMMENTS DATED 11/06/2020	MK	SYC	SGI	PRI	
MD-216-1000-EG-ST-CRT-1001	CIVIL,STRUCTURAL AND ARCHITECTURAL DESIGN CRITERIA	A01	18/05/2020	ISSUED FOR REFERENCE	MK	SYC	SGI	PRI	

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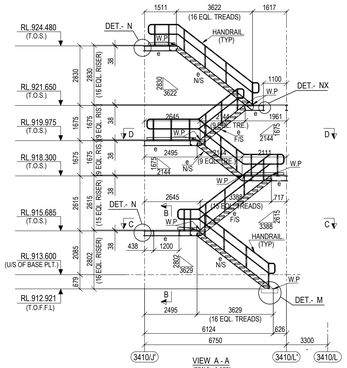
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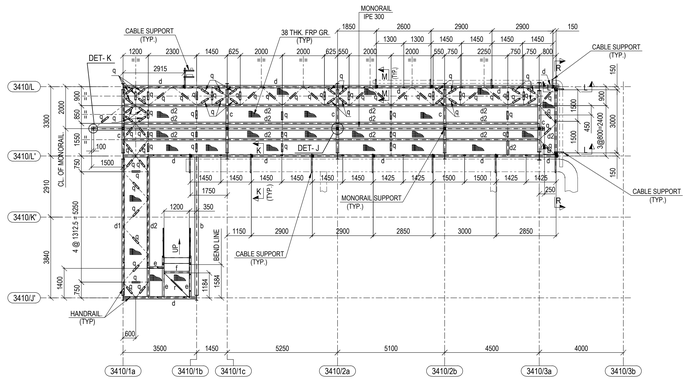
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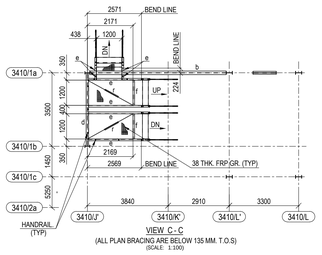
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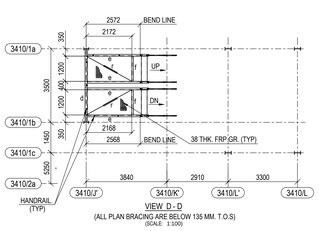
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OUTOTEC DRAWING NUMBER	900707_GAU01_HL15 _MTL01_33805	DOCUMENT ID OU100674668	REVISION 3
DRAWN	MK	14/05/2020	MANSOURAH - MASSARAH GOLD PROJECT
CHECKED	SYC	15/05/2020	GENERAL ARRANGEMENT DRAWING
ENGINEER	SGI	16/05/2020	ACIDULATION BUILDING
APPROVED	PRI	18/05/2020	PLAN, ELEVATIONS, SECTIONS & DETAILS
EMPLOYER			
SCALE	1 : 20, 100 & 150	A0	DRAWING NUMBER MD - 216 - 3380 - EG - ST - GAD - 1001 REV D01

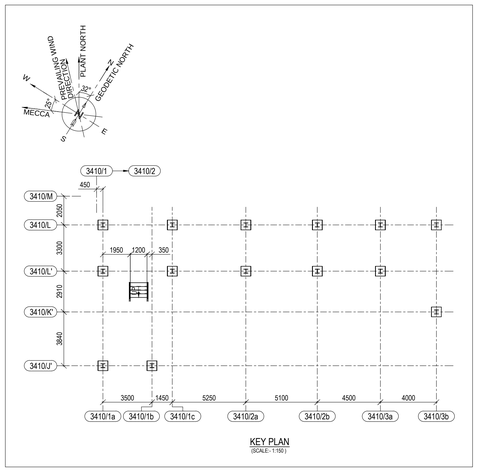
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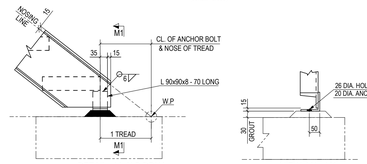


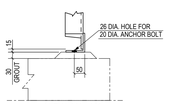


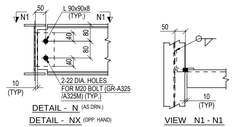


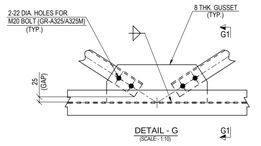


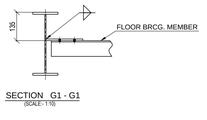


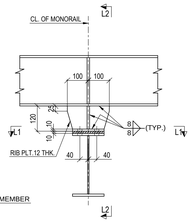


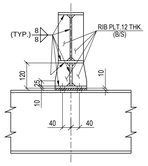


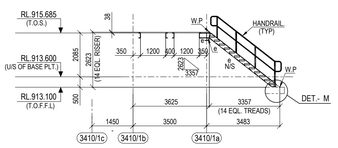


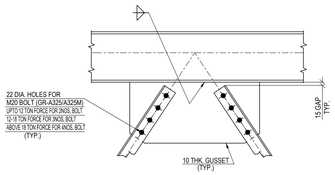


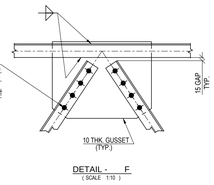


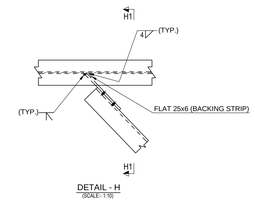


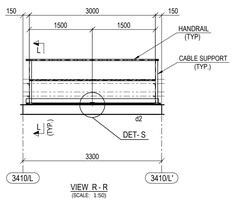


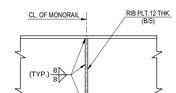


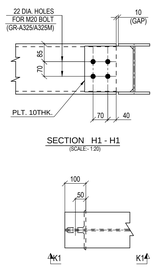


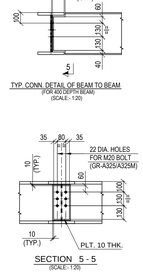


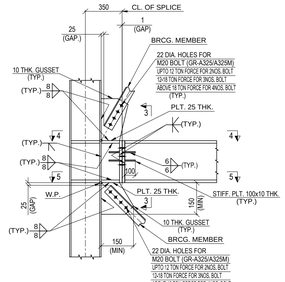


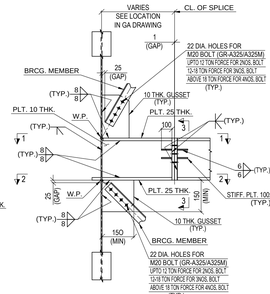


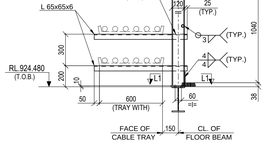


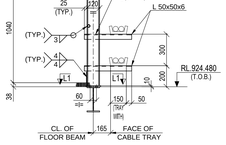


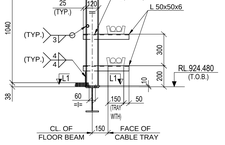


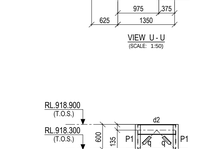


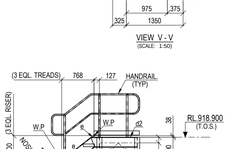


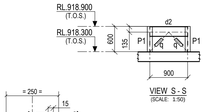


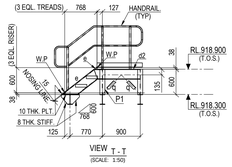


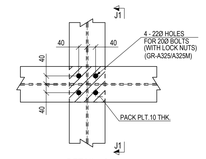


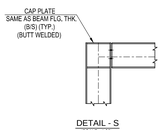


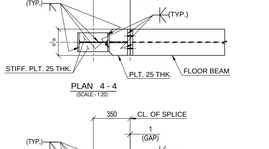


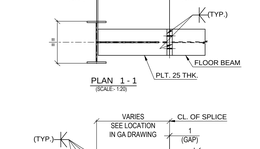


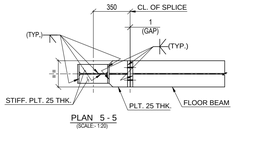


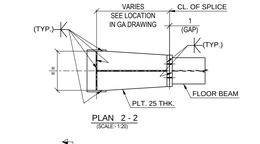


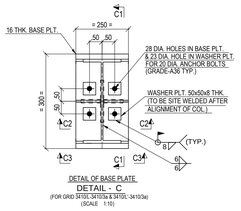


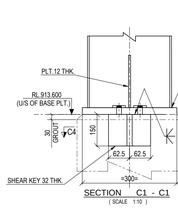


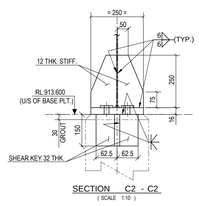


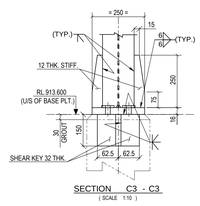


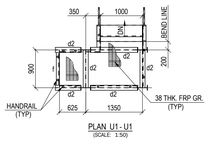


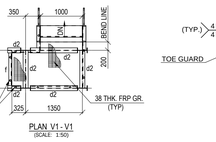


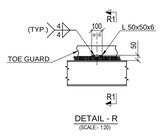


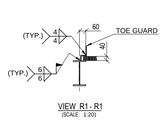


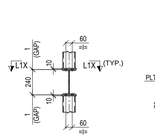


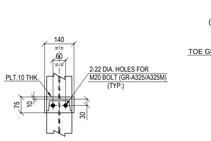


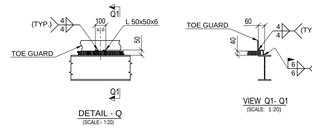


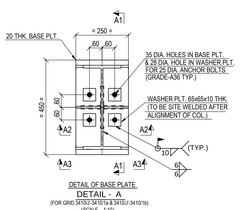


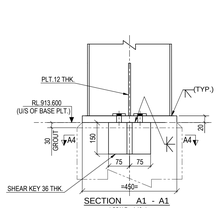


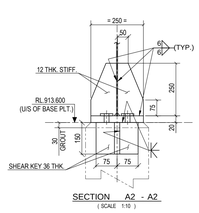


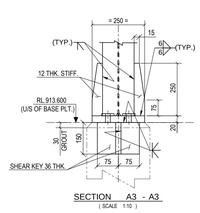


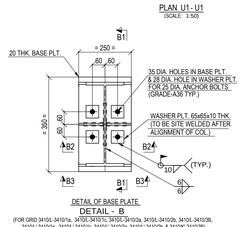


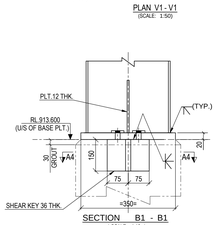


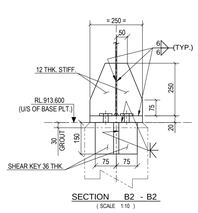


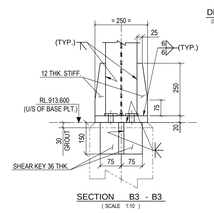


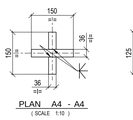


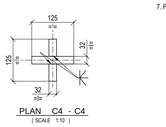












NOTES:

1. ALL DIMENSIONS & CO-ORDINATES ARE IN MILLIMETER AND LEVELS ARE IN METER UNO.
2. ANCHOR BOLT MATERIAL SHALL BE AS PER ASTM F 1554 Gr.36.
3. HANDRAIL PIPE / POST MATERIAL SHALL BE AS PER ASTM A53 Gr.B
4. SIZE OF WELD SHALL BE DECIDED BASED ON LESSER THICKNESS OF THE JOINED MEMBER (UNLESS NOTED OTHERWISE).
5. ALL WELDS SHALL BE OF SHOP QUALITY & TO BE TESTED AS PER QUALITY ASSURANCE PLAN.
6. ALL PERMANENT STRUCTURAL BOLTS CONFORM TO ASTM A 325M.
7. PAINTING ON STRUCTURES WILL BE AS PER DESIGN BASIS REPORT (DBR).

LEGENDS:-

TOS - TOP OF STEEL
U/S - UNDER SIDE
RL - REDUCED LEVEL
T.O.F.F.L - TOP OF FINISHED FLOOR LEVEL
F/S - FAR SIDE
N/S - NEAR SIDE
DN. - DOWN
W.P. WORKING POINT
GR. - GRATING
MC. - MOMENT CONNECTION
TYP. - TYPICAL
PLT. - PLATE
THK. - THICKNESS
B/O - BOTTOM OF

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		D01	18/11/2023	AS BUILT	MK	SYC	SGI	PRI	
		C02	29/08/2020	ISSUED FOR CONSTRUCTION & REVISED AS MKD.	MK	SYC	SGI	PRI	
		C01	17/06/2020	ISSUED FOR CONSTRUCTION	MK	SYC	SGI	PRI	
MD-216-1000-EG-ST-CRT-1001	CIVIL,STRUCTURAL AND ARCHITECTURAL DESIGN CRITERIA	A01	18/05/2020	ISSUED FOR REFERENCE	MK	SYC	SGI	PRI	

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MA'ADEN معادن Saudi Arabian Mining Company شركة معادن العربية السعودية

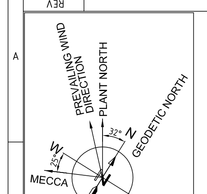
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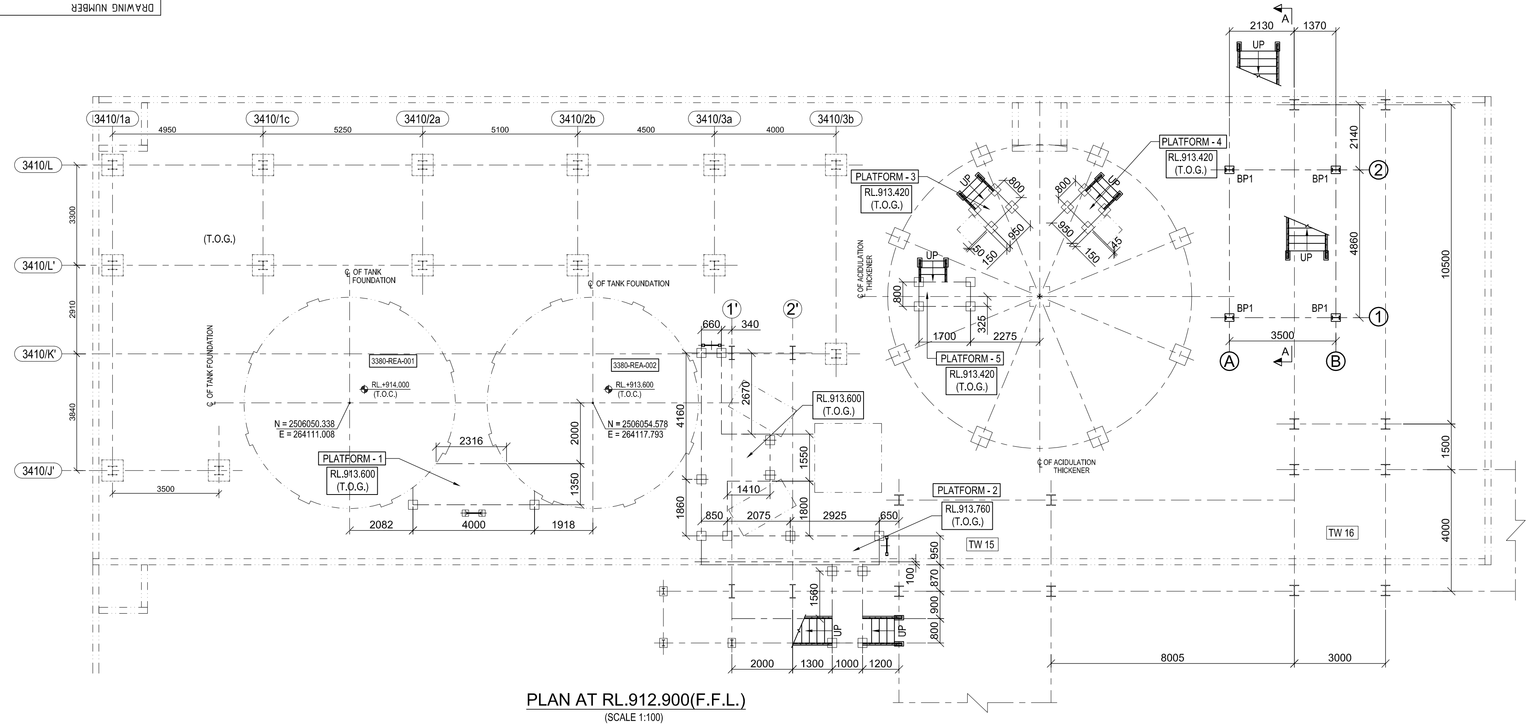
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DOCUMENT ID	OU100674669
REVISION	3
DRAWN	MK 14/05/2020
CHECKED	SYC 15/05/2020
ENGINEER	SGI 16/05/2020
APPROVED	PRI 18/05/2020
EMPLOYER	
SCALE	1 : 10, 20, 100 & 150
	A0

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
ACIDULATION BUILDING
PLAN, ELEVATIONS, SECTIONS & DETAILS

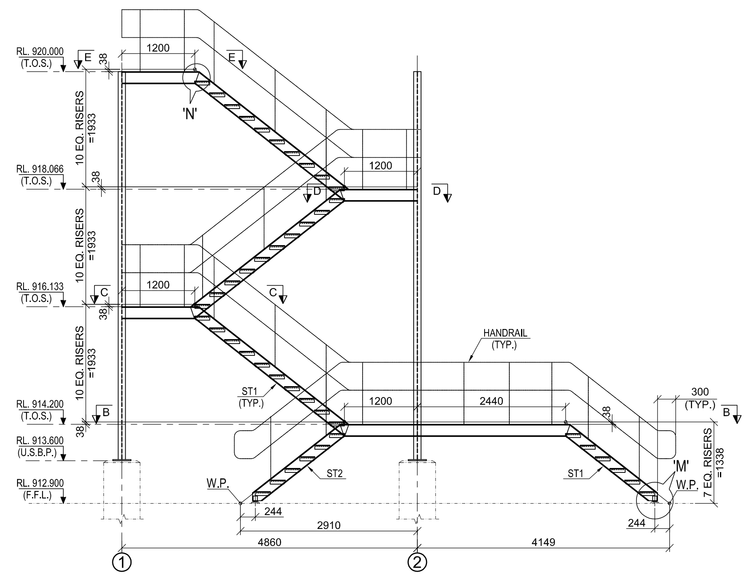
DRAWING NUMBER: MD-216-3380-EG-ST-GAD-1002
REV: D01

DRAWING NUMBER: MD - 216 - 3380 - EG - ST - GAD - 1003 | REV: A01





PLAN AT RL.912.900(F.F.L.)
(SCALE 1:100)

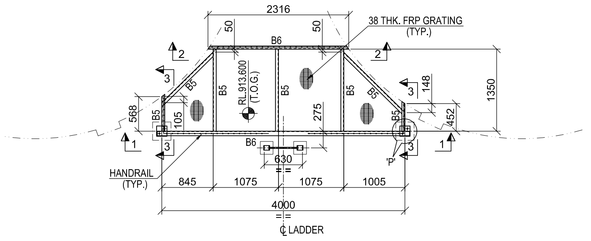


SECTION A-A
(SCALE 1:50)

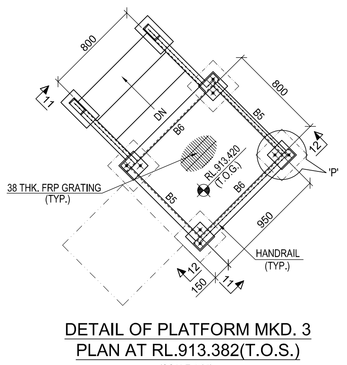
SCHEDULE OF MEMBERS

MEMBER MARKED	MEMBER	SECTON	FACTORED AXIAL FORCE(KN)	FACTORED SHEAR FORCE(KN)	REMARKS
C1	IPE 200	I	80	-	GUSSET PLATE 8 THK.
B1	IPE 200	I	-	50	
B2	UPN 140	[	30	-	
B3	UPN 100	[	25	-	
B4	UPN 140	[	-	40	
B5	UPN 100	[	-	30	
B6	UPN 200	[	-	40	
ST1	UPN 240	[	-	40	
ST2	UPN 200	[	-	40	
a1	L 65X65X6	L	25	-	
b1	L 65X65X6	L	30	-	
b2	L 75X75X8	L	50	-	
P1	UPN 140	[	60	-	

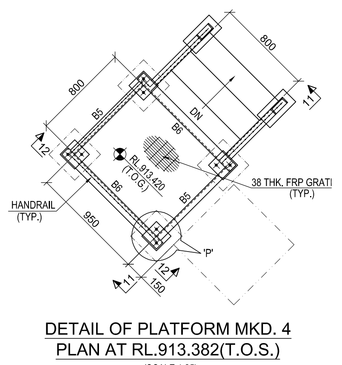
GRADE OF MATERIAL IS S275 JR.
FOR DEVELOPING CONNECTIONS MEMBER END FORCES MENTIONED IN SCHEDULE OF MEMBERS SHALL BE FOLLOWED.
ALL MEMBER CONNECTIONS SHOULD BE DESIGN FOR A MINIMUM FORCE OF 50KN.
MINIMUM 6mm FILLET WELD OF 150 mm LENGTH SHALL BE PROVIDED FOR WELDED CONNECTION. FOR BOLTED CONNECTION, MINIMUM 2 Nos. M20 BOLTS SHALL BE PROVIDED.



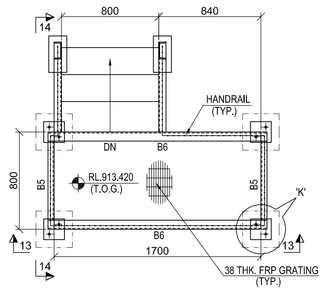
DETAIL OF PLATFORM MKD. 1
PLAN AT RL.913.562(T.O.S.)
(SCALE 1:50)



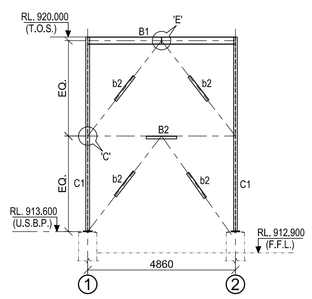
DETAIL OF PLATFORM MKD. 3
PLAN AT RL.913.382(T.O.S.)
(SCALE 1:25)



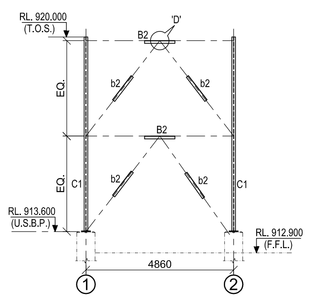
DETAIL OF PLATFORM MKD. 4
PLAN AT RL.913.382(T.O.S.)
(SCALE 1:25)



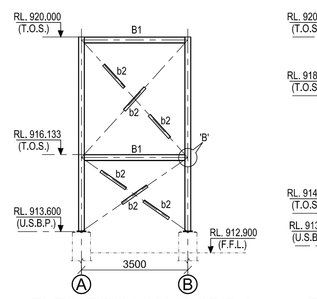
DETAIL OF PLATFORM MKD. 5
PLAN AT RL.913.382(T.O.S.)
(SCALE 1:25)



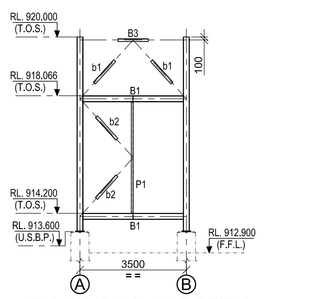
ELEVATION ALONG GRID A
(SCALE 1:100)



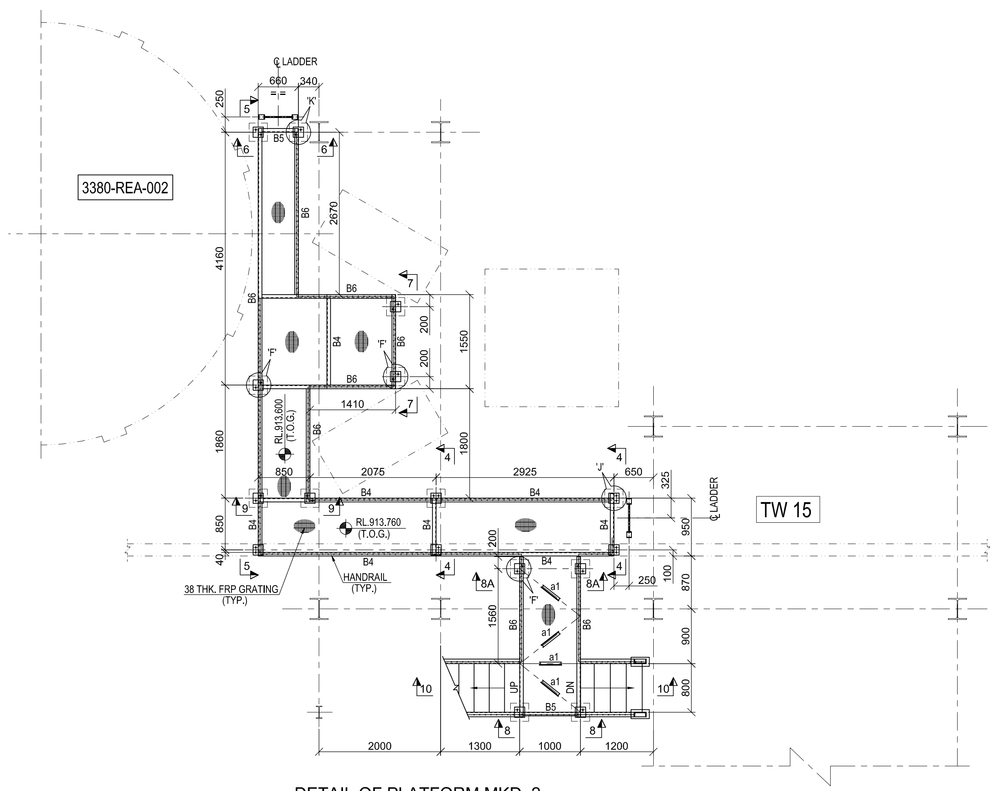
ELEVATION ALONG GRID B
(SCALE 1:100)



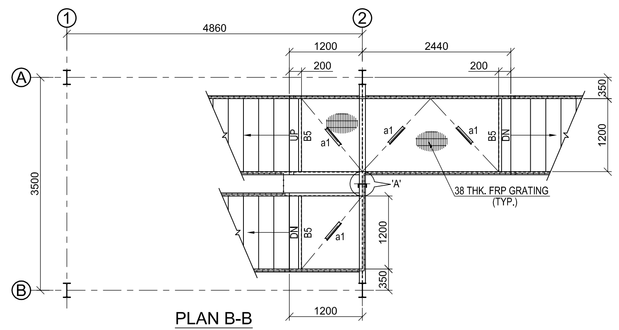
ELEVATION ALONG GRID 1
(SCALE 1:100)

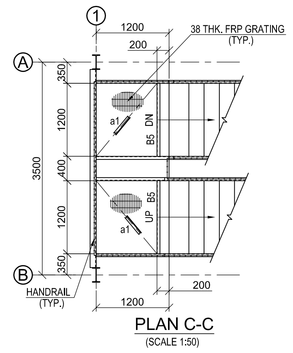


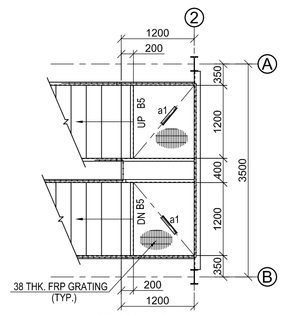
ELEVATION ALONG GRID 2
(SCALE 1:100)

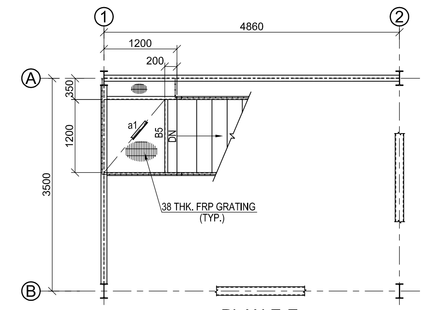


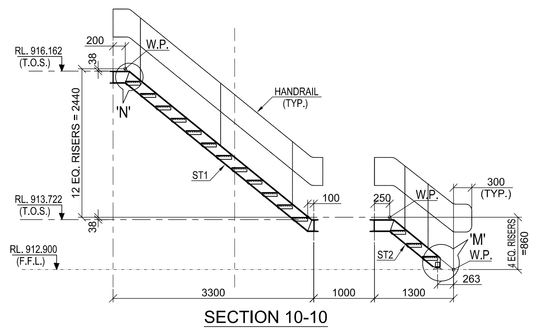
DETAIL OF PLATFORM MKD. 2
PLAN AT RL.913.722(T.O.S.) & RL.913.562(T.O.S.)
(SCALE 1:50)



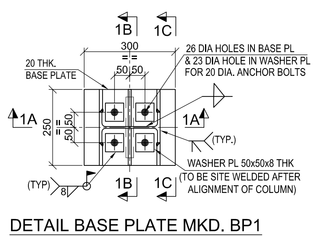




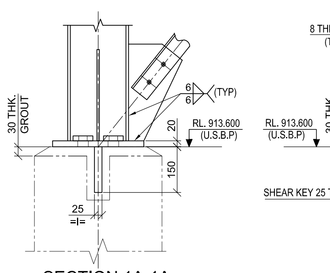




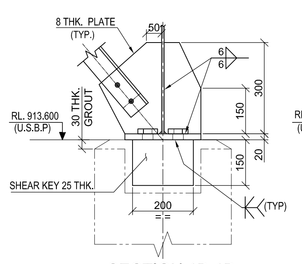
SECTION 10-10
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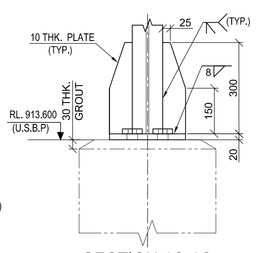
DETAIL BASE PLATE MKD. BP1
(SCALE - 1:10)



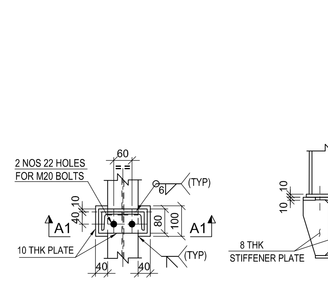
SECTION 1A-1A
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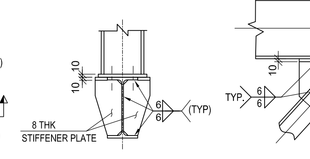
SECTION 1B-1B
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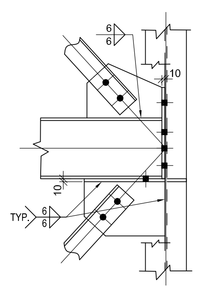
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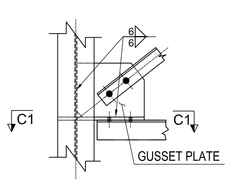
DETAIL `A'
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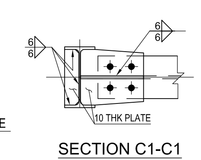
SECTION A1-A1
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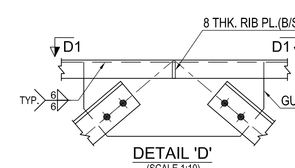
DETAIL `B'
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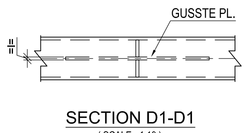
DETAIL `C'
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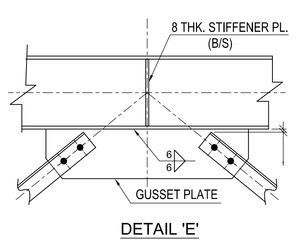
SECTION C1-C1
(SCALE 1:10)



DETAIL 'D'
(SCALE 1:10)



SECTION D1-D1
(SCALE - 1:10)



DETAIL 'E'
(SCALE 1:10)

NOTES:

1. ALL DIMENSIONS ARE IN MILLIMETRE AND LEVELS & COORDINATES ARE IN METRE UNO.
2. ANCHOR BOLT MATERIAL SHALL BE AS PER ASTM F 1554 Gr.36.
3. SIZE OF WELD SHALL BE DECIDED BASED ON LESSER THICKNESS OF THE JOINED MEMBER UNO.
4. ALL WELDS ARE MINIMUM 6 mm LENGTH SIZE CONTINUOUS FILLET UNO.
5. ALL PERMANENT HOLES SHALL BE 21.5 Ø FOR M20 BOLT AS PER (ASTM - A325M) UNO.
6. THREAD TO BE PERMANENTLY DAMAGED AFTER TIGHTENING OF NUT FOR ALL PERMANENT BOLTS BELOW GRIP LENGTH.
7. PROPER ERECTION SEQUENCE TO BE MAINTAINED TO ENSURE STABILITY AND SAFETY OF THE STRUCTURE.
8. GROUTING TO BE DONE AFTER ERECTION AND ALIGNMENT OF STEEL COLUMN(FIRST LIFT).
9. GRADE OF STEEL USED IS S275R AS PER EN10025.
10. ALL STRUCTURAL WORK SHALL CONFORM TO CIVIL STRUCTURAL DESIGN CRITERIA NO MD - 216 - 1000 -EG -ST - CRT - 1001
11. FOR ALL SHEAR CONNECTION DETAILS REFER DRAWING NO. MD-216-1000-EG-ST-DED-1009.
12. FOR LADDER DETAILS REFER DRAWING NO. MD-216-100-EG-ST-DED-1005 & 1006.
13. FOR ALL OTHER GENERAL NOTES REFER DRAWING NO. MD-216-1000-EG-ST-DED-1001.
14. PAINTING ON ALL STRUCTURAL MEMBERS SHALL BE DONE AS PER DESIGN BASIS REPORT (DBR).
15. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH DRAWING NO. MD - 216 - 3380 -EG -ST - GAD - 1004.

LEGENDS :-

R.L. - REDUCED LEVEL
℄ - CENTRE LINE
T.O.S. - TOP OF STEEL
F.G.L. - FINISHED GROUND LEVEL.
F.F.L. - FINISHED FLOOR LEVEL.
TYP. - TYPICAL.
MKD. - MARKED.
THK. - THICK
U.S.B.P. - UNDER SIDE OF BASE PLATE
FRP - FIBERGLASS REINFORCED PLASTIC

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		A01	28/08/2020	ISSUED FOR REFERENCE	SUS	SYC	APL	PRI	

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Saudi Arabian Mining Company شركة المعادن العربية السعودية

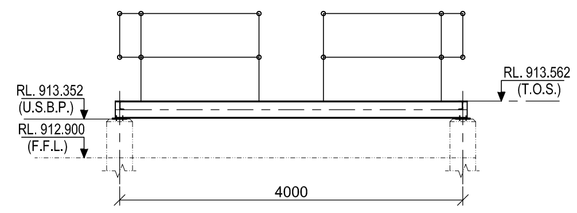
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OUTOTEC DRAWING NUMBER	900707 __ GAU01 __ HL15 __ MTL01 __ 33810
DOCUMENT ID	OU100695736
REVISION	0

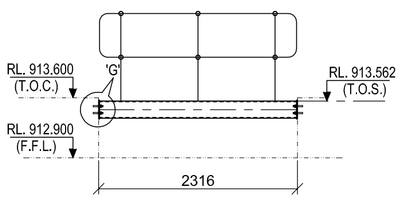
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CHECKED	SYC	29/08/2020
ENGINEER	APL	28/08/2020
APPROVED	PRI	31/08/2020
EMPLOYER		
SCALE	1:100,50,20,10	A0

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
ACIDULATION AREA - ON GROUND PLATFORMS, SHT 1
PLAN, ELEVATIONS & DETAILS

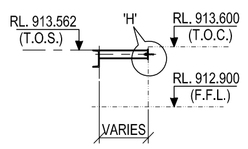
DRAWING NUMBER: MD - 216 - 3380 - EG - ST - GAD - 1003 | REV: A01



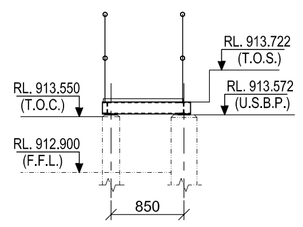
SECTION 1-1
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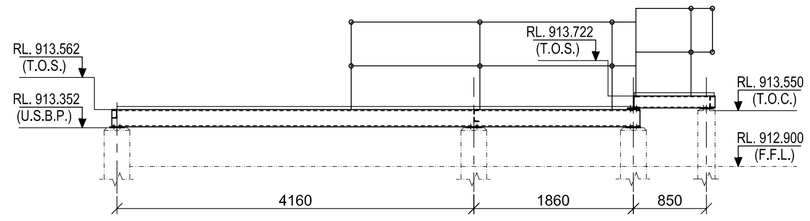
SECTION 2-2
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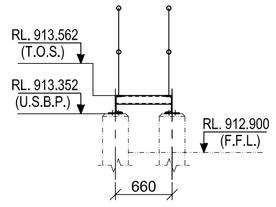
SECTION 3-3
(SCALE 1:50)



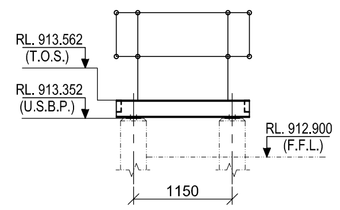
VIEW 4-4
(SCALE 1:50)



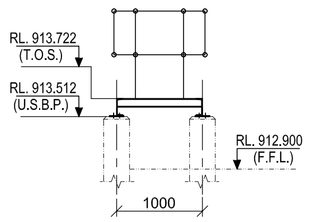
VIEW 5-5
(SCALE 1:50)



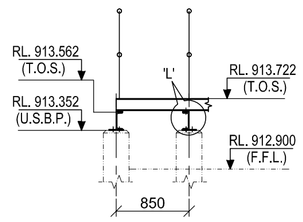
VIEW 6-6
(SCALE 1:50)



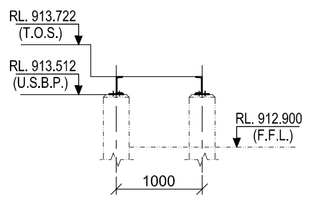
VIEW 7-7
(SCALE 1:50)



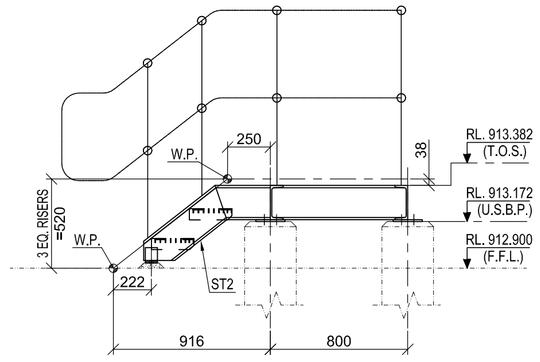
VIEW 8-8
(SCALE 1:50)



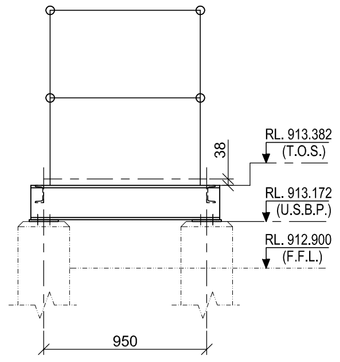
VIEW 9-9
(SCALE 1:50)



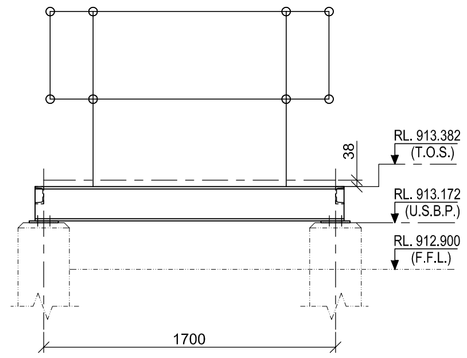
VIEW 8A-8A
(SCALE 1:50)



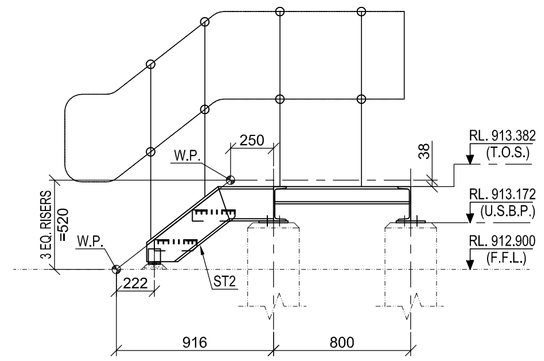
VIEW 11-11
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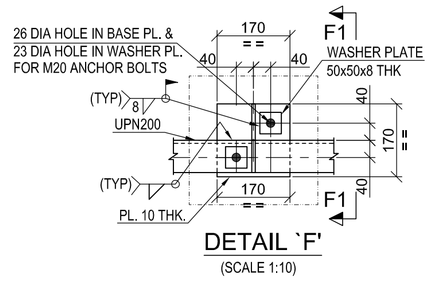
VIEW 12-12
(SCALE 1:25)



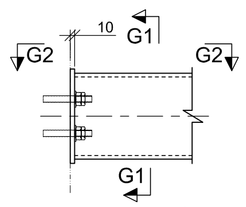
VIEW 13-13
(SCALE 1:25)



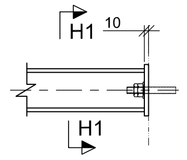
VIEW 14-14
(SCALE 1:25)



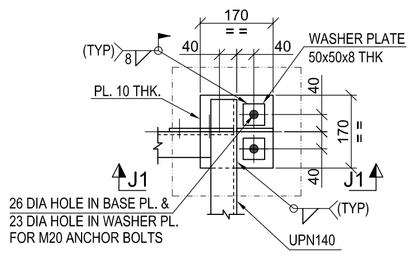
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(SCALE 1:10)



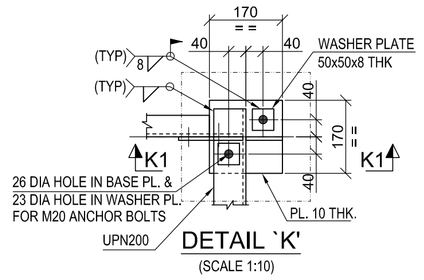
DETAIL `G'
(SCALE 1:10)



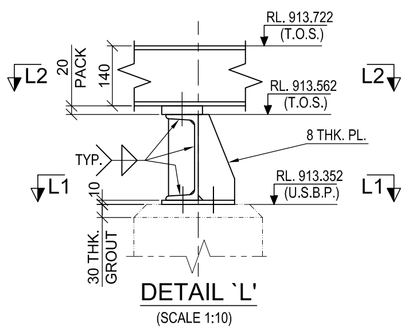
DETAIL `H'
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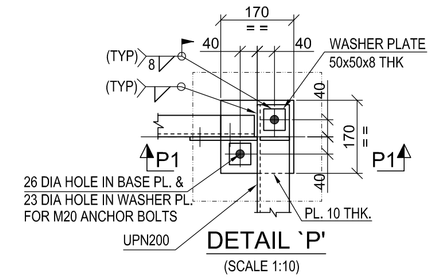
DETAIL `J'
(SCALE 1:10)



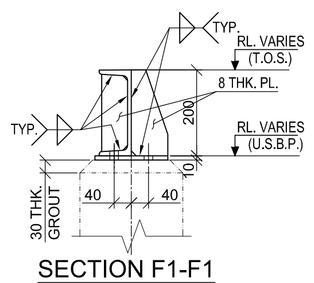
DETAIL `K'
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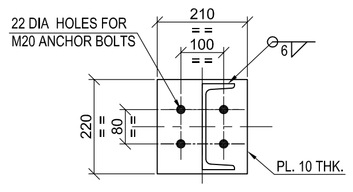
DETAIL `L'
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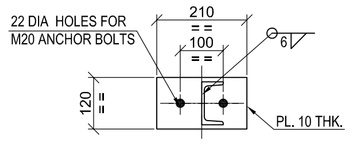
DETAIL `P'
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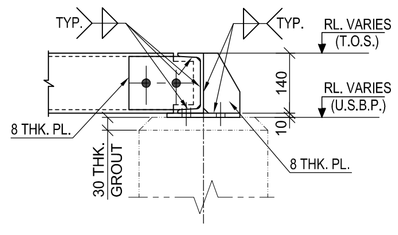
SECTION F1-F1
(SCALE 1:10)



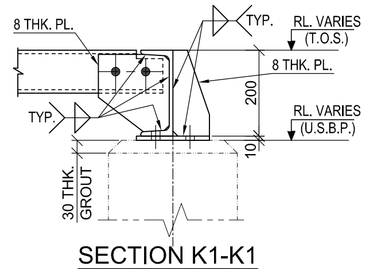
SECTION G1-G1
(SCALE 1:10)



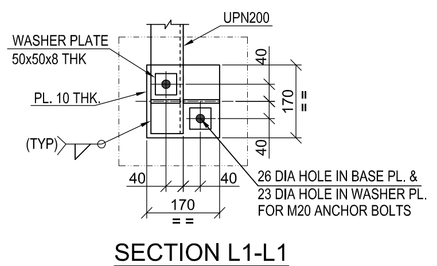
SECTION H1-H1
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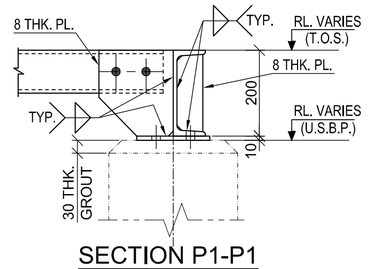
SECTION J1-J1
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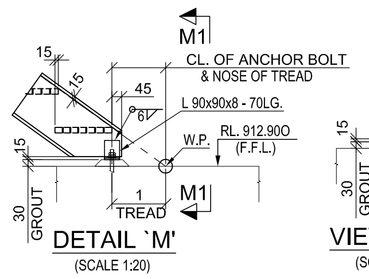
SECTION K1-K1
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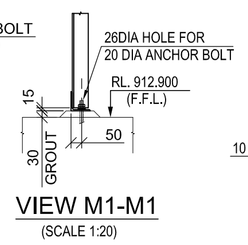
SECTION L1-L1
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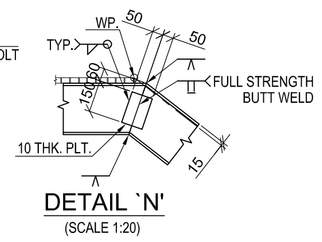
SECTION P1-P1
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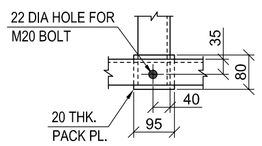
DETAIL `M'
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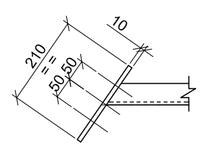
VIEW M1-M1
(SCALE 1:20)



DETAIL `N'
(SCALE 1:20)



SECTION L2-L2
(SCALE 1:10)



SECTION G2-G2
(SCALE 1:10)

LEGENDS :-

R.L. - REDUCED LEVEL
℄ - CENTRE LINE
T.O.S. - TOP OF STEEL.
F.G.L. - FINISHED GROUND LEVEL.
F.F.L. - FINISHED FLOOR LEVEL.
TYP. - TYPICAL.
MKD. - MARKED.
THK. - THICK
U.S.B.P. - UNDER SIDE OF BASE PLATE
FRP - FIBERGLASS REINFORCED PLASTIC

NOTES:

1. ALL DIMENSIONS ARE IN MILLIMETRE AND LEVELS & COORDINATES ARE IN METRE UNO.
2. ANCHOR BOLT MATERIAL SHALL BE AS PER ASTM F 1554 Gr.36,
3. SIZE OF WELD SHALL BE DECIDED BASED ON LESSER THICKNESS OF THE JOINED MEMBER UNO.
4. ALL WELDS ARE MINIMUM 6 mm LENGTH SIZE CONTINUOUS FILLET UNO.
5. ALL PERMANENT HOLES SHALL BE 21.5 Ø FOR M20 BOLT AS PER (ASTM - A325M) UNO.
6. THREAD TO BE PERMANENTLY DAMAGED AFTER TIGHTENING OF NUT FOR ALL PERMANENT BOLTS BELOW GRIP LENGTH.
7. PROPER ERECTION SEQUENCE TO BE MAINTAINED TO ENSURE STABILITY AND SAFETY OF THE STRUCTURE.
8. GROUTING TO BE DONE AFTER ERECTION AND ALIGNMENT OF STEEL COLUMN(FIRST LIFT).
9. GRADE OF STEEL USED IS S275R AS PER EN10025.
10. ALL STRUCTURAL WORK SHALL CONFORM TO CIVIL STRUCTURAL DESIGN CRITERIA NO MD - 216 - 1000 -EG -ST - CRT - 1001
11. FOR ALL SHEAR CONNECTION DETAILS REFER DRAWING NO. MD-216-1000-EG-ST-DED-1009.
12. FOR LADDER DETAILS REFER DRAWING NO. MD-216-1000-EG-ST-DED-1005 & 1006.
13. FOR ALL OTHER GENERAL NOTES REFER DRAWING NO. MD-216-1000-EG-ST-DED-1001.
14. PAINTING ON ALL STRUCTURAL MEMBERS SHALL BE DONE AS PER DESIGN BASIS REPORT (DBR).
15. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH DRAWING NO. MD - 216 - 3380 -EG -ST - GAD - 1003.

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		A01	31/08/2020	ISSUED FOR REFERENCE	SUS	SYC	APL	PRI	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

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Saudi Arabian Mining Company شركة التعدين العربية السعودية

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DRAWN	SUS	28/08/2020
CHECKED	SYC	29/08/2020
ENGINEER	APL	28/08/2020
APPROVED	PRI	31/08/2020
EMPLOYER		
SCALE	1:100,20,15	A1

LARSEN & TOUBRO DRAWING NUMBER: 019045_450_S_G_26208_002 — REVISION: 0
OUTOTEC DRAWING NUMBER: 900707 _GAU01 _HL15 _MTL01 _33811 — DOCUMENT ID: OU100695737 — REVISION: 0

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
ACIDULATION AREA - ON GROUND PLATFORMS, SHT 2
PLAN, ELEVATIONS & DETAILS

DRAWING NUMBER: MD - 216 - 3380 - EG - ST - GAD - 1004 — REV: A01

ATTACHMENT 2 STAAD INPUT – FOUNDATION F1

PROPRIETARY & CONFIDENTIAL

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```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 08/05/2020
JOB NAME FDN. MKD. F1
JOB CLIENT MA'ADEN
JOB NO O19045
JOB REV 0
ENGINEER NAME SOR
END JOB INFORMATION
INPUT WIDTH 79
SET NL 200
SET ITERLIM 15
SET PRINT 1
UNIT METER KN
JOINT COORDINATES
2 0 911.65 3.175; 6 0.75 911.65 3.175; 10 -0.75 911.65 3.175;
34 -0.5 911.65 3.175; 35 -0.25 911.65 3.175; 41 0.25 911.65 3.175;
42 0.5 911.65 3.175; 128 0 913.6 3.175; 302 -0.5 911.65 3.42438;
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```
543 1.25 911.65 2.92438; 545 1.25 911.65 1.42438; 546 1.25 911.65 1.17538;
549 1.25 911.65 1.67438; 551 1.25 911.65 1.92438; 553 1.25 911.65 0.42438;
554 1.25 911.65 0.17538; 557 1.25 911.65 0.67438; 559 1.25 911.65 0.92438;
561 1.25 911.65 -0.62362; 562 1.25 911.65 -0.92462; 565 1.25 911.65 -0.37362;
567 1.25 911.65 -0.12462; 569 -1.25 911.65 3.42438; 570 -1.25 911.65 3.175;
573 -1.25 911.65 3.67438; 575 -1.25 911.65 3.97538; 577 -1.25 911.65 2.42438;
578 -1.25 911.65 2.17538; 581 -1.25 911.65 2.67438; 583 -1.25 911.65 2.92438;
585 -1.25 911.65 1.42438; 586 -1.25 911.65 1.17538; 589 -1.25 911.65 1.67438;
591 -1.25 911.65 1.92438; 593 -1.25 911.65 0.42438; 594 -1.25 911.65 0.17538;
597 -1.25 911.65 0.67438; 599 -1.25 911.65 0.92438; 601 -1.25 911.65 -0.62362;
602 -1.25 911.65 -0.92462; 605 -1.25 911.65 -0.37362;
607 -1.25 911.65 -0.12462; 634 1.625 911.65 3.42438; 635 1.625 911.65 3.175;
636 1.625 911.65 3.67438; 637 1.625 911.65 3.97538; 638 1.625 911.65 2.42438;
639 1.625 911.65 2.17538; 640 1.625 911.65 2.67438; 641 1.625 911.65 2.92438;
642 1.625 911.65 1.42438; 643 1.625 911.65 1.17538; 644 1.625 911.65 1.67438;
645 1.625 911.65 1.92438; 646 1.625 911.65 0.42438; 647 1.625 911.65 0.17538;
648 1.625 911.65 0.67438; 649 1.625 911.65 0.92438; 650 1.625 911.65 -0.62362;
651 1.625 911.65 -0.92462; 652 1.625 911.65 -0.37362;
653 1.625 911.65 -0.12462; 656 -1.625 911.65 3.42438; 657 -1.625 911.65 3.175;
658 -1.625 911.65 3.67438; 659 -1.625 911.65 3.97538;
660 -1.625 911.65 2.42438; 661 -1.625 911.65 2.17538;
662 -1.625 911.65 2.67438; 663 -1.625 911.65 2.92438;
664 -1.625 911.65 1.42438; 665 -1.625 911.65 1.17538;
666 -1.625 911.65 1.67438; 667 -1.625 911.65 1.92438;
668 -1.625 911.65 0.42438; 669 -1.625 911.65 0.17538;
670 -1.625 911.65 0.67438; 671 -1.625 911.65 0.92438;
672 -1.625 911.65 -0.62362; 673 -1.625 911.65 -0.92462;
674 -1.625 911.65 -0.37362; 675 -1.625 911.65 -0.12462;
MEMBER INCIDENCES
185 2 128; 529 522 528;
ELEMENT INCIDENCES SHELL
326 303 302 34 10; 327 302 304 35 34; 328 304 305 2 35; 332 305 306 41 2;
333 306 307 42 41; 334 307 308 6 42; 337 308 313 309 6; 342 320 303 10 316;
343 321 322 302 303; 344 322 323 304 302; 345 323 324 305 304;
346 324 325 306 305; 347 325 326 307 306; 348 326 327 308 307;
349 327 328 313 308; 352 331 321 303 320; 353 332 333 322 321;
354 333 334 323 322; 355 334 335 324 323; 356 335 336 325 324;
357 336 337 326 325; 358 337 338 327 326; 359 338 339 328 327;
362 342 332 321 331; 393 387 388 366 365; 394 388 389 369 366;
395 389 390 371 369; 396 390 391 373 371; 397 391 392 375 373;
398 392 393 377 375; 399 393 394 379 377; 402 397 387 365 384;
403 398 399 388 387; 404 399 400 389 388; 405 400 401 390 389;
406 401 402 391 390; 407 402 403 392 391; 408 403 404 393 392;
409 404 405 394 393; 412 408 398 387 397; 413 409 410 399 398;
414 410 411 400 399; 415 411 412 401 400; 416 412 413 402 401;
417 413 414 403 402; 418 414 415 404 403; 419 415 416 405 404;
422 419 409 398 408; 423 10 34 410 409; 424 34 35 411 410; 425 35 2 412 411;
426 2 41 413 412; 427 41 42 414 413; 428 42 6 415 414; 429 6 309 416 415;
432 316 10 409 419; 433 420 421 422 423; 434 421 424 425 422;
435 424 426 427 425; 436 426 428 429 427; 437 428 430 431 429;
438 430 432 433 431; 439 432 434 435 433; 440 436 420 423 437;
441 438 439 421 420; 442 439 440 424 421; 443 440 441 426 424;
444 441 442 428 426; 445 442 443 430 428; 446 443 444 432 430;
447 444 445 434 432; 448 446 438 420 436; 449 447 448 439 438;
450 448 449 440 439; 451 449 450 441 440; 452 450 451 442 441;
453 451 452 443 442; 454 452 453 444 443; 455 453 454 445 444;
456 455 447 438 446; 457 365 366 448 447; 458 366 369 449 448;
459 369 371 450 449; 460 371 373 451 450; 461 373 375 452 451;
462 375 377 453 452; 463 377 379 454 453; 464 384 365 447 455;
465 456 457 458 459; 466 457 460 461 458; 467 460 462 463 461;
468 462 464 465 463; 469 464 466 467 465; 470 466 468 469 467;
471 468 470 471 469; 472 472 456 459 473; 473 474 475 457 456;
474 475 476 460 457; 475 476 477 462 460; 476 477 478 464 462;
477 478 479 466 464; 478 479 480 468 466; 479 480 481 470 468;
480 482 474 456 472; 481 483 484 475 474; 482 484 485 476 475;
483 485 486 477 476; 484 486 487 478 477; 485 487 488 479 478;
486 488 489 480 479; 487 489 490 481 480; 488 491 483 474 482;
489 423 422 484 483; 490 422 425 485 484; 491 425 427 486 485;
492 427 429 487 486; 493 429 431 488 487; 494 431 433 489 488;
495 433 435 490 489; 496 437 423 483 491; 497 492 493 494 495;
498 493 496 497 494; 499 496 498 499 497; 500 498 500 501 499;
501 500 502 503 501; 502 502 504 505 503; 503 504 506 507 505;
504 508 492 495 509; 505 510 511 493 492; 506 511 512 496 493;
507 512 513 498 496; 508 513 514 500 498; 509 514 515 502 500;
510 515 516 504 502; 511 516 517 506 504; 512 518 510 492 508;
513 519 520 511 510; 514 520 521 512 511; 515 521 522 513 512;
516 522 523 514 513; 517 523 524 515 514; 518 524 525 516 515;
519 525 526 517 516; 520 527 519 510 518; 521 459 458 520 519;
522 458 461 521 520; 523 461 463 522 521; 524 463 465 523 522;
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522 458 461 521 520; 523 461 463 522 521; 524 463 465 523 522;
525 465 467 524 523; 526 467 469 525 524; 527 469 471 526 525;
528 473 459 519 527; 530 313 529 530 309; 532 328 533 529 313;
534 339 535 533 328; 536 394 537 538 379; 538 405 541 537 394;
540 416 543 541 405; 542 309 530 543 416; 544 434 545 546 435;
546 445 549 545 434; 548 454 551 549 445; 550 379 538 551 454;
552 470 553 554 471; 554 481 557 553 470; 556 490 559 557 481;
558 435 546 559 490; 560 506 561 562 507; 562 517 565 561 506;
564 526 567 565 517; 566 471 554 567 526; 568 569 320 316 570;
570 573 331 320 569; 572 575 342 331 573; 574 577 397 384 578;
576 581 408 397 577; 578 583 419 408 581; 580 570 316 419 583;
582 585 436 437 586; 584 589 446 436 585; 586 591 455 446 589;
588 578 384 455 591; 590 593 472 473 594; 592 597 482 472 593;
594 599 491 482 597; 596 586 437 491 599; 598 601 508 509 602;
600 605 518 508 601; 602 607 527 518 605; 604 594 473 527 607;
629 529 634 635 530; 630 533 636 634 529; 631 535 637 636 533;
632 537 638 639 538; 633 541 640 638 537; 634 543 641 640 541;
635 530 635 641 543; 636 545 642 643 546; 637 549 644 642 545;
638 551 645 644 549; 639 538 639 645 551; 640 553 646 647 554;
641 557 648 646 553; 642 559 649 648 557; 643 546 643 649 559;
644 561 650 651 562; 645 565 652 650 561; 646 567 653 652 565;
647 554 647 653 567; 650 656 569 570 657; 651 658 573 569 656;
652 659 575 573 658; 653 660 577 578 661; 654 662 581 577 660;
655 663 583 581 662; 656 657 570 583 663; 657 664 585 586 665;
658 666 589 585 664; 659 667 591 589 666; 660 661 578 591 667;
661 668 593 594 669; 662 670 597 593 668; 663 671 599 597 670;
664 665 586 599 671; 665 672 601 602 673; 666 674 605 601 672;
667 675 607 605 674; 668 669 594 607 675;
START GROUP DEFINITION
ELEMENT
_RAFT 326 TO 328 332 TO 334 337 342 TO 349 352 TO 359 362 393 TO 399 -
402 TO 409 412 TO 419 422 TO 429 432 TO 528 530 532 534 536 538 540 542 544 -
546 548 550 552 554 556 558 560 562 564 566 568 570 572 574 576 578 580 582 -
584 586 588 590 592 594 596 598 600 602 604 629 TO 647 650 TO 668
_RAFT_MOM 326 327 333 334 337 342 TO 349 352 TO 359 362 393 TO 399 -
402 TO 409 412 TO 419 422 TO 424 427 TO 429 432 TO 514 517 TO 522 -
525 TO 528 530 532 534 536 538 540 542 544 546 548 550 552 554 556 558 560 -
562 564 566 568 570 572 574 576 578 580 582 584 586 588 590 592 594 596 598 -
600 602 604 629 TO 647 650 TO 668
_RAFT_SHEAR 326 334 337 342 343 348 349 352 TO 359 362 393 TO 399 402 TO 409 -
412 413 418 419 422 423 428 429 432 TO 465 470 TO 505 510 TO 513 518 TO 521 -
526 TO 528 530 532 534 536 538 540 542 544 546 548 550 552 554 556 558 560 -
562 564 566 568 570 572 574 576 578 580 582 584 586 588 590 592 594 596 598 -
600 602 604 629 TO 647 650 TO 668
MEMBER
END GROUP DEFINITION
ELEMENT PROPERTY
326 TO 328 332 TO 334 337 342 TO 349 352 TO 359 362 393 TO 399 402 TO 409 -
412 TO 419 422 TO 429 432 TO 528 530 532 534 536 538 540 542 544 546 548 -
550 552 554 556 558 560 562 564 566 568 570 572 574 576 578 580 582 584 586 -
588 590 592 594 596 598 600 602 604 629 TO 647 650 TO 668 THICKNESS 0.5
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 25
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
185 529 PRIS YD 0.4 ZD 0.6
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
326 TO 328 332 TO 334 337 342 TO 349 352 TO 359 362 393 TO 399 402 TO 409 -
412 TO 419 422 TO 429 432 TO 528 530 532 534 536 538 540 542 544 546 548 -
550 552 554 556 558 560 562 564 566 568 570 572 574 576 578 580 582 584 586 -
588 590 592 594 596 598 600 602 604 629 TO 647 650 TO 667 -
668 PLATE MAT DIRECT Y SUBGRADE 57600 PRINT
*********************************************************************************************
LOAD 1 LOADTYPE None  TITLE DEAD LOAD ( DL )
SELFWEIGHT Y -1
ELEMENT LOAD
* LOAD DUE TO BACKFILL
326 TO 328 332 TO 334 337 342 TO 349 352 TO 359 362 393 TO 399 402 TO 409 -
412 TO 419 422 TO 429 432 TO 528 530 532 534 536 538 540 542 544 546 548 -
550 552 554 556 558 560 562 564 566 568 570 572 574 576 578 580 582 584 586 -
```



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550 552 554 556 558 560 562 564 566 568 570 572 574 576 578 580 582 584 586 -
588 590 592 594 596 598 600 602 604 629 TO 647 650 TO 668 PR GY -15
* LOAD DUE TO SURCHARGE
326 TO 328 332 TO 334 337 342 TO 349 352 TO 359 362 393 TO 399 402 TO 409 -
412 TO 419 422 TO 429 432 TO 528 530 532 534 536 538 540 542 544 546 548 -
550 552 554 556 558 560 562 564 566 568 570 572 574 576 578 580 582 584 586 -
588 590 592 594 596 598 600 602 604 629 TO 647 650 TO 668 PR GY -20
JOINT LOAD
128 FX 0.002 FY -152 FZ 0.011
528 FX 0.007 FY -115.459 FZ 1.384
*********************************************************************************************
LOAD 2 LOADTYPE None  TITLE LIVE LOAD ( LL )
JOINT LOAD
128 FX 0.017 FY -173.225 FZ 2.323
528 FX 0.013 FY -84.387 FZ 1.87
*********************************************************************************************
LOAD 3 LOADTYPE None  TITLE WIND LOAD +X  ( WLX1 )
JOINT LOAD
128 FX 2.339 FY 4.483 FZ 0.853
528 FX 2.36 FY -3.789 FZ 0.76
*********************************************************************************************
LOAD 4 LOADTYPE None  TITLE WIND LOAD -X  ( WLX2 )
JOINT LOAD
128 FX 2.339 FY -4.483 FZ 0.853
528 FX 2.36 FY 3.789 FZ 0.76
*********************************************************************************************
LOAD 5 LOADTYPE None  TITLE WIND LOAD +Z  ( WLZ1 )
JOINT LOAD
128 FX 0.001 FY 202.574 FZ 51.359
528 FX 0.05 FY -202.574 FZ 49.368
*********************************************************************************************
LOAD 6 LOADTYPE None  TITLE WIND LOAD -Z  ( WLZ2 )
JOINT LOAD
128 FX 0.001 FY -202.395 FZ 51.324
528 FX 0.05 FY 202.395 FZ 49.368
*********************************************************************************************
PERFORM ANALYSIS
CHANGE
SUPPORTS
326 TO 328 332 TO 334 337 342 TO 349 352 TO 359 362 393 TO 399 402 TO 409 -
412 TO 419 422 TO 429 432 TO 528 530 532 534 536 538 540 542 544 546 548 -
550 552 554 556 558 560 562 564 566 568 570 572 574 576 578 580 582 584 586 -
588 590 592 594 596 598 600 602 604 629 TO 647 650 TO 667 -
668 PLATE MAT DIRECT Y SUBGRADE 57600 COMPRESSION
*********************************************************************************************
***FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
LOAD 101 1.4DL
REPEAT LOAD
1 1.4
PERFORM ANALYSIS
CHANGE
LOAD 102 1.4DL + 1.7LL
REPEAT LOAD
1 1.4 2 1.7
PERFORM ANALYSIS
CHANGE
LOAD 103 1.2DL + 1.6WLX1 + 1.0 LL
REPEAT LOAD
1 1.2 3 1.6 2 1.0
PERFORM ANALYSIS
CHANGE
LOAD 104 1.2DL + 1.6WLX2 + 1.0 LL
REPEAT LOAD
1 1.2 4 1.6 2 1.0
PERFORM ANALYSIS
CHANGE
LOAD 105 1.2DL + 1.6WLZ1 + 1.0 LL
REPEAT LOAD
1 1.2 5 1.6 2 1.0
PERFORM ANALYSIS
CHANGE
LOAD 106 1.2DL + 1.6WLZ2 + 1.0 LL
REPEAT LOAD
1 1.2 6 1.6 2 1.0
PERFORM ANALYSIS
CHANGE
LOAD 107 0.9DL + 1.6WLX1
REPEAT LOAD
1 0.9 3 1.6
```

```
1 0.9 3 1.6
PERFORM ANALYSIS
CHANGE
LOAD 108 0.9DL + 1.6WLX2
REPEAT LOAD
1 0.9 4 1.6
PERFORM ANALYSIS
CHANGE
LOAD 109 0.9DL + 1.6WLZ1
REPEAT LOAD
1 0.9 5 1.6
PERFORM ANALYSIS
CHANGE
LOAD 110 0.9DL + 1.6WLZ2
REPEAT LOAD
1 0.9 6 1.6
PERFORM ANALYSIS
CHANGE
************************************************************************************************
* UNFACTORED LOAD COMBINATIONS FOR STABILITY CHECK
LOAD 201 1.0DL
REPEAT LOAD
1 1.0
PERFORM ANALYSIS
CHANGE
LOAD 202 1.0DL + 1.0LL
REPEAT LOAD
1 1.0 2 1.0
PERFORM ANALYSIS
CHANGE
LOAD 203 1.0DL + 1.0WLX1
REPEAT LOAD
1 1.0 3 1.0
PERFORM ANALYSIS
CHANGE
LOAD 204 1.0DL + 1.0WLX2
REPEAT LOAD
1 1.0 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 205 1.0DL + 1.0WLZ1
REPEAT LOAD
1 1.0 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 206 1.0DL + 1.0WLZ2
REPEAT LOAD
1 1.0 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD 207 1.0DL + 0.75WLX1 + 0.75LL
REPEAT LOAD
1 1.0 3 0.75 2 0.75
PERFORM ANALYSIS
CHANGE
LOAD 208 1.0DL + 0.75WLX2 + 0.75LL
REPEAT LOAD
1 1.0 4 0.75 2 0.75
PERFORM ANALYSIS
CHANGE
LOAD 209 1.0DL + 0.75WLZ1 + 0.75LL
REPEAT LOAD
1 1.0 5 0.75 2 0.75
PERFORM ANALYSIS
CHANGE
LOAD 210 1.0DL + 0.75WLZ2 + 0.75LL
REPEAT LOAD
1 1.0 6 0.75 2 0.75
PERFORM ANALYSIS
CHANGE
LOAD 211 0.6DL + 1.0WLX1
REPEAT LOAD
1 0.6 3 1.0
PERFORM ANALYSIS
CHANGE
LOAD 212 0.6DL + 1.0WLX2
REPEAT LOAD
1 0.6 4 1.0
PERFORM ANALYSIS
```



```
PERFORM ANALYSIS
CHANGE
LOAD 213 0.6DL + 1.0WLZ1
REPEAT LOAD
1 0.6 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 214 0.6DL + 1.0WLZ2
REPEAT LOAD
1 0.6 6 1.0
PERFORM ANALYSIS
CHANGE
************************************************************************************************
PERFORM ANALYSIS
LOAD LIST 101 TO 110
START CONCRETE DESIGN
CODE ACI
TRACK 2 ALL
FC 35000 MEMB 185 529
CLB 0.05 MEMB 185 529
CLS 0.05 ALL
CLT 0.05 MEMB 185 326 TO 328 332 TO 334 337 342 TO 349 352 TO 359 362 393 -
394 TO 399 402 TO 409 412 TO 419 422 TO 429 432 530 532 534 536 538 540 542 -
568 570 572 574 576 578 580 629 TO 635 650 TO 656
MAXMAIN 25 ALL
MINMAIN 25 ALL
MINSEC 10 ALL
DESIGN COLUMN 185 529
END CONCRETE DESIGN
FINISH
```

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 08/05/2020
JOB NAME FDN. MKD. F1_STABILITY CHECK
JOB CLIENT MA'ADEN
JOB NO O19045
JOB REV 0
ENGINEER NAME SOR
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
2 0 911.65 3.175; 128 0 913.6 3.175; 522 0 911.65 -0.12462; 528 0 913.6 -0.125;
MEMBER INCIDENCES
185 2 128; 529 522 528;
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 25
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
185 529 PRIS YD 0.4 ZD 0.6
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
2 522 FIXED
*********************************************************************************************
LOAD 1 LOADTYPE None  TITLE DEAD LOAD ( DL )
SELFWEIGHT Y -1
JOINT LOAD
128 FX 0.002 FY -152 FZ 0.011
528 FX 0.007 FY -115.459 FZ 1.384
*********************************************************************************************
LOAD 2 LOADTYPE None  TITLE LIVE LOAD ( LL )
JOINT LOAD
128 FX 0.017 FY -173.225 FZ 2.323
528 FX 0.013 FY -84.387 FZ 1.87
*********************************************************************************************
LOAD 3 LOADTYPE None  TITLE WIND LOAD +X  ( WLX1 )
JOINT LOAD
128 FX 2.339 FY 4.483 FZ 0.853
528 FX 2.36 FY -3.789 FZ 0.76
*********************************************************************************************
LOAD 4 LOADTYPE None  TITLE WIND LOAD -X  ( WLX2 )
JOINT LOAD
128 FX 2.339 FY -4.483 FZ 0.853
528 FX 2.36 FY 3.789 FZ 0.76
*********************************************************************************************
LOAD 5 LOADTYPE None  TITLE WIND LOAD +Z  ( WLZ1 )
JOINT LOAD
128 FX 0.001 FY 202.574 FZ 51.359
528 FX 0.05 FY -202.574 FZ 49.368
*********************************************************************************************
LOAD 6 LOADTYPE None  TITLE WIND LOAD -Z  ( WLZ2 )
JOINT LOAD
128 FX 0.001 FY -202.395 FZ 51.324
528 FX 0.05 FY 202.395 FZ 49.368
*********************************************************************************************
*********************************************************************************************
* UNFACTORED LOAD COMBINATIONS FOR STABILITY CHECK
LOAD COMB 201 1.0DL
1 1.0
LOAD COMB 202 1.0DL + 1.0LL
1 1.0 2 1.0
LOAD COMB 203 1.0DL + 1.0WLX1
1 1.0 3 1.0
LOAD COMB 204 1.0DL + 1.0WLX2
1 1.0 4 1.0
LOAD COMB 205 1.0DL + 1.0WLZ1
1 1.0 5 1.0
LOAD COMB 206 1.0DL + 1.0WLZ2
1 1.0 6 1.0
LOAD COMB 207 1.0DL + 0.75WLX1 + 0.75LL
1 1.0 3 0.75 2 0.75
```



```
1 1.0 3 0.75 2 0.75
LOAD COMB 208 1.0DL + 0.75WLX2 + 0.75LL
1 1.0 4 0.75 2 0.75
LOAD COMB 209 1.0DL + 0.75WLZ1 + 0.75LL
1 1.0 5 0.75 2 0.75
LOAD COMB 210 1.0DL + 0.75WLZ2 + 0.75LL
1 1.0 6 0.75 2 0.75
LOAD COMB 211 0.6DL + 1.0WLX1
1 0.6 3 1.0
LOAD COMB 212 0.6DL + 1.0WLX2
1 0.6 4 1.0
LOAD COMB 213 0.6DL + 1.0WLZ1
1 0.6 5 1.0
LOAD COMB 214 0.6DL + 1.0WLZ2
1 0.6 6 1.0
*********************************************************************************************
PERFORM ANALYSIS
FINISH
```

ATTACHMENT 3 STAAD INPUT – FOUNDATION F2

PROPRIETARY & CONFIDENTIAL

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```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 08/05/2020
JOB NAME FDN. MKD. F2
JOB CLIENT MA'ADEN
JOB NO O19045
JOB REV 0
ENGINEER NAME SOR
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
2 0 911.65 3.175; 6 0.75 911.65 3.175; 10 -0.75 911.65 3.175;
34 -0.5 911.65 3.175; 35 -0.25 911.65 3.175; 41 0.25 911.65 3.175;
42 0.5 911.65 3.175; 128 0 913.6 3.175; 302 -0.5 911.65 3.42438;
303 -0.75 911.65 3.42438; 304 -0.25 911.65 3.42438; 305 0 911.65 3.42438;
306 0.25 911.65 3.42438; 307 0.5 911.65 3.42438; 308 0.75 911.65 3.42438;
321 -0.75 911.65 3.67438; 322 -0.5 911.65 3.67438; 323 -0.25 911.65 3.67438;
324 0 911.65 3.67438; 325 0.25 911.65 3.67438; 326 0.5 911.65 3.67438;
327 0.75 911.65 3.67438; 332 -0.75 911.65 3.92438; 333 -0.5 911.65 3.92438;
334 -0.25 911.65 3.92438; 335 0 911.65 3.92438; 336 0.25 911.65 3.92438;
337 0.5 911.65 3.92438; 338 0.75 911.65 3.92438; 343 -0.75 911.65 4.17438;
344 -0.5 911.65 4.17438; 345 -0.25 911.65 4.17438; 346 0 911.65 4.17438;
347 0.25 911.65 4.17438; 348 0.5 911.65 4.17438; 349 0.75 911.65 4.17438;
354 -0.75 911.65 4.42438; 355 -0.5 911.65 4.42438; 356 -0.25 911.65 4.42438;
357 0 911.65 4.42438; 358 0.25 911.65 4.42438; 359 0.5 911.65 4.42438;
360 0.75 911.65 4.42438; 365 -0.75 911.65 2.17438; 366 -0.5 911.65 2.17438;
367 -0.5 911.65 1.925; 368 -0.75 911.65 1.925; 369 -0.25 911.65 2.17438;
370 -0.25 911.65 1.925; 371 0 911.65 2.17438; 372 0 911.65 1.925;
373 0.25 911.65 2.17438; 374 0.25 911.65 1.925; 375 0.5 911.65 2.17438;
376 0.5 911.65 1.925; 377 0.75 911.65 2.17438; 378 0.75 911.65 1.925;
387 -0.75 911.65 2.42438; 388 -0.5 911.65 2.42438; 389 -0.25 911.65 2.42438;
390 0 911.65 2.42438; 391 0.25 911.65 2.42438; 392 0.5 911.65 2.42438;
393 0.75 911.65 2.42438; 398 -0.75 911.65 2.67438; 399 -0.5 911.65 2.67438;
400 -0.25 911.65 2.67438; 401 0 911.65 2.67438; 402 0.25 911.65 2.67438;
403 0.5 911.65 2.67438; 404 0.75 911.65 2.67438; 409 -0.75 911.65 2.92438;
410 -0.5 911.65 2.92438; 411 -0.25 911.65 2.92438; 412 0 911.65 2.92438;
413 0.25 911.65 2.92438; 414 0.5 911.65 2.92438; 415 0.75 911.65 2.92438;
MEMBER INCIDENCES
185 2 128;
ELEMENT INCIDENCES SHELL
326 303 302 34 10; 327 302 304 35 34; 328 304 305 2 35; 332 305 306 41 2;
333 306 307 42 41; 334 307 308 6 42; 343 321 322 302 303; 344 322 323 304 302;
345 323 324 305 304; 346 324 325 306 305; 347 325 326 307 306;
348 326 327 308 307; 353 332 333 322 321; 354 333 334 323 322;
355 334 335 324 323; 356 335 336 325 324; 357 336 337 326 325;
358 337 338 327 326; 363 343 344 333 332; 364 344 345 334 333;
365 345 346 335 334; 366 346 347 336 335; 367 347 348 337 336;
368 348 349 338 337; 373 354 355 344 343; 374 355 356 345 344;
375 356 357 346 345; 376 357 358 347 346; 377 358 359 348 347;
378 359 360 349 348; 383 365 366 367 368; 384 366 369 370 367;
385 369 371 372 370; 386 371 373 374 372; 387 373 375 376 374;
388 375 377 378 376; 393 387 388 366 365; 394 388 389 369 366;
395 389 390 371 369; 396 390 391 373 371; 397 391 392 375 373;
398 392 393 377 375; 403 398 399 388 387; 404 399 400 389 388;
405 400 401 390 389; 406 401 402 391 390; 407 402 403 392 391;
408 403 404 393 392; 413 409 410 399 398; 414 410 411 400 399;
415 411 412 401 400; 416 412 413 402 401; 417 413 414 403 402;
418 414 415 404 403; 423 10 34 410 409; 424 34 35 411 410; 425 35 2 412 411;
426 2 41 413 412; 427 41 42 414 413; 428 42 6 415 414;
START GROUP DEFINITION
ELEMENT
_RAFT 326 TO 328 332 TO 334 343 TO 348 353 TO 358 363 TO 368 373 TO 378 383 -
384 TO 388 393 TO 398 403 TO 408 413 TO 418 423 TO 428
_RAFT_MOM 326 327 333 334 343 TO 348 353 TO 358 363 TO 368 373 TO 378 383 -
384 TO 388 393 TO 398 403 TO 408 413 TO 418 423 424 427 428
_RAFT_SHEAR 326 334 343 348 353 TO 358 363 TO 368 373 TO 378 383 TO 388 393 -
394 TO 398 403 TO 408 413 418 423 428
MEMBER
END GROUP DEFINITION
ELEMENT PROPERTY
326 TO 328 332 TO 334 343 TO 348 353 TO 358 363 TO 368 373 TO 378 383 TO 388 -
393 TO 398 403 TO 408 413 TO 418 423 TO 428 THICKNESS 0.5
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 25
ALPHA 1e-005
```



```
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
185 PRIS YD 0.4 ZD 0.6
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
326 TO 328 332 TO 334 343 TO 348 353 TO 358 363 TO 368 373 TO 378 383 TO 388 -
393 TO 398 403 TO 408 413 TO 418 423 TO 427 -
428 PLATE MAT DIRECT Y SUBGRADE 57600 PRINT
**************************************************************************************************
LOAD 1 LOADTYPE None  TITLE DEAD LOAD ( DL )
SELFWEIGHT Y -1
ELEMENT LOAD
* LOAD DUE TO BACKFILL
326 TO 328 332 TO 334 343 TO 348 353 TO 358 363 TO 368 373 TO 378 383 TO 388 -
393 TO 398 403 TO 408 413 TO 418 423 TO 428 PR GY -15
* LOAD DUE TO SURCHARGE
326 TO 328 332 TO 334 343 TO 348 353 TO 358 363 TO 368 373 TO 378 383 TO 388 -
393 TO 398 403 TO 408 413 TO 418 423 TO 428 PR GY -20
JOINT LOAD
128 FX -0.001 FY -76.404 FZ 0.282
**************************************************************************************************
LOAD 2 LOADTYPE None  TITLE LIVE LOAD ( LL )
JOINT LOAD
128 FX -0.005 FY -57.023 FZ -1.257
**************************************************************************************************
LOAD 3 LOADTYPE None  TITLE WIND LOAD +X  ( WLX1 )
JOINT LOAD
128 FX 1.334 FY -0.866 FZ 0.111
**************************************************************************************************
LOAD 4 LOADTYPE None  TITLE WIND LOAD -X  ( WLX2 )
JOINT LOAD
128 FX 1.334 FY 0.866 FZ 0.111
**************************************************************************************************
LOAD 5 LOADTYPE None  TITLE WIND LOAD +Z  ( WLZ1 )
JOINT LOAD
128 FX 0.025 FY 9.749 FZ 3.225
**************************************************************************************************
LOAD 6 LOADTYPE None  TITLE WIND LOAD -Z  ( WLZ2 )
JOINT LOAD
128 FX 0.025 FY -9.744 FZ 3.224
**************************************************************************************************
***FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
LOAD COMB 101 1.4DL
1 1.4
LOAD COMB 102 1.4DL + 1.7LL
1 1.4 2 1.7
LOAD COMB 103 1.4DL + 1.6WLX1 + 1.0 LL
1 1.4 3 1.6 2 1.0
LOAD COMB 104 1.4DL + 1.6WLX2 + 1.0 LL
1 1.4 4 1.6 2 1.0
LOAD COMB 105 1.4DL + 1.6WLZ1 + 1.0 LL
1 1.4 5 1.6 2 1.0
LOAD COMB 106 1.4DL + 1.6WLZ2 + 1.0 LL
1 1.4 6 1.6 2 1.0
LOAD COMB 107 0.9DL + 1.6WLX1
1 0.9 3 1.6
LOAD COMB 108 0.9DL + 1.6WLX2
1 0.9 4 1.6
LOAD COMB 109 0.9DL + 1.6WLZ1
1 0.9 5 1.6
LOAD COMB 110 0.9DL + 1.6WLZ2
1 0.9 6 1.6
**************************************************************************************************
* UNFACTORED LOAD COMBINATIONS FOR STABILITY CHECK
LOAD COMB 201 1.0DL
1 1.0
LOAD COMB 202 1.0DL + 1.0LL
1 1.0 2 1.0
LOAD COMB 203 1.0DL + 1.0WLX1
1 1.0 3 1.0
LOAD COMB 204 1.0DL + 1.0WLX2
1 1.0 4 1.0
LOAD COMB 205 1.0DL + 1.0WLZ1
1 1.0 5 1.0
```

```
1 1.0 5 1.0
LOAD COMB 206 1.0DL + 1.0WLZ2
1 1.0 6 1.0
LOAD COMB 207 1.0DL + 0.75WLX1 + 0.75LL
1 1.0 3 0.75 2 0.75
LOAD COMB 208 1.0DL + 0.75WLX2 + 0.75LL
1 1.0 4 0.75 2 0.75
LOAD COMB 209 1.0DL + 0.75WLZ1 + 0.75LL
1 1.0 5 0.75 2 0.75
LOAD COMB 210 1.0DL + 0.75WLZ2 + 0.75LL
1 1.0 6 0.75 2 0.75
LOAD COMB 211 0.6DL + 1.0WLX1
1 0.6 3 1.0
LOAD COMB 212 0.6DL + 1.0WLX2
1 0.6 4 1.0
LOAD COMB 213 0.6DL + 1.0WLZ1
1 0.6 5 1.0
LOAD COMB 214 0.6DL + 1.0WLZ2
1 0.6 6 1.0
********************************************************************************************
PERFORM ANALYSIS
LOAD LIST 101 TO 110
START CONCRETE DESIGN
CODE ACI
TRACK 2 ALL
FC 35000 MEMB 185
CLB 0.05 MEMB 185
CLS 0.05 ALL
CLT 0.05 ALL
MAXMAIN 25 ALL
MINMAIN 25 ALL
MINSEC 10 ALL
DESIGN COLUMN 185
END CONCRETE DESIGN
FINISH
```

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 08/05/2020
JOB NAME FDN. MKD. F2_STABILITY CHECK
JOB CLIENT MA'ADEN
JOB NO O19045
JOB REV 0
ENGINEER NAME SOR
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
2 0 911.65 3.175; 128 0 913.6 3.175;
MEMBER INCIDENCES
185 2 128;
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 25
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
185 PRIS YD 0.4 ZD 0.6
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
2 FIXED
*********************************************************************************************
LOAD 1 LOADTYPE None  TITLE DEAD LOAD ( DL )
SELFWEIGHT Y -1
JOINT LOAD
128 FX -0.001 FY -76.404 FZ 0.282
*********************************************************************************************
LOAD 2 LOADTYPE None  TITLE LIVE LOAD ( LL )
JOINT LOAD
128 FX -0.005 FY -57.023 FZ -1.257
*********************************************************************************************
LOAD 3 LOADTYPE None  TITLE WIND LOAD +X  ( WLX1 )
JOINT LOAD
128 FX 1.334 FY -0.866 FZ 0.111
*********************************************************************************************
LOAD 4 LOADTYPE None  TITLE WIND LOAD -X  ( WLX2 )
JOINT LOAD
128 FX 1.334 FY 0.866 FZ 0.111
*********************************************************************************************
LOAD 5 LOADTYPE None  TITLE WIND LOAD +Z  ( WLZ1 )
JOINT LOAD
128 FX 0.025 FY 9.749 FZ 3.225
*********************************************************************************************
LOAD 6 LOADTYPE None  TITLE WIND LOAD -Z  ( WLZ2 )
JOINT LOAD
128 FX 0.025 FY -9.744 FZ 3.224
*********************************************************************************************
* UNFACTORED LOAD COMBINATIONS FOR STABILITY CHECK
LOAD COMB 201 1.0DL
1 1.0
LOAD COMB 202 1.0DL + 1.0LL
1 1.0 2 1.0
LOAD COMB 203 1.0DL + 1.0WLX1
1 1.0 3 1.0
LOAD COMB 204 1.0DL + 1.0WLX2
1 1.0 4 1.0
LOAD COMB 205 1.0DL + 1.0WLZ1
1 1.0 5 1.0
LOAD COMB 206 1.0DL + 1.0WLZ2
1 1.0 6 1.0
LOAD COMB 207 1.0DL + 0.75WLX1 + 0.75LL
1 1.0 3 0.75 2 0.75
LOAD COMB 208 1.0DL + 0.75WLX2 + 0.75LL
1 1.0 4 0.75 2 0.75
LOAD COMB 209 1.0DL + 0.75WLZ1 + 0.75LL
1 1.0 5 0.75 2 0.75
LOAD COMB 210 1.0DL + 0.75WLZ2 + 0.75LL
1 1.0 6 0.75 2 0.75
LOAD COMB 211 0.6DL + 1.0WLX1
```



```
LOAD COMB 211 0.6DL + 1.0WLX1
1 0.6 3 1.0
LOAD COMB 212 0.6DL + 1.0WLX2
1 0.6 4 1.0
LOAD COMB 213 0.6DL + 1.0WLZ1
1 0.6 5 1.0
LOAD COMB 214 0.6DL + 1.0WLZ2
1 0.6 6 1.0
*********************************************************************************************
PERFORM ANALYSIS
FINISH
```

ATTACHMENT 4 STAAD INPUT – FOUNDATION F3

PROPRIETARY & CONFIDENTIAL

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```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 08/05/2020
JOB NAME FDN. MKD. F3
JOB CLIENT MA'ADEN
JOB NO O19045
JOB REV 0
ENGINEER NAME SOR
END JOB INFORMATION
INPUT WIDTH 79
SET NL 200
SET ITERLIM 15
SET PRINT 1
UNIT METER KN
JOINT COORDINATES
422 -0.5 911.65 1.12538; 423 -0.75 911.65 1.12538; 425 -0.25 911.65 1.12538;
427 0 911.65 1.12538; 429 0.25 911.65 1.12538; 431 0.5 911.65 1.12538;
433 0.75 911.65 1.12538; 435 1 911.65 1.12538; 456 -0.75 911.65 0.42438;
457 -0.5 911.65 0.42438; 458 -0.5 911.65 0.17538; 459 -0.75 911.65 0.17538;
460 -0.25 911.65 0.42438; 461 -0.25 911.65 0.17538; 462 0 911.65 0.42438;
463 0 911.65 0.17538; 464 0.25 911.65 0.42438; 465 0.25 911.65 0.17538;
466 0.5 911.65 0.42438; 467 0.5 911.65 0.17538; 468 0.75 911.65 0.42438;
469 0.75 911.65 0.17538; 470 1 911.65 0.42438; 471 1 911.65 0.17538;
474 -0.75 911.65 0.67438; 475 -0.5 911.65 0.67438; 476 -0.25 911.65 0.67438;
477 0 911.65 0.67438; 478 0.25 911.65 0.67438; 479 0.5 911.65 0.67438;
480 0.75 911.65 0.67438; 481 1 911.65 0.67438; 483 -0.75 911.65 0.92438;
484 -0.5 911.65 0.92438; 485 -0.25 911.65 0.92438; 486 0 911.65 0.92438;
487 0.25 911.65 0.92438; 488 0.5 911.65 0.92438; 489 0.75 911.65 0.92438;
490 1 911.65 0.92438; 492 -0.75 911.65 -0.62362; 493 -0.5 911.65 -0.62362;
494 -0.5 911.65 -0.87362; 495 -0.75 911.65 -0.87362; 496 -0.25 911.65 -0.62362;
497 -0.25 911.65 -0.87362; 498 0 911.65 -0.62362; 499 0 911.65 -0.87362;
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503 0.5 911.65 -0.87362; 504 0.75 911.65 -0.62362; 505 0.75 911.65 -0.87362;
506 1 911.65 -0.62362; 507 1 911.65 -0.87362; 510 -0.75 911.65 -0.37362;
511 -0.5 911.65 -0.37362; 512 -0.25 911.65 -0.37362; 513 0 911.65 -0.37362;
514 0.25 911.65 -0.37362; 515 0.5 911.65 -0.37362; 516 0.75 911.65 -0.37362;
517 1 911.65 -0.37362; 519 -0.75 911.65 -0.12462; 520 -0.5 911.65 -0.12462;
521 -0.25 911.65 -0.12462; 522 0 911.65 -0.12462; 523 0.25 911.65 -0.12462;
524 0.5 911.65 -0.12462; 525 0.75 911.65 -0.12462; 526 1 911.65 -0.12462;
528 0 913.6 -0.125; 546 1.25 911.65 1.12538; 553 1.25 911.65 0.42438;
554 1.25 911.65 0.17538; 557 1.25 911.65 0.67438; 559 1.25 911.65 0.92438;
561 1.25 911.65 -0.62362; 562 1.25 911.65 -0.87362; 565 1.25 911.65 -0.37362;
567 1.25 911.65 -0.12462; 677 1.5 911.65 1.12538; 679 1.75 911.65 1.12538;
681 2 911.65 1.12538; 683 2.25 911.65 1.12538; 684 1.5 911.65 0.42438;
685 1.5 911.65 0.17538; 686 1.75 911.65 0.42438; 687 1.75 911.65 0.17538;
688 2 911.65 0.42438; 689 2 911.65 0.17538; 690 2.25 911.65 0.42438;
691 2.25 911.65 0.17538; 692 1.5 911.65 0.67438; 693 1.75 911.65 0.67438;
694 2 911.65 0.67438; 695 2.25 911.65 0.67438; 696 1.5 911.65 0.92438;
697 1.75 911.65 0.92438; 698 2 911.65 0.92438; 699 2.25 911.65 0.92438;
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703 1.75 911.65 -0.87362; 704 2 911.65 -0.62362; 705 2 911.65 -0.87362;
706 2.25 911.65 -0.62362; 707 2.25 911.65 -0.87362; 708 1.5 911.65 -0.37362;
709 1.75 911.65 -0.37362; 710 2 911.65 -0.37362; 711 2.25 911.65 -0.37362;
712 1.5 911.65 -0.12462; 713 1.75 911.65 -0.12462; 714 2 911.65 -0.12462;
715 2.25 911.65 -0.12462; 717 2.5 911.65 1.12538; 718 2.5 911.65 0.42438;
719 2.5 911.65 0.17538; 720 2.5 911.65 0.67438; 721 2.5 911.65 0.92438;
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725 2.5 911.65 -0.12462; 727 2.75 911.65 1.12538; 729 3 911.65 1.12538;
731 3.25 911.65 1.12538; 733 3.5 911.65 1.12538; 734 2.75 911.65 0.42438;
735 2.75 911.65 0.17538; 736 3 911.65 0.42438; 737 3 911.65 0.17538;
738 3.25 911.65 0.42438; 739 3.25 911.65 0.17538; 740 3.5 911.65 0.42438;
741 3.5 911.65 0.17538; 742 2.75 911.65 0.67438; 743 3 911.65 0.67438;
744 3.25 911.65 0.67438; 745 3.5 911.65 0.67438; 746 2.75 911.65 0.92438;
747 3 911.65 0.92438; 748 3.25 911.65 0.92438; 749 3.5 911.65 0.92438;
750 2.75 911.65 -0.62362; 751 2.75 911.65 -0.87362; 752 3 911.65 -0.62362;
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756 3.5 911.65 -0.62362; 757 3.5 911.65 -0.87362; 758 2.75 911.65 -0.37362;
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762 2.75 911.65 -0.12462; 763 3 911.65 -0.12462; 764 3.25 911.65 -0.12462;
765 3.5 911.65 -0.12462; 767 3.75 911.65 1.12538; 768 3.75 911.65 0.42438;
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775 3.75 911.65 -0.12462; 776 3.5 913.6 -0.125; 778 4 911.65 1.12538;
779 4 911.65 0.42438; 780 4 911.65 0.17538; 781 4 911.65 0.67438;
782 4 911.65 0.92438; 783 4 911.65 -0.62362; 784 4 911.65 -0.87362;
785 4 911.65 -0.37362; 786 4 911.65 -0.12462; 788 4.25 911.65 1.12538;
789 4.25 911.65 0.42438; 790 4.25 911.65 0.17538; 791 4.25 911.65 0.67438;
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804 0 911.65 -1.37362; 805 0.25 911.65 -1.12362; 806 0.25 911.65 -1.37362;
807 0.5 911.65 -1.12362; 808 0.5 911.65 -1.37362; 809 0.75 911.65 -1.12362;
810 0.75 911.65 -1.37362; 811 1 911.65 -1.12362; 812 1 911.65 -1.37362;
813 1.25 911.65 -1.12362; 814 1.25 911.65 -1.37362; 815 1.5 911.65 -1.12362;
816 1.5 911.65 -1.37362; 817 1.75 911.65 -1.12362; 818 1.75 911.65 -1.37362;
819 2 911.65 -1.12362; 820 2 911.65 -1.37362; 821 2.25 911.65 -1.12362;
822 2.25 911.65 -1.37362; 823 2.5 911.65 -1.12362; 824 2.5 911.65 -1.37362;
825 2.75 911.65 -1.12362; 826 2.75 911.65 -1.37362; 827 3 911.65 -1.12362;
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831 3.5 911.65 -1.12362; 832 3.5 911.65 -1.37362; 833 3.75 911.65 -1.12362;
834 3.75 911.65 -1.37362; 835 4 911.65 -1.12362; 836 4 911.65 -1.37362;
837 4.25 911.65 -1.12362; 838 4.25 911.65 -1.37362;
MEMBER INCIDENCES
529 522 528; 759 765 776;
ELEMENT INCIDENCES SHELL
465 456 457 458 459; 466 457 460 461 458; 467 460 462 463 461;
468 462 464 465 463; 469 464 466 467 465; 470 466 468 469 467;
471 468 470 471 469; 473 474 475 457 456; 474 475 476 460 457;
475 476 477 462 460; 476 477 478 464 462; 477 478 479 466 464;
478 479 480 468 466; 479 480 481 470 468; 481 483 484 475 474;
482 484 485 476 475; 483 485 486 477 476; 484 486 487 478 477;
485 487 488 479 478; 486 488 489 480 479; 487 489 490 481 480;
489 423 422 484 483; 490 422 425 485 484; 491 425 427 486 485;
492 427 429 487 486; 493 429 431 488 487; 494 431 433 489 488;
495 433 435 490 489; 497 492 493 494 495; 498 493 496 497 494;
499 496 498 499 497; 500 498 500 501 499; 501 500 502 503 501;
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519 525 526 517 516; 521 459 458 520 519; 522 458 461 521 520;
523 461 463 522 521; 524 463 465 523 522; 525 465 467 524 523;
526 467 469 525 524; 527 469 471 526 525; 552 470 553 554 471;
554 481 557 553 470; 556 490 559 557 481; 558 435 546 559 490;
560 506 561 562 507; 562 517 565 561 506; 564 526 567 565 517;
566 471 554 567 526; 673 553 684 685 554; 674 684 686 687 685;
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684 698 699 695 694; 685 546 677 696 559; 686 677 679 697 696;
687 679 681 698 697; 688 681 683 699 698; 689 561 700 701 562;
690 700 702 703 701; 691 702 704 705 703; 692 704 706 707 705;
693 565 708 700 561; 694 708 709 702 700; 695 709 710 704 702;
696 710 711 706 704; 697 567 712 708 565; 698 712 713 709 708;
699 713 714 710 709; 700 714 715 711 710; 701 554 685 712 567;
702 685 687 713 712; 703 687 689 714 713; 704 689 691 715 714;
706 690 718 719 691; 707 695 720 718 690; 708 699 721 720 695;
709 683 717 721 699; 710 706 722 723 707; 711 711 724 722 706;
712 715 725 724 711; 713 691 719 725 715; 718 718 734 735 719;
719 734 736 737 735; 720 736 738 739 737; 721 738 740 741 739;
722 720 742 734 718; 723 742 743 736 734; 724 743 744 738 736;
725 744 745 740 738; 726 721 746 742 720; 727 746 747 743 742;
728 747 748 744 743; 729 748 749 745 744; 730 717 727 746 721;
731 727 729 747 746; 732 729 731 748 747; 733 731 733 749 748;
734 722 750 751 723; 735 750 752 753 751; 736 752 754 755 753;
737 754 756 757 755; 738 724 758 750 722; 739 758 759 752 750;
740 759 760 754 752; 741 760 761 756 754; 742 725 762 758 724;
743 762 763 759 758; 744 763 764 760 759; 745 764 765 761 760;
746 719 735 762 725; 747 735 737 763 762; 748 737 739 764 763;
749 739 741 765 764; 751 740 768 769 741; 752 745 770 768 740;
753 749 771 770 745; 754 733 767 771 749; 755 756 772 773 757;
756 761 774 772 756; 757 765 775 774 761; 758 741 769 775 765;
761 768 779 780 769; 762 770 781 779 768; 763 771 782 781 770;
764 767 778 782 771; 765 772 783 784 773; 766 774 785 783 772;
767 775 786 785 774; 768 769 780 786 775; 770 779 789 790 780;
771 781 791 789 779; 772 782 792 791 781; 773 778 788 792 782;
774 783 793 794 784; 775 785 795 793 783; 776 786 796 795 785;
777 780 790 796 786; 778 797 798 799 800; 779 798 801 802 799;
780 801 803 804 802; 781 803 805 806 804; 782 805 807 808 806;
783 807 809 810 808; 784 809 811 812 810; 785 495 494 798 797;
786 494 497 801 798; 787 497 499 803 801; 788 499 501 805 803;
789 501 503 807 805; 790 503 505 809 807; 791 505 507 811 809;
792 811 813 814 812; 793 507 562 813 811; 794 813 815 816 814;
795 815 817 818 816; 796 817 819 820 818; 797 819 821 822 820;
798 562 701 815 813; 799 701 703 817 815; 800 703 705 819 817;
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798 562 701 815 813; 799 701 703 817 815; 800 703 705 819 817;
801 705 707 821 819; 802 821 823 824 822; 803 707 723 823 821;
804 823 825 826 824; 805 825 827 828 826; 806 827 829 830 828;
807 829 831 832 830; 808 723 751 825 823; 809 751 753 827 825;
810 753 755 829 827; 811 755 757 831 829; 812 831 833 834 832;
813 757 773 833 831; 814 833 835 836 834; 815 773 784 835 833;
816 835 837 838 836; 817 784 794 837 835;
START GROUP DEFINITION
ELEMENT
_RAFT 465 TO 471 473 TO 479 481 TO 487 489 TO 495 497 TO 503 505 TO 511 513 -
514 TO 519 521 TO 527 552 554 556 558 560 562 564 566 673 TO 704 706 TO 713 -
718 TO 749 751 TO 758 761 TO 768 770 TO 817
_RAFT_MOM 465 TO 471 473 TO 479 481 TO 487 489 TO 495 497 TO 503 505 TO 511 -
513 514 517 TO 519 521 522 525 TO 527 552 554 556 558 560 562 564 566 673 -
674 TO 688 690 TO 704 706 TO 713 718 TO 744 746 TO 748 751 TO 756 761 TO 768 -
770 TO 817
_RAFT_SHEAR 465 470 471 473 TO 479 481 TO 487 489 TO 495 497 TO 503 505 510 -
511 513 518 519 521 526 527 552 554 556 558 560 562 564 566 673 TO 704 706 -
707 TO 713 718 719 722 TO 739 742 743 746 747 752 TO 755 762 TO 765 -
770 TO 788 790 TO 817
MEMBER
END GROUP DEFINITION
ELEMENT PROPERTY
465 TO 471 473 TO 479 481 TO 487 489 TO 495 497 TO 503 505 TO 511 513 TO 519 -
521 TO 527 552 554 556 558 560 562 564 566 673 TO 704 706 TO 713 718 TO 749 -
751 TO 758 761 TO 768 770 TO 817 THICKNESS 0.5
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 25
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
529 759 PRIS YD 0.4 ZD 0.6
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
465 TO 471 473 TO 479 481 TO 487 489 TO 495 497 TO 503 505 TO 511 513 TO 519 -
521 TO 527 552 554 556 558 560 562 564 566 673 TO 704 706 TO 713 718 TO 749 -
751 TO 758 761 TO 768 770 TO 817 PLATE MAT DIRECT Y SUBGRADE 57600 PRINT
*********************************************************************************************
LOAD 1 LOADTYPE None  TITLE DEAD LOAD ( DL )
SELFWEIGHT Y -1
ELEMENT LOAD
* LOAD DUE TO BACKFILL
465 TO 471 473 TO 479 481 TO 487 489 TO 495 497 TO 503 505 TO 511 513 TO 519 -
521 TO 527 552 554 556 558 560 562 564 566 673 TO 704 706 TO 713 718 TO 749 -
751 TO 758 761 TO 768 770 TO 817 PR GY -15
* LOAD DUE TO SURCHARGE
465 TO 471 473 TO 479 481 TO 487 489 TO 495 497 TO 503 505 TO 511 513 TO 519 -
521 TO 527 552 554 556 558 560 562 564 566 673 TO 704 706 TO 713 718 TO 749 -
751 TO 758 761 TO 768 770 TO 817 PR GY -20
JOINT LOAD
528 FX 3.1 FY -35.889 FZ 0.03
776 FX 3.14 FY -26.831 FZ 0.07
*********************************************************************************************
LOAD 2 LOADTYPE None  TITLE LIVE LOAD ( LL )
JOINT LOAD
528 FX 8.7 FY -74.7 FZ 0.05
776 FX 8.743 FY -48.829 FZ 0.176
*********************************************************************************************
LOAD 3 LOADTYPE None  TITLE WIND LOAD +X  ( WLX1 )
JOINT LOAD
528 FX 22.7 FY 81.634
776 FX 22.725 FY -82.271 FZ 0.003
*********************************************************************************************
LOAD 4 LOADTYPE None  TITLE WIND LOAD -X  ( WLX2 )
JOINT LOAD
528 FX 22.721 FY -81.634
776 FX 22.725 FY 82.271 FZ 0.003
*********************************************************************************************
LOAD 5 LOADTYPE None  TITLE WIND LOAD +Z  ( WLZ1 )
JOINT LOAD
528 FX 2.434 FY -26.337 FZ 0.473
776 FX 2.335 FY 14.741 FZ 0.616
```



```
776 FX 2.335 FY 14.741 FZ 0.616
*********************************************************************************************
LOAD 6 LOADTYPE None  TITLE WIND LOAD -Z  ( WLZ2 )
JOINT LOAD
528 FX 2.434 FY 26.3 FZ 0.473
776 FX 2.331 FY -14.716 FZ 0.618
*********************************************************************************************
PERFORM ANALYSIS
CHANGE
SUPPORTS
465 TO 471 473 TO 479 481 TO 487 489 TO 495 497 TO 503 505 TO 511 513 TO 519 -
521 TO 527 552 554 556 558 560 562 564 566 673 TO 704 706 TO 713 718 TO 749 -
751 TO 758 761 TO 768 770 TO 816 -
817 PLATE MAT DIRECT Y SUBGRADE 57600 COMPRESSION
*********************************************************************************************
***FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
LOAD 101 1.4DL
REPEAT LOAD
1 1.4
PERFORM ANALYSIS
CHANGE
LOAD 102 1.4DL + 1.7LL
REPEAT LOAD
1 1.4 2 1.7
PERFORM ANALYSIS
CHANGE
LOAD 103 1.2DL + 1.6WLX1 + 1.0 LL
REPEAT LOAD
1 1.2 3 1.6 2 1.0
PERFORM ANALYSIS
CHANGE
LOAD 104 1.2DL + 1.6WLX2 + 1.0 LL
REPEAT LOAD
1 1.2 4 1.6 2 1.0
PERFORM ANALYSIS
CHANGE
LOAD 105 1.2DL + 1.6WLZ1 + 1.0 LL
REPEAT LOAD
1 1.2 5 1.6 2 1.0
PERFORM ANALYSIS
CHANGE
LOAD 106 1.2DL + 1.6WLZ2 + 1.0 LL
REPEAT LOAD
1 1.2 6 1.6 2 1.0
PERFORM ANALYSIS
CHANGE
LOAD 107 0.9DL + 1.6WLX1
REPEAT LOAD
1 0.9 3 1.6
PERFORM ANALYSIS
CHANGE
LOAD 108 0.9DL + 1.6WLX2
REPEAT LOAD
1 0.9 4 1.6
PERFORM ANALYSIS
CHANGE
LOAD 109 0.9DL + 1.6WLZ1
REPEAT LOAD
1 0.9 5 1.6
PERFORM ANALYSIS
CHANGE
LOAD 110 0.9DL + 1.6WLZ2
REPEAT LOAD
1 0.9 6 1.6
PERFORM ANALYSIS
CHANGE
*********************************************************************************************
* UNFACTORED LOAD COMBINATIONS FOR STABILITY CHECK
LOAD 201 1.0DL
REPEAT LOAD
1 1.0
PERFORM ANALYSIS
CHANGE
LOAD 202 1.0DL + 1.0LL
REPEAT LOAD
1 1.0 2 1.0
PERFORM ANALYSIS
CHANGE
LOAD 203 1.0DL + 1.0WLX1
```

```
LOAD 203 1.0DL + 1.0WLX1
REPEAT LOAD
1 1.0 3 1.0
PERFORM ANALYSIS
CHANGE
LOAD 204 1.0DL + 1.0WLX2
REPEAT LOAD
1 1.0 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 205 1.0DL + 1.0WLZ1
REPEAT LOAD
1 1.0 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 206 1.0DL + 1.0WLZ2
REPEAT LOAD
1 1.0 6 1.0
PERFORM ANALYSIS
CHANGE
LOAD 207 1.0DL + 0.75WLX1 + 0.75LL
REPEAT LOAD
1 1.0 3 0.75 2 0.75
PERFORM ANALYSIS
CHANGE
LOAD 208 1.0DL + 0.75WLX2 + 0.75LL
REPEAT LOAD
1 1.0 4 0.75 2 0.75
PERFORM ANALYSIS
CHANGE
LOAD 209 1.0DL + 0.75WLZ1 + 0.75LL
REPEAT LOAD
1 1.0 5 0.75 2 0.75
PERFORM ANALYSIS
CHANGE
LOAD 210 1.0DL + 0.75WLZ2 + 0.75LL
REPEAT LOAD
1 1.0 6 0.75 2 0.75
PERFORM ANALYSIS
CHANGE
LOAD 211 0.6DL + 1.0WLX1
REPEAT LOAD
1 0.6 3 1.0
PERFORM ANALYSIS
CHANGE
LOAD 212 0.6DL + 1.0WLX2
REPEAT LOAD
1 0.6 4 1.0
PERFORM ANALYSIS
CHANGE
LOAD 213 0.6DL + 1.0WLZ1
REPEAT LOAD
1 0.6 5 1.0
PERFORM ANALYSIS
CHANGE
LOAD 214 0.6DL + 1.0WLZ2
REPEAT LOAD
1 0.6 6 1.0
PERFORM ANALYSIS
CHANGE
*************************************************************************************************
PERFORM ANALYSIS
LOAD LIST 101 TO 110
START CONCRETE DESIGN
CODE ACI
TRACK 2 ALL
FC 35000 MEMB 529 759
CLB 0.05 MEMB 529 759
CLS 0.05 ALL
MAXMAIN 25 ALL
MINMAIN 25 ALL
MINSEC 10 ALL
DESIGN COLUMN 529 759
END CONCRETE DESIGN
FINISH
```

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 08/05/2020
JOB NAME FDN. MKD. F3_STABILITY CHECK
JOB CLIENT MA'ADEN
JOB NO O19045
JOB REV 0
ENGINEER NAME SOR
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
522 0 911.65 -0.12462; 528 0 913.6 -0.125; 765 3.5 911.65 -0.12462;
776 3.5 913.6 -0.125;
MEMBER INCIDENCES
529 522 528; 759 765 776;
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 25
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
529 759 PRIS YD 0.4 ZD 0.6
CONSTANTS
MATERIAL CONCRETE ALL
SUPPORTS
522 765 FIXED
*********************************************************************************************
LOAD 1 LOADTYPE None  TITLE DEAD LOAD ( DL )
SELFWEIGHT Y -1
JOINT LOAD
528 FX 3.1 FY -35.889 FZ 0.03
776 FX 3.14 FY -26.831 FZ 0.07
*********************************************************************************************
LOAD 2 LOADTYPE None  TITLE LIVE LOAD ( LL )
JOINT LOAD
528 FX 8.7 FY -74.7 FZ 0.05
776 FX 8.743 FY -48.829 FZ 0.176
*********************************************************************************************
LOAD 3 LOADTYPE None  TITLE WIND LOAD +X  ( WLX1 )
JOINT LOAD
528 FX 22.7 FY 81.634
776 FX 22.725 FY -82.271 FZ 0.003
*********************************************************************************************
LOAD 4 LOADTYPE None  TITLE WIND LOAD -X  ( WLX2 )
JOINT LOAD
528 FX 22.721 FY -81.634
776 FX 22.725 FY 82.271 FZ 0.003
*********************************************************************************************
LOAD 5 LOADTYPE None  TITLE WIND LOAD +Z  ( WLZ1 )
JOINT LOAD
528 FX 2.434 FY -26.337 FZ 0.473
776 FX 2.335 FY 14.741 FZ 0.616
*********************************************************************************************
LOAD 6 LOADTYPE None  TITLE WIND LOAD -Z  ( WLZ2 )
JOINT LOAD
528 FX 2.434 FY 26.3 FZ 0.473
776 FX 2.331 FY -14.716 FZ 0.618
*********************************************************************************************
* UNFACTORED LOAD COMBINATIONS FOR STABILITY CHECK
LOAD COMB 201 1.0DL
1 1.0
LOAD COMB 202 1.0DL + 1.0LL
1 1.0 2 1.0
LOAD COMB 203 1.0DL + 1.0WLX1
1 1.0 3 1.0
LOAD COMB 204 1.0DL + 1.0WLX2
1 1.0 4 1.0
LOAD COMB 205 1.0DL + 1.0WLZ1
1 1.0 5 1.0
LOAD COMB 206 1.0DL + 1.0WLZ2
1 1.0 6 1.0
LOAD COMB 207 1.0DL + 0.75WLX1 + 0.75LL
1 1.0 3 0.75 2 0.75
```



```
1 1.0 3 0.75 2 0.75
LOAD COMB 208 1.0DL + 0.75WLX2 + 0.75LL
1 1.0 4 0.75 2 0.75
LOAD COMB 209 1.0DL + 0.75WLZ1 + 0.75LL
1 1.0 5 0.75 2 0.75
LOAD COMB 210 1.0DL + 0.75WLZ2 + 0.75LL
1 1.0 6 0.75 2 0.75
LOAD COMB 211 0.6DL + 1.0WLX1
1 0.6 3 1.0
LOAD COMB 212 0.6DL + 1.0WLX2
1 0.6 4 1.0
LOAD COMB 213 0.6DL + 1.0WLZ1
1 0.6 5 1.0
LOAD COMB 214 0.6DL + 1.0WLZ2
1 0.6 6 1.0
*********************************************************************************************
PERFORM ANALYSIS
FINISH
```

ATTACHMENT 5 STAAD OUTPUT – FOUNDATION F1

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```
**************************************************
*                                                *
*          STAAD.Pro V8i SELECTseries5           *
*          Version  20.07.10.64                  *
*          Proprietary Program of                *
*          Bentley Systems, Inc.                 *
*          Date=    OCT 16, 2020                 *
*          Time=    10:30:35                     *
*                                                *
*     USER ID: LNT ECC DIV                       *
**************************************************
```

```
    1. STAAD SPACE
INPUT FILE: F1 DESIGN.STD
    2. START JOB INFORMATION
    3. ENGINEER DATE 08/10/2020
    4. JOB NAME FDN. MKD. F1
    5. JOB CLIENT MA'ADEN
    6. JOB NO O19045
    7. JOB REV 0
    8. ENGINEER NAME SOR
    9. END JOB INFORMATION
   10. INPUT WIDTH 79
   11. SET NL 200
   12. SET ITERLIM 15
   13. SET PRINT 1
   14. UNIT METER KN
   15. JOINT COORDINATES
   16. 2 0 911.65 3.175; 6 0.75 911.65 3.175; 10 -0.75 911.65 3.175
   17. 34 -0.5 911.65 3.175; 35 -0.25 911.65 3.175; 41 0.25 911.65 3.175
   18. 42 0.5 911.65 3.175; 128 0 913.6 3.175; 302 -0.5 911.65 3.57438
   19. 303 -0.75 911.65 3.57438; 304 -0.25 911.65 3.57438; 305 0 911.65 3.57438
   20. 306 0.25 911.65 3.57438; 307 0.5 911.65 3.57438; 308 0.75 911.65 3.57438
   21. 365 -0.75 911.65 2.17438; 366 -0.5 911.65 2.17438; 367 -0.5 911.65 1.925
   22. 368 -0.75 911.65 1.925; 369 -0.25 911.65 2.17438; 370 -0.25 911.65 1.925
   23. 371 0 911.65 2.17438; 372 0 911.65 1.925; 373 0.25 911.65 2.17438
   24. 374 0.25 911.65 1.925; 375 0.5 911.65 2.17438; 376 0.5 911.65 1.925
   25. 377 0.75 911.65 2.17438; 378 0.75 911.65 1.925; 387 -0.75 911.65 2.42438
   26. 388 -0.5 911.65 2.42438; 389 -0.25 911.65 2.42438; 390 0 911.65 2.42438
   27. 391 0.25 911.65 2.42438; 392 0.5 911.65 2.42438; 393 0.75 911.65 2.42438
   28. 398 -0.75 911.65 2.67438; 399 -0.5 911.65 2.67438; 400 -0.25 911.65 2.67438
   29. 401 0 911.65 2.67438; 402 0.25 911.65 2.67438; 403 0.5 911.65 2.67438
   30. 404 0.75 911.65 2.67438; 409 -0.75 911.65 2.92438; 410 -0.5 911.65 2.92438
   31. 411 -0.25 911.65 2.92438; 412 0 911.65 2.92438; 413 0.25 911.65 2.92438
   32. 414 0.5 911.65 2.92438; 415 0.75 911.65 2.92438; 416 1 911.65 3.57438
   33. 417 1 911.65 3.175; 428 1 911.65 2.17438; 429 1 911.65 1.925
   34. 432 1 911.65 2.42438; 434 1 911.65 2.67438; 436 1 911.65 2.92438
   35. 438 -1.25 911.65 3.57438; 439 -1 911.65 3.57438; 440 -1 911.65 3.175
   36. 441 -1.25 911.65 3.175; 450 -1.25 911.65 2.17438; 451 -1 911.65 2.17438
   37. 452 -1 911.65 1.925; 453 -1.25 911.65 1.925; 454 -1.25 911.65 2.42438
   38. 455 -1 911.65 2.42438; 456 -1.25 911.65 2.67438; 457 -1 911.65 2.67438
```



```
   39. 458 -1.25 911.65 2.92438; 459 -1 911.65 2.92438; 460 -0.75 911.65 1.67438
   40. 461 -0.5 911.65 1.67438; 462 -0.5 911.65 1.425; 463 -0.75 911.65 1.425
   41. 464 -0.25 911.65 1.67438; 465 -0.25 911.65 1.425; 466 0 911.65 1.67438
   42. 467 0 911.65 1.425; 468 0.25 911.65 1.67438; 469 0.25 911.65 1.425
   43. 470 0.5 911.65 1.67438; 471 0.5 911.65 1.425; 472 0.75 911.65 1.67438
   44. 473 0.75 911.65 1.425; 474 1 911.65 1.67438; 475 1 911.65 1.425
   45. 478 -1.25 911.65 1.67438; 479 -1 911.65 1.67438; 480 -1 911.65 1.425
   46. 481 -1.25 911.65 1.425; 504 -0.5 911.65 1.175; 505 -0.75 911.65 1.175
   47. 506 -0.25 911.65 1.175; 507 0 911.65 1.175; 508 0.25 911.65 1.175
   48. 509 0.5 911.65 1.175; 510 0.75 911.65 1.175; 511 1 911.65 1.175
   49. 513 -1 911.65 1.175; 514 -1.25 911.65 1.175; 526 -1.5 911.65 3.57438
   50. 527 -1.5 911.65 3.175; 532 -1.5 911.65 2.17438; 533 -1.5 911.65 1.925
   51. 534 -1.5 911.65 2.42438; 535 -1.5 911.65 2.67438; 536 -1.5 911.65 2.92438
   52. 537 -1.5 911.65 1.67438; 538 -1.5 911.65 1.425; 541 -1.5 911.65 1.175
   53. 560 -0.5 911.65 0.925; 561 -0.75 911.65 0.925; 562 -0.25 911.65 0.925
   54. 563 0 911.65 0.925; 564 0.25 911.65 0.925; 565 0.5 911.65 0.925
   55. 566 0.75 911.65 0.925; 567 1 911.65 0.925; 569 -1 911.65 0.925
   56. 570 -1.25 911.65 0.925; 571 -1.5 911.65 0.925; 573 -0.75 911.65 0.67438
   57. 574 -0.5 911.65 0.67438; 575 -0.5 911.65 0.425; 576 -0.75 911.65 0.425
   58. 577 -0.25 911.65 0.67438; 578 -0.25 911.65 0.425; 579 0 911.65 0.67438
   59. 580 0 911.65 0.425; 581 0.25 911.65 0.67438; 582 0.25 911.65 0.425
   60. 583 0.5 911.65 0.67438; 584 0.5 911.65 0.425; 585 0.75 911.65 0.67438
   61. 586 0.75 911.65 0.425; 587 1 911.65 0.67438; 588 1 911.65 0.425
   62. 591 -1.25 911.65 0.67438; 592 -1 911.65 0.67438; 593 -1 911.65 0.425
   63. 594 -1.25 911.65 0.425; 595 -0.5 911.65 0.175; 596 -0.75 911.65 0.175
   64. 597 -0.25 911.65 0.175; 598 0 911.65 0.175; 599 0.25 911.65 0.175
   65. 600 0.5 911.65 0.175; 601 0.75 911.65 0.175; 602 1 911.65 0.175
   66. 604 -1 911.65 0.175; 605 -1.25 911.65 0.175; 606 -1.5 911.65 0.67438
   67. 607 -1.5 911.65 0.425; 608 -1.5 911.65 0.175; 612 -0.5 911.65 -0.075
   68. 613 -0.75 911.65 -0.075; 614 -0.25 911.65 -0.075; 615 0 911.65 -0.075
   69. 616 0.25 911.65 -0.075; 617 0.5 911.65 -0.075; 618 0.75 911.65 -0.075
   70. 619 1 911.65 -0.075; 621 -1 911.65 -0.075; 622 -1.25 911.65 -0.075
   71. 623 -1.5 911.65 -0.075; 625 -0.75 911.65 -0.32562; 626 -0.5 911.65 -0.32562
   72. 627 -0.5 911.65 -0.575; 628 -0.75 911.65 -0.575; 629 -0.25 911.65 -0.32562
   73. 630 -0.25 911.65 -0.575; 631 0 911.65 -0.32562; 632 0 911.65 -0.575
   74. 633 0.25 911.65 -0.32562; 634 0.25 911.65 -0.575; 635 0.5 911.65 -0.32562
   75. 636 0.5 911.65 -0.575; 637 0.75 911.65 -0.32562; 638 0.75 911.65 -0.575
   76. 639 1 911.65 -0.32562; 640 1 911.65 -0.575; 643 -1.25 911.65 -0.32562
   77. 644 -1 911.65 -0.32562; 645 -1 911.65 -0.575; 646 -1.25 911.65 -0.575
   78. 647 -0.5 911.65 -0.825; 648 -0.75 911.65 -0.825; 649 -0.25 911.65 -0.825
   79. 650 0 911.65 -0.825; 651 0.25 911.65 -0.825; 652 0.5 911.65 -0.825
   80. 653 0.75 911.65 -0.825; 654 1 911.65 -0.825; 656 -1 911.65 -0.825
   81. 657 -1.25 911.65 -0.825; 658 -1.5 911.65 -0.32562; 659 -1.5 911.65 -0.575
   82. 660 -1.5 911.65 -0.825; 664 -0.5 911.65 -1.075; 665 -0.75 911.65 -1.075
   83. 666 -0.25 911.65 -1.075; 667 0 911.65 -1.075; 668 0.25 911.65 -1.075
   84. 669 0.5 911.65 -1.075; 670 0.75 911.65 -1.075; 671 1 911.65 -1.075
   85. 673 -1 911.65 -1.075; 674 -1.25 911.65 -1.075; 675 -1.5 911.65 -1.075
   86. 677 -0.75 911.65 -1.32562; 678 -0.5 911.65 -1.32562; 679 -0.5 911.65 -1.575
   87. 680 -0.75 911.65 -1.575; 681 -0.25 911.65 -1.32562; 682 -0.25 911.65 -1.575
   88. 683 0 911.65 -1.32562; 684 0 911.65 -1.575; 685 0.25 911.65 -1.32562
   89. 686 0.25 911.65 -1.575; 687 0.5 911.65 -1.32562; 688 0.5 911.65 -1.575
   90. 689 0.75 911.65 -1.32562; 690 0.75 911.65 -1.575; 691 1 911.65 -1.32562
   91. 692 1 911.65 -1.575; 695 -1.25 911.65 -1.32562; 696 -1 911.65 -1.32562
   92. 697 -1 911.65 -1.575; 698 -1.25 911.65 -1.575; 699 -0.5 911.65 -1.825
   93. 700 -0.75 911.65 -1.825; 701 -0.25 911.65 -1.825; 702 0 911.65 -1.825
   94. 703 0.25 911.65 -1.825; 704 0.5 911.65 -1.825; 705 0.75 911.65 -1.825
```

95. 706 1 911.65 -1.825; 708 -1 911.65 -1.825; 709 -1.25 911.65 -1.825
96. 710 -1.5 911.65 -1.32562; 711 -1.5 911.65 -1.575; 712 -1.5 911.65 -1.825
97. 716 -0.5 911.65 -2.075; 717 -0.75 911.65 -2.075; 718 -0.25 911.65 -2.075
98. 719 0 911.65 -2.075; 720 0.25 911.65 -2.075; 721 0.5 911.65 -2.075
99. 722 0.75 911.65 -2.075; 723 1 911.65 -2.075; 725 -1 911.65 -2.075
100. 726 -1.25 911.65 -2.075; 727 -1.5 911.65 -2.075; 729 -0.75 911.65 -2.32562
101. 730 -0.5 911.65 -2.32562; 731 -0.5 911.65 -2.575; 732 -0.75 911.65 -2.575
102. 733 -0.25 911.65 -2.32562; 734 -0.25 911.65 -2.575; 735 0 911.65 -2.32562
103. 736 0 911.65 -2.575; 737 0.25 911.65 -2.32562; 738 0.25 911.65 -2.575
104. 739 0.5 911.65 -2.32562; 740 0.5 911.65 -2.575; 741 0.75 911.65 -2.32562
105. 742 0.75 911.65 -2.575; 743 1 911.65 -2.32562; 744 1 911.65 -2.575
106. 747 -1.25 911.65 -2.32562; 748 -1 911.65 -2.32562; 749 -1 911.65 -2.575
107. 750 -1.25 911.65 -2.575; 751 -0.5 911.65 -2.825; 752 -0.75 911.65 -2.825
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297. MEMBER INCIDENCES
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STAAD SPACE -- PAGE NO. 17

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STAAD SPACE -- PAGE NO. 18

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1137	0.062500
1138	0.062500
1139	0.062423
1140	0.062500
1141	0.062578
1142	0.062578
1143	0.062500
1144	0.062423
1145	0.081250
1146	0.062500
1147	0.062423
1148	0.081250
1149	0.050000
1150	0.050000
1153	0.049922
1154	0.081250
1156	0.062422
1157	0.062500
1158	0.062500
1159	0.062500
1160	0.062577
1161	0.062500
1162	0.062422
1163	0.062500
1164	0.062577
1165	0.062500
1166	0.062422
1167	0.062500
1168	0.062577
1169	0.062500
1170	0.062422
1171	0.062500

1172	0.062577
1173	0.062500
1174	0.062422
1175	0.062500
1176	0.062577
1177	0.062500
1178	0.062422
1179	0.081250
1180	0.050000
1184	0.049922
1185	0.081250
1187	0.062423
1188	0.062500
1189	0.062500
1190	0.062500
1191	0.062578
1192	0.062500
1193	0.062423
1194	0.062500
1195	0.062578
1196	0.062500
1197	0.062423
1198	0.062500
1199	0.062578
1200	0.062500
1201	0.062423
1202	0.062500
1203	0.062578
1204	0.062500
1205	0.062423
1206	0.062500
1207	0.062578
1208	0.062500
1209	0.062423
1210	0.081250
1211	0.050000
1213	0.049922
1214	0.081250
1216	0.062423
1217	0.062500
1218	0.062500
1219	0.062500
1220	0.062578
1221	0.062500
1222	0.062423
1223	0.062500
1235	0.062578
1237	0.062500
1238	0.062423
1239	0.062500
1243	0.062578
1245	0.062500
1246	0.062423
1247	0.062500
1251	0.062578
1253	0.062500

1254	0.062423
1255	0.062500
1259	0.062578
1261	0.062500
1262	0.062423
1263	0.081250
1267	0.050000
1300	0.049922
1301	0.081250
1302	0.049922
1303	0.081250
1309	0.062422
1310	0.062500
1311	0.062422
1312	0.062500
1315	0.062500
1316	0.062500
1318	0.062500
1319	0.062500
1321	0.062578
1322	0.062577
1332	0.062500
1333	0.062422
1334	0.062500
1335	0.062422
1340	0.062500
1341	0.062500
1344	0.062577
1345	0.062577
1348	0.062500
1349	0.062422
1350	0.062500
1351	0.062422
1356	0.062500
1357	0.062500
1360	0.062577
1361	0.062577
1364	0.062500
1365	0.062423
1366	0.062500
1367	0.062422
1372	0.062500
1373	0.062500
1376	0.062578
1377	0.062577
1380	0.062500
1381	0.062422
1382	0.062500
1383	0.062422
1388	0.062500
1389	0.062500
1392	0.062578
1393	0.062577
1396	0.062500
1397	0.062422
1398	0.062500

1399	0.062422
1404	0.081250
1405	0.081250
1408	0.050000
1409	0.050000
1414	0.049922
1415	0.081250
1417	0.062422
1418	0.062500
1419	0.062500
1420	0.062500
1421	0.062578
1422	0.062500
1423	0.062422
1424	0.062500
1425	0.062577
1426	0.062500
1427	0.062422
1428	0.062500
1429	0.062577
1430	0.062500
1431	0.062423
1432	0.062500
1433	0.062578
1434	0.062500
1435	0.062422
1436	0.062500
1437	0.062578
1438	0.062500
1439	0.062422
1440	0.081250
1441	0.050000
1443	0.049922
1444	0.081250
1446	0.062422
1447	0.062500
1448	0.062500
1449	0.062500
1450	0.062577
1451	0.062500
1452	0.062422
1453	0.062500
1454	0.062577
1455	0.062500
1456	0.062422
1457	0.062500
1458	0.062577
1459	0.062500
1460	0.062422
1461	0.062500
1462	0.062577
1463	0.062500
1464	0.062422
1465	0.062500
1466	0.062577
1467	0.062500

1468	0.062422
1469	0.081250
1470	0.050000
1472	0.024961
1473	0.040625
1475	0.031211
1476	0.031250
1477	0.031250
1478	0.031250
1479	0.031289
1480	0.031250
1481	0.031211
1482	0.031250
1483	0.031289
1484	0.031250
1485	0.031211
1486	0.031250
1487	0.031289
1488	0.031250
1489	0.031211
1490	0.031250
1491	0.031289
1492	0.031250
1493	0.031211
1494	0.031250
1495	0.031289
1496	0.031250
1497	0.031211
1498	0.040625
1499	0.025000
1501	0.024961
1502	0.040625
1504	0.031211
1505	0.031250
1506	0.031250
1507	0.031250
1508	0.031289
1509	0.031250
1510	0.031211
1511	0.031250
1512	0.031289
1513	0.031250
1514	0.031211
1515	0.031250
1516	0.031289
1517	0.031250
1518	0.031211
1519	0.031250
1520	0.031289
1521	0.031250
1522	0.031211
1523	0.031250
1524	0.031289
1525	0.031250
1526	0.031211
1527	0.040625

1528	0.025000
TOTAL	50.315361

```
653. ********************************************************************************
654. LOAD 1 LOADTYPE NONE  TITLE DEAD LOAD ( DL )
655. SELFWEIGHT Y -1
656. ELEMENT LOAD
657. * LOAD DUE TO BACKFILL
658. 326 TO 328 332 TO 334 383 TO 388 393 TO 398 403 TO 408 413 TO 418 423 TO 429 -
659. 439 441 443 445 447 449 450 459 TO 481 483 485 TO 488 509 TO 515 517 518 -
660. 529 534 TO 540 543 561 TO 567 569 TO 571 573 TO 585 587 589 TO 599 -
661. 601 TO 605 609 TO 615 617 TO 619 621 TO 633 635 637 TO 647 649 TO 653 657 -
662. 658 TO 663 665 TO 667 669 TO 681 683 685 TO 695 697 TO 701 705 TO 711 713 -
663. 714 TO 715 717 TO 729 731 733 TO 743 745 TO 749 753 TO 759 761 TO 763 814 -
664. 815 TO 816 820 TO 836 839 TO 864 866 TO 975 982 984 TO 992 994 TO 1035 1038 -
665. 1040 TO 1064 1066 1068 TO 1076 1078 TO 1119 1122 1124 TO 1148 1152 -
666. 1154 TO 1178 1180 1182 TO 1189 1200 1202 TO 1204 1208 1210 TO 1212 1216 1218 -
667. 1219 TO 1220 1224 1226 TO 1228 1232 1264 1265 1270 1271 1273 1274 1276 1277 -
668. 1279 1280 1282 1283 1292 1293 1295 1296 1300 1301 1304 1305 1308 1309 1311 -
669. 1312 1316 1317 1320 1321 1324 1325 1327 1328 1332 1333 1336 1337 1340 1341 -
670. 1343 1344 1348 1349 1352 1353 1356 1357 1359 1360 1364 1365 1368 1369 1374 -
671. 1376 TO 1400 1402 1404 TO 1428 1430 1432 TO 1456 1458 1460 TO 1483 -
672. 1484 PR GY -15
673. * LOAD DUE TO SURCHARGE
674. 326 TO 328 332 TO 334 383 TO 388 393 TO 398 403 TO 408 413 TO 418 423 TO 429 -
675. 439 441 443 445 447 449 450 459 TO 481 483 485 TO 488 509 TO 515 517 518 -
676. 529 534 TO 540 543 561 TO 567 569 TO 571 573 TO 585 587 589 TO 599 -
677. 601 TO 605 609 TO 615 617 TO 619 621 TO 633 635 637 TO 647 649 TO 653 657 -
678. 658 TO 663 665 TO 667 669 TO 681 683 685 TO 695 697 TO 701 705 TO 711 713 -
679. 714 TO 715 717 TO 729 731 733 TO 743 745 TO 749 753 TO 759 761 TO 763 814 -
680. 815 TO 816 820 TO 836 839 TO 864 866 TO 975 982 984 TO 992 994 TO 1035 1038 -
681. 1040 TO 1064 1066 1068 TO 1076 1078 TO 1119 1122 1124 TO 1148 1152 -
682. 1154 TO 1178 1180 1182 TO 1189 1200 1202 TO 1204 1208 1210 TO 1212 1216 1218 -
683. 1219 TO 1220 1224 1226 TO 1228 1232 1264 1265 1270 1271 1273 1274 1276 1277 -
684. 1279 1280 1282 1283 1292 1293 1295 1296 1300 1301 1304 1305 1308 1309 1311 -
685. 1312 1316 1317 1320 1321 1324 1325 1327 1328 1332 1333 1336 1337 1340 1341 -
686. 1343 1344 1348 1349 1352 1353 1356 1357 1359 1360 1364 1365 1368 1369 1374 -
687. 1376 TO 1400 1402 1404 TO 1428 1430 1432 TO 1456 1458 1460 TO 1483 -
688. 1484 PR GY -20
689. JOINT LOAD
690. 128 FX -1.6 FY -123.27 FZ 1.2
691. 833 FX 2.465 FY -137.481 FZ -0.619
692. 1183 FX 1.67 FY -17.22 FZ -0.114
693. 1182 FX -2.386 FY -59.143 FZ 0.113
694. ********************************************************************************
695. LOAD 2 LOADTYPE NONE  TITLE LIVE LOAD ( LL )
696. JOINT LOAD
697. 128 FX -8.421 FY -421.455 FZ -12.193
698. 833 FX 8.864 FY -463.674 FZ -2.402
699. 1183 FX 8.8 FY 43.569 FZ -0.224
700. 1182 FX -8.477 FY -143.572 FZ 0.219
701. ********************************************************************************
702. LOAD 3 LOADTYPE NONE  TITLE WIND LOAD +X  ( WLX1 )
703. JOINT LOAD
704. 128 FX 98 FY -618.25 FZ 15
705. 833 FX 84.846 FY -505.5 FZ -0.56
706. 1183 FX 84.894 FY 531.024 FZ 0.715
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707. 1182 FX 98.08 FY 592.731 FZ -0.715
708. ********************************************************************************
709. LOAD 4 LOADTYPE NONE  TITLE WIND LOAD -X  ( WLX2 )
710. JOINT LOAD
711. 128 FX -98 FY 618.25 FZ -15
712. 833 FX -84.846 FY 505.5 FZ 0.56
713. 1183 FX -84.894 FY -531.024 FZ -0.715
714. 1182 FX -98.08 FY -592.731 FZ 0.715
715. ********************************************************************************
716. LOAD 5 LOADTYPE NONE  TITLE WIND LOAD +Z  ( WLZ1 )
717. JOINT LOAD
718. 128 FX 4.5 FY -166 FZ 71.5
719. 833 FX -0.05 FY 125.6 FZ 2.1
720. 1183 FX -0.018 FY 49.921 FZ 1.424
721. 1182 FX 4.431 FY -9.5 FZ 30.435
722. ********************************************************************************
723. LOAD 6 LOADTYPE NONE  TITLE WIND LOAD -Z  ( WLZ2 )
724. JOINT LOAD
725. 128 FX -4.4 FY 164 FZ -70.6
726. 833 FX 0.03 FY -123.5 FZ -2.1
727. 1183 FX 0.001 FY -50.068 FZ -1.424
728. 1182 FX -4.431 FY 9.65 FZ -30.435
729. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**841**	**NUMBER OF MEMBERS**	**4**
NUMBER OF PLATES	**780**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**837**

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 640/ 45/ 141 DOF
TOTAL PRIMARY LOAD CASES = 6, TOTAL DEGREES OF FREEDOM = 2535
TOTAL LOAD COMBINATION CASES = 0 SO FAR.
SIZE OF STIFFNESS MATRIX = 358 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 20.1/ 142870.2 MB

```
730. CHANGE
731. SUPPORTS
732. 326 TO 328 332 TO 334 383 TO 388 393 TO 398 403 TO 408 413 TO 418 423 TO 429 -
733. 439 441 443 445 447 449 450 459 TO 481 483 485 TO 488 509 TO 515 517 518 -
734. 529 534 TO 540 543 561 TO 567 569 TO 571 573 TO 585 587 589 TO 599 -
735. 601 TO 605 609 TO 615 617 TO 619 621 TO 633 635 637 TO 647 649 TO 653 657 -
736. 658 TO 663 665 TO 667 669 TO 681 683 685 TO 695 697 TO 701 705 TO 711 713 -
737. 714 TO 715 717 TO 729 731 733 TO 743 745 TO 749 753 TO 759 761 TO 763 814 -
738. 815 TO 816 820 TO 836 839 TO 864 866 TO 975 982 984 TO 992 994 TO 1035 1038 -
```



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739. 1040 TO 1064 1066 1068 TO 1076 1078 TO 1119 1122 1124 TO 1148 1152 -
740. 1154 TO 1178 1180 1182 TO 1189 1200 1202 TO 1204 1208 1210 TO 1212 1216 1218 -
741. 1219 TO 1220 1224 1226 TO 1228 1232 1264 1265 1270 1271 1273 1274 1276 1277 -
742. 1279 1280 1282 1283 1292 1293 1295 1296 1300 1301 1304 1305 1308 1309 1311 -
743. 1312 1316 1317 1320 1321 1324 1325 1327 1328 1332 1333 1336 1337 1340 1341 -
744. 1343 1344 1348 1349 1352 1353 1356 1357 1359 1360 1364 1365 1368 1369 1374 -
745. 1376 TO 1400 1402 1404 TO 1428 1430 1432 TO 1456 1458 1460 TO 1483 -
746. 1484 PLATE MAT DIRECT Y SUBGRADE 57600 COMPRESSION
747. ********************************************************************************
748. ***FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
749. LOAD 101 1.4DL
750. REPEAT LOAD
751. 1 1.4
752. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  101

753. CHANGE
754. LOAD 102 1.4DL + 1.7LL
755. REPEAT LOAD
756. 1 1.4 2 1.7
757. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  102

758. CHANGE
759. LOAD 103 1.2DL + 1.6WLX1 + 1.0 LL
760. REPEAT LOAD
761. 1 1.2 3 1.6 2 1.0
762. PERFORM ANALYSIS
**START ITERATION NO.           2
**START ITERATION NO.           3
**START ITERATION NO.           4
**START ITERATION NO.           5

**NOTE-Tension/Compression converged after  5 iterations, Case=  103

763. CHANGE
764. LOAD 104 1.2DL + 1.6WLX2 + 1.0 LL
765. REPEAT LOAD
766. 1 1.2 4 1.6 2 1.0
```

```
 767. PERFORM ANALYSIS
**START ITERATION NO.           2
**START ITERATION NO.           3

**NOTE-Tension/Compression converged after  3 iterations, Case=  104



 768. CHANGE
 769. LOAD 105 1.2DL + 1.6WLZ1 + 1.0 LL
 770. REPEAT LOAD
 771. 1 1.2 5 1.6 2 1.0
 772. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  105



 773. CHANGE
 774. LOAD 106 1.2DL + 1.6WLZ2 + 1.0 LL
 775. REPEAT LOAD
 776. 1 1.2 6 1.6 2 1.0
 777. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  106



 778. CHANGE
 779. LOAD 107 0.9DL + 1.6WLX1
 780. REPEAT LOAD
 781. 1 0.9 3 1.6
 782. PERFORM ANALYSIS
**START ITERATION NO.           2
**START ITERATION NO.           3
**START ITERATION NO.           4
**START ITERATION NO.           5
**START ITERATION NO.           6

**NOTE-Tension/Compression converged after  6 iterations, Case=  107



 783. CHANGE
```



```
 784. LOAD 108 0.9DL + 1.6WLX2
 785. REPEAT LOAD
 786. 1 0.9 4 1.6
 787. PERFORM ANALYSIS
**START ITERATION NO.           2
**START ITERATION NO.           3
**START ITERATION NO.           4
**START ITERATION NO.           5
**START ITERATION NO.           6
**START ITERATION NO.           7

**NOTE-Tension/Compression converged after  7 iterations, Case=  108



 788. CHANGE
 789. LOAD 109 0.9DL + 1.6WLZ1
 790. REPEAT LOAD
 791. 1 0.9 5 1.6
 792. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  109



 793. CHANGE
 794. LOAD 110 0.9DL + 1.6WLZ2
 795. REPEAT LOAD
 796. 1 0.9 6 1.6
 797. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  110



 798. CHANGE
 799. ********************************************************************************
 800. * UNFACTORED LOAD COMBINATIONS FOR STABILITY CHECK
 801. LOAD 201 1.0DL
 802. REPEAT LOAD
 803. 1 1.0
 804. PERFORM ANALYSIS
```

**NOTE-Tension/Compression converged after 1 iterations, Case= 201

805. CHANGE
806. LOAD 202 1.0DL + 1.0LL
807. REPEAT LOAD
808. 1 1.0 2 1.0
809. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 202

810. CHANGE
811. LOAD 203 1.0DL + 1.0WLX1
812. REPEAT LOAD
813. 1 1.0 3 1.0
814. PERFORM ANALYSIS
**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5

**NOTE-Tension/Compression converged after 5 iterations, Case= 203

815. CHANGE
816. LOAD 204 1.0DL + 1.0WLX2
817. REPEAT LOAD
818. 1 1.0 4 1.0
819. PERFORM ANALYSIS
**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 204

820. CHANGE
821. LOAD 205 1.0DL + 1.0WLZ1
822. REPEAT LOAD
823. 1 1.0 5 1.0
824. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 205

825. CHANGE
826. LOAD 206 1.0DL + 1.0WLZ2
827. REPEAT LOAD
828. 1 1.0 6 1.0
829. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 206

830. CHANGE
831. LOAD 207 1.0DL + 0.75WLX1 + 0.75LL
832. REPEAT LOAD
833. 1 1.0 3 0.75 2 0.75
834. PERFORM ANALYSIS
**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 207

835. CHANGE
836. LOAD 208 1.0DL + 0.75WLX2 + 0.75LL
837. REPEAT LOAD
838. 1 1.0 4 0.75 2 0.75
839. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 208

840. CHANGE
841. LOAD 209 1.0DL + 0.75WLZ1 + 0.75LL
842. REPEAT LOAD
843. 1 1.0 5 0.75 2 0.75
844. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 209

845. CHANGE
846. LOAD 210 1.0DL + 0.75WLZ2 + 0.75LL
847. REPEAT LOAD
848. 1 1.0 6 0.75 2 0.75
849. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 210

850. CHANGE
851. LOAD 211 0.6DL + 1.0WLX1
852. REPEAT LOAD
853. 1 0.6 3 1.0
854. PERFORM ANALYSIS
**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5
**START ITERATION NO. 6

**NOTE-Tension/Compression converged after 6 iterations, Case= 211

855. CHANGE
856. LOAD 212 0.6DL + 1.0WLX2
857. REPEAT LOAD
858. 1 0.6 4 1.0
859. PERFORM ANALYSIS
**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5
**START ITERATION NO. 6

**NOTE-Tension/Compression converged after 6 iterations, Case= 212

860. CHANGE
861. LOAD 213 0.6DL + 1.0WLZ1
862. REPEAT LOAD
863. 1 0.6 5 1.0
864. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 213

865. CHANGE
866. LOAD 214 0.6DL + 1.0WLZ2
867. REPEAT LOAD
868. 1 0.6 6 1.0
869. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 214

870. CHANGE
871. **
872. PERFORM ANALYSIS

** ALL CASES BEING MADE ACTIVE BEFORE RE-ANALYSIS. **

**NOTE-Tension/Compression converged after 1 iterations, Case= 1

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5

**NOTE-Tension/Compression converged after 5 iterations, Case= 2

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5
**START ITERATION NO. 6
**START ITERATION NO. 7

***WARNING - INSTABILITY AT JOINT 1472 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 3.8779284E+06 L-MATRIX DIAG= 5.1627401E-05 EQN NO 2535
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 8

***WARNING - INSTABILITY AT JOINT 1472 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 3.8779284E+06 L-MATRIX DIAG= 5.1630661E-05 EQN NO 2535
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 9

***WARNING - INSTABILITY AT JOINT 1472 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 3.8779284E+06 L-MATRIX DIAG= 5.1623676E-05 EQN NO 2535
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 10

***WARNING - INSTABILITY AT JOINT 1472 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 3.8779284E+06 L-MATRIX DIAG= 5.1626004E-05 EQN NO 2535
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 11

***WARNING - INSTABILITY AT JOINT 1472 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.9252917E+06 L-MATRIX DIAG= 2.0463951E-05 EQN NO 2534
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 1472 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 3.8779284E+06 L-MATRIX DIAG= 5.1624142E-05 EQN NO 2535
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

**NOTE-Tension/Compression converged after 11 iterations, Case= 3

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5
**START ITERATION NO. 6
**START ITERATION NO. 7

***WARNING - INSTABILITY AT JOINT 1472 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 3.8779284E+06 L-MATRIX DIAG= -4.8793852E-05 EQN NO 2535
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 8

***WARNING - INSTABILITY AT JOINT 1472 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 3.8779284E+06 L-MATRIX DIAG= -4.3074135E-05 EQN NO 2535
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 9

***WARNING - INSTABILITY AT JOINT 1472 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 3.8779284E+06 L-MATRIX DIAG= -4.6210364E-05 EQN NO 2535
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 10

***WARNING - INSTABILITY AT JOINT 1472 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 3.8779284E+06 L-MATRIX DIAG= -2.6977155E-05 EQN NO 2535
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 11

***WARNING - INSTABILITY AT JOINT 1472 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.9252917E+06 L-MATRIX DIAG= 5.1786192E-06 EQN NO 2534
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 1472 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 3.8779284E+06 L-MATRIX DIAG= -3.7306454E-05 EQN NO 2535
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
**START ITERATION NO. 12

***WARNING - INSTABILITY AT JOINT 1472 DIRECTION = FY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.6079315E+04 L-MATRIX DIAG= 1.4188117E-10 EQN NO 2533
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 1472 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.9252917E+06 L-MATRIX DIAG= 2.0463951E-05 EQN NO 2534
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 1472 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 3.8779284E+06 L-MATRIX DIAG= 5.1624142E-05 EQN NO 2535
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
**START ITERATION NO. 13
**START ITERATION NO. 14
**START ITERATION NO. 15

**WARNING-Tension/Compression did not converge after 15 iterations, Case= 4
** If results are not satisfactory then enter a
** SET ITERLIM i command (i>10) before loads to get more iterations.

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5
**START ITERATION NO. 6

***WARNING - INSTABILITY AT JOINT 1472 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.9252917E+06 L-MATRIX DIAG= 1.7458107E-05 EQN NO 2534
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 7

***WARNING - INSTABILITY AT JOINT 1472 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.9252917E+06 L-MATRIX DIAG= 1.6050413E-05 EQN NO 2534
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 8

***WARNING - INSTABILITY AT JOINT 1472 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.9252917E+06 L-MATRIX DIAG= 2.0375941E-05 EQN NO 2534
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 9

***WARNING - INSTABILITY AT JOINT 1472 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.9252917E+06 L-MATRIX DIAG= 1.9145198E-05 EQN NO 2534
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 10

***WARNING - INSTABILITY AT JOINT 1472 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.9252917E+06 L-MATRIX DIAG= 1.9851141E-05 EQN NO 2534
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 11

***WARNING - INSTABILITY AT JOINT 1472 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.9252917E+06 L-MATRIX DIAG= 2.0463951E-05 EQN NO 2534
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 1472 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 3.8779284E+06 L-MATRIX DIAG= 5.1624142E-05 EQN NO 2535
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
**START ITERATION NO. 12

***WARNING - INSTABILITY AT JOINT 1472 DIRECTION = FY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.6079315E+04 L-MATRIX DIAG= 1.4188117E-10 EQN NO 2533
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 1472 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.9252917E+06 L-MATRIX DIAG= 2.0463951E-05 EQN NO 2534
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 1472 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 3.8779284E+06 L-MATRIX DIAG= 5.1624142E-05 EQN NO 2535
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

**NOTE-Tension/Compression converged after 12 iterations, Case= 5

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5
**START ITERATION NO. 6

***WARNING - INSTABILITY AT JOINT 1472 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.9252917E+06 L-MATRIX DIAG= -1.4159363E-05 EQN NO 2534
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 7

***WARNING - INSTABILITY AT JOINT 1472 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.9252917E+06 L-MATRIX DIAG= -1.5430152E-05 EQN NO 2534
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 8

***WARNING - INSTABILITY AT JOINT 1472 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.9252917E+06 L-MATRIX DIAG= 1.3573095E-05 EQN NO 2534
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 9

***WARNING - INSTABILITY AT JOINT 1472 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.9252917E+06 L-MATRIX DIAG= -2.8880313E-06 EQN NO 2534
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 10

***WARNING - INSTABILITY AT JOINT 1472 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.9252917E+06 L-MATRIX DIAG= -5.7155266E-06 EQN NO 2534
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 11

***WARNING - INSTABILITY AT JOINT 1472 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.9252917E+06 L-MATRIX DIAG= 5.1786192E-06 EQN NO 2534
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 1472 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 3.8779284E+06 L-MATRIX DIAG= -3.7306454E-05 EQN NO 2535
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
**START ITERATION NO. 12

***WARNING - INSTABILITY AT JOINT 1472 DIRECTION = FY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.6079315E+04 L-MATRIX DIAG= 1.4188117E-10 EQN NO 2533
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 1472 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 2.9252917E+06 L-MATRIX DIAG= 2.0463951E-05 EQN NO 2534
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 1472 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 3.8779284E+06 L-MATRIX DIAG= 5.1624142E-05 EQN NO 2535

***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
**START ITERATION NO. 13
**START ITERATION NO. 14
**START ITERATION NO. 15

**WARNING-Tension/Compression did not converge after 15 iterations, Case= 6
** If results are not satisfactory then enter a
** SET ITERLIM i command (i>10) before loads to get more iterations.

**NOTE-Tension/Compression converged after 1 iterations, Case= 101

**NOTE-Tension/Compression converged after 1 iterations, Case= 102

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5

**NOTE-Tension/Compression converged after 5 iterations, Case= 103

**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 104

**NOTE-Tension/Compression converged after 1 iterations, Case= 105

**NOTE-Tension/Compression converged after 1 iterations, Case= 106

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5
**START ITERATION NO. 6

**NOTE-Tension/Compression converged after 6 iterations, Case= 107

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5
**START ITERATION NO. 6
**START ITERATION NO. 7

**NOTE-Tension/Compression converged after 7 iterations, Case= 108

**NOTE-Tension/Compression converged after 1 iterations, Case= 109

**NOTE-Tension/Compression converged after 1 iterations, Case= 110

**NOTE-Tension/Compression converged after 1 iterations, Case= 201

**NOTE-Tension/Compression converged after 1 iterations, Case= 202

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5

**NOTE-Tension/Compression converged after 5 iterations, Case= 203

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4

**NOTE-Tension/Compression converged after 4 iterations, Case= 204

**NOTE-Tension/Compression converged after 1 iterations, Case= 205

**NOTE-Tension/Compression converged after 1 iterations, Case= 206

**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 207

**NOTE-Tension/Compression converged after 1 iterations, Case= 208

**NOTE-Tension/Compression converged after 1 iterations, Case= 209

**NOTE-Tension/Compression converged after 1 iterations, Case= 210

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5
**START ITERATION NO. 6

**NOTE-Tension/Compression converged after 6 iterations, Case= 211

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5
**START ITERATION NO. 6

**NOTE-Tension/Compression converged after 6 iterations, Case= 212

**NOTE-Tension/Compression converged after 1 iterations, Case= 213

**NOTE-Tension/Compression converged after 1 iterations, Case= 214

873. LOAD LIST 101 TO 110
874. START CONCRETE DESIGN

```
875. CODE ACI
876. TRACK 2 ALL
877. FC 35000 MEMB 185 813 1150 1151
878. CLB 0.05 MEMB 185 813 1150 1151
879. CLS 0.05 ALL
880. CLT 0.05 MEMB 185 326 TO 328 332 TO 334 383 TO 388 393 TO 398 403 TO 408 413 -
881. 414 TO 418 423 TO 428 813 TO 816 820 TO 834 1150 1151 1264 1265 1270 1271 -
882. 1273 1274 1276 1277 1279 1280 1282 1283 1374 1376 TO 1380 1430 1432 TO 1436
883. MAXMAIN 25 ALL
884. MINMAIN 25 ALL
885. MINSEC 10 ALL
886. DESIGN COLUMN 185 813 1150 1151
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COLUMN NO. 185 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 35.0 MPA, RECT SIZE - 700.0 X 500.0 MMS, TIED
AREA OF STEEL REQUIRED = 6706.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
16 - 25 MM	2.244	108	STA	0.900

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 400.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
13427.90	10742.32	4222.66	1035.06	245.1
M0	P-tens.	Des.Pn	Des.Mn	e/h
638.77	-3249.05	-958.78	342.84	-0.18338

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
13427.90	10742.32	4464.79	1515.92	339.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
945.41	-3249.05	-958.78	49.66	-0.02656



COLUMN NO. 813 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 35.0 MPA, RECT SIZE - 700.0 X 500.0 MMS, TIED
AREA OF STEEL REQUIRED = 5068.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 25 MM	1.683	108	STA	0.900

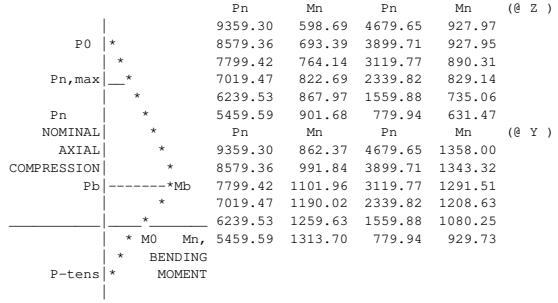
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 400.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12674.05	10139.24	4246.17	940.35	221.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
497.33	-2436.79	-744.12	289.32	-0.19939

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12674.05	10139.24	4465.99	1368.92	306.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
726.79	-2436.79	-744.12	0.73	-0.00051



COLUMN NO. 1150 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 35.0 MPA, RECT SIZE - 700.0 X 500.0 MMS, TIED
AREA OF STEEL REQUIRED = 6433.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
16 - 25 MM	2.244	107	STA	0.900

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 400.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
13427.90	10742.32	4222.66	1035.06	245.1
M0	P-tens.	Des.Pn	Des.Mn	e/h
638.77	-3249.05	-977.54	335.35	-0.17593

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
13427.90	10742.32	4464.79	1515.92	339.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
945.41	-3249.05	-977.54	2.26	-0.00118

==========

COLUMN NO. 1151 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 35.0 MPA, RECT SIZE - 700.0 X 500.0 MMS, TIED
AREA OF STEEL REQUIRED = 6160.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
16 - 25 MM	2.244	103	STA	0.900

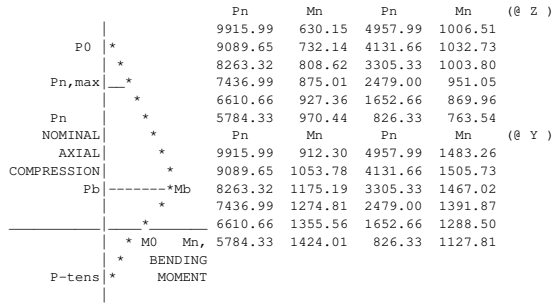
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 400.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
13427.90	10742.32	4222.66	1035.06	245.1
M0	**P-tens.**	**Des.Pn**	**Des.Mn**	**e/h**
638.77	-3249.05	-946.74	317.70	-0.17209

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
13427.90	10742.32	4464.79	1515.92	339.5
M0	**P-tens.**	**Des.Pn**	**Des.Mn**	**e/h**
945.41	-3249.05	-946.74	1.70	-0.00092



	Pn	Mn	Pn	Mn	
	9915.99	630.15	4957.99	1006.51	
P0	9089.65	732.14	4131.66	1032.73	
	8263.32	808.62	3305.33	1003.80	
Pn,max	7436.99	875.01	2479.00	951.05	
	6610.66	927.36	1652.66	869.96	
Pn	5784.33	970.44	826.33	763.54	
NOMINAL	Pn	Mn	Pn	Mn	(@ Y)
AXIAL	9915.99	912.30	4957.99	1483.26	
COMPRESSION	9089.65	1053.78	4131.66	1505.73	
Pb	8263.32	1175.19	3305.33	1467.02	
	7436.99	1274.81	2479.00	1391.87	
	6610.66	1355.56	1652.66	1288.50	
M0 Mn,	5784.33	1424.01	826.33	1127.81	
BENDING					
P-tens MOMENT					

**********END OF COLUMN DESIGN RESULTS**********

887. END CONCRETE DESIGN

888. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= OCT 16,2020 TIME= 10:34: 5 ****

****
* For technical assistance on STAAD.Pro, please visit *
* http://selectservices.bentley.com/en-US/ *
* *
* Details about additional assistance from *
* Bentley and Partners can be found at program menu *
* Help->Technical Support *
* *
* Copyright (c) 1997-2014 Bentley Systems, Inc. *
* http://www.bentley.com *
****

```
            *****************************************************
            *                                                   *
            *           STAAD.Pro V8i SELECTseries5             *
            *           Version  20.07.10.64                    *
            *           Proprietary Program of                  *
            *           Bentley Systems, Inc.                   *
            *           Date=    OCT  7, 2020                   *
            *           Time=    17:10:24                       *
            *                                                   *
            *      USER ID: LNT ECC DIV                         *
            *****************************************************
```

```
    1. STAAD SPACE
INPUT FILE: F1_STABILITY CHECK.STD
    2. START JOB INFORMATION
    3. ENGINEER DATE 08/05/2020
    4. JOB NAME FDN. MKD. F1_STABILITY CHECK
    5. JOB CLIENT MA'ADEN
    6. JOB NO O19045
    7. JOB REV 0
    8. ENGINEER NAME SOR
    9. END JOB INFORMATION
   10. INPUT WIDTH 79
   11. UNIT METER KN
   12. JOINT COORDINATES
   13. 2 0 911.65 3.175; 128 0 913.6 3.175; 522 0 911.65 -0.12462; 528 0 913.6 -0.125
   14. MEMBER INCIDENCES
   15. 185 2 128; 529 522 528
   16. DEFINE MATERIAL START
   17. ISOTROPIC CONCRETE
   18. E 2.17185E+007
   19. POISSON 0.17
   20. DENSITY 25
   21. ALPHA 1E-005
   22. DAMP 0.05
   23. TYPE CONCRETE
   24. STRENGTH FCU 27579
   25. END DEFINE MATERIAL
   26. MEMBER PROPERTY AMERICAN
   27. 185 529 PRIS YD 0.4 ZD 0.6
   28. CONSTANTS
   29. MATERIAL CONCRETE ALL
   30. SUPPORTS
   31. 2 522 FIXED
   32. ******************************************************************************
   33. LOAD 1 LOADTYPE NONE  TITLE DEAD LOAD ( DL )
  **WARNING- THIS STRUCTURE IS DISJOINTED. IGNORE IF
            MASTER/SLAVE OR IF UNCONNECTED JOINTS.
   34. SELFWEIGHT Y -1
   35. JOINT LOAD
   36. 128 FX 0.002 FY -152 FZ 0.011
```



```
   37. 528 FX 0.007 FY -115.459 FZ 1.384
   38. ******************************************************************************
   39. LOAD 2 LOADTYPE NONE  TITLE LIVE LOAD ( LL )
   40. JOINT LOAD
   41. 128 FX 0.017 FY -173.225 FZ 2.323
   42. 528 FX 0.013 FY -84.387 FZ 1.87
   43. ******************************************************************************
   44. LOAD 3 LOADTYPE NONE  TITLE WIND LOAD +X  ( WLX1 )
   45. JOINT LOAD
   46. 128 FX 2.339 FY 4.483 FZ 0.853
   47. 528 FX 2.36 FY -3.789 FZ 0.76
   48. ******************************************************************************
   49. LOAD 4 LOADTYPE NONE  TITLE WIND LOAD -X  ( WLX2 )
   50. JOINT LOAD
   51. 128 FX 2.339 FY -4.483 FZ 0.853
   52. 528 FX 2.36 FY 3.789 FZ 0.76
   53. ******************************************************************************
   54. LOAD 5 LOADTYPE NONE  TITLE WIND LOAD +Z  ( WLZ1 )
   55. JOINT LOAD
   56. 128 FX 0.001 FY 202.574 FZ 51.359
   57. 528 FX 0.05 FY -202.574 FZ 49.368
   58. ******************************************************************************
   59. LOAD 6 LOADTYPE NONE  TITLE WIND LOAD -Z  ( WLZ2 )
   60. JOINT LOAD
   61. 128 FX 0.001 FY -202.395 FZ 51.324
   62. 528 FX 0.05 FY 202.395 FZ 49.368
   63. ******************************************************************************
   64. ******************************************************************************
   65. * UNFACTORED LOAD COMBINATIONS FOR STABILITY CHECK
   66. LOAD COMB 201 1.0DL
   67. 1 1.0
   68. LOAD COMB 202 1.0DL + 1.0LL
   69. 1 1.0 2 1.0
   70. LOAD COMB 203 1.0DL + 1.0WLX1
   71. 1 1.0 3 1.0
   72. LOAD COMB 204 1.0DL + 1.0WLX2
   73. 1 1.0 4 1.0
   74. LOAD COMB 205 1.0DL + 1.0WLZ1
   75. 1 1.0 5 1.0
   76. LOAD COMB 206 1.0DL + 1.0WLZ2
   77. 1 1.0 6 1.0
   78. LOAD COMB 207 1.0DL + 0.75WLX1 + 0.75LL
   79. 1 1.0 3 0.75 2 0.75
   80. LOAD COMB 208 1.0DL + 0.75WLX2 + 0.75LL
   81. 1 1.0 4 0.75 2 0.75
   82. LOAD COMB 209 1.0DL + 0.75WLZ1 + 0.75LL
   83. 1 1.0 5 0.75 2 0.75
   84. LOAD COMB 210 1.0DL + 0.75WLZ2 + 0.75LL
   85. 1 1.0 6 0.75 2 0.75
   86. LOAD COMB 211 0.6DL + 1.0WLX1
   87. 1 0.6 3 1.0
   88. LOAD COMB 212 0.6DL + 1.0WLX2
   89. 1 0.6 4 1.0
   90. LOAD COMB 213 0.6DL + 1.0WLZ1
   91. 1 0.6 5 1.0
   92. LOAD COMB 214 0.6DL + 1.0WLZ2
```

93. 1 0.6 6 1.0
94. **********
95. PERFORM ANALYSIS

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS 4 NUMBER OF MEMBERS 2
NUMBER OF PLATES 0 NUMBER OF SOLIDS 0
NUMBER OF SURFACES 0 NUMBER OF SUPPORTS 2

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 1/ 1/ 6 DOF
TOTAL PRIMARY LOAD CASES = 6, TOTAL DEGREES OF FREEDOM = 12
TOTAL LOAD COMBINATION CASES = 14 SO FAR.
SIZE OF STIFFNESS MATRIX = 1 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 12.0/ 143074.6 MB

96. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= OCT 7,2020 TIME= 17:10:28 ****

****
* For technical assistance on STAAD.Pro, please visit *
* http://selectservices.bentley.com/en-US/ *
* *
* Details about additional assistance from *
* Bentley and Partners can be found at program menu *
* Help->Technical Support *
* *
* Copyright (c) 1997-2014 Bentley Systems, Inc.
* http://www.bentley.com *
****

ATTACHMENT 6 STAAD OUTPUT – FOUNDATION F2

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```
*****************************************************
*                                                   *
*           STAAD.Pro V8i SELECTseries5             *
*           Version  20.07.10.64                    *
*           Proprietary Program of                  *
*           Bentley Systems, Inc.                   *
*           Date=    SEP 25, 2020                   *
*           Time=    12:52:46                       *
*                                                   *
*      USER ID: LNT ECC DIV                         *
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```
     1. STAAD SPACE
INPUT FILE: F2 DESIGN.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 08/05/2020
     4. JOB NAME FDN. MKD. F2
     5. JOB CLIENT MA'ADEN
     6. JOB NO O19045
     7. JOB REV 0
     8. ENGINEER NAME SOR
     9. END JOB INFORMATION
    10. INPUT WIDTH 79
    11. UNIT METER KN
    12. JOINT COORDINATES
    13. 2 0 911.65 3.175; 6 0.75 911.65 3.175; 10 -0.75 911.65 3.175
    14. 34 -0.5 911.65 3.175; 35 -0.25 911.65 3.175; 41 0.25 911.65 3.175
    15. 42 0.5 911.65 3.175; 128 0 913.6 3.175; 302 -0.5 911.65 3.42438
    16. 303 -0.75 911.65 3.42438; 304 -0.25 911.65 3.42438; 305 0 911.65 3.42438
    17. 306 0.25 911.65 3.42438; 307 0.5 911.65 3.42438; 308 0.75 911.65 3.42438
    18. 321 -0.75 911.65 3.67438; 322 -0.5 911.65 3.67438; 323 -0.25 911.65 3.67438
    19. 324 0 911.65 3.67438; 325 0.25 911.65 3.67438; 326 0.5 911.65 3.67438
    20. 327 0.75 911.65 3.67438; 332 -0.75 911.65 3.92438; 333 -0.5 911.65 3.92438
    21. 334 -0.25 911.65 3.92438; 335 0 911.65 3.92438; 336 0.25 911.65 3.92438
    22. 337 0.5 911.65 3.92438; 338 0.75 911.65 3.92438; 343 -0.75 911.65 4.17438
    23. 344 -0.5 911.65 4.17438; 345 -0.25 911.65 4.17438; 346 0 911.65 4.17438
    24. 347 0.25 911.65 4.17438; 348 0.5 911.65 4.17438; 349 0.75 911.65 4.17438
    25. 354 -0.75 911.65 4.42438; 355 -0.5 911.65 4.42438; 356 -0.25 911.65 4.42438
    26. 357 0 911.65 4.42438; 358 0.25 911.65 4.42438; 359 0.5 911.65 4.42438
    27. 360 0.75 911.65 4.42438; 365 -0.75 911.65 2.17438; 366 -0.5 911.65 2.17438
    28. 367 -0.5 911.65 1.925; 368 -0.75 911.65 1.925; 369 -0.25 911.65 2.17438
    29. 370 -0.25 911.65 1.925; 371 0 911.65 2.17438; 372 0 911.65 1.925
    30. 373 0.25 911.65 2.17438; 374 0.25 911.65 1.925; 375 0.5 911.65 2.17438
    31. 376 0.5 911.65 1.925; 377 0.75 911.65 2.17438; 378 0.75 911.65 1.925
    32. 387 -0.75 911.65 2.42438; 388 -0.5 911.65 2.42438; 389 -0.25 911.65 2.42438
    33. 390 0 911.65 2.42438; 391 0.25 911.65 2.42438; 392 0.5 911.65 2.42438
    34. 393 0.75 911.65 2.42438; 398 -0.75 911.65 2.67438; 399 -0.5 911.65 2.67438
    35. 400 -0.25 911.65 2.67438; 401 0 911.65 2.67438; 402 0.25 911.65 2.67438
    36. 403 0.5 911.65 2.67438; 404 0.75 911.65 2.67438; 409 -0.75 911.65 2.92438
    37. 410 -0.5 911.65 2.92438; 411 -0.25 911.65 2.92438; 412 0 911.65 2.92438
    38. 413 0.25 911.65 2.92438; 414 0.5 911.65 2.92438; 415 0.75 911.65 2.92438
```

```
39. MEMBER INCIDENCES
40. 185 2 128
41. ELEMENT INCIDENCES SHELL
42. 326 303 302 34 10; 327 302 304 35 34; 328 304 305 2 35; 332 305 306 41 2
43. 333 306 307 42 41; 334 307 308 6 42; 343 321 322 302 303; 344 322 323 304 302
44. 345 323 324 305 304; 346 324 325 306 305; 347 325 326 307 306
45. 348 326 327 308 307; 353 332 333 322 321; 354 333 334 323 322
46. 355 334 335 324 323; 356 335 336 325 324; 357 336 337 326 325
47. 358 337 338 327 326; 363 343 344 333 332; 364 344 345 334 333
48. 365 345 346 335 334; 366 346 347 336 335; 367 347 348 337 336
49. 368 348 349 338 337; 373 354 355 344 343; 374 355 356 345 344
50. 375 356 357 346 345; 376 357 358 347 346; 377 358 359 348 347
51. 378 359 360 349 348; 383 365 366 367 368; 384 366 369 370 367
52. 385 369 371 372 370; 386 371 373 374 372; 387 373 375 376 374
53. 388 375 377 378 376; 393 387 388 366 365; 394 388 389 369 366
54. 395 389 390 371 369; 396 390 391 373 371; 397 391 392 375 373
55. 398 392 393 377 375; 403 398 399 388 387; 404 399 400 389 388
56. 405 400 401 390 389; 406 401 402 391 390; 407 402 403 392 391
57. 408 403 404 393 392; 413 409 410 399 398; 414 410 411 400 399
58. 415 411 412 401 400; 416 412 413 402 401; 417 413 414 403 402
59. 418 414 415 404 403; 423 10 34 410 409; 424 34 35 411 410; 425 35 2 412 411
60. 426 2 41 413 412; 427 41 42 414 413; 428 42 6 415 414
61. START GROUP DEFINITION
62. ELEMENT
63. _RAFT 326 TO 328 332 TO 334 343 TO 348 353 TO 358 363 TO 368 373 TO 378 383 -
64. 384 TO 388 393 TO 398 403 TO 408 413 TO 418 423 TO 428
65. _RAFT_MOM 326 327 333 334 343 TO 348 353 TO 358 363 TO 368 373 TO 378 383 -
66. 384 TO 388 393 TO 398 403 TO 408 413 TO 418 423 424 427 428
67. _RAFT_SHEAR 326 334 343 348 353 TO 358 363 TO 368 373 TO 378 383 TO 388 393 -
68. 394 TO 398 403 TO 408 413 418 423 428
69. MEMBER
70. END GROUP DEFINITION
71. ELEMENT PROPERTY
72. 326 TO 328 332 TO 334 343 TO 348 353 TO 358 363 TO 368 373 TO 378 383 TO 388 -
73. 393 TO 398 403 TO 408 413 TO 418 423 TO 428 THICKNESS 0.5
74. DEFINE MATERIAL START
75. ISOTROPIC CONCRETE
76. E 2.17185E+007
77. POISSON 0.17
78. DENSITY 25
79. ALPHA 1E-005
80. DAMP 0.05
81. TYPE CONCRETE
82. STRENGTH FCU 27579
83. END DEFINE MATERIAL
84. MEMBER PROPERTY AMERICAN
85. 185 PRIS YD 0.4 ZD 0.6
86. CONSTANTS
87. MATERIAL CONCRETE ALL
88. SUPPORTS
89. 326 TO 328 332 TO 334 343 TO 348 353 TO 358 363 TO 368 373 TO 378 383 TO 388 -
90. 393 TO 398 403 TO 408 413 TO 418 423 TO 427 -
91. 428 PLATE MAT DIRECT Y SUBGRADE 57600 PRINT
```

JOINT	INFLUENCE AREA
2	0.062500
6	0.031250
10	0.031250
34	0.062500
35	0.062500
41	0.062500
42	0.062500
302	0.062422
303	0.031211
304	0.062422
305	0.062422
306	0.062422
307	0.062422
308	0.031211
321	0.031250
322	0.062500
323	0.062500
324	0.062500
325	0.062500
326	0.062500
327	0.031250
332	0.031250
333	0.062500
334	0.062500
335	0.062500
336	0.062500
337	0.062500
338	0.031250
343	0.031250
344	0.062500
345	0.062500
346	0.062500
347	0.062500
348	0.062500
349	0.031250
354	0.015625
355	0.031250
356	0.031250
357	0.031250
358	0.031250
359	0.031250
360	0.015625
365	0.031211
366	0.062422
367	0.031173
368	0.015586
369	0.062422
370	0.031173
371	0.062422
372	0.031173
373	0.062422
374	0.031173

375	0.062422
376	0.031173
377	0.031211
378	0.015586
387	0.031250
388	0.062500
389	0.062500
390	0.062500
391	0.062500
392	0.062500
393	0.031250
398	0.031250
399	0.062500
400	0.062500
401	0.062500
402	0.062500
403	0.062500
404	0.031250
409	0.031289
410	0.062578
411	0.062578
412	0.062578
413	0.062578
414	0.062578
415	0.031289
TOTAL	3.749070

```
 92. ********************************************************************************
 93. LOAD 1 LOADTYPE NONE  TITLE DEAD LOAD ( DL )
 94. SELFWEIGHT Y -1
 95. ELEMENT LOAD
 96. * LOAD DUE TO BACKFILL
 97. 326 TO 328 332 TO 334 343 TO 348 353 TO 358 363 TO 368 373 TO 378 383 TO 388 -
 98. 393 TO 398 403 TO 408 413 TO 418 423 TO 428 PR GY -15
 99. * LOAD DUE TO SURCHARGE
100. 326 TO 328 332 TO 334 343 TO 348 353 TO 358 363 TO 368 373 TO 378 383 TO 388 -
101. 393 TO 398 403 TO 408 413 TO 418 423 TO 428 PR GY -20
102. JOINT LOAD
103. 128 FX -0.001 FY -76.404 FZ 0.282
104. ********************************************************************************
105. LOAD 2 LOADTYPE NONE  TITLE LIVE LOAD ( LL )
106. JOINT LOAD
107. 128 FX -0.005 FY -57.023 FZ -1.257
108. ********************************************************************************
109. LOAD 3 LOADTYPE NONE  TITLE WIND LOAD +X  ( WLX1 )
110. JOINT LOAD
111. 128 FX 1.334 FY -0.866 FZ 0.111
112. ********************************************************************************
113. LOAD 4 LOADTYPE NONE  TITLE WIND LOAD -X  ( WLX2 )
114. JOINT LOAD
115. 128 FX 1.334 FY 0.866 FZ 0.111
116. ********************************************************************************
117. LOAD 5 LOADTYPE NONE  TITLE WIND LOAD +Z  ( WLZ1 )
118. JOINT LOAD
119. 128 FX 0.025 FY 9.749 FZ 3.225
120. ********************************************************************************
121. LOAD 6 LOADTYPE NONE  TITLE WIND LOAD -Z  ( WLZ2 )
```

```
122. JOINT LOAD
123. 128 FX 0.025 FY -9.744 FZ 3.224
124. ********************************************************************************
125. ***FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
126. LOAD COMB 101 1.4DL
127. 1 1.4
128. LOAD COMB 102 1.4DL + 1.7LL
129. 1 1.4 2 1.7
130. LOAD COMB 103 1.4DL + 1.6WLX1 + 1.0 LL
131. 1 1.4 3 1.6 2 1.0
132. LOAD COMB 104 1.4DL + 1.6WLX2 + 1.0 LL
133. 1 1.4 4 1.6 2 1.0
134. LOAD COMB 105 1.4DL + 1.6WLZ1 + 1.0 LL
135. 1 1.4 5 1.6 2 1.0
136. LOAD COMB 106 1.4DL + 1.6WLZ2 + 1.0 LL
137. 1 1.4 6 1.6 2 1.0
138. LOAD COMB 107 0.9DL + 1.6WLX1
139. 1 0.9 3 1.6
140. LOAD COMB 108 0.9DL + 1.6WLX2
141. 1 0.9 4 1.6
142. LOAD COMB 109 0.9DL + 1.6WLZ1
143. 1 0.9 5 1.6
144. LOAD COMB 110 0.9DL + 1.6WLZ2
145. 1 0.9 6 1.6
146. ********************************************************************************
147. * UNFACTORED LOAD COMBINATIONS FOR STABILITY CHECK
148. LOAD COMB 201 1.0DL
149. 1 1.0
150. LOAD COMB 202 1.0DL + 1.0LL
151. 1 1.0 2 1.0
152. LOAD COMB 203 1.0DL + 1.0WLX1
153. 1 1.0 3 1.0
154. LOAD COMB 204 1.0DL + 1.0WLX2
155. 1 1.0 4 1.0
156. LOAD COMB 205 1.0DL + 1.0WLZ1
157. 1 1.0 5 1.0
158. LOAD COMB 206 1.0DL + 1.0WLZ2
159. 1 1.0 6 1.0
160. LOAD COMB 207 1.0DL + 0.75WLX1 + 0.75LL
161. 1 1.0 3 0.75 2 0.75
162. LOAD COMB 208 1.0DL + 0.75WLX2 + 0.75LL
163. 1 1.0 4 0.75 2 0.75
164. LOAD COMB 209 1.0DL + 0.75WLZ1 + 0.75LL
165. 1 1.0 5 0.75 2 0.75
166. LOAD COMB 210 1.0DL + 0.75WLZ2 + 0.75LL
167. 1 1.0 6 0.75 2 0.75
168. LOAD COMB 211 0.6DL + 1.0WLX1
169. 1 0.6 3 1.0
170. LOAD COMB 212 0.6DL + 1.0WLX2
171. 1 0.6 4 1.0
172. LOAD COMB 213 0.6DL + 1.0WLZ1
173. 1 0.6 5 1.0
174. LOAD COMB 214 0.6DL + 1.0WLZ2
175. 1 0.6 6 1.0
176. ********************************************************************************
177. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS 78 NUMBER OF MEMBERS 1
NUMBER OF PLATES 60 NUMBER OF SOLIDS 0
NUMBER OF SURFACES 0 NUMBER OF SUPPORTS 77

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 76/ 12/ 39 DOF
TOTAL PRIMARY LOAD CASES = 6, TOTAL DEGREES OF FREEDOM = 237
TOTAL LOAD COMBINATION CASES = 24 SO FAR.
SIZE OF STIFFNESS MATRIX = 10 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 12.4/ 143875.1 MB

178. LOAD LIST 101 TO 110
179. START CONCRETE DESIGN

```
180. CODE ACI
181. TRACK 2 ALL
182. FC 35000 MEMB 185
183. CLB 0.05 MEMB 185
184. CLS 0.05 ALL
185. CLT 0.05 ALL
186. MAXMAIN 25 ALL
187. MINMAIN 25 ALL
188. MINSEC 10 ALL
189. DESIGN COLUMN 185
```

==========

COLUMN NO. 185 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 35.0 MPA, RECT SIZE - 600.0 X 400.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2400.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 25 MM	1.636	101	END	0.650

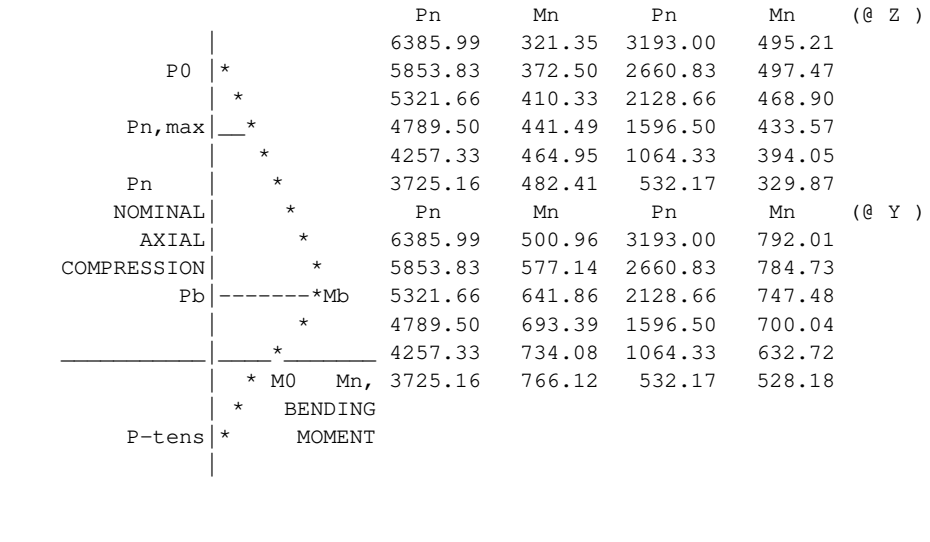
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 400.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
8647.70	6918.16	2791.93	503.22	180.2
M0	P-tens.	Des.Pn	Des.Mn	e/h
257.31	-1624.53	164.56	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
8647.70	6918.16	2974.50	801.47	269.4
M0	P-tens.	Des.Pn	Des.Mn	e/h
407.87	-1624.53	164.56	0.00	0.00000



**********END OF COLUMN DESIGN RESULTS**********

190. END CONCRETE DESIGN
191. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= SEP 25,2020 TIME= 12:52:52 ****

****
* For technical assistance on STAAD.Pro, please visit *
* http://selectservices.bentley.com/en-US/ *
* *
* Details about additional assistance from *
* Bentley and Partners can be found at program menu *
* Help->Technical Support *
* *
* Copyright (c) 1997-2014 Bentley Systems, Inc.
* http://www.bentley.com *
****

```
            **************************************************
            *                                                *
            *          STAAD.Pro V8i SELECTseries5           *
            *          Version  20.07.10.64                  *
            *          Proprietary Program of                *
            *          Bentley Systems, Inc.                 *
            *          Date=    OCT  7, 2020                 *
            *          Time=    17:12:43                     *
            *                                                *
            *     USER ID: LNT ECC DIV                       *
            **************************************************
```

```
     1. STAAD SPACE
INPUT FILE: F2_STABILITY CHECK.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 08/05/2020
     4. JOB NAME FDN. MKD. F2_STABILITY CHECK
     5. JOB CLIENT MA'ADEN
     6. JOB NO O19045
     7. JOB REV 0
     8. ENGINEER NAME SOR
     9. END JOB INFORMATION
    10. INPUT WIDTH 79
    11. UNIT METER KN
    12. JOINT COORDINATES
    13. 2 0 911.65 3.175; 128 0 913.6 3.175
    14. MEMBER INCIDENCES
    15. 185 2 128
    16. DEFINE MATERIAL START
    17. ISOTROPIC CONCRETE
    18. E 2.17185E+007
    19. POISSON 0.17
    20. DENSITY 25
    21. ALPHA 1E-005
    22. DAMP 0.05
    23. TYPE CONCRETE
    24. STRENGTH FCU 27579
    25. END DEFINE MATERIAL
    26. MEMBER PROPERTY AMERICAN
    27. 185 PRIS YD 0.4 ZD 0.6
    28. CONSTANTS
    29. MATERIAL CONCRETE ALL
    30. SUPPORTS
    31. 2 FIXED
    32. ******************************************************************************
    33. LOAD 1 LOADTYPE NONE  TITLE DEAD LOAD ( DL )
    34. SELFWEIGHT Y -1
    35. JOINT LOAD
    36. 128 FX -0.001 FY -76.404 FZ 0.282
    37. ******************************************************************************
    38. LOAD 2 LOADTYPE NONE  TITLE LIVE LOAD ( LL )
```



```
    39. JOINT LOAD
    40. 128 FX -0.005 FY -57.023 FZ -1.257
    41. ******************************************************************************
    42. LOAD 3 LOADTYPE NONE  TITLE WIND LOAD +X  ( WLX1 )
    43. JOINT LOAD
    44. 128 FX 1.334 FY -0.866 FZ 0.111
    45. ******************************************************************************
    46. LOAD 4 LOADTYPE NONE  TITLE WIND LOAD -X  ( WLX2 )
    47. JOINT LOAD
    48. 128 FX 1.334 FY 0.866 FZ 0.111
    49. ******************************************************************************
    50. LOAD 5 LOADTYPE NONE  TITLE WIND LOAD +Z  ( WLZ1 )
    51. JOINT LOAD
    52. 128 FX 0.025 FY 9.749 FZ 3.225
    53. ******************************************************************************
    54. LOAD 6 LOADTYPE NONE  TITLE WIND LOAD -Z  ( WLZ2 )
    55. JOINT LOAD
    56. 128 FX 0.025 FY -9.744 FZ 3.224
    57. ******************************************************************************
    58. * UNFACTORED LOAD COMBINATIONS FOR STABILITY CHECK
    59. LOAD COMB 201 1.0DL
    60. 1 1.0
    61. LOAD COMB 202 1.0DL + 1.0LL
    62. 1 1.0 2 1.0
    63. LOAD COMB 203 1.0DL + 1.0WLX1
    64. 1 1.0 3 1.0
    65. LOAD COMB 204 1.0DL + 1.0WLX2
    66. 1 1.0 4 1.0
    67. LOAD COMB 205 1.0DL + 1.0WLZ1
    68. 1 1.0 5 1.0
    69. LOAD COMB 206 1.0DL + 1.0WLZ2
    70. 1 1.0 6 1.0
    71. LOAD COMB 207 1.0DL + 0.75WLX1 + 0.75LL
    72. 1 1.0 3 0.75 2 0.75
    73. LOAD COMB 208 1.0DL + 0.75WLX2 + 0.75LL
    74. 1 1.0 4 0.75 2 0.75
    75. LOAD COMB 209 1.0DL + 0.75WLZ1 + 0.75LL
    76. 1 1.0 5 0.75 2 0.75
    77. LOAD COMB 210 1.0DL + 0.75WLZ2 + 0.75LL
    78. 1 1.0 6 0.75 2 0.75
    79. LOAD COMB 211 0.6DL + 1.0WLX1
    80. 1 0.6 3 1.0
    81. LOAD COMB 212 0.6DL + 1.0WLX2
    82. 1 0.6 4 1.0
    83. LOAD COMB 213 0.6DL + 1.0WLZ1
    84. 1 0.6 5 1.0
    85. LOAD COMB 214 0.6DL + 1.0WLZ2
    86. 1 0.6 6 1.0
    87. ******************************************************************************
    88. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS 2 NUMBER OF MEMBERS 1
NUMBER OF PLATES 0 NUMBER OF SOLIDS 0
NUMBER OF SURFACES 0 NUMBER OF SUPPORTS 1

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 1/ 1/ 6 DOF
TOTAL PRIMARY LOAD CASES = 6, TOTAL DEGREES OF FREEDOM = 6
TOTAL LOAD COMBINATION CASES = 14 SO FAR.
SIZE OF STIFFNESS MATRIX = 1 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 12.0/ 143074.5 MB

89. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= OCT 7,2020 TIME= 17:12:46 ****

****
* For technical assistance on STAAD.Pro, please visit *
* http://selectservices.bentley.com/en-US/ *
* *
* Details about additional assistance from *
* Bentley and Partners can be found at program menu *
* Help->Technical Support *
* *
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* http://www.bentley.com *
****

ATTACHMENT 7 STAAD OUTPUT – FOUNDATION F3

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```
**************************************************
*                                                *
*          STAAD.Pro V8i SELECTseries5           *
*          Version  20.07.10.64                  *
*          Proprietary Program of                *
*          Bentley Systems, Inc.                 *
*          Date=    SEP 25, 2020                 *
*          Time=    12:53:46                     *
*                                                *
*     USER ID: LNT ECC DIV                       *
**************************************************
```

```
    1. STAAD SPACE
INPUT FILE: F3.STD
    2. START JOB INFORMATION
    3. ENGINEER DATE 08/05/2020
    4. JOB NAME FDN. MKD. F3
    5. JOB CLIENT MA'ADEN
    6. JOB NO O19045
    7. JOB REV 0
    8. ENGINEER NAME SOR
    9. END JOB INFORMATION
   10. INPUT WIDTH 79
   11. SET NL 200
   12. SET ITERLIM 15
   13. SET PRINT 1
   14. UNIT METER KN
   15. JOINT COORDINATES
   16. 422 -0.5 911.65 1.12538; 423 -0.75 911.65 1.12538; 425 -0.25 911.65 1.12538
   17. 427 0 911.65 1.12538; 429 0.25 911.65 1.12538; 431 0.5 911.65 1.12538
   18. 433 0.75 911.65 1.12538; 435 1 911.65 1.12538; 456 -0.75 911.65 0.42438
   19. 457 -0.5 911.65 0.42438; 458 -0.5 911.65 0.17538; 459 -0.75 911.65 0.17538
   20. 460 -0.25 911.65 0.42438; 461 -0.25 911.65 0.17538; 462 0 911.65 0.42438
   21. 463 0 911.65 0.17538; 464 0.25 911.65 0.42438; 465 0.25 911.65 0.17538
   22. 466 0.5 911.65 0.42438; 467 0.5 911.65 0.17538; 468 0.75 911.65 0.42438
   23. 469 0.75 911.65 0.17538; 470 1 911.65 0.42438; 471 1 911.65 0.17538
   24. 474 -0.75 911.65 0.67438; 475 -0.5 911.65 0.67438; 476 -0.25 911.65 0.67438
   25. 477 0 911.65 0.67438; 478 0.25 911.65 0.67438; 479 0.5 911.65 0.67438
   26. 480 0.75 911.65 0.67438; 481 1 911.65 0.67438; 483 -0.75 911.65 0.92438
   27. 484 -0.5 911.65 0.92438; 485 -0.25 911.65 0.92438; 486 0 911.65 0.92438
   28. 487 0.25 911.65 0.92438; 488 0.5 911.65 0.92438; 489 0.75 911.65 0.92438
   29. 490 1 911.65 0.92438; 492 -0.75 911.65 -0.62362; 493 -0.5 911.65 -0.62362
   30. 494 -0.5 911.65 -0.87362; 495 -0.75 911.65 -0.87362; 496 -0.25 911.65 -0.62362
   31. 497 -0.25 911.65 -0.87362; 498 0 911.65 -0.62362; 499 0 911.65 -0.87362
   32. 500 0.25 911.65 -0.62362; 501 0.25 911.65 -0.87362; 502 0.5 911.65 -0.62362
   33. 503 0.5 911.65 -0.87362; 504 0.75 911.65 -0.62362; 505 0.75 911.65 -0.87362
   34. 506 1 911.65 -0.62362; 507 1 911.65 -0.87362; 510 -0.75 911.65 -0.37362
   35. 511 -0.5 911.65 -0.37362; 512 -0.25 911.65 -0.37362; 513 0 911.65 -0.37362
   36. 514 0.25 911.65 -0.37362; 515 0.5 911.65 -0.37362; 516 0.75 911.65 -0.37362
   37. 517 1 911.65 -0.37362; 519 -0.75 911.65 -0.12462; 520 -0.5 911.65 -0.12462
   38. 521 -0.25 911.65 -0.12462; 522 0 911.65 -0.12462; 523 0.25 911.65 -0.12462
```



```
   39. 524 0.5 911.65 -0.12462; 525 0.75 911.65 -0.12462; 526 1 911.65 -0.12462
   40. 528 0 913.6 -0.125; 546 1.25 911.65 1.12538; 553 1.25 911.65 0.42438
   41. 554 1.25 911.65 0.17538; 557 1.25 911.65 0.67438; 559 1.25 911.65 0.92438
   42. 561 1.25 911.65 -0.62362; 562 1.25 911.65 -0.87362; 565 1.25 911.65 -0.37362
   43. 567 1.25 911.65 -0.12462; 677 1.5 911.65 1.12538; 679 1.75 911.65 1.12538
   44. 681 2 911.65 1.12538; 683 2.25 911.65 1.12538; 684 1.5 911.65 0.42438
   45. 685 1.5 911.65 0.17538; 686 1.75 911.65 0.42438; 687 1.75 911.65 0.17538
   46. 688 2 911.65 0.42438; 689 2 911.65 0.17538; 690 2.25 911.65 0.42438
   47. 691 2.25 911.65 0.17538; 692 1.5 911.65 0.67438; 693 1.75 911.65 0.67438
   48. 694 2 911.65 0.67438; 695 2.25 911.65 0.67438; 696 1.5 911.65 0.92438
   49. 697 1.75 911.65 0.92438; 698 2 911.65 0.92438; 699 2.25 911.65 0.92438
   50. 700 1.5 911.65 -0.62362; 701 1.5 911.65 -0.87362; 702 1.75 911.65 -0.62362
   51. 703 1.75 911.65 -0.87362; 704 2 911.65 -0.62362; 705 2 911.65 -0.87362
   52. 706 2.25 911.65 -0.62362; 707 2.25 911.65 -0.87362; 708 1.5 911.65 -0.37362
   53. 709 1.75 911.65 -0.37362; 710 2 911.65 -0.37362; 711 2.25 911.65 -0.37362
   54. 712 1.5 911.65 -0.12462; 713 1.75 911.65 -0.12462; 714 2 911.65 -0.12462
   55. 715 2.25 911.65 -0.12462; 717 2.5 911.65 1.12538; 718 2.5 911.65 0.42438
   56. 719 2.5 911.65 0.17538; 720 2.5 911.65 0.67438; 721 2.5 911.65 0.92438
   57. 722 2.5 911.65 -0.62362; 723 2.5 911.65 -0.87362; 724 2.5 911.65 -0.37362
   58. 725 2.5 911.65 -0.12462; 727 2.75 911.65 1.12538; 729 3 911.65 1.12538
   59. 731 3.25 911.65 1.12538; 733 3.5 911.65 1.12538; 734 2.75 911.65 0.42438
   60. 735 2.75 911.65 0.17538; 736 3 911.65 0.42438; 737 3 911.65 0.17538
   61. 738 3.25 911.65 0.42438; 739 3.25 911.65 0.17538; 740 3.5 911.65 0.42438
   62. 741 3.5 911.65 0.17538; 742 2.75 911.65 0.67438; 743 3 911.65 0.67438
   63. 744 3.25 911.65 0.67438; 745 3.5 911.65 0.67438; 746 2.75 911.65 0.92438
   64. 747 3 911.65 0.92438; 748 3.25 911.65 0.92438; 749 3.5 911.65 0.92438
   65. 750 2.75 911.65 -0.62362; 751 2.75 911.65 -0.87362; 752 3 911.65 -0.62362
   66. 753 3 911.65 -0.87362; 754 3.25 911.65 -0.62362; 755 3.25 911.65 -0.87362
   67. 756 3.5 911.65 -0.62362; 757 3.5 911.65 -0.87362; 758 2.75 911.65 -0.37362
   68. 759 3 911.65 -0.37362; 760 3.25 911.65 -0.37362; 761 3.5 911.65 -0.37362
   69. 762 2.75 911.65 -0.12462; 763 3 911.65 -0.12462; 764 3.25 911.65 -0.12462
   70. 765 3.5 911.65 -0.12462; 767 3.75 911.65 1.12538; 768 3.75 911.65 0.42438
   71. 769 3.75 911.65 0.17538; 770 3.75 911.65 0.67438; 771 3.75 911.65 0.92438
   72. 772 3.75 911.65 -0.62362; 773 3.75 911.65 -0.87362; 774 3.75 911.65 -0.37362
   73. 775 3.75 911.65 -0.12462; 776 3.5 913.6 -0.125; 778 4 911.65 1.12538
   74. 779 4 911.65 0.42438; 780 4 911.65 0.17538; 781 4 911.65 0.67438
   75. 782 4 911.65 0.92438; 783 4 911.65 -0.62362; 784 4 911.65 -0.87362
   76. 785 4 911.65 -0.37362; 786 4 911.65 -0.12462; 788 4.25 911.65 1.12538
   77. 789 4.25 911.65 0.42438; 790 4.25 911.65 0.17538; 791 4.25 911.65 0.67438
   78. 792 4.25 911.65 0.92438; 793 4.25 911.65 -0.62362; 794 4.25 911.65 -0.87362
   79. 795 4.25 911.65 -0.37362; 796 4.25 911.65 -0.12462; 797 -0.75 911.65 -1.12362
   80. 798 -0.5 911.65 -1.12362; 799 -0.5 911.65 -1.37362; 800 -0.75 911.65 -1.37362
   81. 801 -0.25 911.65 -1.12362; 802 -0.25 911.65 -1.37362; 803 0 911.65 -1.12362
   82. 804 0 911.65 -1.37362; 805 0.25 911.65 -1.12362; 806 0.25 911.65 -1.37362
   83. 807 0.5 911.65 -1.12362; 808 0.5 911.65 -1.37362; 809 0.75 911.65 -1.12362
   84. 810 0.75 911.65 -1.37362; 811 1 911.65 -1.12362; 812 1 911.65 -1.37362
   85. 813 1.25 911.65 -1.12362; 814 1.25 911.65 -1.37362; 815 1.5 911.65 -1.12362
   86. 816 1.5 911.65 -1.37362; 817 1.75 911.65 -1.12362; 818 1.75 911.65 -1.37362
   87. 819 2 911.65 -1.12362; 820 2 911.65 -1.37362; 821 2.25 911.65 -1.12362
   88. 822 2.25 911.65 -1.37362; 823 2.5 911.65 -1.12362; 824 2.5 911.65 -1.37362
   89. 825 2.75 911.65 -1.12362; 826 2.75 911.65 -1.37362; 827 3 911.65 -1.12362
   90. 828 3 911.65 -1.37362; 829 3.25 911.65 -1.12362; 830 3.25 911.65 -1.37362
   91. 831 3.5 911.65 -1.12362; 832 3.5 911.65 -1.37362; 833 3.75 911.65 -1.12362
   92. 834 3.75 911.65 -1.37362; 835 4 911.65 -1.12362; 836 4 911.65 -1.37362
   93. 837 4.25 911.65 -1.12362; 838 4.25 911.65 -1.37362
   94. MEMBER INCIDENCES
```

```
 95. 529 522 528; 759 765 776
 96. ELEMENT INCIDENCES SHELL
 97. 465 456 457 458 459; 466 457 460 461 458; 467 460 462 463 461
 98. 468 462 464 465 463; 469 464 466 467 465; 470 466 468 469 467
 99. 471 468 470 471 469; 473 474 475 457 456; 474 475 476 460 457
100. 475 476 477 462 460; 476 477 478 464 462; 477 478 479 466 464
101. 478 479 480 468 466; 479 480 481 470 468; 481 483 484 475 474
102. 482 484 485 476 475; 483 485 486 477 476; 484 486 487 478 477
103. 485 487 488 479 478; 486 488 489 480 479; 487 489 490 481 480
104. 489 423 422 484 483; 490 422 425 485 484; 491 425 427 486 485
105. 492 427 429 487 486; 493 429 431 488 487; 494 431 433 489 488
106. 495 433 435 490 489; 497 492 493 494 495; 498 493 496 497 494
107. 499 496 498 499 497; 500 498 500 501 499; 501 500 502 503 501
108. 502 502 504 505 503; 503 504 506 507 505; 505 510 511 493 492
109. 506 511 512 496 493; 507 512 513 498 496; 508 513 514 500 498
110. 509 514 515 502 500; 510 515 516 504 502; 511 516 517 506 504
111. 513 519 520 511 510; 514 520 521 512 511; 515 521 522 513 512
112. 516 522 523 514 513; 517 523 524 515 514; 518 524 525 516 515
113. 519 525 526 517 516; 521 459 458 520 519; 522 458 461 521 520
114. 523 461 463 522 521; 524 463 465 523 522; 525 465 467 524 523
115. 526 467 469 525 524; 527 469 471 526 525; 552 470 553 554 471
116. 554 481 557 553 470; 556 490 559 557 481; 558 435 546 559 490
117. 560 506 561 562 507; 562 517 565 561 506; 564 526 567 565 517
118. 566 471 554 567 526; 673 553 684 685 554; 674 684 686 687 685
119. 675 686 688 689 687; 676 688 690 691 689; 677 557 692 684 553
120. 678 692 693 686 684; 679 693 694 688 686; 680 694 695 690 688
121. 681 559 696 692 557; 682 696 697 693 692; 683 697 698 694 693
122. 684 698 699 695 694; 685 546 677 696 559; 686 677 679 697 696
123. 687 679 681 698 697; 688 681 683 699 698; 689 561 700 701 562
124. 690 700 702 703 701; 691 702 704 705 703; 692 704 706 707 705
125. 693 565 708 700 561; 694 708 709 702 700; 695 709 710 704 702
126. 696 710 711 706 704; 697 567 712 708 565; 698 712 713 709 708
127. 699 713 714 710 709; 700 714 715 711 710; 701 554 685 712 567
128. 702 685 687 713 712; 703 687 689 714 713; 704 689 691 715 714
129. 706 690 718 719 691; 707 695 720 718 690; 708 699 721 720 695
130. 709 683 717 721 699; 710 706 722 723 707; 711 711 724 722 706
131. 712 715 725 724 711; 713 691 719 725 715; 718 718 734 735 719
132. 719 734 736 737 735; 720 736 738 739 737; 721 738 740 741 739
133. 722 720 742 734 718; 723 742 743 736 734; 724 743 744 738 736
134. 725 744 745 740 738; 726 721 746 742 720; 727 746 747 743 742
135. 728 747 748 744 743; 729 748 749 745 744; 730 717 727 746 721
136. 731 727 729 747 746; 732 729 731 748 747; 733 731 733 749 748
137. 734 722 750 751 723; 735 750 752 753 751; 736 752 754 755 753
138. 737 754 756 757 755; 738 724 758 750 722; 739 758 759 752 750
139. 740 759 760 754 752; 741 760 761 756 754; 742 725 762 758 724
140. 743 762 763 759 758; 744 763 764 760 759; 745 764 765 761 760
141. 746 719 735 762 725; 747 735 737 763 762; 748 737 739 764 763
142. 749 739 741 765 764; 751 740 768 769 741; 752 745 770 768 740
143. 753 749 771 770 745; 754 733 767 771 749; 755 756 772 773 757
144. 756 761 774 772 756; 757 765 775 774 761; 758 741 769 775 765
145. 761 768 779 780 769; 762 770 781 779 768; 763 771 782 781 770
146. 764 767 778 782 771; 765 772 783 784 773; 766 774 785 783 772
147. 767 775 786 785 774; 768 769 780 786 775; 770 779 789 790 780
148. 771 781 791 789 779; 772 782 792 791 781; 773 778 788 792 782
149. 774 783 793 794 784; 775 785 795 793 783; 776 786 796 795 785
150. 777 780 790 796 786; 778 797 798 799 800; 779 798 801 802 799
```



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151. 780 801 803 804 802; 781 803 805 806 804; 782 805 807 808 806
152. 783 807 809 810 808; 784 809 811 812 810; 785 495 494 798 797
153. 786 494 497 801 798; 787 497 499 803 801; 788 499 501 805 803
154. 789 501 503 807 805; 790 503 505 809 807; 791 505 507 811 809
155. 792 811 813 814 812; 793 507 562 813 811; 794 813 815 816 814
156. 795 815 817 818 816; 796 817 819 820 818; 797 819 821 822 820
157. 798 562 701 815 813; 799 701 703 817 815; 800 703 705 819 817
158. 801 705 707 821 819; 802 821 823 824 822; 803 707 723 823 821
159. 804 823 825 826 824; 805 825 827 828 826; 806 827 829 830 828
160. 807 829 831 832 830; 808 723 751 825 823; 809 751 753 827 825
161. 810 753 755 829 827; 811 755 757 831 829; 812 831 833 834 832
162. 813 757 773 833 831; 814 833 835 836 834; 815 773 784 835 833
163. 816 835 837 838 836; 817 784 794 837 835
164. START GROUP DEFINITION
165. ELEMENT
166. _RAFT 465 TO 471 473 TO 479 481 TO 487 489 TO 495 497 TO 503 505 TO 511 513 -
167. 514 TO 519 521 TO 527 552 554 556 558 560 562 564 566 673 TO 704 706 TO 713 -
168. 718 TO 749 751 TO 758 761 TO 768 770 TO 817
169. _RAFT_MOM 465 TO 471 473 TO 479 481 TO 487 489 TO 495 497 TO 503 505 TO 511 -
170. 513 514 517 TO 519 521 522 525 TO 527 552 554 556 558 560 562 564 566 673 -
171. 674 TO 688 690 TO 704 706 TO 713 718 TO 744 746 TO 748 751 TO 756 761 TO 768 -
172. 770 TO 817
173. _RAFT_SHEAR 465 470 471 473 TO 479 481 TO 487 489 TO 495 497 TO 503 505 510 -
174. 511 513 518 519 521 526 527 552 554 556 558 560 562 564 566 673 TO 704 706 -
175. 707 TO 713 718 719 722 TO 739 742 743 746 747 752 TO 755 762 TO 765 -
176. 770 TO 788 790 TO 817
177. MEMBER
178. END GROUP DEFINITION
179. ELEMENT PROPERTY
180. 465 TO 471 473 TO 479 481 TO 487 489 TO 495 497 TO 503 505 TO 511 513 TO 519 -
181. 521 TO 527 552 554 556 558 560 562 564 566 673 TO 704 706 TO 713 718 TO 749 -
182. 751 TO 758 761 TO 768 770 TO 817 THICKNESS 0.5
183. DEFINE MATERIAL START
184. ISOTROPIC CONCRETE
185. E 2.17185E+007
186. POISSON 0.17
187. DENSITY 25
188. ALPHA 1E-005
189. DAMP 0.05
190. TYPE CONCRETE
191. STRENGTH FCU 27579
192. END DEFINE MATERIAL
193. MEMBER PROPERTY AMERICAN
194. 529 759 PRIS YD 0.4 ZD 0.6
195. CONSTANTS
196. MATERIAL CONCRETE ALL
197. SUPPORTS
198. 465 TO 471 473 TO 479 481 TO 487 489 TO 495 497 TO 503 505 TO 511 513 TO 519 -
199. 521 TO 527 552 554 556 558 560 562 564 566 673 TO 704 706 TO 713 718 TO 749 -
200. 751 TO 758 761 TO 768 770 TO 817 PLATE MAT DIRECT Y SUBGRADE 57600 PRINT

     JOINT      INFLUENCE AREA
     -----      --------------

       422          0.025125
       423          0.012563
```

STAAD SPACE -- PAGE NO. 5

425	0.025125
427	0.025125
429	0.025125
431	0.025125
433	0.025125
435	0.025125
456	0.031188
457	0.062375
458	0.068625
459	0.034313
460	0.062375
461	0.068625
462	0.062375
463	0.068625
464	0.062375
465	0.068625
466	0.062375
467	0.068625
468	0.062375
469	0.068625
470	0.062375
471	0.068625
474	0.031250
475	0.062500
476	0.062500
477	0.062500
478	0.062500
479	0.062500
480	0.062500
481	0.062500
483	0.028188
484	0.056375
485	0.056375
486	0.056375
487	0.056375
488	0.056375
489	0.056375
490	0.056375
492	0.031250
493	0.062500
494	0.062500
495	0.031250
496	0.062500
497	0.062500
498	0.062500
499	0.062500
500	0.062500
501	0.062500
502	0.062500
503	0.062500
504	0.062500
505	0.062500
506	0.062500
507	0.062500
510	0.031188
511	0.062375

STAAD SPACE -- PAGE NO. 6

512	0.062375
513	0.062375
514	0.062375
515	0.062375
516	0.062375
517	0.062375
519	0.034313
520	0.068625
521	0.068625
522	0.068625
523	0.068625
524	0.068625
525	0.068625
526	0.068625
546	0.025125
553	0.062375
554	0.068625
557	0.062500
559	0.056375
561	0.062500
562	0.062500
565	0.062375
567	0.068625
677	0.025125
679	0.025125
681	0.025125
683	0.025125
684	0.062375
685	0.068625
686	0.062375
687	0.068625
688	0.062375
689	0.068625
690	0.062375
691	0.068625
692	0.062500
693	0.062500
694	0.062500
695	0.062500
696	0.056375
697	0.056375
698	0.056375
699	0.056375
700	0.062500
701	0.062500
702	0.062500
703	0.062500
704	0.062500
705	0.062500
706	0.062500
707	0.062500
708	0.062375
709	0.062375
710	0.062375
711	0.062375
712	0.068625

713	0.068625
714	0.068625
715	0.068625
717	0.025125
718	0.062375
719	0.068625
720	0.062500
721	0.056375
722	0.062500
723	0.062500
724	0.062375
725	0.068625
727	0.025125
729	0.025125
731	0.025125
733	0.025125
734	0.062375
735	0.068625
736	0.062375
737	0.068625
738	0.062375
739	0.068625
740	0.062375
741	0.068625
742	0.062500
743	0.062500
744	0.062500
745	0.062500
746	0.056375
747	0.056375
748	0.056375
749	0.056375
750	0.062500
751	0.062500
752	0.062500
753	0.062500
754	0.062500
755	0.062500
756	0.062500
757	0.062500
758	0.062375
759	0.062375
760	0.062375
761	0.062375
762	0.068625
763	0.068625
764	0.068625
765	0.068625
767	0.025125
768	0.062375
769	0.068625
770	0.062500
771	0.056375
772	0.062500
773	0.062500
774	0.062375

775	0.068625
778	0.025125
779	0.062375
780	0.068625
781	0.062500
782	0.056375
783	0.062500
784	0.062500
785	0.062375
786	0.068625
788	0.012562
789	0.031187
790	0.034312
791	0.031250
792	0.028187
793	0.031250
794	0.031250
795	0.031187
796	0.034312
797	0.031250
798	0.062500
799	0.031250
800	0.015625
801	0.062500
802	0.031250
803	0.062500
804	0.031250
805	0.062500
806	0.031250
807	0.062500
808	0.031250
809	0.062500
810	0.031250
811	0.062500
812	0.031250
813	0.062500
814	0.031250
815	0.062500
816	0.031250
817	0.062500
818	0.031250
819	0.062500
820	0.031250
821	0.062500
822	0.031250
823	0.062500
824	0.031250
825	0.062500
826	0.031250
827	0.062500
828	0.031250
829	0.062500
830	0.031250
831	0.062500
832	0.031250
833	0.062500

```
       834          0.031250
       835          0.062500
       836          0.031250
       837          0.031250
       838          0.015625
    TOTAL          12.494999
201. ********************************************************************************
202. LOAD 1 LOADTYPE NONE  TITLE DEAD LOAD ( DL )
203. SELFWEIGHT Y -1
204. ELEMENT LOAD
205. * LOAD DUE TO BACKFILL
206. 465 TO 471 473 TO 479 481 TO 487 489 TO 495 497 TO 503 505 TO 511 513 TO 519 -
207. 521 TO 527 552 554 556 558 560 562 564 566 673 TO 704 706 TO 713 718 TO 749 -
208. 751 TO 758 761 TO 768 770 TO 817 PR GY -15
209. * LOAD DUE TO SURCHARGE
210. 465 TO 471 473 TO 479 481 TO 487 489 TO 495 497 TO 503 505 TO 511 513 TO 519 -
211. 521 TO 527 552 554 556 558 560 562 564 566 673 TO 704 706 TO 713 718 TO 749 -
212. 751 TO 758 761 TO 768 770 TO 817 PR GY -20
213. JOINT LOAD
214. 528 FX 3.1 FY -35.889 FZ 0.03
215. 776 FX 3.14 FY -26.831 FZ 0.07
216. ********************************************************************************
217. LOAD 2 LOADTYPE NONE  TITLE LIVE LOAD ( LL )
218. JOINT LOAD
219. 528 FX 8.7 FY -74.7 FZ 0.05
220. 776 FX 8.743 FY -48.829 FZ 0.176
221. ********************************************************************************
222. LOAD 3 LOADTYPE NONE  TITLE WIND LOAD +X  ( WLX1 )
223. JOINT LOAD
224. 528 FX 22.7 FY 81.634
225. 776 FX 22.725 FY -82.271 FZ 0.003
226. ********************************************************************************
227. LOAD 4 LOADTYPE NONE  TITLE WIND LOAD -X  ( WLX2 )
228. JOINT LOAD
229. 528 FX 22.721 FY -81.634
230. 776 FX 22.725 FY 82.271 FZ 0.003
231. ********************************************************************************
232. LOAD 5 LOADTYPE NONE  TITLE WIND LOAD +Z  ( WLZ1 )
233. JOINT LOAD
234. 528 FX 2.434 FY -26.337 FZ 0.473
235. 776 FX 2.335 FY 14.741 FZ 0.616
236. ********************************************************************************
237. LOAD 6 LOADTYPE NONE  TITLE WIND LOAD -Z  ( WLZ2 )
238. JOINT LOAD
239. 528 FX 2.434 FY 26.3 FZ 0.473
240. 776 FX 2.331 FY -14.716 FZ 0.618
241. ********************************************************************************
242. PERFORM ANALYSIS
```



```
          P R O B L E M   S T A T I S T I C S
          -----------------------------------

  NUMBER OF JOINTS        233  NUMBER OF MEMBERS       2
  NUMBER OF PLATES        200  NUMBER OF SOLIDS        0
  NUMBER OF SURFACES        0  NUMBER OF SUPPORTS    231

        SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

  ORIGINAL/FINAL BAND-WIDTH=   153/    19/     63 DOF
TOTAL      PRIMARY LOAD CASES =     6, TOTAL DEGREES OF FREEDOM =     705
TOTAL LOAD COMBINATION  CASES =     0  SO FAR.
  SIZE OF STIFFNESS MATRIX =       45 DOUBLE  KILO-WORDS
  REQRD/AVAIL. DISK SPACE  =     13.6/ 143874.2 MB
```

```
243. CHANGE
244. SUPPORTS
245. 465 TO 471 473 TO 479 481 TO 487 489 TO 495 497 TO 503 505 TO 511 513 TO 519 -
246. 521 TO 527 552 554 556 558 560 562 564 566 673 TO 704 706 TO 713 718 TO 749 -
247. 751 TO 758 761 TO 768 770 TO 816 -
248. 817 PLATE MAT DIRECT Y SUBGRADE 57600 COMPRESSION
249. ********************************************************************************
250. ***FACTORED LOAD COMBINATIONS FOR STRENGTH DESIGN
251. LOAD 101 1.4DL
252. REPEAT LOAD
253. 1 1.4
254. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  101

255. CHANGE
256. LOAD 102 1.4DL + 1.7LL
257. REPEAT LOAD
258. 1 1.4 2 1.7
259. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  102

260. CHANGE
261. LOAD 103 1.2DL + 1.6WLX1 + 1.0 LL
```

```
 262. REPEAT LOAD
 263. 1 1.2 3 1.6 2 1.0
 264. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  103



 265. CHANGE
 266. LOAD 104 1.2DL + 1.6WLX2 + 1.0 LL
 267. REPEAT LOAD
 268. 1 1.2 4 1.6 2 1.0
 269. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  104



 270. CHANGE
 271. LOAD 105 1.2DL + 1.6WLZ1 + 1.0 LL
 272. REPEAT LOAD
 273. 1 1.2 5 1.6 2 1.0
 274. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  105



 275. CHANGE
 276. LOAD 106 1.2DL + 1.6WLZ2 + 1.0 LL
 277. REPEAT LOAD
 278. 1 1.2 6 1.6 2 1.0
 279. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  106



 280. CHANGE
 281. LOAD 107 0.9DL + 1.6WLX1
 282. REPEAT LOAD
 283. 1 0.9 3 1.6
 284. PERFORM ANALYSIS
```



```
**START ITERATION NO.          2

**NOTE-Tension/Compression converged after  2 iterations, Case=  107



 285. CHANGE
 286. LOAD 108 0.9DL + 1.6WLX2
 287. REPEAT LOAD
 288. 1 0.9 4 1.6
 289. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  108



 290. CHANGE
 291. LOAD 109 0.9DL + 1.6WLZ1
 292. REPEAT LOAD
 293. 1 0.9 5 1.6
 294. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  109



 295. CHANGE
 296. LOAD 110 0.9DL + 1.6WLZ2
 297. REPEAT LOAD
 298. 1 0.9 6 1.6
 299. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  110



 300. CHANGE
 301. ******************************************************************************
 302. * UNFACTORED LOAD COMBINATIONS FOR STABILITY CHECK
 303. LOAD 201 1.0DL
 304. REPEAT LOAD
 305. 1 1.0
 306. PERFORM ANALYSIS
```

**NOTE-Tension/Compression converged after 1 iterations, Case= 201

307. CHANGE
308. LOAD 202 1.0DL + 1.0LL
309. REPEAT LOAD
310. 1 1.0 2 1.0
311. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 202

312. CHANGE
313. LOAD 203 1.0DL + 1.0WLX1
314. REPEAT LOAD
315. 1 1.0 3 1.0
316. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 203

317. CHANGE
318. LOAD 204 1.0DL + 1.0WLX2
319. REPEAT LOAD
320. 1 1.0 4 1.0
321. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 204

322. CHANGE
323. LOAD 205 1.0DL + 1.0WLZ1
324. REPEAT LOAD
325. 1 1.0 5 1.0
326. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 205

327. CHANGE
328. LOAD 206 1.0DL + 1.0WLZ2
329. REPEAT LOAD
330. 1 1.0 6 1.0
331. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 206

332. CHANGE
333. LOAD 207 1.0DL + 0.75WLX1 + 0.75LL
334. REPEAT LOAD
335. 1 1.0 3 0.75 2 0.75
336. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 207

337. CHANGE
338. LOAD 208 1.0DL + 0.75WLX2 + 0.75LL
339. REPEAT LOAD
340. 1 1.0 4 0.75 2 0.75
341. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 208

342. CHANGE
343. LOAD 209 1.0DL + 0.75WLZ1 + 0.75LL
344. REPEAT LOAD
345. 1 1.0 5 0.75 2 0.75
346. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after 1 iterations, Case= 209

```
 347. CHANGE
 348. LOAD 210 1.0DL + 0.75WLZ2 + 0.75LL
 349. REPEAT LOAD
 350. 1 1.0 6 0.75 2 0.75
 351. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  210



 352. CHANGE
 353. LOAD 211 0.6DL + 1.0WLX1
 354. REPEAT LOAD
 355. 1 0.6 3 1.0
 356. PERFORM ANALYSIS
**START ITERATION NO.           2
**START ITERATION NO.           3

**NOTE-Tension/Compression converged after  3 iterations, Case=  211



 357. CHANGE
 358. LOAD 212 0.6DL + 1.0WLX2
 359. REPEAT LOAD
 360. 1 0.6 4 1.0
 361. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  212



 362. CHANGE
 363. LOAD 213 0.6DL + 1.0WLZ1
 364. REPEAT LOAD
 365. 1 0.6 5 1.0
 366. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  213



 367. CHANGE
 368. LOAD 214 0.6DL + 1.0WLZ2
```



```
 369. REPEAT LOAD
 370. 1 0.6 6 1.0
 371. PERFORM ANALYSIS

**NOTE-Tension/Compression converged after  1 iterations, Case=  214



 372. CHANGE
 373. ******************************************************************************
 374. PERFORM ANALYSIS

  ** ALL CASES BEING MADE ACTIVE BEFORE RE-ANALYSIS. **

**NOTE-Tension/Compression converged after  1 iterations, Case=    1



**NOTE-Tension/Compression converged after  1 iterations, Case=    2



**START ITERATION NO.           2
**START ITERATION NO.           3
**START ITERATION NO.           4
**START ITERATION NO.           5
**START ITERATION NO.           6

***WARNING -  INSTABILITY AT JOINT   838   DIRECTION = MZ
 PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
 K-MATRIX DIAG=  1.1619854E+06 L-MATRIX DIAG= -1.3883226E-05 EQN NO     705
 ***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

   **NOTE** STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
            DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
            BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
            STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
            IS   1.000E-09

            THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
            ELEMENTS AS WELL AS TRUE SINGULARITIES.

 **START ITERATION NO.           7

***WARNING -  INSTABILITY AT JOINT   838   DIRECTION = MZ
 PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
 K-MATRIX DIAG=  1.1619854E+06 L-MATRIX DIAG= -1.3882760E-05 EQN NO     705
 ***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
```

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 8

***WARNING - INSTABILITY AT JOINT 838 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1619854E+06 L-MATRIX DIAG= -1.3882760E-05 EQN NO 705
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**NOTE-Tension/Compression converged after 8 iterations, Case= 3

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5
**START ITERATION NO. 6

***WARNING - INSTABILITY AT JOINT 838 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1619854E+06 L-MATRIX DIAG= -1.6512349E-06 EQN NO 705
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 7

***WARNING - INSTABILITY AT JOINT 838 DIRECTION = FY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.6401225E+04 L-MATRIX DIAG= -5.7116267E-10 EQN NO 703
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 838 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1619839E+06 L-MATRIX DIAG= -1.2405217E-06 EQN NO 704
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 838 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1619854E+06 L-MATRIX DIAG= -1.3882760E-05 EQN NO 705
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
**START ITERATION NO. 8
**START ITERATION NO. 9
**START ITERATION NO. 10
**START ITERATION NO. 11
**START ITERATION NO. 12
**START ITERATION NO. 13

***WARNING - INSTABILITY AT JOINT 838 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1619854E+06 L-MATRIX DIAG= -1.6512349E-06 EQN NO 705
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 14

***WARNING - INSTABILITY AT JOINT 838 DIRECTION = FY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.6401225E+04 L-MATRIX DIAG= -5.7116267E-10 EQN NO 703
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 838 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1619839E+06 L-MATRIX DIAG= -1.2405217E-06 EQN NO 704
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 838 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1619854E+06 L-MATRIX DIAG= -1.3882760E-05 EQN NO 705
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY
**START ITERATION NO. 15

**WARNING-Tension/Compression did not converge after 15 iterations, Case= 4
** If results are not satisfactory then enter a
** SET ITERLIM i command (i>10) before loads to get more iterations.

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5
**START ITERATION NO. 6

***WARNING - INSTABILITY AT JOINT 838 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1619854E+06 L-MATRIX DIAG= -1.6512349E-06 EQN NO 705
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**NOTE-Tension/Compression converged after 6 iterations, Case= 5

**START ITERATION NO. 2
**START ITERATION NO. 3
**START ITERATION NO. 4
**START ITERATION NO. 5
**START ITERATION NO. 6

***WARNING - INSTABILITY AT JOINT 838 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1619854E+06 L-MATRIX DIAG= -1.3883226E-05 EQN NO 705
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

**START ITERATION NO. 7

***WARNING - INSTABILITY AT JOINT 838 DIRECTION = FY
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.6401225E+04 L-MATRIX DIAG= -5.7116267E-10 EQN NO 703
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

NOTE STAAD DETECTS INSTABILITIES AS EXCESSIVE LOSS OF SIGNIFICANT DIGITS
DURING DECOMPOSITION. WHEN A DECOMPOSED DIAGONAL IS LESS THAN THE
BUILT-IN REDUCTION FACTOR TIMES THE ORIGINAL STIFFNESS MATRIX DIAGONAL,
STAAD PRINTS A SINGULARITY NOTICE. THE BUILT-IN REDUCTION FACTOR
IS 1.000E-09

THE ABOVE CONDITIONS COULD ALSO BE CAUSED BY VERY STIFF OR VERY WEAK
ELEMENTS AS WELL AS TRUE SINGULARITIES.

***WARNING - INSTABILITY AT JOINT 838 DIRECTION = MX
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1619839E+06 L-MATRIX DIAG= -1.2405217E-06 EQN NO 704
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

***WARNING - INSTABILITY AT JOINT 838 DIRECTION = MZ
PROBABLE CAUSE SINGULAR-ADDING WEAK SPRING
K-MATRIX DIAG= 1.1619854E+06 L-MATRIX DIAG= -1.3882760E-05 EQN NO 705
***NOTE - VERY WEAK SPRING ADDED FOR STABILITY

**NOTE-Tension/Compression converged after 7 iterations, Case= 6

**NOTE-Tension/Compression converged after 1 iterations, Case= 101

**NOTE-Tension/Compression converged after 1 iterations, Case= 102

**NOTE-Tension/Compression converged after 1 iterations, Case= 103

**NOTE-Tension/Compression converged after 1 iterations, Case= 104

**NOTE-Tension/Compression converged after 1 iterations, Case= 105

**NOTE-Tension/Compression converged after 1 iterations, Case= 106

**START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 107

**NOTE-Tension/Compression converged after 1 iterations, Case= 108

**NOTE-Tension/Compression converged after 1 iterations, Case= 109

**NOTE-Tension/Compression converged after 1 iterations, Case= 110

**NOTE-Tension/Compression converged after 1 iterations, Case= 201

**NOTE-Tension/Compression converged after 1 iterations, Case= 202

**NOTE-Tension/Compression converged after 1 iterations, Case= 203

**NOTE-Tension/Compression converged after 1 iterations, Case= 204

**NOTE-Tension/Compression converged after 1 iterations, Case= 205

**NOTE-Tension/Compression converged after 1 iterations, Case= 206

**NOTE-Tension/Compression converged after 1 iterations, Case= 207

**NOTE-Tension/Compression converged after 1 iterations, Case= 208

**NOTE-Tension/Compression converged after 1 iterations, Case= 209

**NOTE-Tension/Compression converged after 1 iterations, Case= 210

**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 211

**NOTE-Tension/Compression converged after 1 iterations, Case= 212

**NOTE-Tension/Compression converged after 1 iterations, Case= 213

**NOTE-Tension/Compression converged after 1 iterations, Case= 214

375. LOAD LIST 101 TO 110
376. START CONCRETE DESIGN

```
377. CODE ACI
378. TRACK 2 ALL
379. FC 35000 MEMB 529 759
380. CLB 0.05 MEMB 529 759
381. CLS 0.05 ALL
382. MAXMAIN 25 ALL
383. MINMAIN 25 ALL
384. MINSEC 10 ALL
385. DESIGN COLUMN 529 759
```


==========

COLUMN NO. 529 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 35.0 MPA, RECT SIZE - 600.0 X 400.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2400.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 25 MM	1.636	101	END	0.650

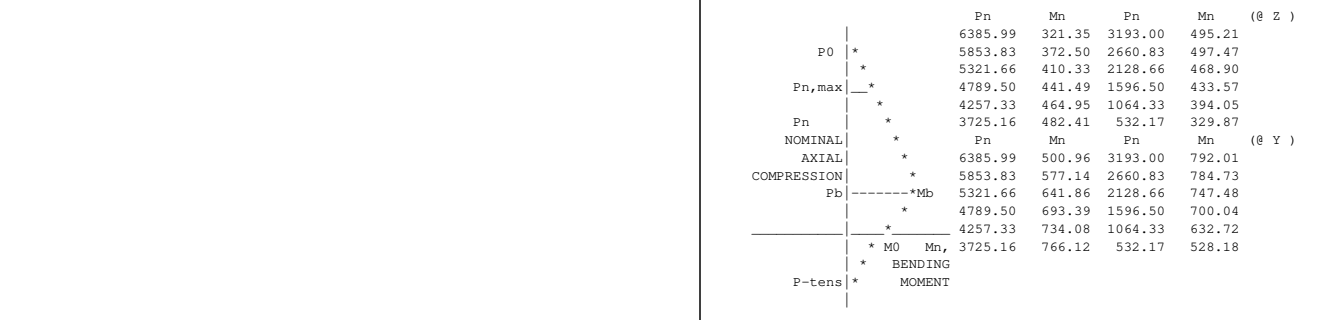
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 400.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
8647.70	6918.16	2791.93	503.22	180.2
M0	P-tens.	Des.Pn	Des.Mn	e/h
257.31	-1624.53	77.30	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
8647.70	6918.16	2974.50	801.47	269.4
M0	P-tens.	Des.Pn	Des.Mn	e/h
407.87	-1624.53	77.30	0.00	0.00000



	Pn	Mn	Pn	Mn	
	6385.99	321.35	3193.00	495.21	
P0	5853.83	372.50	2660.83	497.47	
	5321.66	410.33	2128.66	468.90	
Pn,max	4789.50	441.49	1596.50	433.57	
	4257.33	464.95	1064.33	394.05	
Pn	3725.16	482.41	532.17	329.87	
NOMINAL	Pn	Mn	Pn	Mn	(@ Y)
AXIAL	6385.99	500.96	3193.00	792.01	
COMPRESSION	5853.83	577.14	2660.83	784.73	
Pb	5321.66	641.86	2128.66	747.48	
	4789.50	693.39	1596.50	700.04	
	4257.33	734.08	1064.33	632.72	
M0 Mn,	3725.16	766.12	532.17	528.18	

==========

COLUMN NO. 759 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 35.0 MPA, RECT SIZE - 600.0 X 400.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2400.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 25 MM	1.636	101	END	0.650

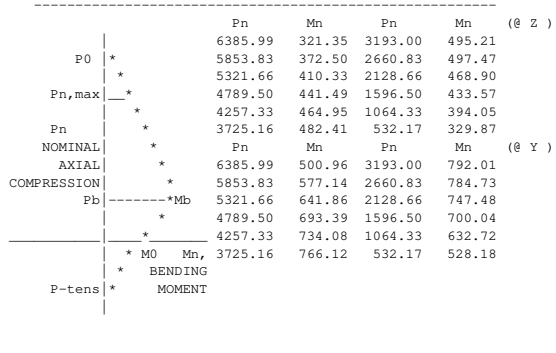
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 400.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
8647.70	6918.16	2791.93	503.22	180.2
M0	P-tens.	Des.Pn	Des.Mn	e/h
257.31	-1624.53	57.79	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
8647.70	6918.16	2974.50	801.47	269.4
M0	P-tens.	Des.Pn	Des.Mn	e/h
407.87	-1624.53	57.79	0.00	0.00000



**********END OF COLUMN DESIGN RESULTS**********

386. END CONCRETE DESIGN
387. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= SEP 25,2020 TIME= 12:55:34 ****

****
* For technical assistance on STAAD.Pro, please visit *
* http://selectservices.bentley.com/en-US/ *
* *
* Details about additional assistance from *
* Bentley and Partners can be found at program menu *
* Help->Technical Support *
* *
* Copyright (c) 1997-2014 Bentley Systems, Inc. *
* http://www.bentley.com *
****

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            **************************************************
            *                                                *
            *          STAAD.Pro V8i SELECTseries5           *
            *          Version  20.07.10.64                  *
            *          Proprietary Program of                *
            *          Bentley Systems, Inc.                 *
            *          Date=    OCT  7, 2020                 *
            *          Time=    17:15:16                     *
            *                                                *
            *     USER ID: LNT ECC DIV                       *
            **************************************************
```

```
    1. STAAD SPACE
INPUT FILE: F3_STABILITY CHECK.STD
    2. START JOB INFORMATION
    3. ENGINEER DATE 08/05/2020
    4. JOB NAME FDN. MKD. F3_STABILITY CHECK
    5. JOB CLIENT MA'ADEN
    6. JOB NO O19045
    7. JOB REV 0
    8. ENGINEER NAME SOR
    9. END JOB INFORMATION
   10. INPUT WIDTH 79
   11. UNIT METER KN
   12. JOINT COORDINATES
   13. 522 0 911.65 -0.12462; 528 0 913.6 -0.125; 765 3.5 911.65 -0.12462
   14. 776 3.5 913.6 -0.125
   15. MEMBER INCIDENCES
   16. 529 522 528; 759 765 776
   17. DEFINE MATERIAL START
   18. ISOTROPIC CONCRETE
   19. E 2.17185E+007
   20. POISSON 0.17
   21. DENSITY 25
   22. ALPHA 1E-005
   23. DAMP 0.05
   24. TYPE CONCRETE
   25. STRENGTH FCU 27579
   26. END DEFINE MATERIAL
   27. MEMBER PROPERTY AMERICAN
   28. 529 759 PRIS YD 0.4 ZD 0.6
   29. CONSTANTS
   30. MATERIAL CONCRETE ALL
   31. SUPPORTS
   32. 522 765 FIXED
   33. ******************************************************************************
   34. LOAD 1 LOADTYPE NONE  TITLE DEAD LOAD ( DL )
  **WARNING- THIS STRUCTURE IS DISJOINTED. IGNORE IF
            MASTER/SLAVE OR IF UNCONNECTED JOINTS.
   35. SELFWEIGHT Y -1
   36. JOINT LOAD
```



```
   37. 528 FX 3.1 FY -35.889 FZ 0.03
   38. 776 FX 3.14 FY -26.831 FZ 0.07
   39. ******************************************************************************
   40. LOAD 2 LOADTYPE NONE  TITLE LIVE LOAD ( LL )
   41. JOINT LOAD
   42. 528 FX 8.7 FY -74.7 FZ 0.05
   43. 776 FX 8.743 FY -48.829 FZ 0.176
   44. ******************************************************************************
   45. LOAD 3 LOADTYPE NONE  TITLE WIND LOAD +X  ( WLX1 )
   46. JOINT LOAD
   47. 528 FX 22.7 FY 81.634
   48. 776 FX 22.725 FY -82.271 FZ 0.003
   49. ******************************************************************************
   50. LOAD 4 LOADTYPE NONE  TITLE WIND LOAD -X  ( WLX2 )
   51. JOINT LOAD
   52. 528 FX 22.721 FY -81.634
   53. 776 FX 22.725 FY 82.271 FZ 0.003
   54. ******************************************************************************
   55. LOAD 5 LOADTYPE NONE  TITLE WIND LOAD +Z  ( WLZ1 )
   56. JOINT LOAD
   57. 528 FX 2.434 FY -26.337 FZ 0.473
   58. 776 FX 2.335 FY 14.741 FZ 0.616
   59. ******************************************************************************
   60. LOAD 6 LOADTYPE NONE  TITLE WIND LOAD -Z  ( WLZ2 )
   61. JOINT LOAD
   62. 528 FX 2.434 FY 26.3 FZ 0.473
   63. 776 FX 2.331 FY -14.716 FZ 0.618
   64. ******************************************************************************
   65. * UNFACTORED LOAD COMBINATIONS FOR STABILITY CHECK
   66. LOAD COMB 201 1.0DL
   67. 1 1.0
   68. LOAD COMB 202 1.0DL + 1.0LL
   69. 1 1.0 2 1.0
   70. LOAD COMB 203 1.0DL + 1.0WLX1
   71. 1 1.0 3 1.0
   72. LOAD COMB 204 1.0DL + 1.0WLX2
   73. 1 1.0 4 1.0
   74. LOAD COMB 205 1.0DL + 1.0WLZ1
   75. 1 1.0 5 1.0
   76. LOAD COMB 206 1.0DL + 1.0WLZ2
   77. 1 1.0 6 1.0
   78. LOAD COMB 207 1.0DL + 0.75WLX1 + 0.75LL
   79. 1 1.0 3 0.75 2 0.75
   80. LOAD COMB 208 1.0DL + 0.75WLX2 + 0.75LL
   81. 1 1.0 4 0.75 2 0.75
   82. LOAD COMB 209 1.0DL + 0.75WLZ1 + 0.75LL
   83. 1 1.0 5 0.75 2 0.75
   84. LOAD COMB 210 1.0DL + 0.75WLZ2 + 0.75LL
   85. 1 1.0 6 0.75 2 0.75
   86. LOAD COMB 211 0.6DL + 1.0WLX1
   87. 1 0.6 3 1.0
   88. LOAD COMB 212 0.6DL + 1.0WLX2
   89. 1 0.6 4 1.0
   90. LOAD COMB 213 0.6DL + 1.0WLZ1
   91. 1 0.6 5 1.0
   92. LOAD COMB 214 0.6DL + 1.0WLZ2
```

```
93. 1 0.6 6 1.0
94. ********************************************************************************
95. PERFORM ANALYSIS


        P R O B L E M   S T A T I S T I C S
        -----------------------------------

  NUMBER OF JOINTS          4  NUMBER OF MEMBERS       2
  NUMBER OF PLATES          0  NUMBER OF SOLIDS        0
  NUMBER OF SURFACES        0  NUMBER OF SUPPORTS      2


       SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

  ORIGINAL/FINAL BAND-WIDTH=     1/     1/      6 DOF
TOTAL      PRIMARY LOAD CASES =      6, TOTAL DEGREES OF FREEDOM =      12
TOTAL LOAD COMBINATION  CASES =     14  SO FAR.
  SIZE OF STIFFNESS MATRIX =         1 DOUBLE  KILO-WORDS
  REQRD/AVAIL. DISK SPACE  =     12.0/ 143074.2 MB



96. FINISH


          *********** END OF THE STAAD.Pro RUN ***********

            **** DATE= OCT  7,2020   TIME= 17:15:20 ****
```



```
      ************************************************************
      *   For technical assistance on STAAD.Pro, please visit    *
      *   http://selectservices.bentley.com/en-US/               *
      *                                                          *
      *   Details about additional assistance from               *
      *   Bentley and Partners can be found at program menu      *
      *   Help->Technical Support                                *
      *                                                          *
      *      Copyright (c) 1997-2014 Bentley Systems, Inc.
      *               http://www.bentley.com                     *
      ************************************************************
```